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# From the Editor Desk

This issue of Biospectra is in your hands in a crucial period of the second wave of Covid-19 pandemic gripping the nation severely. It has had a huge impact on social, economical, educational, healthcare, productivity and lives. After its first identification in Wuhan (China), the whole world made a consented effect to combat the infection. Biologists all over the globe initiated massive efforts by research community, which have greatly enhanced understanding of the properties and the host responses to the virus. India too took up it as a priority task and successfully launched two vaccines- Covaxin & Covishield. Hats off to all Bioresearches evolved in the project. This issue of Biospectra is published to mark the "International Day of Immunology" on 29th April 2021.

We all know, healthcare is directly dependent upon the advances in Biomedical Research in its broadest aspects, as it's a multidisciplinary work. Rapid advances in Biological Research, Biomedical Technology and Physical Science Research are in progress in many laboratories of the world, focused on Covid-19 Epidemiology and Vaccine development. Severe Acute Respiratory Syndrome Corona Virus-2 (SARS-CoV-2), which has caused Covid-19 pandemic, enters the target cells through the interaction of its envelope spike protein with primary host cell receptor-angiotensin in converting enzyme-2 (ACE2), which is then cleaved by a serine protease (TMPRSS-2) to allow viral fusion and entry across the cell membrane. The much needed researches in the areas of Basic Science, Technology Development in Healthcare System, Molecular Biology Immunology, Lasers & Ultrasonic, Nanotechnology, Microbiology etc may contribute significantly in diagnosis, patient treatment or prevention of disease, resulting in better healthcare of large population. A consented effort in network model in emerging area of Bioscience will result in appropriate healthcare system.

Bioscientists need to concentrate more and more on research on COVID-19 related issues viz. origin, epidemiology and containment strategies of SARS- CoV-2, its complete biology, genome and replications, interaction of COVID-19 with innate and adaptive immune system, the thromoboembolic events associated with COVID-19 and its effect on vital organs. COVID-19 virus is a single stranded RNA genome and has a high mutation rate, which may complete the development of vaccine and treatment protocols. Structural biology research can too play a significant role, which includes macromolecular crystallography and cryoelectron microscopy, intersect with in vitro, in vivo and in silica studies of SARS CoV-2 and COVID-19.

Through this editorial, I call upon all members of MSET/ICCB to get precise information on best global practices of Covid research and then identify important research questions in the national and global context through consultation with Epidemiologists, Clinicians, Scientists, Sociologists and Economists and then generate prioritized list of research topics. ICMR has recently released a Covid specific priority over of biological and biomedical research too.Net working and coordinated system of research should be taken in order to formulate long term strategy to prevent future pandemics of Zoonatics Origin. It may be challenging but certainly not impossible. For the welfare of mankind by ensuring effective healthcare system researches to be done the right earnest at any cost. Our members will accept this challenge and do meaning project in this direction.

**31.03.2021**

**Mahendra Prasad**
(Editor-in-Chief)

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***Animal Science***

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# Eco friendly management of banana scarring beetle, *Basilepta subcostatum* (Jacoby) in North Bihar

**Anshu Prabha, Arun Kumar* & Bhuwan Bhaskar Mishra**
University Department of Zoology, B.N.M.University, Madhepura, Bihar, India

*Received : 12th November, 2020 ; Revised : 14th December, 2020*

**Abstract:-** A field survey was conducted during 2017-18 in North Bihar, to study the status of Banana growers and their cultivation practices. It has been found that the infestation of *Basilepta subcostatum* (Jacoby) beetle is one of the most important beetles damaging the crop in the region. The study indicated that different chemical and biological pesticides are used in the different stage of infestation. The weather condition also plays an important factor in the intensity of the infestation. It has been found the Entomopathogenic fungus (*Beauveria bassiana*) and three neem products, chemical pesticides azadirachtin (5ml/L) were highly effective. The fruit damage, beetle population, number of scars, pulp-skin ratio, TSS and benefit-cost ratio was measured. The findings suggested that an eco- friendly management strategy can be developed by including these products.

**Key words: Banana, scarring beetle, *Basilepta subcostatum*, neem**

## INTRODUCTION

Banana (*Musa sapientum*) is an important fruit crop in India. Bananas are grown in more than 150 countries, producing 105 million tones of fruit per year. The global production of banana is around 10,20,28,17 thousand tons of which India contributes 29.19%. India has been predominantly agrarian economy since time immemorial. Development efforts over the last four decades have doubtless strengthened our industrial base. However, agriculture continues to be mainstay of our economy even today. Agriculture generates around 16 per cent of national income and provides livelihood to over 64.5 per cent of rural manpower. In agriculture, the production of fruits and vegetables are of so vital importance that it provide three to four time more income than cereals per unit of land.

The fruit crops hold a great promise for accelerating income of the farmers. Realizing the importance of fruit cultivation many farmers are diverting their resources towards plantation of fruit crops. Area under fruit crops is, therefore, increasing day by day. In the scenario of declining land and water resources, climate change and increasing population, the food and livelihood security are the challenge which have received worldwide attention. Among them Banana has emerged as an important versatile Fruit crop providing nourishment to millions of people, source of medicinal value, and generation of livelihood

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through production, processing and marketing. Bihar state has achieved dominance in banana production which has been possible through concerted efforts by the research, farmers and government policies.

In Bihar state, there are two distinct banana growing areas viz. old Vaishali region and new North- Eastern (Kosi) region, both of the regions are endowed with congenial climatic conditions favorable for growth and development of banana. It is an important source of income for small as well as large farmer who produce it either on small or large areas.

Recently, the demand for banana is increasing and area under the crop in the state has undergone large expansion is last 20 years. There are number pests on Bananas have been found associated with banana in India from planting to harvesting.[1] Of these, banana scarring beetle, *Basilepta subcostatum* Jacoby has been considered as the most serious one in different parts of the country. It causes extensive damage to leaves as well as fruits during summer and kharif season.[2] The extent of damage has been reported to be approximately 30 per cent of the banana bunches during rainy season in Bihar[3-6] but no systematic efforts have been made to study the efficacy of insecticides and biopesticides applied through foliar and whorl application against this pest on banana in Bihar. Hence, the present investigations were undertaken to find out a suitable eco-friendly measure against this insect.

## MATERIALS & METHODS

A field survey was conducted during 2017-18 to study the cultivation practices adopted by the farmers and method to control the scarring beetle, *Basilepta subcostatum* Jacoby. Population of the beetle was recorded before and after application of the pesticides on plants. Similarly number of scars on leaves in treated and untreated plants was also recorded from 5 sq. cm. on youngest leaf at three different locations. Observations pertaining to the number of healthy and infested fingers/bunch data were used for computing the mean (%) finger infestation. The efficiency of different treatments was also recorded in terms of pulp-skin ratio, TSS ($^{o}$B) and benefit- cost ratio.

## RESULTS & DISCUSSION

**Table 1. Relative efficacy of biopesticides and neem products against scarring beetle (*Basilepta subcostatum*) on banana**

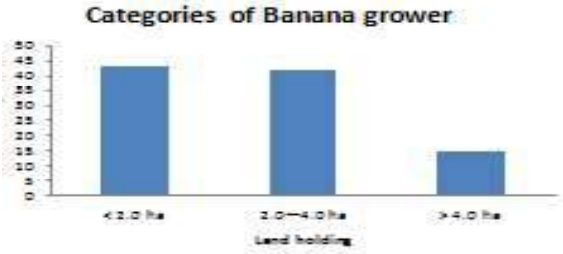
| Treatment | No. of scarring beetles / plant | No. of scars/ 5 $cm^2$ leaf surface | Mean fruit infestation (%) | Bunch weight (Kg/ plant) |
|---|---|---|---|---|
| Azadirachtin (0.15%) @ 5ml per litre of water spray | 9.89 | 11.89 | 18.58 | 14.74 |
| Azadirachtin (0.15%) @ 5ml per litre of water whorl application banana heart | 10.64 | 13.56 | 21.49 | 14.33 |
| Neem oil (2%) spray | 10.44 | 13.31 | 21.01 | 15.54 |
| Neem oil (2%) @ 20ml whorl application per banana heart | 11.14 | 13.67 | 21.88 | 13.69 |
| NSKE (5%) spray | 12.35 | 14.81 | 23.33 | 13.63 |
| NSKE (10%) spray | 11.40 | 14.56 | 22.51 | 14.44 |
| *Beauveria bassiana* 5g/litre spray | 7.44 | 9.75 | 14.68 | 16.40 |
| Carbaryl (0.3%) spray | 5.48 | 7.65 | 9.71 | 17.42 |
| Untreated | 19.01 | 21.21 | 78.67 | 12.14 |

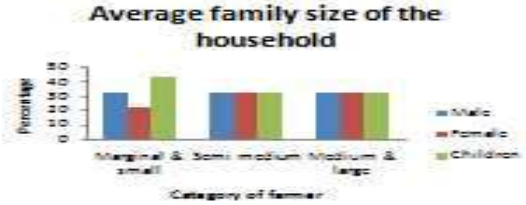
**Table 2. Influence of different eco-friendly measures on pulp: skin ratio, TSS (0B) and benefit; cost ratio of banana**

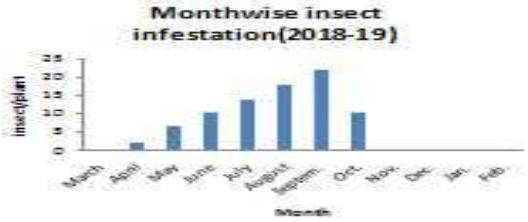
| Treatment | Pulp-Skin ratio | TSS ($^{o}$B) | Benefit-Cost ratio |
|---|---|---|---|
| Azadirachtin (0.15%) @ 5ml per litre of water spray | 3.26 | 20.85 | 3.25 |
| Azadirachtin (0.15%) @ 5ml per litre of water whorl application banana heart | 3.22 | 20.15 | 3.13 |
| Neem oil (2%) spray | 3.24 | 20.21 | 3.48 |
| Neem oil (2%) @ 20ml whorl application per banana heart | 3.19 | 20.02 | 2.95 |
| NSKE (5%) spray | 3.17 | 19.74 | 2.93 |
| NSKE (10%) spray | 3.18 | 19.91 | 3.17 |
| *Beauveria bassiana* 5g/litre spray | 3.32 | 21.08 | 3.73 |
| Carbaryl (0.3%) spray | 3.42 | 22.11 | 4.02 |
| Untreated | 3.06 | 19.56 | 2.50 |

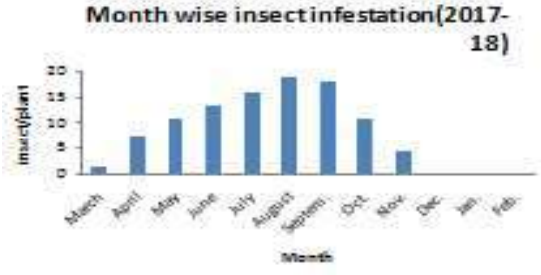
The result have been presented according to size group of banana grower; i.e. marginal & small (with less than 2.0 ha of operational land), semi-medium (with land between 2.0 to 4.0 ha of operational land) and medium & large (with more than 4.0 ha of operational land) because of their distinctly different farm resources and decision making capabilities. Though the farmers have been classified into five categories; viz. marginal (less than 1.0 ha), small (1.0-2.0 ha), semi-medium (2.0-4.0 ha), medium (4.0-10.0 ha) and large (more than 10.0 ha) farmers. The present analysis has been conducted by grouping marginal & small (less than 2.0 ha) and medium & large (more than 4.0 ha) as one group due to similar nature of other resources at their hand and decision making.

The classification of sample banana grower reveals that the sample (60) includes 26 (43.33 per cent) of the marginal and small category of farmers, while semi-medium and medium & large category includes 25 (41.67 per cent) and 9 (15.00 percent) respectively.



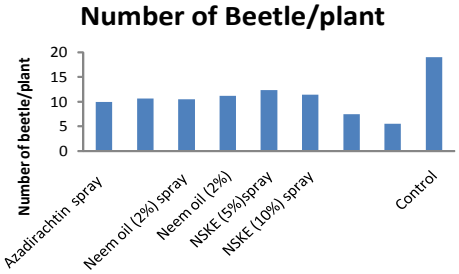



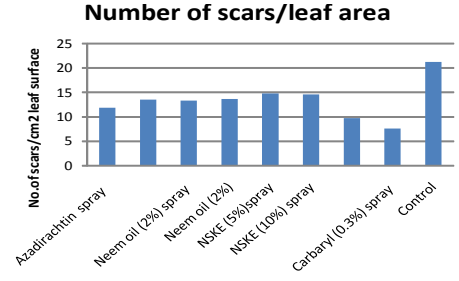

The composition of the farm family and the education level was found that the average size of the family of three categories of farm families was 9, 9 and 12, respectively. In terms of the sex-wise composition of the family, it includes 3 male members, while female and children member included 3 and 4 on an overall basis with a total number of members as 10.

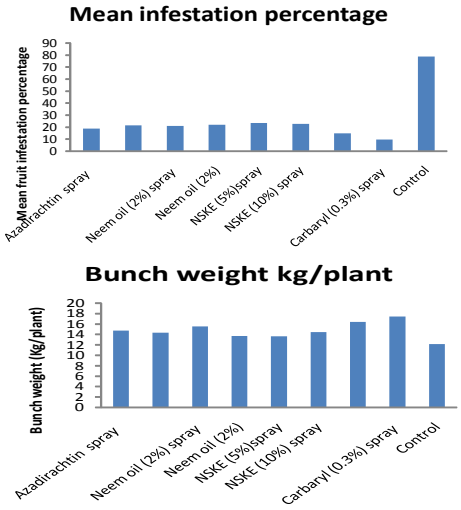
The illiteracy (11.54 per cent) was found in marginal & small category growers only. In terms of graduate & above level of education, medium size growers had highest (33.33 per cent) proportion, while the semi-medium farmers with 24.0 per cent. The higher secondary level was highest (36.0 per cent) under semi-medium category, while high school (38.46 per cent) on marginal and small category.

The operational land holding size ranged between 1.52, 3.14, and 8.55 ha among three categories of farms with overall mean as 3.16 ha. The table reveals that a few marginal & small farmers took land on lease, while semi-medium and medium & large farmers found leasing out their land, with mean value as 0.03 and 0.04 ha, respectively.

The percentage area under banana cultivation and variety/cultivar reveals that marginal & small farmers devoted maximum (76.64 per cent) area under the banana, while semi-medium and medium & large category growers devoted 56.49 and 42.28 per cent area, respectively, while the reverse order was found with respect to the allocation of the area under other crops. This indicates that because of cash needs & labour intensive nature of crop (banana) marginal & small farmers allotted more area under this crop.






The mean number of beetles on banana after treatment varied significantly in different treatments. All the treatments under the survey found significantly lower level of pest population compared to before treatments. Among types of different treatments, spraying with carbaryl (0.3%) and *Beauveria bassiana* (1.15%) were most effective in minimizing the pest population, recording 5.48 and 7.44 adults/plant, respectively. The use of azadirachtin (0.15%), neem oil (2%) (Pouring and spray), neem seed kernel extract (NSKE) (5%) and NSKE (10%) spray were found effective in reducing the beetle population recording 9.89, 10.64, 10.44, 11.14, 12.35 and 11.40/plant, respectively. All the neem products viz., azadirachtin, neem oil, NSKE applied either as spray or poured at their test doses showed more or less similar effect recording between 9.89 to 12.35 plant. The neem products proved superior over untreated and at par with each other.

More are less similar observations were recorded in case of number of scars /5 sq. cm leaf surface areas. It reveals that all the treatments recorded significantly lower number of scars (7.75 to 14.81/5 sq. cm leaf surface) compared to untreated plant (21.21). Spray with carbaryl (0.3%) recorded significantly lower number of scars (7.75/5 sq. cm leaf surface) which was statistically at par with spraying *B.bassiana* (1.15%) recording 9.75/5 sq. cm leaf surface.

Spraying of *B.bassiana* was observed to be the second best treatment with pulp-skin ratio 3.32:1. All the neem products were superior over untreated control. Effect of different treatments on TSS (0B) and benefit-cost ratio followed similar trend with carbaryl (0.3%) giving highest value (22.11and 4.02:1) followed by *B. bassiana* (21.08 and 3.73:1, respectively). The neem products also gave higher TSS. The present findings are in agreement with the results of earlier worker.[7,4,6]

It may be concluded that use of eco- friendly pest management strategies including biopesticides and neem products are quite effective against B. subcostatum on banana and are comparable with chemical treatment. Further, the present study has given the possibilities of minimizing the frequency of application of synthetic insecticides and encouraging the use of bio-pesticides and phyto-products in IPM strategies in banana.

## REFERENCES

1. **Padmanaban, B., Rajeswari, R. and Satiamoorthy, S. 2002.** Integrated management of insect pests on banana and plantation. Souvenir Global Conference on Banana and Plantation, 28-31.
2. **Singh, P. P., Singh, S. P. and Mondal, S. S. 1997.** Incidence and seasonal variation of banana scarring beetle, *Nodostoma subcostatum* Jacoby on banana in North Bihar. Natl. Seminar on Orchard Management for sustainable Production of Tropical Fruits held on March 10-11, 1997 at R.A.U. Bihar, Pusa (India) pp. 77.
3. **Ahmad, M.A., Singh, P. P. and Singh, B. 2003.** Efficacy of certain synthetic insecticides and plant products used as foliar and whorl application against the scarring beetle (*Nodostoma subcostatum* Jacoby) on banana. *Journal of Enomological Research*. **27 (4):** 325-328.
4. **Mukherjee, U. 2004.** Management of scarring beetle, *Basilepta subcostatum* through synthetic insecticides. *Proc. Natl. Seminar Banana Industry- Present Scenario and Future Srtategies held at BCKV, Kalyani* 11th to 13th June 2004: 179-181.
5. **Samui,G., Maiti, B. and Bandopadhyay, B. 2004.** Field evaluation of some insecticidal treatments to control banana scarring beetle (*Nodostoma viridipenne* Motsch.). Proc. Natl. Seminar Banana Industry - Present Scenario and Future Strategies held at BCKV, Kalyani, 11th to 13th June, 2004: 194-202.
6. **Mukherjee, U. 2006.** Evaluation of insecticides and some eco-friendly approaches to manage scarring beetle, Basilepta subcostatum in banana. *Journal of Applied Zoological Research*. **17(1):** 54-56.
7. **Ahmad, M.A. 2002.** Seasonal abundance and management of scarring beetle, *Basilepta* (=*Nodostoma*) *subcostatum* Jacoby on banana under North Bihar conditions. M.Sc. (Ag.) thesis submitted to R.A.U. Pusa (Bihar), Samastipur.

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# Some important plants used in the festivals of santhal tribe of Dumka district of Jharkhand

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*Received : 13th November, 2020 ; Revised : 15th December, 2020*

**Abstract:-** The santhal tribe is the major tribe of the state of Jharkhand. The present study focuses on the plants which are used in the festivals of santhal tribe. The rituals and festivals of santhal tribe are unique to them. The santhals of Dumka district observe various festivals in different seasons. They use plants as an essential component in festivals. The present paper is an attempt to analyse the use of plants with respect to their use in various festivals by the santhal tribe in Dumka district of Jharkhand. The use of plants in various festivals throughout the year represents their religious culture and folklore.

**Key words: Santhal tribe, Festivals, Religious culture, Dumka district**

## INTRODUCTION

As one of the largest "Scheduled Tribe" (ST) of India the santhal number about 6.5 million.[1] Santhals are the prevailing tribal community in India who inhabited in the states of Bihar, Jharkhand, Orissa, West Bengal and Assam.[2] Santhal minority is also present in the neighbouring countries like Bangladesh and Nepal.[3] Jharkhand is home to many tribal communities among them are Santhal, Oraon, Munda and Ho. The santhal tribe is the leading tribe of the state of Jharkhand. They are mainly concentrated in the areas of Dumka, Godda, Pakur, Sahebgunj districts of Santhal Pargana of Jharkhand.[4] Santhal Pargana is an integral part of the lower Gangetic plain which is rich in medicinal plants. This tribe exhibit a long history of migration from one place to another. The santhal community use plants for their medicinal and religious rites.[5] Plants play a very significant role. These people roam around the forest for their day to day requirements, cultural activities, beliefs and for performing religious rituals.[6] They are mainly dependent on their traditional system for their information which is passed on from generation to generation orally.[7] The santhals of Dumka district observes many festivals in different seasons.

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**Festivals**

The main festival of santhals was Sohrai. It is celebrated in poush (Dec-Jan). It is celebrated after the rice crop of the year has been harvested and it is of five days. The next important festival is Baha Parab.[8] It is held in the month of Phagun (Feb-March). It indicates commencement of New Year. Like Sohrai it is celebrated with dancing, singing and drinking.[7] It is celebrated when

Sal (*Shorea robusta* Gaertn.) and Mahua (*Madhuca longifolia* (J.Koing) J.F.Macbr.) flowers are in full glory.[9,10] The entire land looks like forest fire due to the flowering of Palash (*Butea monosperma* Lam.).[11] Sal and Mahua flowers are used during Baha Parv.[12] Sakrat is held in the month of Poush (Dec-Jan). The previous day of Sakrat they go for catching fish and on the day of Sakrat, men go for hunting, while the women make pressed rice (chuda) and rice cake. The same afternoon Jog Manjhi cuts a Banana stalk and places it on the end of village and calls men with bow and arrow to shoot the banana tree.[13] This game is called "Bheja Tun". Erok Sim is celebrated in Asar (Jun-Jul). This is also a festival which is celebrated with the whole village.[14] It is a sowing festival. Hariar sim is a feast of the sprouting of the rice is held in Sraban (Jul-Aug). Irigundli-nawai is the offering of the first fruits of millets.[15] It is held in the month of Bhadra (Aug-Sep). Mak sim it is celebrated when the jungle grass is cut during the month of Magh (Jan - Feb). This festival marks the end of santhal year.

## MATERIAL & METHODS

### Study Area

Dumka is located at 24.27°N, 87.25°E. Dumka district has an area of 3716 $Km^2$. The district has a population of 13,21,096 (2011 census). Dumka has a pre-dominantly undulating terrain with hard rock's under the ground. The entire district has topography with high ridges and valleys bounded by mountains and rivers. It was made the headquarters of the Santhal Pargana region, which was carved out of Bhagalpur and Siuri districts after the Santhal Hool 1855. The Santhal revolt started in 1855. Dumka is the sub - capital of the state of Jharkhand. The east Santhal Pargana zone gets rain in summer with an average around 13.5 cm. This area comes under the highlands of Rajmahal range like Sahebgunj and Pakur. The Rajmahal ranges plays very significant role in becoming a roadblock between warm westerly winds and the moisture loaded winds of Bay of Bengal.

Ethnobotanical survey was conducted in different regions of Dumka (Kodokicha, Kukurtoppa, Sujan) during February 2019 - April 2019. During survey interview of "Manjhi hadam" (Mukhya), Naike (Pujari), Vaidyas (Herbal Practitioner), Heads of tribe, Elderman, Elderwomen, Knowledgeable persons by the help of questionnaire about ethnobotanical plants and their relation to rituals of santhal tribe and noted down their santhali name, common name and took photographs. During survey work plants were collected and were identified with the help of Flora of Bihar and Orissa by H.H. Haines in University Department of Botany, Ranchi University, Ranchi. The plants were also preserved in the form of Herbarium.

## RESULT & DISCUSSION

**Table 1: Ethnobotanical plants used in the festivals of santhal tribes**

| Sl. No | Botanical Name | Common Name | Santhali Name | Family | Habit | Parts Used | Name of Festival | Seasons |
|---|---|---|---|---|---|---|---|---|
| 1. | ***Shorea robusta* Gaertn.** | Sal Tree | Sarjom Baha | Dipterocarpaceae | Tree | Flowers | Baha Festival | Phagun (Feb-Mar) |
| 2. | ***Terminalia tomentosa* Roxb ( ex DC ) Wight & Arn .** | Asan Tree | Aatna | Combertaceae | Tree | Leaves | Sohrai | Poush (Dec-Jan) |
| 3. | ***Musa* L.** | Banana Tree | Kaira | Musaceae | Tree | Trunk | Sakrat | Poush (Dec-Jan) |
| 4. | ***Oryza sativa* L.** | Rice | Horo | Poaceae | Grass | Seeds | Hariar Sim | Saraban (Jul-Aug) |
| 5. | ***Madhuca longifolia* (J.Koing ) J.F.Macbr.** | Mahua Tree | Matkom | Sapotaceae | Tree | Flowers | Baha Festival | Phagun (Feb-Mar) |
| 6. | ***Echinochloa frumentaceae* Link.** | Sawa Millet | Gundli | Poaceae | Grass | Seed | Irigundli -nawai | Saraban (Jul-Aug) |
| 7. | ***Cyanodon dactylon* L.** | Doob Grass | Dhoobi Ghas | Poaceae | Grass | Entire plant | Mak Sim | Magh (Jan-Feb) |
| 8. | ***Panicum miliacum* L.** | Common Millet | Iri | Poaceae | Grass | Seeds | Irigundli -nawai | Saraban (Jul-Aug) |
| 9. | ***Triticum aestivum* L.** | Wheat | Guhum | Poaceae | Grass | Seeds | Sohrai | Poush (Dec-Jan) |

The present study attempted to analyze the use of plants with respect to their use in various festivals by the local inhabitants of santhal tribe of Dumka district of Jharkhand. The result of the survey (study) reported that 9 plant species were used in the festivals. In the study 5 plants (*Oryza sativa* L., *Triticum aestivum* L., *Echinochloa frumentacea* Link., *Panicum miliaceum* L., *Cynodon dactylon* L.) belong to the same family Poaceae (grass family). They are mainly used in the festivals. The rest 4 plants belong to different families *Shorea robusta* Gaertn. belongs to the family Dipterocarpaceae, *Terminalia tomentosa* Roxb.(ex DC) Wight & Arn. belongs to the family Combertaceae, *Musa* belongs to the family Musaceae and *Madhuca longifolia* (J.Koing) J.F.Macbr. belongs to the family Sapotaceae, these all are trees.



**A. Jaherthan**



**B. *Shorea robusta***



**C. Preparation for the festival**



**D. Singing and Dancing during festival**

**Fig 1: Picture (A, B, C and D) showing the celebration of Baha Festival during the month of Phagun (Feb-March) at Kukurtoppa village**

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## REFERENCES

1. **Archer,W.G. 2007.** "The Hill of flutes: Life, Love, Poetry in Tribal India. Delhi:Isha Books" (Originally published 1974) pp.1-27.
2. **Murmu, N.C and Hasda. 2010.** R.C.A Portal for Santhals. Wesanthals E-Group. http://www.wesanthals.org.
3. **Acharya, D. and Shrivastava, A. 2008.** Indigenous Herbal Medicines "Tribal Formulations and Traditional Herbal Practices". pp: 440. Aavishkar Publishers Distributor, Jaipur.
4. **Singh, Meena. 2018.** "Study on women costume of Santhal Tribe of Jharkhand"; Textiles and clothing research centre-*Journal*. **2(3):**21-28.
5. **Rana, Shafkat; Sharma, Kumar Dilip; Paliwal, P.P. 2016.** "Rituals plants used by indigenous and ethnic societies of district Banswara (South Rajasthan), India: *American Journal of Ethnomedicine*. **3(1):** 26-34
6. **Jha, H.K; Singh, S.Bijay and Varshney, K. Anil. 2016.** "Ethnomedicinal plants used by Paharia tribe in Santhal Pargana division of Jharkhand". *Indian Botanical Society*. **95(3&4):**297-300.

7. **Kujur, Salomy and Kandir, Kunul. 2015.** "Studies of Ethnomedicinal plants used in Rice-Beer (Handia) by Ho tribe of Jharkhand"; *International Journal for exchange of knowledge*. **2(1):**104-108.

8. **Doresch-Schulte Lea. 2014**. "Fertility or indignity? Two versions of the Santhal Flower Festival"; *Asian Ethnology.* Nov.**1-2:**165-180.

9. **Barla, Sonly. 2020.** "Form Factors of an economically valuable Sal tree (*Shorea robusta*) of Nepal. *Forests*. **11:**574 pp. 1-12

10. **Mishra, Alok (Mar 2019).** "Mahua (*Madhuca longifolia*) flowers: Review on processing and biological properties. *Nutrition & food science*. **49(6):**1153-1163.

11. **Tiwari, Prashan; Jean Sushmita and Sahu, Kumar, Prsatap (April 2019).** "*Butea monosperma*: Phytochemestry and Pharmacology. *Acta scientific pharmaceutical science*. **3(4):**19-26.

12. **Roy, Manadev. 2016.** "Life style culture of the Santhals of Jalpaiguri district in the twentieth century: A historical outline. *International Journal of multidisciplinary research and development*. **3(4):** 221-228.

13. **Baski, Boro. 2018.** "A brief introduction of Santhal life and culture and our approach to development" *Tribal cultural heritage in India foundation*. 1-6.

14. **Maji, A and Pathak. S. 2016.** "A original traditional method a used by the Santhal tribe in Hoogly district of West Bengal, India". *An International Journal of Environmental and Biodiversity*. **6(1):**27-29.

15. **Gope, Laxmiram, Kumar, Santosh and Roy, Raajashri. 2017. "**Identification of indigenous knowledge components for sustainable development among the Santhal community". *American Journal of Educational Research*. **5(8):** 887-893.

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# Studies on the season-wise prevalence of specific intestinal parasite in the rural areas of Saran, Bihar, India

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*Received : 15th November, 2020 ; Revised : 04th January, 2021*

**Abstract:-** Prevalence percentage of total and specific intestinal parasites of rural areas of Saran district of Bihar, India was studied and for which 15732 single stool sample from 15 C.D. block, comprising 9079 males and 6653 females, were collected on random sampling basis and microscopically examined. Highest infestation of intestinal parasites was recorded during SWMS (74.18%), followed by HWS (69.03%), PMS (66.60%) and WS (63.25%) respectively.

**Key words: Season-wise, prevalence, developmental stages, sanitation, transmission, enteric, specimen, centrifuge**

## INTRODUCTION

Humans are subjected to numerous, protozoan, worm and insect related parasites. Parasites vary in the ways they use their hosts. In order to survive from one generation to the next, Parasites have a series of distinct developmental stages and hosts collectively known as a life cycle. The prevalence of parasitic infection varies with the level of sanitation and is generally higher in the tropics and sub tropics than in temperate climates. Two-thirds of the world's populations live in the less-developed countries that lack proper sanitary facilities and a safe drinking water supply, which leads to transmission of enteric pathogens. Most people are not aware of the danger to their health posed by parasites. In the Saran district the rural population constitutes more than 90% of the total population of the district.

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## MATERIAL & METHODS

Stool specimens of 15732 human subjects, obtained by random sampling basis from different rural area of Saran district and belonging to different sex, food-habits, religion and economic strata between January 2002 to December 2004 and on an average 25 to 35 stools were collected and examined every day. The total collected specimens comprised about 0.67 percent of the total rural population of the district. The collected faecal samples were subjected to microscopic examination, with a view to identify the larvae, cysts, tape worm proglottids, adult worms, mucus, blood-stains and consistency of the sample.

Three sides of each stool sample were made by the three different methods for their microscopic examination. For protozoan trophozoites, saline preparation under Direct Smear method was used. In this process a loopful of stool sample was taken on a swab stick and diluted in a drop or two of 0.9 percent normal saline. Under Iodine stain preparation of loopful of stool

sample was first diluted and emulsified in 1 or 2 drops of 0.9% normal saline on a very clean glass slide and then on drop of aqueous Iodine solution (Lugol's Iodine) was put on the saline emulsion and spread in a thin film.

The least expensive and most satisfactory results for helminth eggs and embryos was obtained by employing simple floatation technique. For concentration of faecal cysts, ova and larvae, the formol ether concentration method was found to be the best of all three methods employed. It is the modified version of Ritche's Formol-Ether technique[1,2] Commercial Formalin Solution (10%), Ether, Centrifuge (3000 rpm), glass test tube, swab stick, rubber bands and plastic wire gauze of 7 inch in square (40 mesh per inch) were required in this technique.

## RESULT & OBSERVATION

The total examined samples comprised of 9079 male and 6653 female subjects. 10793 persons (68.60%) were found to be infected with one or more species of protozoan and helminth parasites.

To study the season-wise prevalence rate of total intestinal parasites, the season was divided into four seasonal groups. Hot weather season (HWS) from March to May, South west monsoon season (SWMS) from June to August, Post monsoon season (PMS) from September to November and Winter Season (WS) from December to February respectively. In hot weather season 4007 samples of both the sexes of people were examined and 2766 samples (69.03%) were found to be infected with one or more species of intestinal parasites. During hot weather season, 1728 stool samples of male individual were microscopically examined and 1604 sample (70.38%) were found to contain one or more species of protozoan and helminth intestinal parasites. In female 1162 (67.24%) samples were found to be contaminated, from June to August, 4419 samples were examined, out of which 3278 samples (74.18%) contained one or more species of Intestinal parasites. During Post Monsoon Season the prevalence rate of intestinal parasites was found to be lower. It was 2548 samples (66.60%), out of 3826 examined samples. In the winter season out of 3480 samples examined, only 2201 samples (63.25%) were found to be infected with one or more species of Intestinal parasites.

## DISCUSSION

Considering the poor living and sanitary conditions of the surrounding, illiteracy, density of population, low per-capita income and seasonal variations, it was proposed to study the prevalence pattern of protozoan and helminth intestinal parasites in the rural population of Saran district of Bihar state. Prior to this work, Prasad (1983)[3] and Chaurasia (1993)[4] in Bihar, east of river Narayani had done such investigation. It is the first such epidemiological survey to be undertaken in western Bihar. The persual of relevant literature reveals the noteworthy contributions of Maddex (1904)[5], Lane (1917)[6] Mhaskar (1917-24)[7], Korke (1925-26)[8], Chandler (1925-28)[9], Mukherjee and Mathew (1950)[10], Srivastava (1953)[11], Anand (1954)[12], Bagchi and Prasad (1961)[13], Amin *et al.* (1962)[14], Prakash and Tandon (1966)[15], Choudhary and Schillar (1968)[16], Verma (1985)[17], Bariar (1987)[18], Bhatia and Juyal (1989)[19], Chaurasia (1993)[4], Gagandeep, K. *et al.* (1998)[20], Mirdha and Samantray (2002)[21], Kim (2003)[22] and Singh (2004)[23].

Season wise, highest prevalence rate was observed in the south west monsoon season (74.18%), followed by hot weather season (69.03%), Post-Monsoon Season (66.60%) and the lowest in the winter season (63.25%) respectively. It is in the conformity with the findings of Sinha (1980)[24], Chaurasia (1993)[4.] who also reported the highest rate of incidence in the South West monsoon season. It may be attributed to the period being that of heat, high humidity and rainfall, when the drinking water, fruits, vegetables and other eatables become highly contaminated and start perishing earlier, compared to other season. Ali *et al.* (1989)[25] in Jordan observed parasitic incidence to be higher in summer than in winter season. Sex wise, higher infestation was observed in the males, than in the females in all the seasons while sex-wise frequency of specific, Protozoan and helminth in different season were slightly different. In hot weather season, 0.23% incidence was observed in the females in case of other helminthes, as against 0.22% in the males. In south west monsoon season, the rate of incidence was equal in both the sexes (3.10%) winter season exhibited the lowest prevalence percentage and it was higher in the females.

## REFERENCES

1. **Ridley, O.S., Hawgood, BC. 1956.** The value of formol ether concentration of faecal cysts and ova. *J. Clin Parasit.* **9:** 974-76

2. **Allen and Ridely. 1976.** Further Observations on the formol-ether concentration technique for faecal parasites. *J. Clin Path.* **23:** 545-546.

3. **Prasad, R. 1983.** Prevalence of human intestinal parasites in Hajipur Municipal Area. Ph.D. Thesis Under P.U., Bihar.

4. **Chauraisa. 1993.** Prevalence and Pattern of Intestinal parasitic infection in Man in Rural areas of Vaishali district. Ph.D. Thesis, Bihar Unversity, Muzaffarpur.

5. **Maddex, R.H. 1904.** A note on occurrence of intestinal parasites in Ranchi, Chotanagpur. *Ind. Med. Gaz.* **39 :** 136-137.

6. **Lane. 1917.** Final report on the Ankylostomiasis Enquiry in the Darzeeling district (India). *Ind. J. Med. Res.* **5(3):** 350-385.

7. **Mhaskar, K.S. 1917-24.** The prevalence of ankylostomiasis in Madras Presidency. *Ind. J. Med. Res.* **4(4)**.

8. **Korke, V.T. 1925-26.** Observation on the life history of Hookworm in nature. *Ind. J. Med. Res.* **13(2):** 301-320.

9. **Chandler, A.C. 1925-28.** Part-VIII, Bihar & Orissa, *Ibid*. **14(3):** 745-774.

10. **Mukherjee and Mathew. 1950.** Survey of hookworm infection in the Jharia Coal Field settlement area. *Ind. J. Med. Res.* **38(1):** 318.

11. **Srivastava, J.B. 1953.** A survey of intestinal parasites in the human population in Bombay, with special reference to Entamoeba histolytica. *Ind. J. Med. Res.* **41:** 397-416.

12. **Anand, B.R. 1954.** A survey of Carrier of intestinal parasites in Rajasthan with special reference to *E. histolytica. J. Ind. Ed. Res.* **23 :** M/398-400.

13. **Bagachi, S.C. & Prasad B.G. 1961.** A general health survey in a group of village in Rural Health Centre, Sarojini Nagar, Lucknow. *Jour. Ind. Med. Assoc.* **36 :** 348-353.

14. **Amin *et al.* 1962.** Non symptomatic intestinal parasitism in Baroda. A survey with particular reference to *E. histolytica. Ind. J. Med. Res.* **50(3):**348-353.

15. **Prakash and Tandon. 1966.** Intestinal parasites with reference to *Entamoeba histolytica* Complex as revealed by Routine concentration and cultural examinations from patient with gastrintestinal symptoms. *Ind. J. Med. Res.* **54(1):** 10-14

16. **Choudhary and Schiller. 1968.** A survey of parasitic infection in rural community near Calcutta. *Amer. J. Epiderm.* **87(2):** 299-312.

17. **Verma, K.K. 1985.** Studies on the Human Intestinal Parasites of west Patna, Bihar. Ph.D. Thesis, P.U.

18. **Bariar. 1987.** Studies on Human Intestinal Parasites of Danapur, Patna. Ph.D. Thesis, P.U.

19. **Bhatia, BB and Juyal P.D. 1989.** Incidence of Parasitic infection in human population of tarai area of Uttar Pradesh. *Ind. J. Parasitol.* **13(3):** 323-4.

20. **Gagandeep, K. 1998.** Prevalence of intestinal parasites in rural southern Indians. *Trop. Med. & Intl. Hith.* **3(1):** 70.

21. **Mirdha and Samantray. 2002.** Hymenolepis nana: A common cause of Paediatric Diarrhoea in Urban Slum Dwellers in India. *J. Trop. Paed,* **48:** 331-334.

22. **Kim, B.J., Oek, M.S. 2003.** The intestinal parasite infection status of inhabitants in the Roxas city. The Philippines. *Korean J. Parasit.* **41(2):** 113-5.

23. **Singh Kashinath. 2004.** Studies on human intestinal parasites of Chapra (urban) - Ph.D. Thesis, J.P. University, Chapra.

24. **Sinha, L.K. 1980.** Studies on the human intestinal parasites of central Patna, Bihar and Possible sources of its function. Ph.D. Thesis, Patna University.

25. **Ali-Shtayeh, M.S., Hamidan, A.H.Y.1989.** Prevalence and seasonal fluctuations of Intestinal parasitic infections in the Nabulus area, West Bank of Jordan. *Ann. Trop. Med.& Parasitol.* **83(1):** 67-72.


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# A study on the diversity of bryophytes in Madhepura with reference to pollution

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*Received : 18th December, 2020 ; Revised : 07th January, 2021*

**Abstract:-** Understanding diversity patterns and community structure of bryophytes will help integrate nature conservation at multiple biotic-group levels. We conducted a survey of ground bryophytes in Madhepura along an edge-to-interior gradient. We recorded several species out of which 11 species from 10 genera of 7 families were liverworts, while 26 species from 23 genera of 16 families were mosses. A two-way cluster analysis detected the environmental gradient between the forest edge and forest interior for bryophytes with habitat specificity. Functional diversity of bryophytes differed significantly across an edge-to-interior gradient. Seven species were detected with a significant indicator value for indicating environmental conditions in the forest edge, while only one such species was found indicative of the intermediate transect. Our results demonstrate that remarkable edge effects exist for species composition and functional diversity patterns, and the edge is a marginal habitat with high biotic heterogeneity. Furthermore, functional diversity metrics are more sensitive to the edge effect than species diversity.

**Key words:bryophytes, liverworts, mosses, edge effect, diversity, indicator**

## INTRODUCTION

The term bryophyte is a combination of two Greek words; Bryon = moss and phyton = plant. These plants occupy an intermediary position or form a link between algae and pteridophytes. Often regarded as the plant amphibians, they need water for fertilization. Their lineages can be traced in the tertiary beds of the Cenozoic era.

Bryophytes, known as liverworts, mosses, and hornworts, are the earliest land plants in the phylogenetic systematics of the plant kingdom. They occur widely in the global terrestrial ecosystem, often as dominants in the floor layer of the moist tropical and subtropical broadleaved forest biomes.[1] However, bryophytes appear to have been neglected in many ecological studies, where only vascular plants or even woody plants were investigated such as in the emerging fields of community ecology to explore the role of ecological processes and biotic diversity in maintaining ecosystem function.[2] Until now, little has been known about the diversity patterns of bryophytes, their spatial heterogeneity, their role in forest community assembly, and their biotic and abiotic interactions to maintain the ecosystem function as a whole. The knowledge gap surrounding bryophyte community function is even greater for tropical and subtropical forest ecosystems, thus hampering our steps towards a better understanding of the ecosystem functions as a whole.[3]

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Botanists have argued that as the earliest land plants, bryophytes reflect the dispersal history of plants of various evolutionary stages in the terrestrial ecosystem, and their physiological adaptation, community structure, and the ecological functions in response to environmental change are much more complicated than previously imagined.[4] Disseminated by spores, bryophytes have an outstanding capability for dispersal and will respond sensitively to environmental change. Previous studies have demonstrated that bryophytes are good bio-indicators for environmental pollution due to the special leaf architecture of the plant organism.[5] However, to gain a whole picture of the bryophytes' distribution, diversity patterns, community structure, and their response to ecological factors along environmental gradients, extensive studies in the community ecology of bryophytes should be carried out in natural ecosystems, especially in the forest ecosystem.[6] In this study, we conducted a survey of the understory ground bryophytes using a quadrate sampling method in Madhepura along an edge-to-interior gradient, followed by multivariate statistical analysis of the field data. We aimed to address the following question that is there species with high habitat specificity that can act as bio-indicators for significant species-habitat association.

## MATERIALS & METHODS

To assess whether bryophyte species have a specific habitat association and to detect an indicator value of different species for indicating environmental gradient, we performed indicator species analysis (ISA) using Dufrêne and Legendre's method. ISA calculates an indicator value for each species and provides a p-value for each indicator value using permutation.

## RESULTS

Out of the total number, 11 species from 10 genera of 7 families were liverworts, while 26 species from 23 genera of 16 families were mosses (Table 1). No hornworts were found. High ecological dominance existed in the bryophyte community.

### Indicator Species

Seventeen species had an indicator value>10 for indicating a particular transect habitat, but only eight indicator species were detected to have a significant indicator value (Table 2). Seven such species with a significant indicator value were found confined to the forest edge. These were *Haplocladium microphyllum*, *Fissidens oblongifolius, Homalia trichomanoides* var. *japonica, Entodon schleicheri, Taxiphyllum taxirameum, Homalia trichomanoides* and *Radula obscura*. Only one species, *Leucobryum juniperoideum*, was significantly indicative to the intermediate transect with the maximum indicator value.

**Table 1- Taxonomic composition of bryophytes and community structural attributes**.

| Family | Species | F | AC | RF | RC | IV |
|---|---|---|---|---|---|---|
| **Liverworts** | | | | | | |
| Lepidoziaceae | *Bazzania tridens* | 23 | 28.69 | 7.80 | 6.93 | 14.72 |
| Lophocoleaceae | *Chiloscyphus latifolius* | 28 | 15.62 | 9.49 | 3.77 | 13.26 |
| Lophocoleaceae | *Chiloscyphus profundus* | 1 | 0.10 | 0.34 | 0.02 | 0.36 |
| Lophocoleaceae | *Heteroscyphus zolliingeri* | 3 | 15.30 | 1.02 | 3.69 | 4.71 |
| Lepidoziaceae | *Kurzia gonyotricha* | 33 | 39.98 | 11.19 | 9.65 | 20.84 |
| Lejeuneaceae | *Lejeunea eifrigii* | 8 | 5.94 | 2.71 | 1.43 | 4.15 |
| Metzgeriaceae | *Metzgeria conjugata* | 5 | 5.65 | 1.69 | 1.36 | 3.06 |
| Pallaviciniaceae | *Pallavicinia lyellii* | 1 | 0.05 | 0.34 | 0.01 | 0.35 |
| Plagiochilaceae | *Plagiochila flexuosa* | 4 | 12.30 | 1.36 | 2.97 | 4.33 |
| Radulaceae | *Radula obscura* | 3 | 9.80 | 1.02 | 2.37 | 3.38 |
| Lejeuneaceae | *Spruceanthus polymorphus* | 4 | 3.30 | 1.36 | 0.80 | 2.15 |
| **Mosses** | | | | | | |
| Meteoriaceae | *Aerobryopsis wallichii* | 5 | 6.40 | 1.69 | 1.54 | 3.24 |
| Pylaisiadelphaceae | *Brotherella henonii* | 10 | 6.51 | 3.39 | 1.57 | 4.96 |
| Leucobryaceae | *Campylopus atrovirens* | 1 | 0.20 | 0.34 | 0.05 | 0.39 |
| Hypnaceae | *Ectropothecium dealbatum* | 6 | 13.00 | 2.03 | 3.14 | 5.17 |
| Entodontaceae | *Entodon schleicheri* | 5 | 7.78 | 1.69 | 1.88 | 3.57 |
| Heterocladiaceae | *Fauriella tenuis* | 1 | 0.40 | 0.34 | 0.10 | 0.44 |
| Fissidentaceae | *Fissidens laxus* | 25 | 13.72 | 8.47 | 3.31 | 11.79 |
| Fissidentaceae | *Fissidens nobilis* | 2 | 2.53 | 0.68 | 0.61 | 1.29 |
| Fissidentaceae | *Fissidens oblongifolius* | 8 | 17.89 | 2.71 | 4.32 | 7.03 |
| Thuidiaceae | *Haplocladium microphyllum* | 9 | 30.15 | 3.05 | 7.28 | 10.33 |
| Anomodontaceae | *Herpetineuron toccoae* | 5 | 23.40 | 1.69 | 5.65 | 7.34 |
| Neckeraceae | *Homalia trichomanoides* | 3 | 0.90 | 1.02 | 0.22 | 1.23 |
| Neckeraceae | *Homalia trichomanoides var. japonica* | 8 | 9.43 | 2.71 | 2.28 | 4.99 |
| Neckeraceae | *Homaliodendron flabellatum* | 1 | 0.20 | 0.34 | 0.05 | 0.39 |
| Hypnaceae | *Hypnum fauriei* | 3 | 8.20 | 1.02 | 1.98 | 3.00 |
| Leucobryaceae | *Leucobryum juniperoideum* | 14 | 32.65 | 4.75 | 7.88 | 12.63 |
| Orthotrichaceae | *Macromitrium schmidii var. macroperichaetialium* | 1 | 0.20 | 0.34 | 0.05 | 0.39 |
| Neckeraceae | *Neckeropsis calcicola* | 1 | 1.80 | 0.34 | 0.43 | 0.77 |
| Mniaceae | *Plagiomnium rhynchophorum* | 1 | 3.60 | 0.34 | 0.87 | 1.21 |
| Polytrichaceae | *Pogonatum inflexum* | 1 | 0.30 | 0.34 | 0.07 | 0.41 |
| Hypnaceae | *Pseudotaxiphyllum pohliaecarpum* | 47 | 41.68 | 15.93 | 10.06 | 25.99 |
| Brachytheciaceae | *Rhynchostegium pallidifolium* | 3 | 10.50 | 1.02 | 2.53 | 3.55 |
| Sematophyllaceae | *Sematophyllum subpinnatum* | 2 | 9.00 | 0.68 | 2.17 | 2.85 |
| Calymperaceae | *Syrrhopodon prolifer* | 3 | 1.90 | 1.02 | 0.46 | 1.48 |
| Hypnaceae | *Taxiphyllum taxirameum* | 5 | 12.90 | 1.69 | 3.11 | 4.81 |
| Thuidiaceae | *Thuidium pristocalyx* | 12 | 22.30 | 4.07 | 5.38 | 9.45 |

Abbreviations: F = Frequency;
AC = Average Cover;
RF = Relative Frequency;
RC = Relative Cover;
IV = Importance Value.

**Table 2- Indicator species of bryophytes with an indicator value >10 across transects**

| Species | Observed indicator value | Indicator value from randomization | | *p* |
|---|---|---|---|---|
| | | Mean | Standard deviation | |
| *Leucobryum juniperoideum* | 37 | 17.4 | 5.45 | 0.007 |
| *Fissidens laxus* | 32.9 | 24.9 | 5.08 | 0.078 |
| *Haplocladium microphyllum* | 28.4 | 13.3 | 5.1 | 0.015 |
| *Fissidens oblongifolius* | 25.4 | 12.9 | 5.38 | 0.031 |
| *Homalia trichomanoides var. japonica* | 24.5 | 12.5 | 4.98 | 0.022 |
| *Entodon schleicheri* | 22.7 | 9.9 | 4.77 | 0.036 |
| *Thuidium pristocalyx* | 21.8 | 15.8 | 5.26 | 0.129 |
| *Taxiphyllum taxirameum* | 21.3 | 9.9 | 4.72 | 0.042 |
| *Homalia trichomanoides* | 20 | 7.7 | 4.27 | 0.046 |
| *Radula obscura* | 20 | 7.7 | 4.24 | 0.047 |
| *Brotherella henonii* var. *henonii* | 20 | 14.2 | 5.12 | 0.121 |
| *Ectropothecium dealbatum* | 19.9 | 10.7 | 4.81 | 0.063 |
| *Plagiochila flexuosa* | 16.2 | 8.8 | 4.27 | 0.069 |
| *Lejeunea eifrigii* | 13.6 | 12.7 | 5.11 | 0.369 |
| *Aerobryopsis wallichii* | 13 | 9.7 | 4.59 | 0.249 |
| *Herpetineuron toccoae* | 12.6 | 9.8 | 4.58 | 0.273 |
| *Metzgeria conjugata* | 12.3 | 9.8 | 4.66 | 0.284 |

## DISCUSSION

Bryophyte diversity patterns and community structure changed in response to environmental gradients. Except for the intermediate transect, high habitat specificity for the bryophyte species was found in both the forest-edge and forest-interior transects.[3] High interspecific association existed in both common species that had co-occurrence in a number of sample units, and rare species which were found in only one or two sample units. Two bryophyte species, *Pseudotaxiphyllum pohliaecarpus* and *Chiloscyphus latifolius*, occurred in almost all sample units, reflecting their robust adaptability to heterogeneous habitats. Although three common species, *Fissidens laxus, Kurzia gonyotricha* and *Bazzania tridens,* were restricted to the intermediate and the forest-interior transects, they occurred in most sample units of the two transects, indicating that the forest-edge habitat might act as an environmental filter for their distribution. The nine bryophyte species detected as rare species with unique occurrence in the sample units demonstrated high habitat specificity, and a very high species association was exhibited in these species with coincidence in the same sample unit.

Biodiversity is regarded as the essential property characteristic of an ecosystem or biotic community.[6]

Eight species (*Leucobryum juniperoideum, Haplocladium microphyllum, Fissidens oblongifolius, Homalia trichomanoides var. japonica, Entodon schleicheri, Taxiphyllum taxirameum, Homalia trichomanoides*, and *Radula obscura*) were detected to have a significant indicator value. Seven such species with a significant indicator value were found with a restricted occurrence in the forest edge, while only one species (*Leucobryum juniperoideum*) was significantly indicative of the intermediate transect with the maximum indicator value. The significant indicator species were the representations of both species habitat association and habitat specificity.

## CONCLUSION

In conclusion, the spatial patterns of bryophyte species diversity and community structure in response to the edge-to-interior gradient were determined by habitat characteristics as well as the bryophytes' biological properties. Our results demonstrated that the conspicuous diversity patterns, community structure, and interspecific association in ground bryophytes are related to the extent of change of diversity patterns under climate change, thus acting as indicators.

The data sets generated during the current study are available from the corresponding author on reasonable request.

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## REFERENCES

1. **Gensel, P. G. 2008.** The earliest land plants. *Annual Review of Ecology, Evolution, and Systematics*. **39:**459-477
2. **Hsu, C. C., Horng, F. W. & Kuo, C. M. 2002.** Epiphyte biomass and nutrient capital of a moist subtropical forest in north-eastern Taiwan. *J.Trop. Ecol.* **18:**659-670.
3. **Liu, W. Y., Fox, J. E. D. & Xu, Z. F. 2002.** Nutrient fluxes in bulk precipitation, throughfall and stemflow in montane subtropical moist forest on Ailao Mountains in Yunnan, south-west China. *J. Trop. Ecol.* **18:**527-548
4. **Song, L., Liu, W. Y. & Nadkarni, N. M. 2012.** Response of non-vascular epiphytes to simulated climate change in a montane moist evergreen broad-leaved forest in southwest China. *Biol. Conserv.* **152:** 127-135.
5. **Loreau, M. & de Mazancourt, C. 2013.** Biodiversity and ecosystem stability: a synthesis of underlying mechanisms. *Ecol. Lett.* **16:**106-115
6. **Alam, A & S.C. Srivastava. 2009.** Liverwort diversity in Palni hills, Tamil Nadu, India. A Checklist. NELUMBO (Botanical Survey of India). **51:**99-122

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# Ethnobotanical study of sacred groves (jaher) of Sri Amra of Dumka block, Dumka district, Jharkhand

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*Received : 20th December, 2020 ; Revised : 09th January, 2021*

**Abstract :** Sacred groves popularly known as Jaher by Santals serve as major contribution for in-situ conservation of nature. A significant change has been noticed regarding its conservation in recent years. In the present paper an attempt has been made for ethnobotanical study of sacred groves of Santals of Dumka block. Field observation, secondary data and several interviews have been conducted with the prominent village people (Pradhan, Naike) and other knowledgeable tribals to gather information. Various ethnobotanical uses of economically important plants used by Santal tribes have been discussed in this paper.

**Keywords : Sacred groves, Ethnobotany, Dumka, Santals**

## INTRODUCTION

Sacred groves are group of trees considered sacred for a particular community. It is popularly known as Jaher by Santals. They are the forest patches communally protected by local people due to their religious association with them. A significant change has been noticed regarding its conservation in recent years. Due to modernization, the natives (Santals) are drawn far away from their cultures and taking less active participation for its conservation. The ethnic groups present here are mostly Santals then Hindus, Muslims, Bengalis and paharias are rare. Worship of sacred groves is the traditional practice of various group of the society. In recent years traditional ethnobotanical studies have received much attention due to their wide local acceptability and clues for new or less known medicinal plants.[1] It is mandatory to compile the ethnobotanical knowledge presently existing among the diverse community before its values are completely vanished.[2] Thus, there is now urgency for ethnobotanical research amongst aboriginal people.[3] There are many studies entitled to further quantify these ethics which leads to biodiversity conservation and sustainable ecosystem.[2] Considering previous studies and the present exploration clearly indicate that such kind of ethnobotanical studies may be beneficial to human race in conserving these sacred groves as depository for many rare endangered floras holding numerous faunas.

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## MATERIALS & METHODS

In Jharkhand, the Savar is one of the primitive tribal group and they are most deprived section in our country.[4] Another tribal group is that of Santal which is mostly centred in Santal Pargana. The Santals have a culture of their own which they have preserved unchanged from time immemorial.[5] Dumka covers most of the area of Santal Pargana and is also a second capital of Jharkhand. Dumka district of Jharkhand state comprises 10 blocks namely-

Dumka, Gopikander, Jama, Jarmundi, Kathikund, Masaliya, Ramgarh, Ranishwer, Saraiyahat, Shikaripara. Here, the study area taken is Dumka block which comprises around 277 villages. The entire block of Dumka is divided into 25 panchayats. Each panchayat consists of 1-26 villages. Here a nearby village has been taken into consideration for the study, i.e., Sri Amra present in the north-east region.

The present study considers the species diversity of particular area and the ethnobotanical study of vegetation inside the sacred grove, Jaher. Regular visits and surveys were made to collect plant specimens both in flowering and fruiting stages. Field observation on habit, habitat, pollination, seed dispersal, medicinal and socio-cultural uses have been recorded in the field at the time of collection.

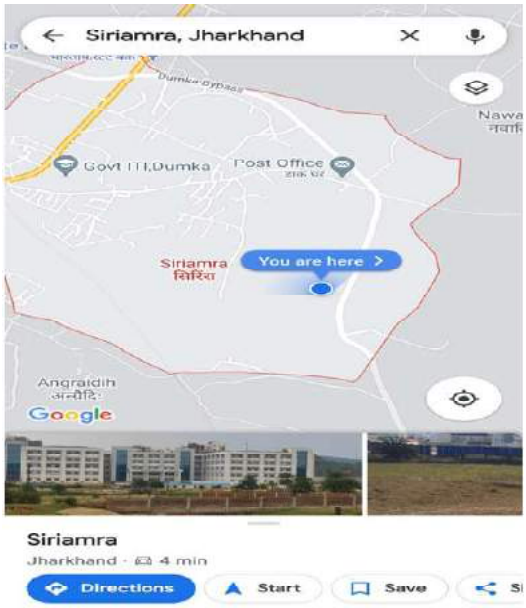



**Figure 1- GPS Location of Sri Amra Jaherthan**

## OBSERVATION

| | | |
|---|---|---|
| Topography | - | Plain |
| Boundary | - | Not present |
| Humus | - | poor |
| Soil type | - | Sandy and rocky |
| Date | - | 10.09.2019 |

**Quantitative structure**

Quantitative structure of plant communities includes-

**1. Abundance**-It is the number of individuals of any species per sampling unit of occurrence.

Abundance= Total no. of individual of species / No of quadrant per unit in which they occur

**2. Density-** It is the numerical strength of a species in the community. It gives an idea of degree of competition.

Density = Total no. of individual of the species ÷ No of quadrant per units studied

**3. Relative Density, RD**E

RDE = (No. of individuals of species ÷ No. of individuals of all species) X 100

**4. Frequency-** It is the number of sampling units in which the particular species occurs.

Frequency (%) = (No. of units in which species occurred ÷Total no. of units studied) X 100

**5. Relative frequency, RFR**

RFR of Species = (No. of occurrences of species ÷ No. of occurrences of all species) X 100

**6. Frequency class-** Raunkiaer's (1934) grouped five frequency classes

Frequency, A= 1-20 %, B= 21-40%, C= 41-60%, D= 61-80%, E= 81-100%

**Table 1-List of plants recorded in Sri Amra Jaher, Dumka**

| Sl. No. | Species name | Family | English name | Hindi name | Local Name | Total no. of Individuals |
|---|---|---|---|---|---|---|
| | **Upper storey** | | | | | |
| 1 | *Shorea robusta* | Dipterocarpaceae | Sal | Sakhu, Sakhua | Sarjom | 11 |
| 2 | *Madhuca indica* | Sapotaceae | Indian butter tree | Mahua | Matkom | 1 |
| | | | | | | |
| | **Middle storey** | | | | | |
| 3 | *Butea monosperma* | Fabaceae | Bastard teak | Palash | Muruk' | 2 |
| 4 | *Phoenix dactylifera* | Arecaceae | Date palm | Khajur | Khijur | 1 |
| | | | | | | |
| | **Shrub layer** | | | | | |
| 5 | *Lantana camara* | Verbenaceae | Sage | Raimuni, Panchphuli | Putus | 3 |
| | | | | | | |
| | **Ground flora** | | | | | |
| 6 | *Tridax procumbens* | Asteraceae | Coatbutton | Kanphuli | Jormot | 6 |
| 7 | *Athyrium filix-femina* | Athyriaceae | Lady fern | Kashiphul, Parnang | Kokro arak' | 10 |
| 8 | *Miscanthus sinensis* | Poaceae | Silver grass | Chandighaas | Kharan'/ Khadan' | 8 |

**Table 2- Quantitative analysis of vegetation structure of Sri Amra Jaher**

| Sl. No | Name of Species | No of individuals in each quadrant | | | | | Total no of individuals of each species (X) | Total no of quadrant of occurrences (Y) | Total no. of quadrant studied (Z) | Frequency Y/Z * 100 | Frequency class | Density (X/Z) | Abundance (X/Y) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | | | | | | | |
| **I** | **UPPER STOREY** | | | | | | | | | | | | |
| **1** | *Shorea robusta* | 3 | | 5 | 2 | 1 | 11 | 4 | 5 | 80 | D | 2.2 | 2.8 |
| **2** | *Madhuca indica* | | 1 | | | | 1 | 1 | 5 | 20 | A | 0.2 | 1 |
| | | | | | | | | | | | | | |
| **II** | **MIDDLE STOREY** | | | | | | | | | | | | |
| **3** | *Butea monosperma* | | 1 | | 1 | | 2 | 2 | 5 | 40 | B | 0.4 | 1 |
| **4** | *Phoenix dactylifera* | | | 1 | | | 1 | 1 | 5 | 20 | A | 0.2 | 1 |
| | | | | | | | | | | | | | |
| **III** | **SHRUB LAYER AND CLIMBERS** | | | | | | | | | | | | |
| **5** | *Lantana camara* | | | 2 | | 1 | 3 | 2 | 5 | 40 | B | 0.6 | 1.5 |
| | | | | | | | | | | | | | |
| **IV** | **GROUND FLORA** | | | | | | | | | | | | |
| **6** | *Tridax procumbens* | 1 | | 2 | | 3 | 6 | 3 | 5 | 60 | C | 1.2 | 2 |
| **7** | *Athyrium filix - femina* | 3 | 2 | 2 | 3 | | 10 | 4 | 5 | 80 | D | 2 | 2.5 |
| **8** | *Miscanthus sinensis* | 2 | 3 | 1 | 2 | | 8 | 4 | 5 | 80 | D | 1.6 | 2 |

## DISCUSSION

Ethnobotanical significance of plants listed in Sri Amra Jaher

### 1. Religious plants:

***Shorea robusta*** was the dominant species found which has close association with the native Santals. Flowers are used during worship in festivals like Sohrai, celebrated in the month of January, also known as Harvest festival and Baha, meaning flower in Santali, is a festival which falls in the month of March simultaneously with the Hindu festival, Holi (festival of colours).As we know that spring season which is in March is the month when new leaves arise and flowers bloom, so Baha festival is celebrated after Sohrai, and tribals traditional belief system prohibits them to cut or pluck any plant parts, thus promoting successful flowering process.

***Phoenix dactylifera*** has significance role worldwide since Christians carry it on Palm Sunday to churches. Palm Sunday is celebrated a week before Easter Sunday.

***Butea monosperma*** also known as 'Flame of the Forest' is believed to be the tree form of Agnidev, 'God of Fire'. The leaves are used in many religious rituals of Hindus. Flowers are used in making colours in the festival of Holi.

***Madhuca indica*** is worshipped by many tribal communities because of its usefulness. Seed oil is used in diyas (small mud bowls) during Diwali festival of Hindus. Barter system is still prevalent with this plant product as it acts as goods exchanger in many tribal and rural areas.

### 2. Medicinal Plants:

***Shorea robusta*** is used to treat various diseases such as leucorrhoea, gonorrhoea, diarrhoea, piles, dysentery, etc.

***Madhuca indica*** or ***Madhuca latifolia*** is used to treat anaemia, diabetes, ulcer, uterus disorders, infertility, etc.

***Butea monosperma*** has widespread use in Unani and Homoeopathic medicines. Its flower juice treats diarrhoea, leprosy, skin diseases, thirst sensation, etc.

***Phoenix dactylifera*** treats gastrointestinal troubles, counteract alcohol intoxication, sore throat, colds, bronchial asthma, etc.

***Lantana camara*** treats cancers, chickenpox, measles, ulcers, tumors, high blood pressure, tetanus, rheumatism, etc.

***Tridax procumbens*** has antifungal, anticoagulant and insect repellent properties. Leaf juice treats wounds and skin diseases.

***Athyrium filix-femina*** treats cough, rheumatism, sores, burns and scalds, improves lactation after child birth, as an antiparasitic, anthelmintic and carminative.

***Miscanthus sinensis*** is used to disperse poison, anticoagulant, anti-inflammatory, diuretic and refrigerant.

### 3.Edible plants:

***Shorea robusta*** fruits are occasionally consumed as food. Its seed oil is used in the manufacture of food and non-food industries. Oil can be used as cooking oil and alternative for cocoa butter in chocolate industry.

***Madhuca indica*** plays significant role in tribal life. Its fruit is used as vegetables by native people.Seed oil

can be used as cooking oil. Dried flowers are boiled and consumed as important nutritive food.

***Phoenix dactylifera*** fruits are consumed as dry fruits which is rich in minerals, vitamins and fibres.

**4.Timber yielding plants:**

***Shorea robusta*** is an excellent timber yielding plant.

***Madhuca indica*** is another most common timber.

***Butea monosperma*** is durable under water, thus used in boats, well curbs and water scoops.

**5.Economically important plants:**

***Shorea robusta*** is a highly economic plant since dried leaf plates have commercial importance. Its timber is hard and durable and yields best quality furniture.

***Butea monosperma*** is used for timber, reisn, fodder, medicine, dye and bidis.

***Madhuca indica*** is used for timber. Seed oil is used in making soaps and chocolates. Flowers and fruits are highly nutritive. Flowers are fermented and distilled producing spirituous liquor known as 'country beer' or 'handi' in Santali. It is one of the common sources of earning by poor tribal women.

***Athyrium filix-femina*** is used for decorative purposes and making flower bouquets.

***Lantana camara*** is known as Honey plant, since much nectar collection is possible from this plant. Thus, significant apiculture plant.

***Phoenix dactylifera*** fruits are mostly sold by tribal women. The dried leaves are used for making mats. The local alcoholic beverage known as 'Tari' obtained from the tree trunk is famous among tribal. The unfermented palm juice is delicious and nutritious. Jaggery is made by boiling palm juice in a large container which is rich in calcium and iron.

## CONCLUSION

The species diversity of Sri Amra Jaher is comparatively poor due to dry, sandy and rocky soil condition. However, the presence of a shallow trench which holds and allows water flow temporarily provides home for ferns which is rare in most sacred groves. Communal belief of native Santals has restored the vegetation of this Jaher despite its poor-quality soil since unnecessary human activities is restricted in this area and hence promotes nature conservation.

## ACKNOWLEDGEMENT

We express our sincere gratitude towards our prominent tribals (Manjhi, Naeke) and native villagers who supported us in identification and collection of ethnobotanical and socio-cultural use of the prevalent flora and its fauna.

## REFERENCES

1. **Tripathi, Y.C. 2000.** Ethnomedicinal treasure of tribal Rajasthan. *J. Non timber for products*.**7(1-2):** 77-84.
2. **Rao, R. Raghavendra. 1996.** Traditional knowledge and Sustainable development, Key role of Ethnobiologist. *J. Ethnobotany.* **8:** 14-24.
3. **Maheshwari, J.K. 1983.** Development in ethnobotany *Editorial. J Econ Taxon Bot.* **4(1):** 1-4.
4. **Oraon, P.C. 2003.** Land and people of Jharkhand, Jharkhand Tribal Welfare Research Institute, Welfare Department, Government of Jharkhand, Ranchi. 6-7.
5. **Sinha, V.N.P. and Singh, L.K.P. 2003.** Jharkhand: Land and people. Rajesh Publications, New Delhi, pxii.

## ADDITIONAL REFERENCES

1. **www.researchgate.net**
2. **www.isca.in**
3. **wgbis.ces.iisc.ernet.in**
4. **bioone.org**
5. **www.phytojournal.com**

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# A study of monthly variation in biomass and primary productivity of macrophyte of Hardia wetland of Saran District of North Bihar, India

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*Received : 21st December, 2020 ; Revised : 10th January, 2021*

**Abstract :** The wetland has profuse growth of macrophyte *Potamogeton* sp., *Polygonum* sp., *Nelumbo* sp. were observed as floating and among submerged *Chara* sp., *Nitella* sp., *Ceratophyllum* sp. were dominant. Macrophyte form an important component of wetland ecosystem as it increases the surface area for food and shelter to invertebrates, fishes and birds. The macrophytes are considered as producer of water bodies as they receive radiant energy in the form of food upon which all aquatic life depends. So it helps in stabilizing the ecosystem. They contributed to the major chunks of biomass production of the wetland. *Hydrilla* sp. was the most dominant among submerged weeds. It was present almost throughout the year. During the first year of study the dry biomass of *Hydrilla* sp. ranged from 24 g/m$^2$ in Aug'12 to 172 g/m$^2$ in March'12. The intensity of light available to macrophytes per unit area is greater in the tropical water and its physico-chemical characteristics are vital factors besides temperature for the growth of macrophytes. In the present study low pH has been found for favourable growth of macrophytes. A fall in N, P, K, Ca and Mg in the water during growing period relates to increase in macrophytic biomass.

**Keywords : Growth of macrophytes, growing period, profuse growth, radiant energy, producer, dominant, intensity of light, physico-chemical characteristics, vital factor, favourable growth**

## INTRODUCTION

The rate at which the green plant produces the biomass or store energy is referred as primary product. The primary productivity of aquatic ecosystem gives the quantitative details regarding energy fixation and its availability to support the bioactivity of the total ecosystem.[1] Biomass is associated with its productivity, biologically efficiency and with other community having same type and different type of communities. Macrophyte form an important component of wetland ecosystem as it increases the surface area for food and shelter to invertebrates, fishes and birds.

The macrophytes are considered as producer of water bodies as they receive radiant energy in the form of food upon which all aquatic life depends. So it helps in stabilizing the ecosystem.

Primary productivity of wetlands has been studied extensively abroad and in our country.[2-11]

The present study deals with the biomass and primary production of Hardia Wetland.

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## MATERIAL & METHODS

**PRIMARY PRODUCTIVITY OF MACROPHYTES:**

It was assessed by the Harvest method[12] and was measured as changed in the Biomass Values. The biomass was estimated as the dry matter/unit area. The productivity was calculated on the basis of increase in the biomass in a unit area in a particular time.

**Calculation:**

For the calculation of productivity the biomass per unit area was estimated at suitable intervals.

Productivity= $(B_2-B1)/D$ dry wt./$m^2$/day

Where; $B_1$=Biomass g/$m^2$ at time $t_1$

$B_2$=Biomass g/$m^2$ at time $t_2$

D=Days interval between $t_1$ and $t_2$.

## RESULTS

Several varieties of macrophytic species were observed in the wetland of which *Potamogeton* sp., *Hydrilla* sp. and *Chara* sp. among submerged were found to be dominant forms. They contributed to the major chunks of biomass production of the wetland. *Hydrilla* sp. was the most dominant among submerged weeds. It was present almost throughout the year. During the first year of study the dry biomass of *Hydrilla* sp. ranged from 24 g/$m^2$ in Aug'12 to 172 g/$m^2$ in March'12. The average standing crop production was 101.58g dry wt/$m^2$. The annual net and daily net production was 143.0g dry wt/$m^2$ and 0.40g dry wt/$m^2$. The dry biomass of *Potamogeton* sp. ranged from 20 g/$m^2$ to 97 g/$m^2$ in the first year and from 20 g/$m^2$ to 96 g/$m^2$ in the second year. The dry biomass of *Chara* sp. varied from 19 g/$m^2$ to 154 g/$m^2$ in the first year and from 20 g/$m^2$ to 112 g/$m^2$ in the second year. The average standing crop production indicate that the mean value are of *Hydrilla* sp. 101.58 g dry wt/$m^2$ in the first year and minimum was that of *Chara* 47.75 g dry wt/$m^2$ during second year of study. The annual net production was minimum of *Potamogeton* sp. 68 g dry wt./$m^2$ in the second year and maximum was of *Hydrilla* sp. 143.0 g dry wt/$m^2$ in the first year of study. The daily net production was minimum 0.26 g dry wt./$m^2$ of *Potamogeton* sp. and maximum 40 g dry wt./$m^2$ of both plants *Chara* sp. and *Hydrilla* sp. during first year of study. (Table: 1,2, fig:1,2)

**Table 1- Monthly variation in biomass and primary productivity of macrophyte (2011-13) values in g/$m^2$**

| Month | Macrophyte | | |
|---|---|---|---|
| | ***Potamogeton*** | ***Hydrilla*** | ***Chara*** |
| November | 97 | 164 | 104 |
| December | 84 | 166 | 127 |
| January | 57 | 168 | 130 |
| February | 34 | 154 | 154 |
| March | 25 | 172 | 147 |
| April | 10 | 134 | 104 |
| May | 8 | 74 | 87 |
| June | * | * | * |
| July | 2 | 32 | 48 |
| August | 27 | 24 | 19 |
| September | 46 | 67 | 44 |
| October | 54 | 64 | 87 |
| November | 72 | 76 | 96 |
| December | 96 | 98 | 112 |
| January | 84 | 107 | 84 |
| February | 64 | 74 | 66 |
| March | 56 | 67 | 55 |
| April | 58 | 42 | 51 |
| May | * | * | * |
| June | * | * | * |
| July | 20 | 17 | 20 |
| August | 38 | 36 | 22 |
| September | 57 | 44 | 41 |
| October | 88 | 67 | 46 |

Note: "*" Site was dry.

**Table 2- Primary productivity value of macrophyte (2011-13) values in g dry wt./ $m^2$**

| Macrophytes | Average Standing Crop | | Annual Net Production | | Daily Net Production | |
|---|---|---|---|---|---|---|
| | 2011-12 | 2012-13 | 2011-12 | 2012-13 | 2011-12 | 2012-13 |
| *Potamogeton* sp. | 37 | 51.91 | 95 | 68 | 0.26 | 0.18 |
| *Hydrilla* sp. | 101.58 | 52.33 | 143 | 90 | 0.4 | 0.24 |
| *Chara* sp. | 87.58 | 47.75 | 135 | 90 | 0.4 | 0.24 |
| Total | 226.16 | 151.99 | 373 | 248 | 1.06 | 0.66 |

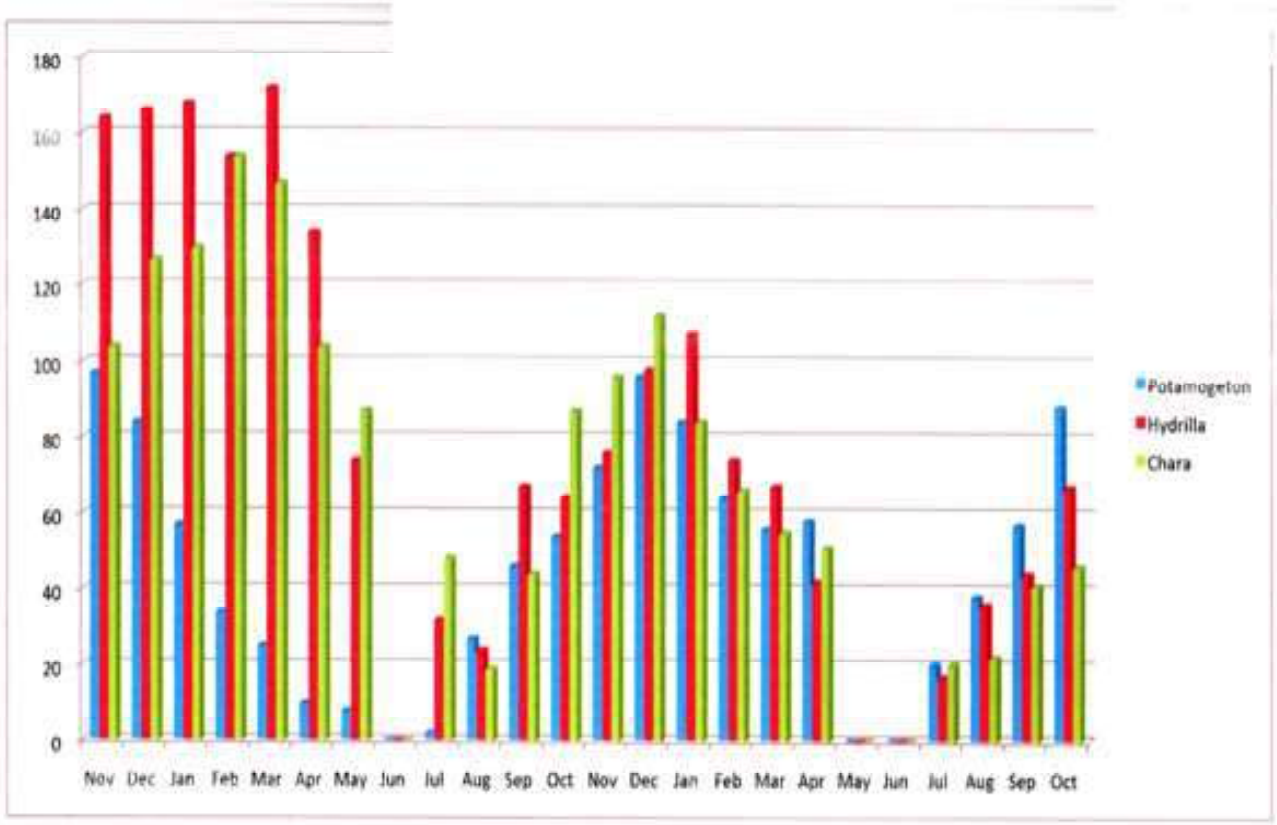



Fig.-1: Monthly variation in Biomass and primary Productivity of Macrophyte (2011-13) values in g/m$^2$.

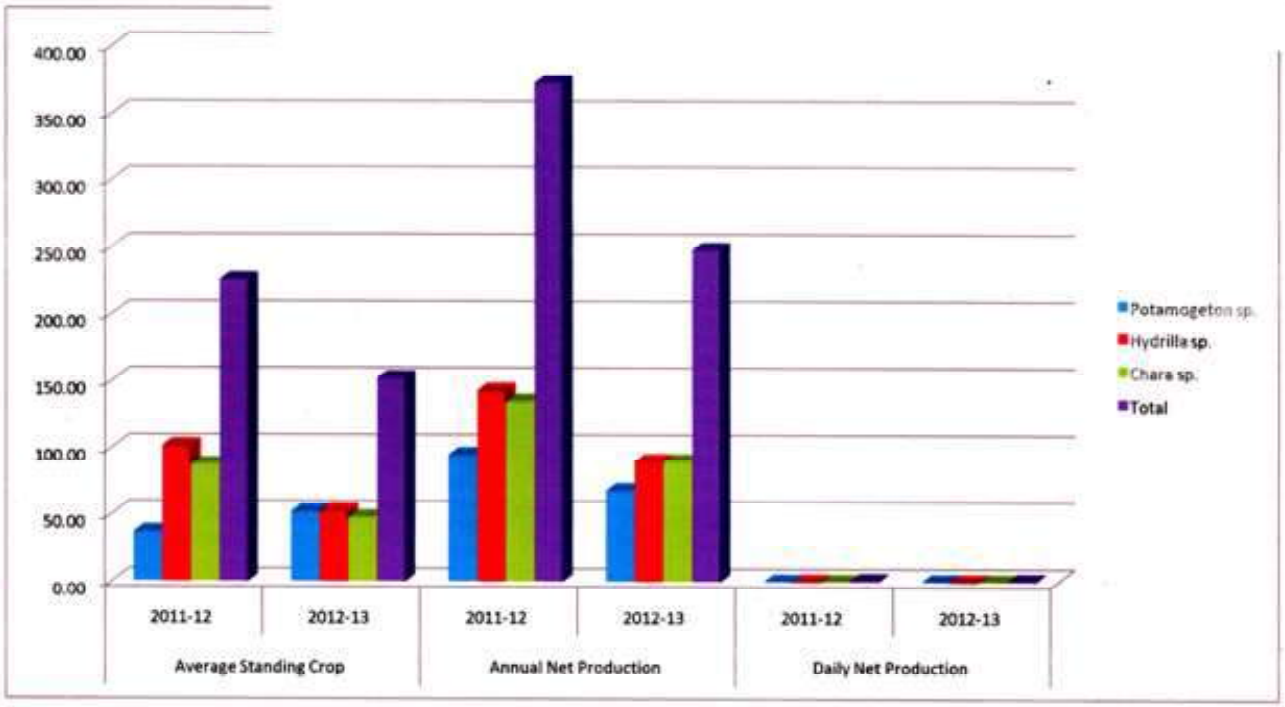



Fig.-2: Primary Productivity value of Macrophyte (2011-13) values in g dry wt./ m$^2$

## DISCUSSION

The productions of hydrophytes are governed by various hydrological parameters. Light is the most important factor regulating the growth of submerged aquatic plants.[13] The intensity of light available to macrophytes per unit area is greater in the tropical water and its physico-chemical characteristics are vital factors besides temperature for the growth of macrophytes. Sen and Chaterjee (1960)[14] observed higher pH for the better growth of hydrophytes but in the present study low pH has been found for favourable growth of macrophytes. Our result also matches with that of Sharma and Munshi (1995)[8] in Kawar lake wetland. Nutrient especially phosphorus is the single nutrient regulating production.[15] The biomass and productivity was quite good due to large amount of nutrients available in the sediment. The site receives large amount of nutrients from surface runoff and agricultural practices during dry season. The lowest value of nitrates and phosphate were observed during the period of maximum vegetation. Both calcium and magnesium has vital role in productivity in the wetland. They are in good quality. A fall in N, P, K, Ca and Mg in the water during growing period relates to increase in macrophytic biomass. Alkalinity and conductivity value indicates rich productivity capacity of the wetland.

## REFERENCES

1. **Goldman, C.R. 1968.** Aquatic primary production, Amer. *Zooplankton.* **B(1):**31-42

2. **Kavet, J. 1971.** Growth analysis approach to the production ecology of reed swamp plant communities, *Hydrobiol*. **12:**15-40

3. **Conner, W.H. and Day, J.W. Jr. 1976.** Productivity and comparison of a bold cypress water tubelo site and a bottom land hard wood site in a Louisiana Swamp. *Amer. J.Bot.* **63:**1354-1364.

4. **Brown, S.L. 1981.** A comparison of the structure, primary productivity and transpiration of cypress ecosystem in Florida. *Ecol. Monogr*. **51:**403-427.

5. **Kaul, V., Trishal,. C.L. and Handoo, J.K. 1978.** Distribution and production of macrophytes in some water bodies of Kashmir. In: Glimpses of Ecology (ed J.S. Singh and B. Gopal). *International Scientific Publ*., Jaipur, India. 313-334.

6. **Trishal, C.H. and Kaul, S. 1983.** Sediment composition, mud water interchanges and role of macrophytes in the Dal Lake, Kashmir. Int. rev. *Hydrobiol*. **68:**671-682.

7. **Vyas, L.N., Sharma, K.P., Sankla, S.K. and Gopla, B. 1990.** Primary production and energetic. In: *Ecology and Management of Aquatic Vegetation in the Indian Sub continent*. (ed. B. Gopal). 149-174.

8. **Sharma, U.P. and Munshi, J.S.D. 1995.** Ecology, Conservation and Management of Kawar Lake. *A major tropical wetland of South East Asia*. 189p.

9. **Eliska Rejmankova. 2011.** The role of macrophytes in wetland Ecosystem. *J. Ecol. Field Biol.* **34(4):**333-345.

10. **Singh, K.K. and Sharma, B.M. 2012.** Ecological productivity studies of the macrophytes in Kharungput Lake, Manipur, North East India. *Int. J. Geo. Earth and Environ. Sci*. **2(2):**58-71

11. **Atkinson, R.B., Perry, J.E., Noe, G.B., Daniels, W.L., Cairns Jr., J. 2010.** Primary productivity in 20 year old created wetland in Southwestern Virginia. *Wetlands*. **30:**200-210.

12. **Wood, R.D. 1975.** *Hydrobiological Methods*. University Park Press, Baltimore, London.173p.

13. **Martin, J.B., Bradford, B.N. and Kennedy, H.G. 1969.** Factors affecting the growth of Najas in Pickwick, Resorvoir, *National Fert. Development centre. Tenn, Valley Author, Muscle Shoals, Ala.* 47.

14. **Sen, D.N. and Chatterjee, U.N. 1960.** A study on the plant community of Gorakhpur in relation to the pH of the medium *Proc. Nat. Acad. Sci*. **B(30):**187-191

15. **Hutchinson, G.E. 1957.** *A Treatise on Limnology*. Vol.1. *Willey, N.Y.* 1015p.

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# *In vitro* antibacterial activity of chitosan from crab, *Maydelliathelphusa masoniana* (Henderson, 1893) against *Staphylococcus aureus* and *Pseudomonas aeruginosa*

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*Received : 23rd December, 2020 ; Revised : 14th January, 2021*

**Abstract :** In the present study, chitosan has been extracted from *M.masoniana.* Further the antibacterial activity of different concentrations (25%, 50%, 75%, 100%) of chitosan was tested against Gram-positive bacteria, *Staphylococcus aureus* (ATCC 6538) and Gram-negative bacteria, *Pseudomonas aeruginosa* (ATCC10145) by Agar Well Diffusion method. The result of the present study suggests that as the concentration of chitosan increased the antibacterial effect was strengthened against both bacterial strains indicating its possibility of using as drug.

**Keywords :** ***M.masoniana,*** **antibacterial, chitosan,** ***S.aureus, P.aeruginosa***

## INTRODUCTION

Nature has played an instrumental role in providing effective therapeutic entities.[1] In the present era, people are suffering from several bacterial diseases and the most common medicine suggested are synthetic antibiotics. The antibiotics are compound that inhibit the growth of undesirable microorganisms. Thus to minimize these effects, more emphasis is given on alternative source which are naturally derived antimicrobial agents. Natural antimicrobial are derived from many, including animals (chitosan).[2]

Chitosan is a copolymer of glucosamine and N-acetyl glucosamine units linked by 1, 4 glycosidic bonds and it is obtained through the alkaline hydrolysis of chitin.[3] Chitosan is derived from chitin, which is found in the shells of crustaceans like freshwater and marine crabs and shrimps. Chitosan has gained attention of the scientific community due to its functional properties such as film-forming capabilities, mineral-binding properties, hydrolipidemic activity, biodegradability, antimicrobial activity, immunoadjuvant activity, acceleration of wound healing, and eliciting of phytalexins.[4]

The goal of this study was to evaluate antibacterial activity of chitosan, extracted from shell of freshwater crab, *Maydelliathelphusa masoniana* against food borne and UTI strains, Gram-positive, *Staphylococcus aureus* and Gram negative, *Pseudomonas aeruginosa.*

## MATERIALS & METHODS

**Collection of the animals:**

*Maydelliathelphusa masoniana* were purchased from local market of Ranchi, Jharkhand. Shells were scraped

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free of loose tissues from the crab wastes in laboratory, washed thoroughly with tap water to remove impurities. They were dried at 60°C and pulverized using pestle and mortar for further analysis.

**Preparation of chitosan:**

Chitin and Chitosan were prepared from *Maydelliathelphusa masoniana* shell waste according to *Takiguchi*[5, 6] with some modifications[7] for purification of chitosan. The production of chitosan from crustaceans shell generally consists of three basic steps demineralization, deproteinization and deacetylation.

**Preparation of stock solution:**

In the preparation of chitosan solutions (15mg) chitosan was dissolved in 3ml of 0.2% (w/v) aqueous acetic acid solution. From this 0.25,0.50,0.75 and 1.0 ml was taken and made up to 1.0 ml by adding 0.2% acetic acid to prepare various concentrations containing 1.25 mg, 2.50 mg, 3.75 mg and 5 mg of chitosan sample corresponding to 25,50,75 and 100% respectively.

**Bacterial Strains:**

The antibacterial activity of the prepared chitosan from *Maydelliathelphusa masoniana* was tested against two strains, *Staphylococcus aureus* (ATCC 25923) and *Pseudomonas aeruginosa* (ATCC 27853), obtained from the Department of Microbiology, Yugantar Bharti Analytical and Environmental Engineering Laboratory, Sidroul, Namkum, Ranchi, Jharkhand. *Staphylococcus aureus*, a Gram positive bacterium and *Pseudomonas aeruginosa,* a Gram negative bacterium were chosen because both are commonly associated with food products and Urinary tract infection.

**Preparation of bacterial culture:**

Nutrient broth medium was prepared and sterilized in an autoclave at 15 lbs pressure. Two bacterial species were incubated in the nutrient broth and incubated at 34°C for 24 hours. Nutrient agar medium was also prepared, autoclaved and transferred aseptically in to sterile Petri dishes. On this, 24 hours bacterial broth cultures were inoculated by using a sterile cotton swab.

**Antibacterial assay:**

The antibacterial activity of the individual bacterial strains was tested using Agar Well Diffusion method.[8] Well of 6mm diameter were made aseptically in the plates. 24 hours old bacterial broth cultures were inoculated using a sterile cotton swab on sterile Nutrient Agar plates. Using micropipette, solution of different concentration of chitosan and 0.2% acetic acid as negative control was loaded in the respective wells. Ciprofloxacin (5µg) disc was placed using sterile forceps, as positive control. The plates were incubated at 34°C for 24 hours in upright position. The antibacterial assay was carried out in triplicate. After incubation at 34°C for 24 hours, zone of inhibition was measured in millimeters.

## RESULT

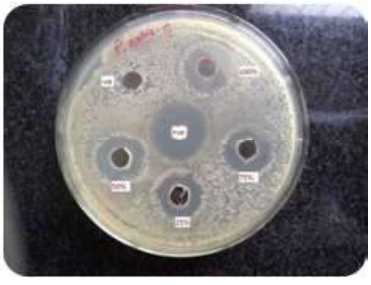
**Table 1: Antibacterial activity of chitosan of *M.masoniana***

| Sl. No. | Microorganisms | Concentration of chitosan | | | | | |
|---|---|---|---|---|---|---|---|
| | | (5mg) 100% | (3.75mg) 75% | (2.5mg) 50% | (1.25mg) 25% | Positive control | Negative control |
| 1. | *Staphylococcus aureus* (ATCC 6538) | 15.13 ±0.15 | 14.16 ±0.15 | 13.03± 0.05 | 12.0 ±0.2 | 25.03 ±0.05 | - |
| 2. | *Pseudomonas aeruginosa* (ATCC 10145) | 19.83± 0.25 | 18.8 ±0.20 | 17.10± 0.26 | 15.26± 0.25 | 24.96 ±0.15 | - |

**( - ) = No zone of inhibition, Results indicate zone of inhibition in mm, Values are given as mean ±SD of three experiments, Positive control (Ciprofloxacin 5µg), Negative control (0.2% acetic acid)**



***Staphylococcus aureus***



***Pseudomonas aeruginosa***

**Fig. 1. Antibacterial activity of chitosan from *M. masoniana***

The chitosan from *M. masoniana* showed good antibacterial activity against both pathogenic bacteria. (Fig 1). The antibacterial activity was found to be concentration dependent. At the same time the activity was absent in negative control (Table 1). Effect of chitosan on pathogenic bacteria revealed that, the highest activity (19.83±.25mm) was observed against *Pseudomonas aeruginosa* with highest (100%) concentration. At the same concentration, *Staphylococcus aureus* showed 15.13±0.15 mm zone of inhibition. Regarding at 25, 50, 75% concentration maximum activity (15.26±0.25, 17.10±0.26, 18.8±0.20 mm) was found against *Pseudomonas aeruginosa.* The lowest activity (12.20±0.2 mm) was found with 25% concentration against *Staphylococcus aureus.*

## DISCUSSION & CONCLUSION

In this study, antibacterial activity of chitosan prepared from crab shell waste was tested against two strains *Staphylococcus aureus* and *Pseudomonas aeruginosa.*

The present investigation showed a higher antibacterial activity in 100% concentration with the maximum inhibition zone of 19.83±0.25 mm clear zone recorded against *Pseudomonas aeruginosa* and in 25% concentration with 12.0±0.2 mm inhibition zone against *Staphylococcus aureus*. When compared to this finding, chitosan from *Sepia kobiensis* reported the maximum inhibition of above 16mm against *Pseudomonas aeruginosa* at the highest concentration of 5mg/ml and minimum inhibition between 11-15 mm against *Staphylococcus aureus*.[9] Chitosan from, *Podophthalmus vigil* reported to have inhibition zone of 8.17±1.21 mm against *Staphylococcus aureus* at only highest concentration of 500µg/ml and no zone of inhibition against *Pseudomonas aeruginosa* at all concentrations.[1]

According to literature[10] chitosan possesses antimicrobial activity against a number of Gram-negative and Gram-positive bacteria. The antibacterial mechanisms of chitosan suggested being, the positive charge of the amino group at C-2 resulted in a polycationic structure which can be expected to interact with the predominantly anionic component of the microorganism surface.[11] The interaction resulted in great alteration of the structure of outer membrane[12], which caused release of major proportion of proteinaceous material from the cells.[13] According to our results, as the concentration of chitosan increased, the antibacterial activity was strengthened. This result is consistent with the work of Jeon *et al.*[10] who have also reported the increased antibacterial activity with increase in the concentration of chitosan.

This study demonstrate that chitosan from fresh water crab, *Maydelliathelphusa masoniana* have excellent antibacterial activity against Gram positive (*Staphylococcus aureus*) and Gram negative (*Pseudomonas aeruginosa*) bacterium. Chitosan may be used against bacterial infection.

## REFERENCES


1. **K.Prabu, E.Natarajan. 2012.** In vitro antimicrobial an antioxidant activity of chitosan isolated from *Podophthalmus vigil. Journal of Applied Pharmaceutical Science*. **2(9):**075-085.
2. **Tiwari BK, Valdramidis VP, O'Donnell CP, Muthukumarappan K, Bourke P and Cullen PJ. 2009.** Application of natural antibiotics for food preservation. *J Agric Food Chem.* **57:**5987-6000.
3. **No,H.K. & Meyer, S.P. 1997.** Preparation of chitin and chitosan. In R.A.A. By Muzzarelli, & M.G. Peter (Eds), *Chitin handbook* (pp.475-485). Grottammare, AP, Italy:European Chitin Society.
4. **Md.Monarul Islam, Shah Md.Masum, Khandaker Rayhan Mahbub and Md.Zahurul Haque. 2011.** In vitro antibacterial activity of shrimp chitosan against *Salmonella paratyphi* and *Staphylococcus aureus. Jour. of Bangladesh Chemical Society.* **24(2):**185-190.
5. **Takiguchi Y. 1991a.** Physical properties of chitinous materials In: R.H.Chen and H.C.Chen (eds.), Advances in Chitin Science. Vol.III. Proceedings from the third Asia –Pacific Chitin. Chitosan Jikken manual chapter 1.*Gihodou Shupan Kaisha*. Japan:1-7
6. **Takiguchi Y. 1991b.** Preparation of chitosan and partially deacetylated chitin. In:A.Otakara and M.Yabuki (eds). Chitin, Chitosan Jikken Manual Chapter-2,*Gihodou Shupan Kaisha*. Japan: 9-17.
7. **Yateendra Shanmukha Puvvada, Saikishore Vankayalapati, Sudheshnababu Sukhavasi. 2012.** Extraction of chitin from chitosan from exoskeleton of shrimp for application in the pharmaceutical industry. *International Current Pharmaceutical Journal.* **1(9):**258-263.

8. **N.Ramasamy, A.Subhaptabha, S.srinivasan, K.Jayalakshmi, A.Shanmugam. 2011.** In vitro evaluation of antimicrobial activity of methanolic extract from selected species of cephalopods on clinical isolates. *Afr.J.Microbiol.Res*. **5(23):**3884-3889.

9. **Annaian Shamugam, Kandasmy Kathiresan, Lakshman Nayak. 2016.** Preparation, characterization and antibacterial activity of chitosan and phosphorylated chitosan from cuttlebone of *Sepia kobiensis. Biotechnology Reports*. **9:**25-30

10. **Jeon YI. Park PJ. and Km SK. 2001.** Antimicrobial effect of chitooligosaccharides produced by bioreactor. *Carbohydrate Polymers*. **44:**71-76.

11. **Helander IM, Nurmiaho-Lassila EL,Ahvenainen R,Rhoades J, and Roler S. 2002.** Chitosan distrupts the barrier properties of the outer membrane of Gram negative bacteria. *International Journal of Food Microbiology*. **71:**235-244.

12. **Helander IM, Latva-kala K. and Lounatmaa K. 1998.** Permeabilizing action of polyethyeneimine on *Salmonella typhimurium* involves destruction of the outer membrane and interactions with lipopoly saccharide. *Microbiology*. **144:**385-390.

13. **Vaara M and Vaara T. 1983.** Polycatio as outer membrane disorganizing agents. Anti microbiology Agents. *Chemotherapeutant*. **24:**114-122.

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# Aquatic Hemipteran biodiversity of Ranchi region of Jharkhand

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**Abstract :** The survey was under taken to investigate the aquatic Hemipteran biodiversity of Ranchi, Jharkhand, during September, 2019-February, 2020. The aquatic insects were collected by using net from the study area. The aquatic Hemipteran community was represented by seven families. The highest population of order aquatic Hemiptera was recorded during six month. Aquatic Hemiptera holds an important place in the ecology of freshwater and non freshwater ecosystem. They are important food for many organism including fish, amphibian and other animals. The chemical properties and the distinct taxa found in the water suggest that the water bodies are polluted and may be dangerous to the health of people around the reservoir.

**Keywords : Aquatic Hemipteran biodiversity, organism, freshwater ecosystem, chemical properties**

## INTRODUCTION

Biodiversity refers to the variety and variability among living organism (number of different items/species and there relative frequency) in which they occur. During past twenty years on so biodiversity has attracted attention to numerous workers expediting fauna and flora of the world.[1] India is the one of the world's twelve mega diversity countries.[2] Insect contribute over half of all the recorded species and over three fourth of the estimated species in the globe.[3] Aquatic biodiversity is one of the most essential characteristics of an aquatic ecosystem for maintaining stability and a means of copying with any environmental change.[4] Aquatic insects are a group of arthropods that lives or spend part of their life cycle in water bodies.[5] They are the great important to water bodies where they are found and their presence in water serve various purpose, some serve as a food for fishes and other invertebrates, other acts as vector through which disease pathogens are transmitted to both human and animals.[6] Most importantly aquatic insects are very good indicators of water qualities since they have various environmental disturbances tolerant levels.[7] Aquatic Hemipteran stand out as an important group of aquatic insects, which are considered important in environment reclamation of aquatic habitats and are after used to gauge toxins in an environment.[8] The Hemipteran insect is a very successful aquatic group because of its adoptive nature in diverse ecological and geographical area and is especially economically important.[9] Majority of hemipteran are terrestrial but there are some species adapted to aquatic or semi aquatic life, occur in all type of fresh water habitat. They can be distinguish from other aquatic insect order by presence of piercing and sucking beak like structure of mouthparts and

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leathery anterior pair of wings at the base of membranous apically and completely second pair. They are highly predaceous some species are as veracious predators of fish, fry and fingerlings.[10] Not only they are most beautiful but they are most stable specimens that can be collected.[11]

## MATERIALS & METHODS

**Study Area**

The field work is carried out on the two ponds and a river selected in Ranchi city, Jharkhand. The pond ecosystems taken under study are RIMS talab, Jora talab and Boreya River. RIMS talab is a rectangular in shape with cemented wall all around. Jora talab is also rectangular in shape, participation into two parts. Boreya River is situated in Boreya area with slow water flow.

**Sampling Method**

The present study is based on the studies carried out for a period of six month from September 2019 to February 2020. For sampling the two ponds area is arbitrarily divided in four zones North, South, East and West and for sampling of Boreya River in river edge is taken.

Collection are done using hand operated net and bar net from the above mentioned zones and river edge. After collecting the aquatic insects, it was identified by using taxonomic key and also by comparing the collected specimens through internet's photo of the specific insects was collected to know the abundances and fluctuations of each species.

## RESULTS & DISCUSSION

The aquatic insects especially Hemipteran were collected in six month. Each aquatic net sample consists of ten separate sweeps done continuously. Abundance of different aquatic Hemipteran insect in pond and river of six month in table-1

**Table 1- Abundances (total number) of Aquatic Hemipteran in ponds and river**

| Name of Insect (Family and Order) | No. of Aquatic Hemipteran in Pond / River | | |
|---|---|---|---|
| **Hemiptera** | **Pond-1 (Rims Talab) total no. of insects in six month** | **Pond-2 (Jora Talab) total no. of insects in six month** | **River (Boreya River) total no. of insects in six month** |
| Water Strider (Gerrididae) | Present in Swarm | Present in Swarm | Present in Swarm |
| Broad Shoulder Water Strider (Viliidae) | Present in Swarm | Present in Swarm | Present in Swarm |
| Water Scorpion (Rantra Sp.) | 38 | 73 | 125 |
| Water Scorpion (Napidae) | 138 | 175 | 150 |
| Back Swimmers (Notonectidae) | 36 | 62 | 57 |
| Water Boatmen (Corixidae) | 73 | 55 | 38 |
| Giant Water Bugs (Belostomatidae) | 15 | 17 | 20 |

**Abundances (total no) of different aquatic Hemipteran in different ecosystem-**

**Aquatic Hemipteran insects**

**Water striders-**

Water striders were found on the water surface of Rims talab, Jora talab and Boreya River were impossible to collect by swiping net. It was found all over season. They remain scattered throughout the water surface but some time they form swarm and they were about 500-1000 insect in a swarm. (Table-1)

**Broad shoulder water striders-**

Broad shoulder water striders were found in the Rims talab, Jora talab and Boreya River. There were 200-500 Broad shoulder water striders insect in each swarm.Table-1

**Water scorpion-**

Abundances of total number of Water scorpion *Ranatra* sp in Rims talab, Jora talab and Boreya River were 38, 73 and 125 respectively. Highest number of Water scorpion *Ranatra* sp were found in Boreya River followed by Jora talab and lowest number were in Rims talab. (Table-1)

**Water scorpion-**

These were found in Rims talab, Jora talab and Boreya River. Abundances of total number of water scorpion *Nepidea* in Rims talab, Jora talab and Boreya River were 138, 175 and 150 respectively. (Table-1)

**Back swimmer-**

Abundances of total number of Back swimmer in Rims talab, Jora talab and Boreya River were 36, 62 and 57 respectively. Highest numbers of Back swimmer were found in Jora talab followed by Boreya River and lowest numbers were in Rims talab. (Table-1)

**Water boatmen-**

Abundances of total number of water boatmen in RIMS talab, Jora talab and Boreya River were 73, 55, and 38 respectively. Highest number of water boatmen were found in RIMS talab followed by Jora talab and lowest number were in Boreya River. (Table-1)

**Gaint water bug-**

Abundances of total number of Giant water bugs in RIMS Talab, Jora Talab and Boreya River were 15, 17 and 20 respectively. Highest numbers of Giant water bug were found Boreya River followed by Jora talab and lowest numbers were in Rims talab. (Table-1)

In this research insect of 7 (seven) families of Aquatic Hemiptera order were collected such as water striders (Gerrididea), broad shoulder water striders (Veliidea),water scorpion (*Napidea* and *Ranatra* sp.), Back swimmers (Notonectidea),water boatmen (Corixidea) and Giant water bugs (Belostomatidea) from the ponds and river in different number. (Table-1)

Besides this, Anonymous (1980a)[12] took an experiment 'Aquatic hemiptera and Ephemeroptera in Dhaka city' and collected insect seven (7) families such as water boatmen (Corixidea) and Giant water bugs (Belostomatidea), water striders (Gerrididea), broad shoulder water striders (Vellidea), water scorpion (*Napidea* and *Ranatra* sp.), Back swimmers (Notonectidea) from the pond and the lake of Dhaka city. In the similar study of aquatic Hemiptera Goswami & Hazarika (2010)[13] recorded 14 species from two ponds of Gawhati University. Kalita (2008)[14] recorded 9 species from the deepor beel. Das & Gupta (2010)[15] recorded 12 and 10 species of aquatic Hemipteran respectively, they found highest density during post monsoon in agriculture field. Gupta & Narzary (2013)[16] reported 5 aquatic Hemiptera from Anua Lake of Cochar. Barman & Baruah (2014)[17] revealed 11 Hemipteran species from kapla beel of Barpeta district.

## CONCLUSION

Study on biodiversity of aquatic insect is valuable research of the concerning area which gives on idea about the diversified environment of the ponds and the river. Aquatic insects are integral part of the aquatic ecosystem as they both ecological and economical value. The result obtained in the present study indicates that the aquatic Hemipteran collected from two ponds and river order Hemiptera represents the highest number of species. Aquatic hemipteran insect was found to vary in river and pond.

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## REFERENCES

1. **Kumar, U. & A Sija, M. 2005.** Biodiversity: Principal and conservation. *Agrobios India*. 234.
2. **Gadgil M. 1996.** Developing student power to monitor India's lifescape. *Cur Sci*. **71:**689-697
3. **Chapman and Hall. 1992.** Hammond PM species inventory in global diversity status of the earth's living resources. In groombridge B. 17-39
4. **Vinson, M.R & C.P Hookines. 1998.** Biodiversity of streams: variation at local, basin and regional scales. *Annual review of entomology.* **43:**271-293.
5. **Pennak, R, W. 1978.** Freshwater invertebrates of the United States. 2nd Edn., John Wiley and Sons, NewYork, pp:810.
6. **Chae, S.J., N. Purstela, E Johnson, E. Derock, S.P. Lawler and J.E.Madigam. 2006.** Infection of aquatic insects with trematode metacercariae carrying *Ehrilichia risticii*, the case of the potomae house fever. *J.Med. Entomol.* **37:**619-625.
7. **Aremono, F.O. & R.B.I Koni. 2008.** Ecological Integriedy of upper Warri River, Niger delta using Aquatic insects as bioindicators, *Ecol. Indic.,***395:**1-7
8. **Papacek, M. 2001.** Small aquatic and ripicolous bugs (Heteropter: Nepomonpha) as predations and prey: The question of economic importance. *European Journal of entomology.* **98:**1-12

9. **Batzer, D.P. & Wissrger, S.A.1996.** Ecology on insect communities in non-tidal wetlands. *Annual review of entomology*. **41:**75-100

10. **Comstock, J.H. 1984.** An introduction to entomology. Satish Book Enterprise India, 1063

11. **Khan, R.A. & Ghosh, L.K. 2001.** Faunal diversity of Aquatic insect in fresh water wetlands of south eastern West Bengal, occasional paper No.194, records of zoological survey of India.

12. **Anonymois. 1980a.** Aquatic Hemipteran and Ephemeroptera in Dhaka city. M.Sc. Thesis, Dept. of Zoology, Dhaka University, Dhaka 109 (call No.196.80)

13. **Goswami, R. & Hazarika, M.M. 2010.** Aquatic Hemiptera of Gawhati University, Gawhati, Assam, India. *Journal of Threatened Taxa*. **2(3):**778-782.

14. **Kalita, G. 2008.** Ecology and Distribution of macro invertebrate enmeshed fauna in deeper wetland of Assam, India. Ph.d. Thesis Gauhati University, Gauhati Assam, India.

15. **Das, K. & Gupta, S. 2010.** Aquatic Hemiptera community of agricultural fields and rain pools in Cachar District, Assam, North East India, *Assam University Journal of Science and Technology: Biological and Environmental Science.* **5(1):**123-128.

16. **Gupta, S. & Narzary, R. 2013.** Aquatic insect community of lake, Phulbari anua in Cachar, Assam. J*ournal of Environmental Biology.* **34:**591-597.

17. **Barman, A. & Baruah, B.K. 2014.** Aquatic Hemipteraof Kapla beel, a resh water wetland of Barpeta District of Assam, India. Abstract, *National Seminar on Recent Trends of Research in Science and Technology.* 29th March, 2014 pp. 143-144

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# Variation in population density of aquatic coleopteran in a freshwater body of Ranchi District, Jharkhand with special reference to pH, temperature and dissolved oxygen

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**Abstract :** Present work deals with the variation in population of aquatic coleoptera species in a freshwater body, i.e. the Joda Talab of Bariatu area in Ranchi, Jharkhand on account of changes in physico-chemical parameters such as pH, temperature and dissolved oxygen. The study was conducted from October-2019 to February 2020 i.e. the post monsoon to winter season. During the study period total 3 families of order coleopteran i.e. Gyrinidae, Dytiscidae and Hydrophilidae were recorded from the two sites of the Talab. Among these families, *Gyrinus* species of Gyrinidae was found in high number at both the sites as they prefer cooler environment for breeding and development. Except these, species of other families declined from October to December and then increased till February. Site 1 with greater number of individual was recorded with low pH values. Similarly decline in temperature from October to January was seen with rise in DO values which was accompanied with increase in number of insect species. Whereas site 2 with high pH, low dissolved oxygen and seasonal changes of temperature, was accompanied with less number of individuals. Thus the study revealed that pH, temperature and DO play an important role in difference of population density of aquatic insects.

**Keywords : Aquatic coleoptera, freshwater pond, physico-chemical parameters, variation in population**

## INTRODUCTION

Aquatic insects belonging to group of Arthropods, spend their life in water bodies and about 45,000 species are known to inhabit diverse freshwater ecosystems.[1] In order to maintain the stability of aquatic ecosystem its aquatic biodiversity is very essential.[2] These insects form an important component of food chain and energy flow pathway in the aquatic ecosystem.[3] The presence or absence of aquatic insects indicates that a particular water body is healthy or polluted. Some of them are very sensitive to pollution and prefer good water quality.[4] Hence these insects are considered as good indicator of water quality.[5] The changes in the physico-chemical properties of water, adversely affect the diversity and distribution of aquatic insects.[6] Thus these are used for water quality assessment to provide appropriate information in regard to the state and quality of water bodies. Physico-chemical parameters such as pH, Temperature, Dissolved oxygen etc have been widely used to assess the quality of water.[7] The aquatic beetle communities are also affected by these factors. However the vegetation and size of the water body also affect the population.

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Thus the present study was undertaken with an aim to investigate the changes and fluctuations in the population of aquatic beetle families in a pond of Ranchi in context of these parameters.

## MATERIALS & METHODS

**Study Area:**

For this research work the Joda Talab of Bariatu area in Ranchi is taken. This Talab is situated in a densely populated area and is also subjected to various anthropogenic interferences. Two sites were selected from the pond, one which is rich in vegetation and the other site which is disturbed by the domestic uses.

**Sampling Method:**

Present study was conducted during the post monsoon & winter season i.e. from October 2019 to February 2020. Water and insect samples were collected from both the sites of the pond. The insects were collected by using hand operated nets. At each site at least three drags were made as they constituted a sample.[8] Collected insects were sorted and taken to the laboratory. They were preserved in 70% ethyl alcohol. Each of them was later identified with the help of standard keys of Imms[9], Mani (1994)[10], Borror and Delong (2005)[11]. Water samples from each site was also collected and brought to the laboratory. All these collections were made during early hours of the day between 8 to 11 a.m. as the insects migrate to deeper water during late hours of the day.

**Water-Parameter Analysis:**

Physico-chemical parameters such as temperature, pH and dissolved oxygen were analyzed by standard titrimetric methods.[12]

**Data Analysis:**

Species diversity index was calculated using Shanon-Weiner diversity index to understand the aquatic community of insects at both the sites.

## RESULTS

During the study period of 5 months, i.e. from October 2019 to February 2020, monthly data of water parameters and insects availability was recorded from both the sites. The monthly water parameter data is shown in Table 1.

**Table 1. Water quality parameters at the two sites of Joda Talab, Ranchi**

| Sl. No. | Parameters | Site 1 | | | | | Site 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Oct | Nov | Dec | Jan | Feb | Oct | Nov | Dec | Jan | Feb |
| 1. | pH | 7.79 | 7.84 | 7.92 | 7.22 | 7.32 | 8.32 | 8.20 | 7.89 | 7.58 | 7.94 |
| 2. | Water temperature (°C) | 27.2 | 25.2 | 21.4 | 19.2 | 25.9 | 27.8 | 25.9 | 21.9 | 19.3 | 26.1 |
| 3. | Dissolved Oxygen (mg/l) | 4.91 | 5.21 | 5.39 | 5.46 | 5.31 | 4.62 | 4.88 | 4.91 | 5.02 | 4.93 |

Variation in pH and Temperature of water was seen at both the sites during the study period. Highest temperature was recorded in the month of October. It was 27.2°C, 27.8°C respectively at both the sites. Whereas, the lowest temperature was recorded in the month of January i.e. 19.21°C & 19.88°C respectively at both sites.

pH at both sites, ranged between 7.22 to 8.32. The maximum pH was recorded in the month of October and minimum in the month of January at both the sites. Dissolved oxygen is also an important physico-chemical parameter in water quality studies. High DO value indicated good aquatic life.[13] The amount of dissolved oxygen recorded in site 1 of Joda Talab ranged between 4.9mg/l to 5.4mg/l. Maximum value was recorded in the month of January (5.46 mg/l) and minimum was recorded in the month of October (4.91 mg/l) at site 1. Similarly, highest dissolved oxygen was seen in the month of January (5.02 mg/l) and lowest value was in the month of October (4.62 mg/l) at site 2.

Monthly insect collection is shown in Table 2. A total of 71 insects of order coleopteran were collected from site 1. Among these, highest number was seen in month of October which was followed by February. December and January were seen with few individuals. Total 42 insects were collected from site 2. October was seen with highest number of insects. Throughout the study period Gyrinus species of family Gyrinidae was collected in highest number from both the sites. Insects belonging to Hydrophilidae were found least. Two taxa of family Dytiscidae were recorded from both sites with highest number in month of October and least in December. Shanon-Weiner diversity index of different species were determined for each site and is shown in Table 3.

**Table 2. List of Coleopteron species found at each site during October 2019 to February 2020 at Joda Talab, Ranchi.**

| | | Site 1 | | | | | | Site 2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Months** | **Oct** | **Nov** | **Dec** | **Jan** | **Feb** | **Total** | **Oct** | **Nov** | **Dec** | **Jan** | **Feb** | **Total** |
| Sl.No. | **Family**/Species | | | | | | | | | | | | |
| | **Gyrinidae** | | | | | | | | | | | | |
| **1.** | *Gyrinus* | 7 | 4 | 2 | 5 | 7 | **25** | 5 | 4 | 3 | 3 | 5 | **20** |
| | **Dytiscidae** | | | | | | | | | | | | |
| **2.** | *Cybister* | 5 | 1 | - | 1 | 3 | **10** | 4 | 1 | - | - | 3 | **8** |
| **3.** | *Dytiscus* | 6 | 4 | - | 2 | 6 | **18** | 4 | 2 | - | - | 4 | **10** |
| | **Hydrophilidae** | | | | | | | | | | | | |
| **4.** | *Hydrophilus* | 6 | 4 | 1 | 2 | 5 | **18** | 2 | 1 | - | - | 1 | **4** |
| **Total number of Insects at Site 1 = 71** | | | | | | | | **Total number of Insects at Site 2 = 42** | | | | | |

**Table 3: Shannon-Weiner Diversity Index**

| Sl.no. | Species | Site 1 | Site 2 |
|---|---|---|---|
| **1.** | *Gyrinus* | 0.1516 | 0.1534 |
| **2.** | *Cybister* | 0.1198 | 0.1371 |
| **3.** | *Dytiscus* | 0.1510 | 0.1483 |
| **4.** | *Hydrophilus* | 0.1510 | 0.0972 |

## DISCUSSION

Present study on aquatic coleopteran at the two sites of Joda Talab showed that the population of aquatic insects is greatly regulated by biotic and abiotic factors. Abiotic factors such as water temperature, dissolved oxygen and pH seem to be responsible for the variation seen in the aquatic coleopteran population at the two sites.

The number of insects declined from October to December and after that increased till February. The minimum and maximum numbers of Coleoptera were recorded in the month of December and February respectively. Fewer individuals were available in December which might be due to hibernation or due to low temperature which might have affected the development process of insects. Total collection comprised mainly of 3 families i.e., Gyrinidae, Dytiscidae and Hydrophilidae. Out of these, individuals of Gyrinidae were abundant as this group of insects increase in abundance with decreasing temperature because they prefer cooler water for their feeding rate, metabolism and reproduction. However, population of species of Dytiscidae & Hydrophilidae decreased with decreased temperature. More individuals were collected at relatively high temperature than when there was a drop in temperature. Ajao (1990)[14] and Oben (2000)[15] recorded similar observation during their studies. But at site 2 less number of individuals of Gyrinidae was recorded even in winter season. Site 1 showed high number of individuals which might be also due to presence of luxuriant vegetation which provided better environment for the growth of insects. Similar readings have also been recorded by Korkeamaki and Subomen (2000)[16].

The variation in pH values at both the sites has been found to affect the population of aquatic insects. However, in India, water bodies are generally alkaline in nature. Lower number of insects was seen at site 2, as high value of pH was recorded there, where as greater number of insects were recorded at site 1 having low value of pH. The factors like photosynthesis, respiratory activity, temperature, exposure to air, disposal of wastes etc. bring out changes in the pH.

Dissolved oxygen showed an inverse relationship with water temperature. This observation agreed with Arimoro and Ikomi (2008)[5] findings, who reported that increase in water temperature brings about a decrease in dissolved oxygen and it may be due to respiration and other process such as breakdown of organic matters. More population was seen at site 1 where DO values were high. The low values of DO at site 2 indicated detoriation of water quality as a result of anthropogenic activities in this site as observed. Similar findings were seen by Yakub (2004)[17].

Shanon-Weiner diversity Index helps to know the relative species abundance. The diversity of entamofauna at site 1 was higher than in site 2. Out of the three families of order Coleoptera, Gyrinidae was dominant family at both the sites.

## CONCLUSION

The result obtained during the study revealed that the physico-chemical parameters greatly influenced the

aquatic coleopteran species composition and abundance. The result also showed that temperature and dissolved oxygen had strongest influence on the community structure at both the sites of Joda Talab. The lower values of Shanon-Weiner diversity index at site 2 pointed towards the disturbances, caused due to anthropogenic activities. Low vegetation was also a reason for lower number of aquatic insects at this site.

## REFERENCES

1. **Balram, P. 2005.** Insect of tropical streams. *Current science*. **89:** 914
2. **Sharma, M.R. and Gupta, A.B., 2004.** Seasonal variation of physico – chemical parameters of Hathi stream in outer Himalayas, *Pollution Research*. **23(2):** 265-270.
3. **Barman A, Baruah BK. 2015.** Macrophyte preference and aquatic entomofaunal diversity of Kapla Beel, A fresh water wetland of Barpeta District of Assam, India. *Asian Journal of Science and Technology*. **6(3):**1205-1210.
4. **Mohd. Rasdi, Z., Fauziah, I., Ismail, R., Mohd Hafezen, S., Fairuz, K., Hazmi, A.D & Che Samah, M.R. 2012.** Diversity of Aquatic Insects in Keniam River, National Park, Pahang, Malaysia. *Asian Journal of Agriculture and Rural Development*. **2(3):** 312-328.
5. **Arimoro FO, Ikoni RB. 2008.** Ecological integrity of upper Warri River, Niger Delta using aquatic insects as bioindicators, *Ecological Indicators*. **395:**1 -7
6. **Majumdar J, Das K, Majumdar P, Ghosh D, Agarwala BK. 2013.** Aquatic Insect fauna and diversity in urban fresh water lakes of Tripura, Northeast India, *Middle-East Journal of Scientific Research*. **13(1):** 25-32.
7. **Zarei, H and Bilondi, M.P. 2013.** Factor analysis of chemical composition in the Karoon River basin, southeast of Iran. *Applied Water Science.* **3:** 753-761.
8. **Subramanian KA, Sivaramakrishnan KG. 2007.** Aquatic insects for Biomonitoring Freshwater ecosystem-A Methodology Manual. Bangalore, India: Asoka Trust for Research in Ecology and Environment.
9. **Imms** : A general textbook of Entomology.
10. **Mani, M.S. 1994.** General Entomology (3rd Edn.). Oxford and IBH publishing co Pvt. Ltd. New Delhi-Bombay-Calcutta, 525.
11. **Borror, D.J., Delong DM. 2005.** Introduction to the study of insects. Edn 7. Norman F. Johnson and Charles A. Triplehorn. Publisher: Peter Marshal, Thomson, Brooks code, 2005.
12. **American Public Health Association (APHA). 2005.** Standard method for the Examination of water and wastewater, 21st edn. Washington, D.C. USA.
13. **Vijayan V.S. 1980-1991.** Keoladeo National park Ecology Study. Final report BNHS, Bombay.
14. **Ajao, E.A. 1990.** The influence of domestic and Industrial effluents on the population of sessile and benthic organisms in Lagos Lagoon. Ph.D. Thesis, University of Ibadan, Ibadan, pp: 411.
15. **Oben, B.O. 2000.** Limnological assessment of the impact of agricultural and domestic effluents of three man-made lakes in Ibadan, Nigeria. Ph.D Thesis, University of Ibadan, Nigeria, pp : 344.
16. **Korkeamaki, E. and Suhonen, J. 2002.** Distribution and habitat specialization of species affect local extinction in dragonfly odonata populations. *Ecography*. **25:** 459-465
17. **Yakub, A.S. 2004.** Assessment of water quality and Plankton of Effluent receiving lower Awba stream and Reservoir. Ibadan. *Afr. I. Appl. Zool. Environ. Biol*. **6:** 107-110.

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# Study of fluorosis problems in Hardia Sector D (Jajpur Village) of Rajauli Block of Nawada District, Bihar

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**Abstract :** Water contamination is a major problem in the present scenario. The groundwater of many sources have become polluted. The contamination is due to both natural reasons as well as due to human activities. The groundwater of some villages of Rajauli block of Nawada district, Bihar is highly contaminated with fluoride - the intake of which has resulted into a severe bone deformity called fluorosis. The maximum permissible limit of fluoride in groundwater as per the WHO guidelines is 1.5 mg/L. However many groundwater sources of Hardia Sector D (Jajpur) have fluoride level more than the permissible limit and in one of the sources it has reached the value of 3.59 mg/L. Many visible cases of fluorosis have been found in the village.

**Keywords : contamination, deformity, fluoride, fluorosis**

## INTRODUCTION

Stress on fresh water is one of the major problems faced these days. Only 0.5% of the global water is available for human use. Major part of the water is present in seas and oceans which is not suitable for domestic consumption.[1] Water pollution is another great challenge to face. Water in many sources-both surface water and groundwater has got polluted. Human beings have played a major role in polluting the water sources. Surface water is exposed to toxic contaminants. The groundwater in many sources too is highly contaminated.[2] Several toxic chemicals including heavy metals have entered the groundwater sources.[3,4] Fluoride ($F^-$) ions beyond the permissible limits have also been found in many groundwater sources in different parts of the country.[5]

Groundwater contaminated by fluoride has been found in 20 states of India including Bihar and Jharkhand. 11 districts of Bihar have high groundwater fluoride contamination.[6] The district of Nawada is one among the 11 districts of Bihar where groundwater is contaminated by fluoride ions.The Rajauli block of Nawada district is one of the worst fluorosis endemic areas. Hardia Sector D (Jajpur) located 5 km away from sub-district headquarter Rajauli is affected by fluorosis. It is located at 24.60°N latitude and 85.51°E longitude.[7]

The intake of fluoride rich water leads to a medical condition called fluorosis. The maximum permissible limit

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of fluoride in drinking water as per the WHO guidelines is 1.5 mg/L.[8,9] But maximum groundwater sources have their fluoride level higher than the maximum permissible limit of 1.5 mg/L and in one source it has touched the value of 3.59 mg/L. Clean and fluoride free water is an essential requirement in the village. However this facility has recently been started. The water of Phulwaria Dam is now being made fluoride free and supplied to the villagers. Certain steps must be taken by both governmental and non-governmental organisations in providing nutritious food to the people.

Fluorosis can be divided into four categories- dental fluorosis, skeletal fluorosis, non-skeletal fluorosis and muscular fluorosis.

**Dental fluorosis**: The initial symptoms of this disease are the appearance of white patches in the teeth. Brown stains appear in mild conditions. Severe cases include the degradation of the brown stained teeth and in some cases complete breaking of the affected teeth.[10]

**Skeletal fluorosis**: It is a bone disease caused by excessive deposition of fluoride ions in the bones. It results in deformity of bones. Some people suffering from skeletal fluorosis have knock knee while others have bow legs.[11,12]

**Non-skeletal fluorosis**: In this type of fluorosis pain in the joints occur which causes painful movement near the joints. Abdominal problems also results in many cases. Non-ulcer dyspepsia, viz. nausea, vomiting, pain in the stomach, bloated feeling/gas formation in the stomach, constipation followed by diarrhoea, may be due to fluoride toxicity.[13,14]

**Muscular fluorosis**: Symptoms include stiffness and pain in the muscle and loss of muscle power.[15,16]

## MATERIALS & METHODS

The method of sampling and testing includes the following processes:

1. Collection of water samples from different groundwater sources of Hardia Sector D.

2. Testing of the samples for fluoride contamination.

**1. Collection of water samples from different groundwater sources of Hardia Sector D**

The different groundwater sources from Hardia Sector D (Jajpur) where the groundwater is contaminated by fluoride was identified by sampling and testing the groundwater sources by the 'Aquasol Kit' (Rakiro Biotech Sys Private Limited). Water from different groundwater sources were collected during June 2019. Water was collected from hand pumps and bore wells- the depth of which ranged from 80 ft to 130 ft.

**2. Testing of the samples for fluoride contamination**

Water samples were collected in 1 L size plastic bottles. A total of 20 samples of different drinking water sources were analysed. The samples were then analysed in the Centre for Fluorosis Research, Department of Chemistry, A.N. College, Patna. The samples were analysed by the Ion Selective Electrode Method.[17,18]



**Fig. 1**



**Fig. 2**

**Fig .1 and 2- Skeletal Fluorosis in Hardia Sector D**

## RESULTS & DISCUSSION

On testing the samples for fluoride it was found that maximum groundwater sources of the village were contaminated with fluoride. Most of the groundwater sources had the fluoride level more than the permissible limit of 1.5 mg/L. out of 20 samples taken from 20 groundwater sources only 1 source had the fluoride level within the permissible limit. All the remaining sources were fluoride contaminated. In one of the sources the fluoride was found to be as high as 3.59 mg/L.

**Table 1- The level of fluoride in various water samples of Hardia Sector D (Jajpur) are as follows**

| Sample no | Concentration of fluoride (mg/L) |
|---|---|
| 01 | 3.51 |
| 02 | 1.64 |
| 03 | 2.61 |
| 04 | 2.31 |
| 05 | 2.54 |
| 06 | 2.38 |
| 07 | 1.88 |
| 08 | 1.13 |
| 09 | 3.58 |
| 10 | 1.85 |
| 11 | 3.07 |
| 12 | 1.75 |
| 13 | 2.21 |
| 14 | 3.59 |
| 15 | 1.98 |
| 16 | 1.95 |
| 17 | 3.35 |
| 18 | 3.35 |
| 29 | 3.13 |
| 20 | 2.68 |

Most of the people during the health survey reported dental problems, muscular pains, joint pains, tooth discolouration, bone deformities and many more ailments related to fluorosis. Some cases of knock knee and bow legs were also observed. Lack of proper diet has also added to the severity of the disease.

## CONCLUSION

Maximum groundwater sources in Hardia Sector D (Jajpur) are highly contaminated by fluoride which has resulted into fluorosis. All the four types of fluorosis affected people were observed in the village.

## REFERENCES


1. **Durack, P. J. 2015.** Ocean salinity and the global water cycle. *Oceanography*. **28(1):**20-31.
2. **Ojo, O. I., Otieno, F. A., & Ochieng, G. M. 2012.** Groundwater: Characteristics, qualities, pollutions and treatments: An overview. *International Journal of Water Resources and Environmental Engineering*. **4(6):**162-170.
3. **Dutta, Anjali; Dutta, Abhijit & Mukherjee, P. 2017.** Physico-Chemical Analysis of water in and around IISCO Mining area of Kiriburu. *The Biobrio*. **4(3&4):** 293-298.
4. **Paul, D. K., & Mukherjee, P. 2006.** A preliminary study of physicochemical characteristics of a perennial fish pond. *Journal of Hematology and Ecotoxicology*. **1(1):**4-56.
5. **Ali, S., Fakhri, Y., Golbini, M., Thakur, S. K., Alinejad, A., Parseh, I. & Bhattacharya, P. 2019.** Concentration of fluoride in groundwater of India: a systematic review, meta-analysis and risk assessment. Groundwater for Sustainable Development, **9:**100224.
6. **Adimalla, N., Venkatayogi, S., & Das, S. V. G. 2019.** Assessment of fluoride contamination and distribution: a case study from a rural part of Andhra Pradesh, India. *Applied Water Science*. **9(4):**1-15.
7. **https://nawada.nic.in/about-district/district-profile/**
8. **Tiwari, A. K., Singh, A. K., & Mahato, M. K. 2017.** GIS based evaluation of fluoride contamination and assessment of fluoride exposure dose in groundwater of a district in Uttar Pradesh, India. Human and Ecological Risk Assessment: *An International Journal*. **23(1):**56-66.
9. **WHO. 1996.** Guideline for drinking water quality, second edition, vol. 2, Health criteria and other supporting information,World Health Organization, Geneva.
10. **Revelo-Mejía, I. A., Hardisson, A., Rubio, C., Gutiérrez, Á. J., & Paz, S. 2020.** Dental fluorosis: the risk of misdiagnosis-a Review. *Biological Trace Element Research*. 1-9.
11. **Ramesh, M., Malathi, N., Ramesh, K., Aruna, R. M., & Kuruvilla, S. 2017.** Comparative evaluation of

dental and skeletal fluorosis in an endemic fluorosed district, Salem, Tamil Nadu. *Journal of Pharmacy & Bioallied Sciences.* **9(Suppl 1):** S88.

12. **Mondal, D., Dutta, G., & Gupta, S. 2016.** Inferring the fluoride hydrogeochemistry and effect of consuming fluoride-contaminated drinking water on human health in some endemic areas of Birbhum district, West Bengal. *Environmental Geochemistry and Health.* **38(2):** 557-576.
13. **Shruthi, M. N., & Anil, N. S. 2018.** A comparative study of dental fluorosis and non-skeletal manifestations of fluorosis in areas with different water fluoride concentrations in rural Kolar. *Journal of Family Medicine and Primary Care,* **7(6):**1222.
14. **Patil, M. M., Lakhkar, B. B., & Patil, S. S. 2018.** Curse of fluorosis. *The Indian Journal of Pediatrics*. **85(5):** 375-383.
15. **Susheela A.K. 1998.** Proceedings of the 1st International Specialized Conference on Water Quality and its Management, New Delhi, 292-300.
16. **Kumar A. 2018.** *Journal of Applicable Chemistry*. **7(6):**1795-1804
17. **Mendes, A. L. G., Nascimento, M. S., Picoloto, R. S., Flores, E. M., & Mello, P. A. 2020.** A sample preparation method for fluoride detection by potentiometry with ion-selective electrode in medicinal plants. *Journal of Fluorine Chemistry*. **231:**109459.
18. **Jaworska, E., Michalska, A., & Maksymiuk, K. 2018.** Fluorimetric readout of ion-selective electrode potential changes. *Electrochimica Acta*. **284:**321-327.

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# Impact of *Citrus limonum* seed on sperm parameters of male albino rats

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**Abstract :** Evaluation of herbs has been in progress worldwide for several decades to identify effective and safe substances for fertility regulation. This approach proved to be a good alternative to synthetic drugs. The present study was carried out to find out the antifertility activity of ethanolic extract of *Citrus limonum* seeds. The extract was administered orally for 30 days. The extract showed a significant reduction in sperm count. ($p<0.001$) at a dose level of 500mg/kg b.wt. Therefore it is suggested that lemon seeds can be a good alternative for regulating sperm parameters.

**Keywords : *Citrus limonum*, sperm count, normality**

## INTRODUCTION

Exploration of drugs having anti-fertility activity is the need of current time, and many time plant extracts have been investigated for their anti-fertility effect in female and male albino mice. *Citrus limonum* commonly known as lemon, belongs to the citrus-Linn (Ruteaceae) family. Lemon peel is used as stomachic and carminative. Anti-pneumonia factor is said to be present in lemon juice and it is also used in the treatment of acute tropical dysentery and diarrhoea. Citrus fruits are rich in vitamin C and in mineral salts.[1] *Citrus limonum* seeds have shown significant antifertility activity in female albino mice.[2,3] In the present study ethanolic extract of seed of the native species *Citrus limonum* is investigated for its anti-fertility activity in male wistar rats.

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## MATERIALS & METHOD

Fresh seeds of *Citrus limonum* were obtained from Birsa Agriculture College and were authenticated in the Department of Botany, Ranchi University, Ranchi. Ethanol alcoholic extract which prepared by taking 50 gm of powdered sample that was extracted in soxhlet by (400) mL of ethanol (95%) solvent. then evaporated at 45°C by rotary evaporator, and further transferred into sterile bottles and refrigerated at 4°C until used.

Adult albino mice of Wistar strain weighing between 150-200 gms housed in clean environment under 12 hour light and 12 hour dark cycle, having free access to food pellet and water *ad-libitum*, were used after approval of the protocol by the Institutional Animal Ethical Committee.

**Treatment phase**

30 male albino rats were divided into three groups (n=10). Each group was kept in a separate cage. Group I served as control. Group II and group III were treated with

lemon seed extract at a dose level of 250mg/kg and 500mg/kg b.w. respectively for 30 days.

**Assessment of sperm count and morphology**

Sperm from the cauda epididymis were released by cutting and was suspended into 2mL of medium (Hams F10) containing 0.5% bovine serum albumin.[4] After 5min incubation at 37ºC the cauda epididymis sperm reserves were determined using the standard hemocytometric method.[5] Sperm motility was analysed using a microscope (Olympus IX7) at 10 successive fields for each sample and reported as the percentage mean of the motile sperm of each sample according to the World Health Organisation (WHO) method. Sperm morphology was determined by the method explained in WHO method manual.

The data was statistically analysed using Student's't' test.

## RESULT & DISCUSSION

Table 1 shows the result of the study on semen parameters. Sperm count significantly decreased from 42.0±4.1 to 31.3±4 and 27.9±5 in group II (low dose animals) and group III animals (high dose animals) respectively. However there were minor differences observed in sperm morphology.

**Table 1: Effect of *Citrus limonum* seed extract on sperm parameters of adult male albino rats**

| Parameters | GROUP I (Control) | GROUP II (250mg/kg b.w) | GROUP III (500mg/kg b.w) |
|---|---|---|---|
| Sperm count (million/ml) | 42.0±4.1 | 31.3±4** | 27.9±5** |
| Sperm normality (%) | 48.7±2.3 | 47.12±1.9 | 45.4±3.1ns |

Values are expressed as mean ± S.D, n =10.
where, **p<0.001 & ns = non significant

The decrease in sperm count may be due to the interference of extract with spermatogenesis and epididymal functions. These decreases are suggestive of alternation in sperm maturation and sperm production.[6] The present result is in agreement with studies related to various other plants extract.[7-9] The sperm count is considered as one of the important parameters to assess the effect of any agents on spermatogenesis.[10]

## REFERENCES

1. Wealth of India Raw materials Council of Scientific and Industrial Research, New Delhi 1950: Vol. II. p. 77–82.
2. **Patil, S. J. & Patil, S. B. 2010.** Pre-clinical toxicity studies of orally administered petroleum ether extract of *Citrus medica* seeds on the reproductive organs of female mice. *International Journal for contempery Reserach. Review*. **1:**1-6.
3. **Patil, S. J. & Patil, S. B. 2011.** Toxicity studies on Hepatic, Nephric and Endocrine Organs of *Citrus medica* seeds extract on female Albino Mice. *Global Journal of Pharma Technology.* **3:** 14-21.
4. **Wang, R. S., Ohtani, K., Suda, M. & Nakajima, T. 2006.** Inhibitory effect of ethylene glycol monoethyl ether on rat sperm motion. *Ind. Health.* **44:** 665-668.
5. **Aman, R. P. and Almquist, J. O. 1961.** Reproductive capacity of Dairy Bulls. I. Technique for direct measurement of gonadal and extra-gonadal sperm reserves. *J. Dairy Sci*. **44:** 1537-1545.
6. **Sarkar, M., Gangopadhyay, P., Basak, B., Chakrabarthy, K., Banerji, J., Adhikari, P. & Chatterjee, A. 2000.** The reversible antifertility effect of Piper betle Linn. on Swiss albino male mice. *Contraception*. **62:** 271-174.
7. **Njar, V. C., Alao, T. O, Okogun, J. I., Raji, Y., Bolarinwa, A. F. & Nduka, E. V. 1995.** Antifertility activity of *Quassia amara*: quassin inhibits the setroidogenesis in rat Leydig cells in vitro. *Planta Med*., **61:**180-182.
8. **Raji, Y. & Bolarinwa, A. F. 1997.** Antifertility Activity of *Quassia amara* in Male Rats-in Vivo Study. *Life Sci*. **61:**1067-1074.
9. **Parveen, S., Das, S., Kundra, C. P. & Pereira, B. M. 2002.** A comprehensive evaluation of the *reproductive* toxicity of *Quassia amara* in male rats. *Reprod. Toxicol*. **17:** 45-50.
10. **Reddy, P. S. Pushpalatha, T. and Reddy, P. S. 2006.** Reduction of spermatogenesis and steroidogenesis in mice after fentin and fenbutation administration. *Toxicol Lett.* **166:** 53-59.

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# Ethology and Phenology of Orchids of Jharkhand with special reference to *Vanda*

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**Abstract :** The present paper deals with three species of *Vanda* from the Jharkhand which are ethno medicinal and ornamental. The detail ethology and reproductive phenology are discussed. Comparative study and characteristics between different *Vanda* species have been discussed. Orchids *Vanda* species are *Vanda tesselata, Vanda testasis* and *Vanda nana*. These three species are the main species of *Vanda* which are found in Jharkhand. Vegetative characters reveal sizeable variation among *Vanda* Orchids which can be differentiated from one another. The length of the epiphytic root ranges from 10 cm to 15 cm. The length was recorded in stem of *Vanda tesselata* 30-80 cm, *Vanda testasis* 25-70 cm and *Vanda nana* 15-30 cm. The maximum length was recorded in *Vanda tesselata*. The sessile lanceolate leaves with maximum length the raceme inflorescence found in these *Vanda* families. The flowers were exquisite and showy which exhibit different shapes and colors, the largest flower in *Vanda* are of *Vanda testasis*. Most of the species began to sprout new shoots during rainy season to winter season, the high temperature to low temperature rainfall and humidity promotes the flower bud initiation. The longevity of flowers ranges 3 to 5 Weeks.

**Keywords :** ***Vanda*, ethnomedicinal, ethnology, phenology.**

## INTRODUCTION

The study of periodically occurring phenomena in plants in relation to the climate and changes of season is known as Phenology. Leith (1970)[1], Leith and Radford (1971)[2] first discussed the concept of phenology. Here the phenological reproductive study deals with vegetative and phase corresponding to the climate and seasonal changes in particular area and it determines the degree of reproductive synchrony with other plant species (Rathcke,1988a,b)[3,4]. Synchrony among species might be advantageous in which the presence of one species facilitates increase in pollinators.[5-7] Phenological study is important in plant management and combating afforestation, the flowering phenology differs from species to species in accordance with the ecosystem types.[8,9] The floral initiation in orchids is determined by the genotypes and its interactions with the environmental conditions such as temperature, humidity, light and photoperiod.[10] The phenological studies reflect the daily occurrence of plants and animals in total response to environment. In plants it includes vegetative and reproductive such as bud formations flowering, fruiting and seed germination, along with vegetative phase like leaf flushing and shedding. These are based on observation of periodic phenomena occurring at a given location over a period of several years.

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The micro climatic study and get actual conditions that influence plant response, the external factors modify the internal factors, among these factors are rainfall, temperature, humidity, soil moisture at different depth, light and its duration, reserve food materials etc.

## MATERIALS & METHODS

*Vanda* belongs to Orchidaceae family there are about 80 species *Vanda* Orchids is found large scale in Jharkhand naturally. The taxonomic history of genus *Vanda* is convoluted and genus has been described most species in *Vanda* genus are (Epiphytes) meaning they grow on other plants without harming the host plants by clinging to trees. Three species are *Vanda tesselata, Vanda testasis* and *Vanda nana*. These *Vanda* species are collected from natural habitat found in the shadow of tree. *Vanda* has access to more sunlight in forest. There are variations in the size of *Vanda. Vanda* can be spread 10 to 15 feet upto half of the tree. *Vanda* grows in cluster and has unique characters. Survey work and data collection for their local abundance, conditions, habitat and flower have been done. The plants were collected from their natural habitats for the experimental studies, 3 healthy plants of each species were maintained in the forest, the choronological events of each species, the phenological detailed studies of both vegetative and phenological observation were made for two years.

**A. Vegetative morphological characters :**

1. Shoot height : The height of the shoot were measured from the basal to the stem and recorded in a standard unit
2. Leaf : (a) shape, (b) apex (c) size
   The length and breadth of the leaves of three *Vanda* species were measured.
3. Inflorescence (a) type (b) position and length measurements of inflorescence were taken in cm from the point of emergence from the nodes to the end of the axis.
4. Floral parts: measurements of each part of flower were taken.
5. Seed pod: length measurements were taken from the point of attachment to the stalk to the tip of the capsule.

**(B) Phenological characters:**

The phenological observations were made from Ranchi in three phases: - phenophases, prepollination and post pollination. The prepollination observations were made from three fully open flowers and three full grown plants of each species. The meteorological parameters of the study period are as follows.

**Meteorological Data**

**Table 1. Meteorological data of the 1st year**

| Sl. No. | Month | Temperature in °C monthly average | | Humidity monthly average | | Rainfall average (mm) |
|---|---|---|---|---|---|---|
| | | Minimum | Maximum | Morning | Evening | |
| 1 | January | 19 | 30 | 31 | 50 | 30 |
| 2 | February | 25 | 30 | 32 | 70 | 40 |
| 3 | March | 30 | 35 | 34 | 54 | 30 |
| 4 | April | 35 | 37 | 31 | 54 | 20 |
| 5 | May | 40 | 45 | 37 | 57 | 20 |
| 6 | June | 35 | 40 | 43 | 63 | 70 |
| 7 | July | 38 | 45 | 47 | 67 | 80 |
| 8 | August | 33 | 40 | 45 | 57 | 60 |
| 9 | September | 30 | 35 | 39 | 49 | 70 |
| 10 | October | 25 | 30 | 31 | 51 | 80 |
| 11 | November | 23 | 30 | 32 | 50 | 90.4 |
| 12 | December | 22 | 30 | 45 | 75 | 53.5 |

**Table 2. Meterological data of the 2nd year**

| Sl. No. | Month | Temperature in °C monthly average | | Humidity monthly average | | Rainfall average (mm) |
|---|---|---|---|---|---|---|
| | | Minimum | Maximum | Morning | Evening | |
| 1 | January | 18 | 30 | 30 | 60 | 40 |
| 2 | February | 24 | 30 | 31 | 70 | 50 |
| 3 | March | 31 | 35 | 35 | 60 | 30 |
| 4 | April | 32 | 35 | 32 | 55 | 20 |
| 5 | May | 38 | 40 | 35 | 60 | 70 |
| 6 | June | 30 | 35 | 40 | 50 | 80 |
| 7 | July | 35 | 40 | 45 | 60 | 85 |
| 8 | August | 30 | 35 | 45 | 70 | 75 |
| 9 | September | 35 | 40 | 40 | 65 | 60 |
| 10 | October | 25 | 30 | 30 | 50 | 50 |
| 11 | November | 20 | 25 | 35 | 45 | 40 |
| 12 | December | 25 | 30 | 40 | 50 | 30 |

## RESULT AND DISCUSSION

**Table 3. Important vegetative and floral characters.**

| Sl. no. | Name of the species | Shoot Nature | Shape | Apex | Type | Color | Nature | Shape |
|---|---|---|---|---|---|---|---|---|
| 1 | *Vanda tesselata* | Tuberous Shoot branching Aerial roots | Ovoid | T,Vandariner Vium | Racemes | Blue or purple/Dark Brown | Epiphyte | Rambling |
| 2 | *Vanda testasis* | Tuberous Shoot | Ovoid | *trinervium* | Raceme panicle | Yellow | Epiphyte | Rambling |
| 3 | *Vanda nana* | Tuberous Shoot | Ovoid | *trinervium* | Raceme | Pink purple | Epiphyte | Rambling |

**Table 4. Measurement of vegetative and floral characters**

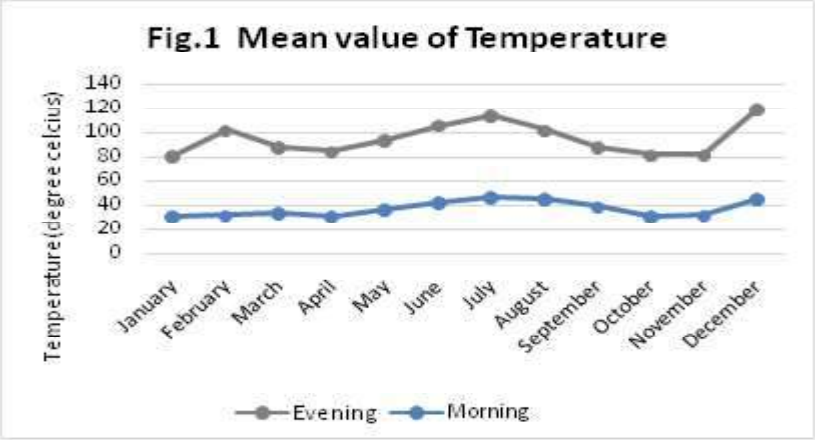
| Sl. No. | Name of the Species | Stem length | Leaf Measurement | | | Flora bract cm | | | Number of flower per inflorescence |
|---|---|---|---|---|---|---|---|---|---|
| | | | L | B | L/B | L/B | B | L/B | |
| **1** | *Vanda tesselata* | 30-80cm | 20 | 10 | 20 | 2 | 2 | 10 | 6-10 |
| **2** | *Vanda testasis* | 25-70cm | 10 | 30 | 15 | 2 | 1.5 | 6.6 | 6-20 |
| **3** | *Vanda nana* | 15-30cm | 5 | 40 | 20 | 2 | 1.5 | 3.3 | 8-10 |

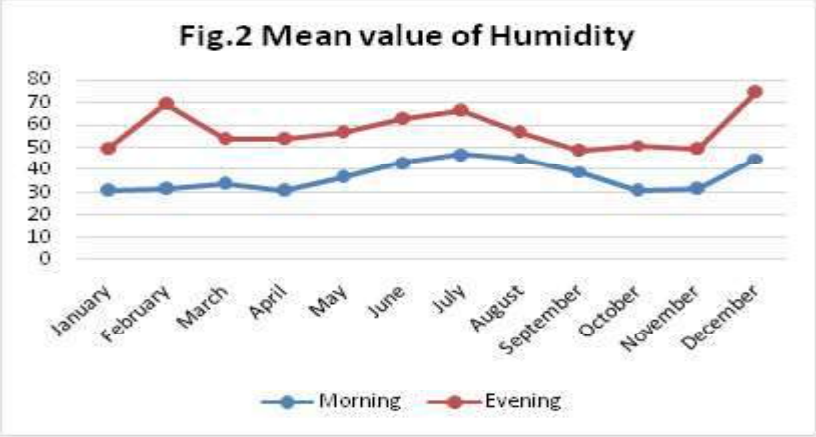
Abbreviations:- L=Length, B=Breadth, L/B=Length/Breadth Ratio

**Table 5. Phonological Observations of Reproductive & Vegetative Phenophase**

| Sl. no. | Name of species | *A Month | *B Month | *C Days | *D Days | *E Month | *F Days | *G Days | *H Month | *I Month | *J Month |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | *Vanda tesselata* | May-June | July-Aug | Aug-Sept | Sept-Oct | Oct-Dec | Nov-Dec | Dec-Jan | Jan-Feb | Feb-Mar | Mar-Apr |
| 2. | *Vanda testasis* | June-July | Aug-Sept | Sept-Oct | Oct-Nov | Nov-Dec | Dec-Jan | Jan-Feb | Feb-Mar | Mar-Apr | Apr-May |
| 3. | *Vanda nana* | April-May | May-June | June-July | July-Aug | Aug-Oct | Oct-Nov | Nov-Dec | Dec-Jan | Jan-Feb | Jan-Mar |

*A=Initiation of Shoot, *B=Initiation of flower buds, *C= Initiation of flower buds of Anthesis, *D=Longevity of individual flower, *E=Flowering Seasons, *F= Initiation of seed pod after of pollination, *G=Maturation of seed pod *H=Brovoning of seed pod, *I=Brusting of seed pod, *J=Dehiscence of seed pod



**Reproductive Phenology Observation**:-

All these three *Vanda* species varies with respect to time. In *Vanda tesselata* shoot initiates in month of May to July flower initiates in month from July to August. Flower buds of anthesis initiates from month of August to September. Flower grows from September to October. Flowering blooms from October to December, initiation of seed pod in December to January. Browning of seed pod is from January to February. Bursting of seed pod is from February to March. Dehiscence of seed pod is from March to April. In *Vanda testasis* initiation of shoot is from June to July due to heavy rain fall and climatic condition. Initiation of flowering buds in August to September. Initiation of flower buds of anthesis. Flower grows from October to November, in some cases in December also. Initiation of seed pod after pollination is from December to January. Maturation of seed pod is from January to February. Browning of seed pod is from February to March. Bursting of seed pod is from March to April. Dehiscence of seed pod is from March to April. In *Vanda nana* shoot initiates in month of April to May flower initiates in month from June. Flower buds of anthesis initiates from month of June to July. Flower grows from September to October. Initiation of seed pod is from December to January. Bursting of seed pod is from January to February. Dehiscence of seed pod is from January to March.

## CONCLUSION

*Vanda* is important Orchids and can adapt easily in the environment of Jharkhand. The environmental factor plays a major role in the Phenological studies. Tribes of Jharkhand use *Vanda* in eye drop and ear drop. Data obtained in the study are useful for conservation of *Vanda* orchids for medicinal and floriculture value.

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## REFERENCES

1. **Leith, H. 1970.** Phenology in productivity studies. In: D. Reichle (eds.). Analysis of Temperate Forest Ecosystems. Heidelberg. Spriner-Verlag.
2. **Leith, H. and J.S. Radford. 1971.** Phenology, resource management and synagraphis computer mapping. *Bio. Sci.* **21:**62-70.
3. **Rathcke, B. and E.P Lacey. 1988a.** Interactions for pollination among coflowering shrubs. *Ecology*. **69:**446-446.
4. **Rathcke, B. and E.P Lacey. 1988b.** Flowering phenologies in a shrub community: competition and constraints. *J. Ecology*. **76:** 975-975.
5. **Thomson, J.D. 1980.** Skewed flowering distributions and pollinator attraction. *Ecology*. **61:**572-572.
6. **Thomson, J.D. 1982.** Patterns of visitation by animal pollinators. *Oikos*. **39:**241-241.
7. **Rathcke, B. and E.P Lacey. 1985.** Phenological patterns of terrestrial plants. *Annu. Rev. Ecol. Syst*. **16:**179-214.
8. **Pojar. 1974.** Reproductive dynamic of four plant communities of Southwestern British Columbia. *Can. J. Bot.* **52:**1819-1834.
9. **Heinrich, B. 1976.** Flowering phenologies: bog, woodland, and disturbed habitats. *Ecology*. **57:** 890-899.
10. **Bose, T.K. and L.P. Yadav. 1992.** Physiology of flowering of orchids. In: S.P.Vij (eds.). Biology, conservation and culture of Orchids. East West Press. Delhi. 79-82.

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# Determination of chlorpyrifos residue in water and liver and muscle of *Clarias batrachus* by HPLC with UV detection

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**Abstract :** The present study was conducted for the determination of chlorpyrifos residues in water and tissue (liver and muscle) of the fish *Clarias batrachus* by HPLC with UV detection. The fish were exposed to 1/5th sublethal concentration of chlorpyrifos i.e., 0.258 mg/L for 96 hr. Water was collected at 0, 24, 48, 72 and 96 hr after the addition of toxicant to correlate the relationship between the concentration added and the concentration left in water. The liver and muscle tissues were collected after 24, 48, 72 and 96 hr for determination of chlorpyrifos residue accumulated in the tissue. Samples were extracted, cleaned up and purified. There was a decrease of chlorpyrifos residue in water and increase in concentration of chlorpyrifos in liver and muscle with increase in time of exposure, which is a sign of bioaccumulation of the toxicant in the fish through uptake. Also, the accumulation was more in the liver than muscle of the fish.

**Keywords : Chlorpyrifos, HPLC, *Clarias batrachus*, bioaccumulation**

## INTRODUCTION

Pesticides are global pollutant of the aquatic environment and have detrimental effects on the living resources. The pesticides which are discharged into the aquatic ecosystem can contaminate fish and in turn man.[1] Aquatic organism's response to insecticides depends on the compounds, exposure time, water quality and species.[2-4] Many pesticides have been banned in India and other advanced countries due to their persistence and toxicity towards non-targeted organisms. Hence, production and use of permitted insecticides could increase considerably.[5] Mortality is obviously not the only endpoint to consider and there is a growing interest in the estimation of residues of the pesticides in the non-target beneficial organism to assess the sub-lethal effects of toxicant. Residue analysis helps to provide information on the action and effects of this toxicant on aquatic life especially in fish. In the present study, an attempt has been made to quantitively evaluate the residue of chlorpyrifos, O, O-diethyl-O-(3,5,6-trichloro-2-pyridyl) phosphorothioate; an organophosphate insecticide on physiologically important tissues of fish, viz. liver and muscle, exposed to sublethal concentration (0.258mg/L; 1/5th of $LC_{50}$) and also to study the residue of chlorpyrifos left in water after its addition. The fish selected for the present investigation was the walking catfish, *Clarias batrachus* L., an edible fish which is economically important in pisciculture.

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## MATERIALS & METHODS

### Fish collection and acclimatization

Healthy walking catfish, *C. batrachus* measuring 35.7±7.7 g in weight and 17.1±1.4 cm in length were bought from the local fishermen from Ranchi district, Jharkhand India. The fishes were subjected to a prophylactic treatment by bathing twice in 0.05% $KMnO_4$ for two minutes to avoid dermal infection. The specimens were kept in glass aquaria containing 20L dechlorinated tap water and were left for acclimatization for two weeks under laboratory condition in semi-static system. The experiment was set under the photoperiod of 13hr light and 11hr dark along with temperature of 24.93±1.8ºC, pH 7.07±0.33 and dissolved oxygen 8.08±1.6 mg/L. Fishes were fed with commercial diet twice daily and feeding was stopped 24hr before the experiment.

### Test solution

For the present study both the technical grade pesticide and commercial formulation of chlorpyrifos (20% EC) with trade name ELDRIN$^{TC}$ manufactured by Crystal Crop Protection Ltd. was purchased from the local pesticide market from Ranchi and used.

### Experimental design

Previous study conducted by Kujur *et al*. (2019)[6] reported $LC_{50}$ 96hr value of commercial grade of chlorpyrifos in *C. batrachus* to be 1.29 mg/L. In the present study 1/5 of $LC5_0$ 96hr value (0.258 mg/L) was taken and added to the aquarium for determination of residue analysis. Liver and muscle were collected after addition of toxicant at 24, 48, 72 and 96hr for conducting the experiment. Water was analyzed for the concentration of chlorpyrifos at 0, 24, 48, 72 and 96 hr.

### Residue analysis

**Water analysis-** Water samples were collected in 500mL amber glass bottles. The amount of chlorpyrifos residue was analyzed using HPLC following the method described by Hasanuzzaman *et al.* (2017)[7] with some modifications.

**Tissue analysis-** The experimental fish were rinsed with distilled water prior to the dissection to avoid the external pesticide residue. Fish were dissected and liver and muscle were collected. At a time, three fishes were taken and their tissues were pooled for the analysis. Extraction of chlorpyrifos from fish tissue was performed by the method of Rao *et al*., (2003)[5].

### HPLC condition

The amount of chlorpyrifos residues in water, liver and muscle tissues were analyzed using HPLC (Shimadzu class VP.V6.10) fitted with UV detector. The mobile phase used was a water- acetonitrile gradient (70% acetonitrile +30% water in 0.1% 1M acetic acid). The flow rate was 1.5 mL/minute and the run time was 15 minutes. $C_{18}$ column (Chiralpak IB- 250 mm x 4.6 mm, with particle size of 5µm) programmed at 25°C was used. The eluents were monitored by UV detection at wavelength of 280 nm.

### Identification and quantification

The compound was identified by comparing its retention time with respect to technical grade chlorpyrifos standard. Quantification of chlorpyrifos residues was done by calibration curve drawn from chromatographic experiments with standard solution of the pesticide.

## RESULTS

Figure 1 shows the standard HPLC chromatogram of chlorpyrifos. The retention time of chlorpyrifos was 4.992 minute. The calibration curve was generated using linear regression of the measured peak areas vs. the concentrations of the calibration standards (in µg/mL) (Fig 2). The correlation coefficient ($R^2$) was 0.9949

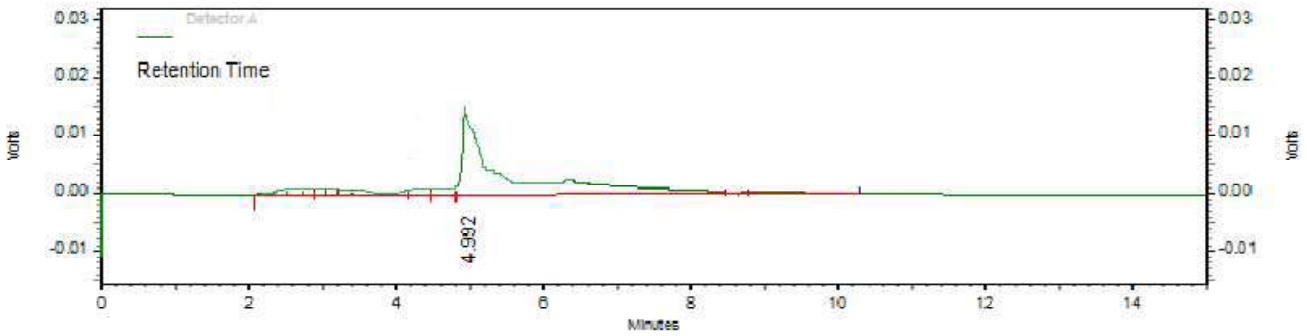



**Figure 1: Standard HPLC chromatogram of chlorpyrifos.**

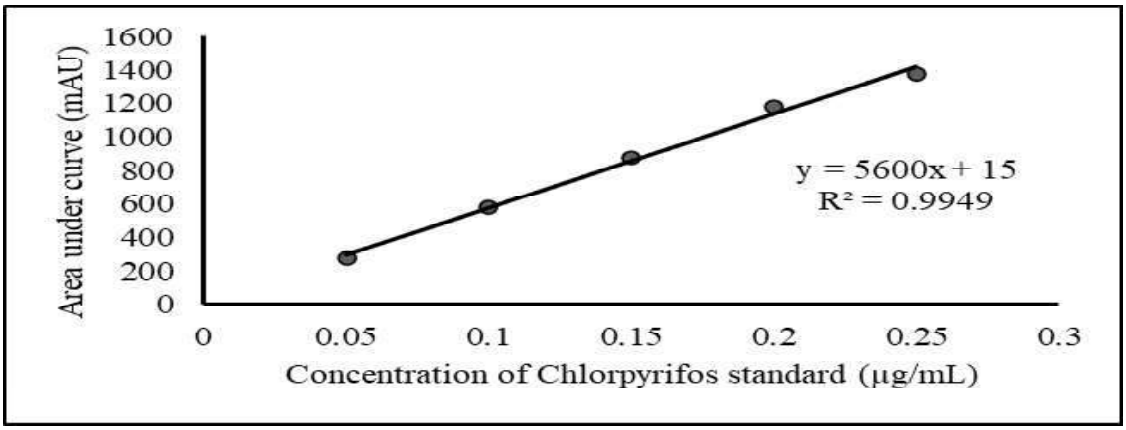



**Figure 2: Calibration curve of standard chlorpyrifos.**

**Table 1: Bio-accumulation of chlorpyrifos in water, liver and muscle of fish *Clarias batrachus*, during sublethal exposure to 0.258 mg/L (1/5 of $LC_{50}$) for a period of 96 hr.**

| Exposure Period (Hours) | Bioaccumulation in different fish tissue | | | | | | Residue present in water | |
|---|---|---|---|---|---|---|---|---|
| | LIVER | | | MUSCLE | | | | |
| | Area (mAU) | Concentration (ug/g) | BCF* | Area (mAU) | Concentration (ug/g) | BCF | Area (mAU) | Concentration (ug/mL) |
| **0 hr** | | | | | | | 1147025 | 204.82 |
| **24 hr** | 47025 | 8.394 | 0.032 | 3379 | 0.6007 | 0.002 | 967025 | 172.6380 |
| **48 hr** | 87025 | 15.5375 | 0.060 | 3711 | 0.66 | 0.002 | 747025 | 133.3946 |
| **72 hr** | 957025 | 170.89 | 0.662 | 5084 | 0.9051 | 0.003 | 477275 | 85.225 |
| **96 hr** | 997025 | 178.035 | 0.690 | 6332 | 1.1280 | 0.004 | 21786 | 3.88 |
| **Average BCF values ± SD** | | | 0.361 ±0.364 | | | 0.002 ±0.0009 | | |

The presence of chlorpyrifos in water was measured after 0, 24, 48, 72 and 96 hr and in liver and muscle chlorpyrifos residues was measured after 24, 48, 72 and 96 hr (Table 1). The amount of chlorpyrifos added to the water initially was 1/5 of $LC_{50}$ 96 hr (0.258 mg/L). Chlorpyrifos residues in water, liver and muscle of fish *Clarias batrachus* is given in Table 1. Table 1 clearly shows that the pesticide residue in the water decreases, while in liver and muscle of fish pesticide residue increases with the lapse of time during the experiment. It is also clear from the results that the accumulation of chlorpyrifos is more in the liver than in muscle.

## DISCUSSION

The present study reports the chlorpyrifos residues detected in water, liver and muscle of *C. batrachus* after sublethal exposure of 1/5 of $LC_{50}$ 96 hr (0.258 mg/L) for 96 hr. chlorpyrifos residue in water refers to its concentration left in the water after it has been added to the water. Concentration of chlorpyrifos remained is analysed to correlate the relationship between the concentration added and the concentration remained.[8] The presence of chlorpyrifos residue in water after 0, 24, 48, 72 and 96 hr were shown in Table 1. A quick degradation of the pesticide was noticed. Manjunatha *et al*. (2015)[8] also observed a gradual decrease in the chlorpyrifos residues in water during the length of exposure. Saad *et al*. (1982)[9] also showed that organophosphorus compounds quickly degrade in the aquatic environment. According to Dow Agro Sciences (2008)[10] chlorpyrifos degrade by hydrolysis and photolysis occurring at moderate rates under neutral conditions with half -lives of about a month at a neutral pH and 25ºC.

According to Tilak *et al.* (2004)[11] residues of organophosphate insecticides in fish species and water depend on the physicochemical characteristics of water, time of consumption, pH of the water and ambient temperature. Various studies have been done indicating the direct toxic effects of chlorpyrifos on various tissues of non- target organisms especially fish. However, chlorpyrifos also indirectly effects by its accumulation in various tissues. In the present study accumulation of chlorpyrifos in liver and muscle of *C. batrachus* was observed after addition of toxicant at 24, 48, 72 and 96 hr. Continuous exposure of fish to chlorpyrifos resulted in bioaccumulation of pesticide in the body of the fish and accumulation was found to be more in liver than in muscle (Table 1). Variation in residue analysis is attributed to factors such as differences in the uptake rate and lipid content of the respective animal tissue. Liver serves as the main detoxifying organ containing relatively high levels of detoxifying enzymes. It is also the first organ to face the effect of pesticides being carried through the portal circulation,[12] which might have been the cause of the greater accumulation of chlorpyrifos in liver than in the muscle.

Liver rapidly synthesizes monooxygenase enzymes during exposure to organic contaminants. Monooxygenase enzymes are found in high concentration in liver, gonads, kidney, intestine, gill and heart.[13] The monooxygenase enzyme facilitates excretion of the pollutants by decreasing the lipid solubility of organic contaminants.[14] The study conducted by Begum *et al.* (1994)[15] on dimethoate bioaccumulation in liver and muscle tissue on *Clarias batrachus* by GC-MS also observed a similar result of higher accumulation in liver than in muscle. In the study conducted by Nagaraju and Rathnama (2014)[12] on residue analysis of carbamate in *Labeo rohita* by GLC-FID, they found that order of carbosulfan residue in different tissues is gill> kidney> liver> muscle at the end of 15 days.

The factor of bioconcentration in liver and muscle were calculated based on the concentration added in the medium (water) and noticed gradual increase in the length of exposure. It may be due to constant vulnerability of fish to chlorpyrifos and subsequent reduction of chlorpyrifos residues in water and their uptake by the fish tissues.

## CONCLUSION

The present study correlated the relationship between the amount of chlorpyrifos added to the water and the amount remained or accumulated in the tissue. The decrease of chlorpyrifos concentration in water and the relative increase in liver and muscle with increase in time of exposure is an indicator of the accumulation of the toxicant in the organism through uptake. This could be hazardous as it could make its way into the food chain. In the present study, the individual secondary metabolites were not considered and further studies are necessary for quantification of the metabolites. A regular surveying must be done on fishes and other aquatic species before the extensive use of this pesticide is permitted to provide a hazard free environment to the aquatic biota and to provide a safe and healthy supply of fish for human consumption.

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## REFERENCES

1. **Metelev VV, Kanaev AI, Dzawkhiva NG. 1983.** In water Toxicology, Amerind Pub.Co Pvt Ltd, New Delhi p.3.
2. **Eisler R. 1970.** Acute toxicities of organochlorine and organophosphorus insecticides to estuarine fishes. *US Bur Sport Fish Wild Tech Pap.* **46:**12.
3. **Fisher SW. 1991.** Changes in the toxicity of the three pesticides as a function of environmental pH and temperature. *Bull Environ Contam Toxicol.* **46:**197-202.
4. **Richmonds CR, Dutta HM. 1992.** Effects of malathion on the brain acetylcholinesterase activity of blue gill sun fish *Lepomis macrochirus*. *Bull Environ Toxicol* **49:**431-435.
5. **Rao JV, Rani CHS, Kavitha P, Rao RN, Madhavendra SS. 2003.** Toxicity of chlorpyrifos to the fish *Oreochromis mossambicus*. *Bull. Environ. Contam. Toxicol*. **70:**985-992.

6. **Kujur P, Samdershi D, Tigga K, Besra S. 2019.** Determination of $LC_{50}$ and behavioral changes in walking catfish, *Clarias batrachus*, exposed to chlorpyrifos. *Journal of experimental Zoology India*. ISSN 0972-0030. **22(2):**1059-1061.

7. **Hasanuzzaman M, Rahman MA, Salam MA. 2017.** Identification and quantification of pesticide residues in water samples of Dhamrai Upazila, Bangladesh. *Appl water Sci.***7:**2681-2688.

8. **Manjunatha B, Tirado JO, Philip GH. 2015.** Determination of chlorpyrifos residues in water and liver tissue of zebrafish (*Danio rerio*) by high performance liquid chromatography (HPLC) with UV detection. *J. of Chemical and Pharmaceutical Research.* **7(6):**721-726.

9. **Saad MAH, Abu El-Amayem MM, El-Sebae AH, Sharaf IF. 1982.** Occurrence and distribution of chemical pollutants in lake Mariut, Egypt. *Water Air Soil Poll.* **17:**245-252.

10. **Dow Agro Sciences. 2008.** www.dowagro.com/chlorp/na/about.

11. **Tilak KS, Veeriah K, Rao DK. 2004.** Toxicity and bioaccumulation of chlorpyrifos in Indian Carp *Catla catla* (Hamilton), *Labeo rohita* (Hamilton) and *Cirrhinus mrigala* (Hamilton). *Bull. Environ, Contam. Toxicol.* **73:**933-941.

12. **Nagaraju B, Rathanamma VV. 2014.** Gas liquid chromatography- flame ionization detector (GLC-FID) residue analysis of carbamate pesticide in freshwater fish *Labeo rohita*. *Toxicol. Res.* **3:**177.

13. **Lindstom-Seppa P, Koivusaari U, Hannien O. 1981.** *Comp. Biochem. Physiol.* **69 C:** 249-53.

14. **Jimener BD, Stegeman JJ. 1990.** Detoxication enzymes as indicators of environmental stress on fish. *American Fisheries Society Symposium.* **8:**67-79.

15. **Begum G, Vijayaraghavan S, Sarma PN, Husain S. 1994.** Study of dimethoate bioaccumulation in liver and muscle tissues of *Clarias batrachus* and its elimination following cessation of exposure. *Pestic. Sci.* **40:**201-205.

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# Demogenetic study on ABO blood group distribution and allelic frequency in tribal population of District Kishanganj, Bihar, India

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**Abstract :** The district Kishanganj is situated in north eastern part of Bihar. It represents 4% tribal population in its total of 16.9 lakhs population. A random stratified sampling was conducted in three administrative blocks to collect blood samples from three major tribe populations namely Santhal, Oraon and Munda. The total tribal population represents phenotypic and genotypic frequencies as B (40.31%)>A(29.61%)>O(18.74%)>AB(11.34%) and $I^0$(0.4553)>$I^B$(0.3094)>$I^A$(0.2351) respectively. The goodness of fit chi square test represents that Santhals of the area are non-homogeneous in ABO blood group distribution ($5\pi^2$ =10.8192, df=1,0.01>p>0.001, significant)whereas Oraon ($5\pi^2$=0.0463, df=1, 0.08>p>0.70, non - significant) and Munda ($5\pi^2$= 0.0131, df = 1, 0.95>p>0.90, non-significant) are homogeneous. The intra-population and inter-population variations are analysed statistically and probable causes are discussed.

**Keywords : ABO Blood group, allelic frequency, distribution, tribal population, Kishanganj**

## INTRODUCTION

In India blood group B is common in northern and western part whereas O is commonly prevalent in eastern, southern and central region. As a whole, O blood group is considered most prevalent among population of Indian subcontinent.[1] The ABO blood group is a multi-allelic character represents co-dominance. The gene responsible for antigenic differentiation is present on chromosome 9 (9p34.1-q34.2) and synthesized ABO glycosyltransferase. There are three hypothesis for genesis of ABO blood groups; primary race hypothesis, O blood group hypothesis and AB blood group hypothesis. The first one represents that all polymorphic forms were originated independently in ancestral population of world isolated geographically like in Europe (A), Asia (B) and South America (O) whereas second and third hypothesis represent that A or AB blood group was present in ancestral stock.[2] The scientist and researchers are not concluded finally that which hypothesis is correct but they are agreed to accept that the human were evolved from African Pliocene fossil *Australopithecus afarensis* like stock and migrated to other parts of globe including India. During course of migration different evolutionary and environmental factors like migration, extra specific struggle, mutation, isolation,

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genetic drift, founder effect, gene flow, survivorship fitness, natural selection etc were responsible for creation of selective pressure on gene pool and consequently changed its composition and nature.[3-5]

The Hardy-Weinberg Principle (HWP) is applicable for a locus carrying two allele p and q. But Bernstein proposed expanded version of the HWP for study of genetic system involve multi alleles. He also presented method for the study of ABO blood group based on the work of Hirschfeld and Hirschfeld.[3,6] A number of research papers have been available on the study of ABO and Rh (D) blood groups.[7-11] On the basis of these studies, it has been suggested that the human gene pool is mostly differ quantitatively in allelic frequency but not in qualitatively in kind. However a few alleles represent prevalence in one population but absent in other. It is well known that gene frequency of any population is regulated by hereditary and counterbalanced by evolutionary forces. The three primary modes of immediate changes in gene frequency have been considered. The first is change due to systematic pressure like recurrent mutation, intrapopulation selection and gene flow; the second is change due to fluctuations like genetic drift and third one is unique events like favourable mutation, unique selective incident, unique reduction in number etc.[3,4,12]

The present paper is based on the study of ABO and Rh(D) phenotypic and allelic frequency in three tribal populations selected from Kishanganj district of Bihar (India). The paper is first report from the district on three populations. The interpopulation and intrapopulation variation in phenotypic and allelic frequencies are analysed and discussed.

## MATERIALS & METHODS

### STUDY AREA

The district Kishanganj is situated in north-eastern part of Bihar and extends from 25°21' to 26°30' north latitude and from 87°7' to 88°19' east longitude. It is boarded by district Araria in the west, Purnea in south west, Uttar Dinajpur (West Bengal) in east and Darjeeling in north east and Nepal in the North. As per census 2011 the total population of the district is 16.9 lakhs in which roughly 4% population is constituted by schedule tribe population. The three blocks Pothia, Bahadurganj and Kishanganj represent 6.6%, 2.4% and 2.7% respectively are selected for present study.

### SAMPLING AND BLOOD GROUPING

The total tribal population as per census 2011 in study area is 64,224 in which the total number of male is 32,506 and total number of female is 31,718. The ¾ tribal population of the district is Santhal followed by Oraon and Munda. All these three endogamous populations were selected for study. The Santhal again represented nine exogamous groups, known as gotra. The sample size of Santhal was 478 where as samples size of Oraon and Munda were 99 and 58 respectively. Santhal population was present in all the three areas selected whereas Oraon and Munda were observed only in Pothia. During sampling it was considered to avoid inclusion of more persons in sample belonged to same lineage for neutralization of sampling biasness.

The samples were collected by stratified random sampling method from selected blocks. The blood was collected by venepuncture from persons of 8 years to 70 years after obtaining consent from person or his/her guardian in the case of minor and carried to laboratory. The personal information, like name, sex, caste, gotra etc, was collected in a pre-formed questionnaire. The blood groups were determined by slide agglutination technique. The reagent used was monoclonal diagnostic test kit.[13,14]

### STATISTICAL ANALYSIS

The ABO blood group allelic frequencies were calculated on the method suggested by Bernstein whereas frequency of Rh(D) was calculated on the method suggested by HWP.[3,12] All calculations were carried in MS-Excel of Microsoft 10 programme. The degree of significance level in ABO blood group distribution in both intra-population (goodness of fit) and inter-population (2x2 contingency table) was analysed by $\pi^2$ values at respective degree of freedom[14]

## RESULT & DISCUSSION

The tribal population of Kishanganj district represents highest frequency of blood group B (40.31%) and lowest frequency of blood group AB (11.34). The blood group frequency is distributed as B>A>O>AB. The population of Santhal and Munda represent same pattern but frequency of blood groups in Oraon is B>O>A>AB (Table-1). If we consider the allelic frequencies ($I^A$, $I^B$, $I^0$) of total tribal population of Kishanganj district as well as Santhal, Oraon and Munda population, the pattern is same and observed as O>B>A (Table-2). It has been reported that tribal

populations were migrated to this area during 18th century from Chhotanagpur plateau of Jharkhand and Hill area of West Bengal as agricultural labourer and tea garden workers. Santhal and Munda were migrated from Chhotanagpur plateau and West Bengal whereas Oraon migrated from Chhotanagpur plateau only. The populations under study were isolated from stem population to a new geographical region and their allelic frequency might be modified due to balancing selection with respect to environmental and evolutionary factors.[3,4,15,16] The goodness off it chi-square test represents that the Oraon and Munda of the district are homogeneous in ABO distribution. The $\pi^2$ value for Oraon (0.0463, df =1, 0.08>p>0.70) and Munda (0.0131 df = 1, 0.95>p>0.90) is statistically non-significant.

Santhals represent phenotypic frequency as 31.17%, 40.78%, 16.95% and 11.09% for A, B, O and AB blood groups. The maximum frequency of B and minimum frequency of AB as well as allelic frequency ($I^A$, $I^B$, $I^0$) of blood groups are same as reported by other workers from Chhotanagpur plateau and West Bengal.[17-20] But the respective frequency of other blood groups A and O represents variations. The Santhals reported from Palamu and Santhal Pargana represent more frequency of O blood group than A whereas Santhals reported from Midnapore, Bihar (Trai) region and present area have more A blood group than O (Table-3). The goodness of fit $\pi^2$ test represents that the Santhals of present area are non-homogeneous in ABO blood group distribution and the variation is statistically significant ($\pi^2$=10.8192, df=1, 0.01>p>0.001). When we analyse the homogeneity between Santhals of present area with other region by 2×2 contingency table we interestingly observe that ABO distribution of Santhals of present area represent statistically significant variation to all other regions except terai region of Bihar (Table-4). The findings suggested that ABO blood groups of Santhals of Kishanganj do not represent homogeneous nature due to non-equilibrium condition of gene frequency with respect to existing environmental and evolutionary factors. Some factors might be operative on particular locus and it required some more work for final conclusion. It has been suggested that the blood groups are not selectively neutral, although the selective forces demonstrated are mostly very weak.[21]

The Oraons of the area represent ABO frequency as B>O>A>AB with their respective values of 38.38 %, 28.28 %, 23.23% and 10.1 %. The goodness of fit $\pi^2$ value represents that Oraons are homogeneous in ABO blood group distribution and variation is statistically non-significant ($\pi^2$=0.0463, df=1, 0.08>p>0.70). The distribution pattern and frequency of ABO blood group of Oraons of present area are same as reported from Palamu, Ranchi and Jalpaiguri[22-24] but differ from Ambikapur (M.P) and Purnea[15,16,25] (Table-3). When we analyse the Oraon of present area and reported from other region by 2×2 contingency table we find that ABO distribution of Oraons of present area represent statistically non-significant variation (Table-4) except with the population reported from Purnea ($\pi^2$ = 12.0949, df = 3, significant at p<0.05) in 1993.[15] In Purnea Oraon represents maximum frequency of blood group $A^-$ and minimum value of blood group AB.

The more frequency of blood group A in Oraon population in Purnea was explained as a mechanism of isolation from parental stock of Chhotanagpur plateau and genetic drift with phenotytic frequency as A>O>B>AB[15] and A>B>O>AB[16,26]. Here it is necessary to mentioned that not only the drift or founder effect but selection is also responsible for change in gene frequency or allelic frequency by increase of reproductive fitness. If the reproductive fitness is non-operative in small population, allele in the population tends towards zero.[27] The allele frequency is an important parameter for the compare of two populations not the phenotypic character.[28] The 95 % confidence level of ABO allelic frequencies for Oraon reported from different regions is calculated as 24.54 ± 8.11, 26.02 ±9.54 and 49.46 ± 7.09 for $I^A$, $I^B$ and $I^o$ respectively. It has been also reported that samples dominated by a single lineage can also lead to deviation from HWP and may be caused some other inferential problems.[29] Hence, a detail study of present variation is necessary for final conclusion otherwise it is considered as a product of sampling error.

The Munda population of the Kishanganj represent distribution of ABO blood group as B (39.66 %) > A (27.59 %) > O (17.24 %) > AB (15.52 %). The goodness of fit $\pi^2$ value (0.0131, df =1, 0.95>p>0.90, non-significant) represents that they are homogeneous in ABO group distribution. The phenotypic frequencies of present population represent similarity except from Mundas of Singhbhum and Ranchi (Table-3). Mundas of present area represent non-significant variation at 0.05 level from Munda of Singbhum, Ranchi, Purnea and Terai of

Bihar[16,30,31] (Table-4) hence the variations are considered as sampling fluctuations.

## CONCLUSION

The tribal population of present area represent the same pattern of ABO blood group distribution as represented by tribal population of north east India like Jharkhand, West Bengal and rest parts of Bihar**.** It suggests that Santhal and Munda represent common origin from Austro-Asiatic group whereas Oraon originated from Dravidian common stock. The variations observed in ABO blood groups in tribal population are considered as chance variation or due to sampling fluctuation. Althrough the variations in Santhals; statistically significant, suggest that the allelic frequencies are not established naturally due to some environmental or evolutionary factor. It requires further study for final conclusion.

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No financial support from any agency and no conflict.

**Table 1- Frequency of ABO and Rh blood group in selected tribal population of Kishanganj District (Bihar)**

| Tribe | Sample Size | A | B | O | AB | Rh +ve | Rh -ve | Phenotypic Frequency |
|---|---|---|---|---|---|---|---|---|
| Santhal | 478 | 149 | 195 | 81 | 53 | 477 | 1 | B > A > O > AB |
| | | 31.17% | 40.78% | 16.95% | 11.09% | 99.80% | 0.20% | |
| Oraon | 99 | 23 | 38 | 28 | 10 | 97 | 1 | B > O > A > AB |
| | | 23.23% | 38.38% | 28.28% | 10% | 99% | 1% | |
| Munda | 58 | 16 | 23 | 1 | 9 | 57 | 1 | B > A > O > AB |
| | | 27.59% | 39.66% | 17.24% | 15.52% | 98.30% | 1.70% | |
| Total | 635 | 188 | 256 | 119 | 72 | 632 | 3 | B > A > O > AB |
| | | 29.61% | 40.31% | 18.74% | 11.34% | 99.52% | 0.48% | |

**Table 2- Allele frequency of ABO group in selected tribal population of Kishanganj District (Bihar)**

| | Santhal | Oraon | Munda | Total |
|---|---|---|---|---|
| N | 478 | 99 | 58 | 635 |
| A ($I^A$) | 0.2452 | 0.1837 | 0.2467 | 0.2351 |
| B ($I^B$) | 0.3128 | 0.2826 | 0.3319 | 0.3094 |
| O ($I^0$) | 0.4417 | 0.5337 | 0.4213 | 0.4553 |
| Frequency | O > B > A | O > B > A | O > B > A | O > B > A |

**Table 3 - Represents phenotype and allele distribution of ABO blood group among tribal population of other area with reference to District Kishanganj (Bihar)**

| Place | N | Phenotype | | | | Allelee | | | Phenotype Frequency | Allele Frequ. | References |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A | B | O | AB | A | B | O | | | |
| **Santhal** | | | | | | | | | | | |
| Kishanganj | 478 | 31.17 | 40.78 | 19.95 | 11.09 | 24.52 | 31.28 | 44.17 | B>A>O>AB | O>B>A | Present Study |
| Palamu | 199 | 20.6 | 35.2 | 32.2 | 12.1 | 17 | 26.3 | 56.7 | B>O>A>AB | O>B>A | Sarkar, 1949 |
| Midanapur | 336 | 27.98 | 36.31 | 25.89 | 9.82 | 21.3 | 26.8 | 51.9 | B>A>O>AB | O>B>A | Choudhury *et al.* 1967 |
| Santhal Paragana | 407 | 21.37 | 35.63 | 31.69 | 11.3 | 17.83 | 26.97 | 55.2 | B>O>A>AB | O>B>A | Sarkar and Sen, 1952 |
| Santhal Paragana | 400 | 23.5 | 38 | 26.25 | 12.25 | 20.2 | 29.8 | 50 | B>O>A>AB | O>B>A | Bhattachrjee, 1968 |
| Bihar (Terai) | 350 | 27.4 | 40.3 | 19.3 | 13 | 22.9 | 31.8 | 45.3 | B>A>O>AB | O>B>A | Pandey *et al.* 2013 |
| **Oraon** | | | | | | | | | | | |
| Kishanganj | 99 | 23.23 | 38.38 | 28.28 | 10 | 18.37 | 28.26 | 53.37 | B>O>A>AB | O>B>A | Present Study |
| Palamu | 115 | 27.83 | 33.91 | 26.09 | 12.17 | 22.52 | 26.55 | 50.93 | B>A>O>AB | O>B>A | Sarkar, 1949 |
| Andamand | 100 | 25 | 38 | 19 | 18 | 24.29 | 33.37 | 42.34 | B>A>O>AB | O>B>A | Lehmann and Ikin, 1953 |
| Ranchi | 125 | 28.4 | 38.2 | 26.3 | 7.1 | 20 | 27 | 53 | B>A>O>AB | O>B>A | Kirk *et al.* 1963 |
| Ambikapur | 478 | 34.51 | 30.54 | 25.94 | 8.99 | 26.3 | 23.7 | 49.9 | A>B>O>AB | O>A>B | Bhatia *et al.* 1986 |
| Jalpaiguri | 119 | 22.72 | 40.34 | 21.1 | 10.92 | 23.43 | 31.76 | 44.81 | B>A>O>AB | O>B>A | Piplai *et al.* 1985 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Purnea** | 117 | 44.44 | 22.22 | 23.93 | 9.4 | 32.34 | 17.46 | 50.19 | A>O>B>AB | O>A>B | Pandey *et al.* 1993 |
| **Purnea** | 202 | 34.65 | 28.71 | 24.76 | 11.88 | 27 | 23 | 50 | A>B>O>AB | O>A>B | Pandey *et al.* 1999 |
| **Bihar (Terai)** | 202 | 34.6 | 29.7 | 24.8 | 10.9 | 26.3 | 23.1 | 50.6 | A>B>O>AB | O>A>B | Pandey *et al.* 2013 |
| **Munda** | | | | | | | | | | | |
| **Kishanganj** | 58 | 27.59 | 39.66 | 17.24 | 15.52 | 24.67 | 33.13 | 42.12 | B>A>O>AB | O.B>A | Present Study |
| **Ranchi** | 105 | 33.33 | 26.67 | 30.48 | 9.52 | 24.44 | 20.16 | 55.4 | A>O>B>AB | O>A>B | Tyagi |
| **Singhabhum** | 120 | 30 | 29.17 | 33.33 | 7.5 | 21.03 | 20.51 | 58.46 | O>A>B>AB | O>B>A | Macfarlane 1941 |
| **Purnea** | 214 | 28.03 | 43.92 | 12.14 | 15.88 | 26 | 37 | 37 | B>A>AB>O | O=B>A | Pandey *et al.* 1999 |
| **Bihar (Terai)** | 202 | 25.7 | 44.5 | 17.9 | 11.9 | 21.3 | 34.5 | 44.3 | B>A>O>AB | O>B>A | Pandey *et al.* 2013 |

**Table 4- Represents degree of homogeniety between tribal populations of District Kishanganj (Bihar) and tribal population of other region. The df is 3 and $\pi^2_c = 7.815$**

| Present Population | Population Compared | Reference | ² Values | Þ - Values | Significance at Þ < 0.05 |
|---|---|---|---|---|---|
| **Oraon** | Palamu | Sarkar, 1949 | 1.0308 | 0.793788 | Non Significant |
| | Andamand | Lahmenn and Ikin, 1953 | 4.0875 | 0.252166 | Non Significant |
| | Ranchi | Kirk *et al.* 1963 | 1.105 | 0.775856 | Non Significant |
| | Ambikapur | Bhatia *et al.* 1986 | 5.0669 | 0.16696 | Non Significant |
| | Purnea | Pandey *et al.* 1993 | 12.0949 | 0.007065 | Significant |
| | Purnea | Pandey *et al.* 1999 | 5.2592 | 0.153774 | Non Significant |
| | Bihar (Terai) | Pandey *et al.* 2013 | 4.7012 | 0.195027 | Non Significant |
| **Santhal** | Palamu | Sarkar, 1949 | 22.0289 | 0.000064 | Significant |
| | Santhal Paragna | Sarkar and Sen, 1952 | 29.6019 | <0.00001 | Significant |
| | Midanapur | Chaudhuri *et al.* 1967 | 9.6468 | 0.02182 | Significant |
| | Santhal Paragna | Bhattacharjee, 1968 | 14.2135 | 0.002628 | Significant |
| | Bihar (Terai) | Pandey *et al.* 2013 | 2.1962 | 0.532695 | Non Significant |
| **Munda** | Singhbhum | Macfarlane, 1941 | 7.488 | 0.057867 | Non Significant |
| | Ranchi | Bhattacharjee and Kumar, 1969 | 6.1001 | 0.106841 | Non Significant |
| | Purnea | Pandey *et al.* 1999 | 1.0946 | 0.778373 | Non Significant |
| | Bihar (Terai) | Pandey *et al.* 2013 | 0.7854 | 0.852958 | Non Significant |

## REFERENCES


1. **Shekhar H, Kaur A, Jadeja P, Parihar P.M and Mangukiya K.K. 2014.** Frequency and distribution of ABO blood group and Rh (D) factor in southern Rajasthan. *International Journal of Science and Nature*, **5:** 494-497.
2. **Farhud D.D and Yeganeh M.Z. 2013.** A brief history of Human blood groups. *Iranian J Publ Health*. **42(1):** 1-6.
3. **Morris L.N. 1973.** *Human Populations, Genetic Variation and Evolution.* International Text book Company Limited, 24 Market Square, Aylesbury, Bucks (USA), HP201TL.
4. **Laure S, Thompson E.E, Flutre T, Lovsted J, Venkat A, Margulis S.W. *et al.* 2012.** The ABO blood group is a transpecies polymorphism in primates. *PNAS*, **109(45):** 18493-98.
5. **Anstee D.J. 2010.** The relationship between blood group and disease. *Blood*, **115(23):** 4636-4643.
6. **Hirschfeld, L and Hirschfeld H. 1919.** Serological differences between the blood of different races. The result of researches on the Mecedonian front, pp-32-43. In Boyer, S.H (Ed) 1963. *Papers on Human Genetics*, Englewood Cliffs, N.J. Prentice Hall.
7. **Macfarlane E.W.E. 1941.** Blood group distribution in India with special reference to Bengal. *Jour of Genetics*, **XXXVI:** 225-237.
8. **Giri A.K, Datta S, Gajra B, Raychaudhury A, Datta S, Talukdar G and Sharma A. 1981.** Relative frequency of some genetic markers in Tribal and Non-tribal populations from the Midnaporedistric, West Bengal. *J Cytol. Genet*, **16:** 179-184.
9. **Majumdar P.P and Roy J. 1982.** Distribution of ABO blood groups in Indian Subcontinent: A cluster Analytic approach. *Current Anthropology*, **23(5):** 152-161.

10. **Kumar V and Reddy B.M. 2003.** Status of Austro-Asiatic groups in the peopling of India : An exploratory study based on available prehistoric, linguistic and biological evidences. *J. Biosci*, **28(4):** 507-522.

11. **Dewan G. 2015.** Comparative frequency and allelic distribution of ABO and Rh(D) blood groups of major tribal communities of southern Bangladesh with general population and their determinant. *The Ezyptian Journal of Medical Human Genetics*, **16:**141-147.

12. **Wright, S. 1949.** Population structure in evolution. *Proc. Amer. Phil. Soc,* **93:** 471-487.

13. **Bhasin M.K and Chahal. 1996.** *A laboratory manual for Human blood Analysis*. Kamla-Raj Enterprises, Delhi- 110006, India.

14. **Srivastava Narendra. 2009.** *Introductory Biometry: Statistic and Research Methodology*; Pointer Publishers, Jaipur, Rajasthan, India.

15. **Pandey B.N, Das P.K.L and Jha A.K. 1993.** A blood group survey in the Oraons of Purnia District, Purnia, India.*J . Hum Ecol,* **4(4):** 291-294.

16. **Pandey B.N, Mishra S.K, Yadav S and Sharma P.D. 1999.** Study of blood groups in migrant Oraons and Mundas of Purnia district of Bihar, India. *Anthropologist*, **1(2):** 143-146.

17. **Bhattacharjee P.N. 1969.** A genetical study of the Santal of Santalpraganas. *Anthropologist* (special Volume), Delhi University, Delhi.

18. **Chaudhuri S, Ghose J, Mukherjee B and Raychaudhury, A.K. 1967.** Study of blood groups and haemoglobin variants among the Santal tribes in the district of Midnapore of West Bengal. *Amer. J. Phys. Anthrop.* **26:** 307-311.

19. **Sarkar S.S and Ser D.K. 1952.** Further blood group investigation in Santhal Praganas. *Bull. Dept. Anthrop*. **1:**8-13.

20. **Sarkar S.S. 1936-37.** Blood grouping variation in India with reference to Santhal Praganas. *Trans. Bose. Res. Insti*: **12:**89-101.

21. **Boyd W.C. 1963.** Four achievements of the genetical method in physical anthropology. *American Anthropologist*, **65:** 243-252

22. **Sarkar S.S. 1949.** ABO blood groups from Palamu, Bihar, India. *Am. J.Phys. Anthrop.* **7:**559-563.

23. **Krik, R.L, Lai L.Y.C, Vos G.H and Vidyarthi L.P. 1963.** A genetical study of the Oraon of Chhotanagpur Platue (Bihar, India). *Am.J.Phys. Anthrop*. **20:**375-385.

24. **Piplai C, Vasantha, k Gorakshakar A.C, Bhatia h.M and Bhattacharya, S.K. 1985.** A note on ABO, MN, Rh and ln (a) blood groups of the Tamang and Oraon tea labourers of Jalpaiguri district, West Bengal. *J. Indian Anthrop. Soc*. **20:** 66-77

25. **Bhatia H.M, Rao V.P, Vasantha K, Sathi M.S and Mishra, R.P. 1986.** A1, A2, BO, MN, Rh blood groups among the Gend, Oraon and Kawar tribal groups of Ambikapur district, Madhya Pradesh. *J. Indian Anthrop. Soc*. **21:**73-77.

26. **Pandey B.N, Jahangeer Md, Kumar D, Vatsa S and Verma, D.K. 2013.** Genetic structure and microgenetic differentiation among population of Terai belt of Bihar, India. *Euro. J. Exp. Bio.* **9(6):** 80-87.

27. **Huneman P. 2015.** Inscrutability and the Opacity of Natural Selection and Random Genetic Drift: Distinguishing the epistemic and metaphysical aspects. *Erkenn.* **80:**491-578

28. **Waples R.S. 2015.** Testing of Hardy-Weinberg proportions: Have we lost the Plot? *Journal of Heredity*. **106(1):** 1-19.

29. **Hansen M.M, Nielsen E.E andMensberg K.L.D. 1997.** The problem of sampling families rather than population: relatedness among individuals in samples in juvenile brown trout Salmotrutta. L. *Mol. Ecol*; **6:**469-474.

30. **Bhattacharjee P.N and Kumar N. 1969.** A blood group genetic survey in the dudhKharias of the Ranchi District (Bihar, India). *Human Heredity*, **19:** 385-397.

31. **Mourant, A.E, Kopec, A.C and Domantewska-Sobezak, K. 1958.** *The ABO Blood Groups*, Blackwell, Oxford.

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# Studies on effects of aqueous leaf extract of *Achyranthes aspera* on thyroid function in male albino rats

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**Abstract :** The aqueous leaf extract of *Achyranthes aspera* was evaluated for its ameliorative effect in the regulation of hypothyroidism induced by methimazole in male albino rats. Rats were divided into four groups, three for each. Group I (control), group II (hypothyroid control) normal rats administered orally with methimazole 25 (mg/kg bwt.) for five successive days, treated group III and group IV with aqueous leaf extract of *Achyranthes aspera* 400 (mg/kg bwt.) for 14 and 28 days respectively, after hypothyroidism. Statistical analysis was done using student's t test. The obtained results revealed that hypothyroidism induced group II showed significant decrease in serum $T_3$ and $T_4$ ($p<0.001$), a significant increase in serum TSH was also observed ($p<0.001$), when compared to control group I. It was also observed that treated group III and group IV with aqueous leaf extract of Achyranthes *aspera* 400 (mg/kg bwt.) for 14 and 28 days respectively, after hypothyroidism showed significant increase in serum $T_3$ and $T_4$ ($p<0.001$), a significant decrease in serum TSH content was also observed, when compared with hypothyroid control group II. To conclude, aqueous extract of *Achyranthes aspera* leaf could be a potential therapeutic alternative for the management of hypothyroidism.

**Keywords :** ***Achyranthes aspera,*** **hypothyroidism, T3, T4 ,TSH**

## INTRODUCTION

Thyroid hormones play an important role in the normal activity of all body tissues. Decrease in the thyroid hormone levels is known as hypothyroidism.[1] It can occur because of the deficiency in iodine intake, thyroid gland lesions, autoimmune disorders and impaired activity of pituitary gland.[2,3] The untreated hypothyroidism leads to a wide range of abnormalities including loss of energy, fatigue sensation, weight gain, dry skin, depression, behavioural fluctuations, hair loss, hands and face swelling and increases cholesterol.[4]

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*Achyranthes aspera* L. (Family: Amaranthaceae), an erect and much branched diffuse herb is a medicinal plant, frequently found in tropical and warmer regions as weed.[5] The review reveals that wide numbers of phytochemical constituents have been isolated from the plant which possesses activities like antiperiodic, antiasthmatic, hepatoprotective, anti-allergic, expectorant, stomach tonic, laxative, antihelmintic, diuretic, linthontriptic, sudorific, demulcent, anti-inflammatory, anticataract, antifungal, antibacterial, hypoglycaemic, antihyperlipidemic and haematinic and various other important medicinal properties.[5] The present study is going to evaluate the effect of aqueous extract of *Achyranthes aspera* leaf on the level of thyroid hormones in methimazole-induced hypothyroid in rats.

## MATERIALS & METHODS

### Experimental Animals

12 adult male *albino rats* of the same age (6 weeks) and weight 110-150 g were used in the study. Animals were maintained under standard condition of ventilation temperature (25± 2ºC) and light/dark condition (12/12 h). Rats were housed in stainless steel cages and were provided with free access of food and drinking water *ad libitum*.

### Preparation of *Achyranthes aspera* leaf extract

Fresh *Achyranthes aspera* leaves were collected from the Lalpur locality of the city Ranchi (Jharkhand). The leaves were thoroughly washed with tap water followed by distilled water. The leaves were shade dried and powdered by using mechanical grinder. The powder was subjected to a series of maceration in distilled water and the extract decanted at an interval of 24 hours. The filtrate was then evaporated to dryness in a water bath at 40ºC giving a dark green solid yield. Powdered extract was refrigerated for future use.[6]

### Determination of doses

**(i) Dose for inducing hypothyroidism with methimazole**

Hypothyroidism was induced by daily oral administration of (25 mg/kg bwt.) methimazole for five successive days.[7]

**(ii) Dose of aqueous extract of *Achyranthes aspera* leaf**

The rats were treated with aqueous extract of *Achyranthes aspera* leaf at the dose of (400 mg/kg bwt.) for a period of 14 and 28 days after induction hypothyroidism.[8]

### Experimental Design

Animals were randomly divided into four groups of 3 rats each.

**Group I** served as the normal control and received rat feed and distilled water.

**Group II** (Hypothyroid Control): Orally administered with methimazole (25 mg/kg bwt.) for five successive days.

**Group III** (Treated group): Received aqueous extract of *Achyranthes aspera* leaf (400 mg/kg bwt.) orally for a period of 14, after induction of hypothyroidism.

**Group IV** (Treated group): Received aqueous extract of *Achyranthes aspera* leaf (400 mg/kg bwt.) orally for a period of 28 days, after induction of hypothyroidism.

### Sample collection

Blood serum was collected through cardiac puncture after anaesthetizing with ether.

### Estimation of hyroid hormones

Estimation of serum $T_3$, $T_4$ and TSH was done by chemiluminescence immunoassay method.[9]

### Statistical analysis

The data were analyzed as mean ± S.D. Statistical analysis were carried out by using student's 't' test.

## RESULTS

Table 1 showed that serum $T_3$, $T_4$ and TSH level of control group albino rats (G I) was 141.86 ±12.65 ng/dl, 7.63 ± 1.10 ug/dl and 0.77 ±0.48 ulU/mL respectively. Administration of methimazole (25 mg/kg bwt.) for five successive days caused hypothyroidism to group (G II). Statistical analysis was done using student's 't' test. Serum $T_3$ level was decreased significantly to 24.43 ± 3.30 ng/dl in group II rats (P<0.001).

Similarly, serum $T_4$ level also decreased significantly to 1.36±0.45 ug/dl after methimazole treatment (P<0.001). It was also noted that serum TSH level increased significantly to 3.77±0.34 ulU /mL in hypothyroid control group (G II) (P<0.001).

**Table 1: Effect of aqueous extract treatment of *Achyranthes aspera* leaf (400 mg/kg bwt.) for 14 and 28 days on serum thyroid profile ($T_3$, $T_4$ and TSH). Their percentage increase (+) or decrease (-) in relation to hypothyroid control and experimental group values in male rats.**

| PARAMETERS | Control Group I | Hypothyroid Control Group II | 14 Days of Leaf Extract Group III | % Change From Group II, (+) Increase, (-) Decrease | 28 Days of Leaf Extract Group IV | % Change From Group II, (+) Increase, (-) Decrease |
|---|---|---|---|---|---|---|
| Triiodothyronine $T_3$ (ng/dl) | 141.86±12.65 | *** 24.43±3.30 | ###189.86±4.36 | + 677.15 | ###191.06±4.53 | +682.07 |
| Thyroxine $T_4$ (ug/dl) | 7.63±1.10 | *** 1.36±0.45 | ###11.80±1.22 | +767.64 | ###10.13±0.28 | + 644.85 |
| TSH (ulU/mL) | 0.77±0.48 | *** 3.77±0.34 | ##0.80±0.03 | -78.77 | ### 0.70±0.04 | -81.43 |

**Values are expressed as mean ±S.D**

*****P<0.001, when compared between control and hypothyroid control group. ### P<0.001, ##p<0.01, when compared between hypothyroid control, 14 and 28 days of aqueous leaf extract of *Achyranthes aspera.***

In group (G III) rats treated with aqueous extract of *Achyranthes aspera* leaf (400 mg/kg bwt.) for 14 days showed significant increase in serum $T_3$ and $T_4$ content that was 189.86±4.36 ng/dl & 11.80±1.22 ug/dl respectively (P<0.001). A significant decrease in serum TSH content to 0.80±0.03 ulU/mL was also observed, when compared with hypothyroid control rats (G II) (P<0.01).

The treated group (G IV) in which rats were administered with aqueous extract of *Achyranthes aspera* leaf (400 mg/kg bwt.) for 28 days showed significant increase in serum $T_3$ and $T_4$ content to 191.06±4.53 ng/dl & 10.13±0.28 ug/dl respectively (P<0.001). A significant decrease in serum TSH content to 0.70±0.04 ulU/mL was observed, when compared with hypothyroid control rats (G II) (P<0.001).

## DISCUSSION & CONCLUSION

The thyroid gland is one of the important endocrine glands that play an essential vital role in the metabolism and energy expense in the body.[10] This gland is responsible for the production, storage and release of thyroid hormones triiodothyronine ($T_3$) and thyroxine ($T_4$).[11] These hormones are essential for cell growth and development.[12] One of the thyroid disorder is hypothyroidism which is defined as a deficiency in thyroid hormones production as a result of dysfunction of the synthesis and secretion of hormones.[13] The current investigation aimed to evaluate the hormonal changes in hypothyroid- model, also to assess whether *Achyranthes aspera* has potential ameliorating effects against hypothyroidism. Biochemically, methimazole induced the hypothyroidism that is reflected in the recorded hormonal levels of $T_3$, $T_4$ and TSH. A significant decrease in the serum $T_3$ and $T_4$ concentration was observed; also the TSH recorded a significant increase compared to the control values.

Methimazole prevents the thyroid peroxidase enzyme from coupling and iodinating the tyrosine residues on thyroglobulin, hence reducing the production of the thyroid hormones $T_3$ and $T_4$,[14] with subsequent increase in TSH. Methimazole was chosen to induce hypothyroidism in the rats, as it is easily absorbed by the gastrointestinal tract and concentrated in the thyroid gland.[15] It produces hypothyroidism by inhibiting the production of thyroid hormone.[16] Methimazole acts as a false substrate for thyroid peroxidase, thus inhibiting the incorporation of iodide into tyrosine residues on thyroglobulin catalyzed by thyroid peroxidise.[17]

Treated groups (G III and G IV) with aqueous extract of *Achyranthes aspera* leaf (400 mg/kg) for 14 and 28 days respectively, after methimazole-induction hypothyroidism, showed ameliorating effect. The extract showed significant increase in serum $T_3$ and $T_4$ content; a decrease in the level of TSH suggesting its thyroid hormone enhancing effects. Similar results have been reported in different plant extracts. In the same context Dehghani *et al*. (2016)[18] observed that on administration of *caraways* extract serum $T_3$ and $T_4$ levels were increased and TSH levels decreased significantly.

In agreement with present findings Mridul Yadav *et al*. (2016)[19] suggested that supplementation of *Ocimum sanctum* leaves significantly (P) increased levels of $T_3$ and $T_4$, whereas TSH levels were significantly reduced as compared to control group. They explained that increase in the levels of $T_3$ and $T_4$ may be due to increase in the activity of thyroid gland by some compound in *Ocimum sanctum* which affects the physiology and function of follicular cells.[20] They also reported that *Ocimum sanctum* may also cause enhancement of the transport of sodium-iodide transport and increased absorption of iodide resulting in increased production of $T_3$ and $T_4$ and effect on iodotyrosine deiodinases.[21]

Corroborating present investigation Tabassum *et al*. (2013)[20] reported a significant increase in $T_3$ and $T_4$, whereas a significant i(P<0.05) decrease in TSH as compared to those in the control group, while working on *Moringa oleifera.*

Similar results were reported by Abbas Abed Shahan *et al.* (2018)[22], with working on effect of methanolic leaf extract of *Ficus carica* on thyroid hormones. They suggested the stimulatory effect of leaf extract on thyroid follicles to elevate synthesis of high amount of $T_3$ and $T_4$ when used to treat hypothyroid animals.

Although further investigation are required to reveal the exact mechanism of action(s) of thyroid hormone regulation by *Achyranthes aspera* leaf extract, the present findings clearly indicates that this extract is stimulant to thyroid functions. However, the authors emphasize that further studies are required to observe the effect of leaf extract which might be effective and safe in ameliorating hypothyroidism.

## REFERENCES

1. **Kar A, Panda S, Bharti S. 2002.** Relative efficacy of three medicinal plant extracts in the alteration of thyroid hormone concentrations in male mice. *J Ethnopharmacol.* **81(2):**281-5.
2. **Ott J, Promberger R, Kober F, Neuhoid N, Tea M, Huber JC. 2011.** Hashimoto's thyroiditis affects symptom load and quality of life unrelated to hypothyroidism: a prospective case-control study in women undergoing thyroidectomy for bening goiter. *Thyroid*. **21(2):**161-7.
3. **Ozturk BT, Kerimoglu H, Dikbas O, Pekel H, Gonen MS. 2009.** Ocular changes in primary hypothyroidism. *BMC Res Notes.* **2:**266.
4. **Koumourou R. 2004.** Running on Empty: hypothyroidism, introduction to an underactive thyroid gland. Australia: Cocoon Books.
5. **Srivastava KP. 2014.** *Achyranthes aspera*: A potent immunostimulating plant for traditional medicine. *International journal pharmaceutical science and research.* E-ISSN:0975-8232; P-ISSN: 2320-5148.
6. **Timothy, S.Y; Wazis, C.H; Adati, R.G & Maspalma. 2012.** Antifungal activity of aqueous and ethanol leaf extracts of *Alata* Linn. *Journal of Applied Pharmaceutical Science*. **02(07):** 182-185.
7. **Francavilla, S; Cordeschi ,Properzi, G; Di cicco, L; B and D' Armiento. 1991.** Effect of thyroid hormone on the pre- and post-natal development of the rat testis. *J. Endocrinol.,* **129:** 35-42.
8. **Aboul-Enein HY, Al-Badr AA. 1979.** Methimazole. In Forey K. 18th edition. Analytical profiles of drug substance. Academic press, New York. 351-371 p.
9. **Aletrari M, Kanari P, Partassides D, Loizou E. 1979.** Study of the British pharmacopeia method on methimazole (thiamazole) content in carbimazole tablets. *Journal of Pharmaceutical and Biomedical Analysis* **16:**785-792.
10. **Bandyopadhyay U, Biswas K, Banerjee RK. 2002.** Extra thyroidal actions of anti-thyroid thionamides. *Toxicological Letter.* **128:**117-127.
11. **Dehghani, F., Panjehshahin, M.R. & V, Z. 2010.** Effect of hydroalcoholic extract of *caraway* on thyroid gland structure and hormones in female rat. *Iranian Journal of Veterinary Research, Shiraz University.* **11(4):**337-41
12. **Demers, L.M and Spencer, C. 2006.** The thyroid pathophysioligy and thyroid function testing. Tietz text book of clinical chemistry and molecular diagnostics. 4th edition. New Delhi: Elsevier's. pp.2063-2073.
13. **Gereben B Zavacki A M, Ribich S. 2008.** Cellular and molecular basis of deiodinase-regulated thyroid hormone signalling. *Endocr.Rev.* **29:**898-938.
14. **Goldman L. & Ausiello DA eds.Cecil medicine. 2008.** Philadelphia: Saunders Elsevier. (pp. 1212-1216).
15. **Nayak Bindu MD & Burman Kenneth MD. 2006.** Thyrotoxicosis and thyroid storm. EndocrinolMetaClin N Am **35:**663-86.
16. **Rajab, Njia M.Ali, Mirela Ukropina,& Maja Cakic-Milosevic. 2017.** "Histological and ultrastructural alterations of rat thyroid gland after short-term treatment with high doses of thyroid hormones*." Saudi Journal of Biological Sciences* **24(6):**1117-1125.
17. **Sharhan Adeb Abbas, Rasheed H.K. 2018.** Role of *Ficus carica* leaves extract in treatment of hypothyroidism. *Journal of University of Babylon for Pure and Applied Science*. **26(10):** 257-272.
18. **Stathatos, Nikolaos. 2019.** "Anatomy and physiology of the thyroid gland." The thyroid and its diseases. *Springer, Cham*. 3-12
19. **Tabassum. W, Kullu. A.R, and Sinha. M.P. 2013.** Effects of leaf extracts of *Moringa oleifera o*n regulation of hypothyroidism and lipid profile. *The Bioscan.***8(2):**665-9.
20. **Talukder, F.Z; Khan, K.A; Uddin, R; Jahan, N; Alam, A. 2012.** In *vitro* free radical scavenging and anti-hyperglycemic activities of *Achyranthes aspera* extract in alloxan- induced diabetic mice. *Drug discoveries & therapeutics*. **6(6):**298-305
21. **Vargas-Uricoechea, Hernando, and Anilza Bonelo-Perdomo. 2017.** "Thyroid dysfunction and heart failure:mechanisms and associations". *Current heart failure reports*. **14(1):** 48-58.
22. **Yadav Mridul, Sethi Jyoti, Dahiya Kiran, Singh Veena. 2016.** Effect of *Ocimum Sanctum* on serum concentration of thyroid hormones and atherogenic profile in rabbits. *International Journal of Health Sciences and Research*. ISSN:2249-9571.

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# Land degradation due to black stone mining in Pakur: Need Eco restoration

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**Abstract:-** Pakur is a small district of Jharkhand and is well known for its Asiatic fame black stone mine. The history of mining dates back to the British period in Pakur. Probably the black stone mining activities initiated here in 1912. Owing to the finest quality of black stone, the boulders and stone chips were being sent to different part of country including present day Bangladesh (The old British India). It is also sent to different parts of India for major construction project. Since 1912 till date approximately 775 black stone mines have been used for the extraction of black stones. At present 114 mines are operational. This is important to note that only two years back the number of black stone mines were 145. But due to reasons like failing to meet the compliance criteria of environmental low, land lease etc. The number of mines has come down. Some of the closed mines are restarted after fulfilling the required documents or meeting the compliance criteria whereas some of the mines are still inoperative/closed or turned as abandoned. If the average mines area considered as 5.5 acres. The total mining area comes to 5.5 hectare x 775 = 4262. 5 hectare, out of that area presently the mine activity going on in 5.5 hectare x 114 = 627 hectare, where as rest of the mines where mining activity has been ceased are abundant and these closed mines poising a great threat to environment as well as human life. There is enough scope to eco restore these closed mines, so that local enjoinment can be replenished, beside it will open and avenue for income generation activities. There are ample scope of fisheries, aquaculture, and water sports in these abundant mines of the district. The large amount of water remains stored in these abundant mines, and so far is not being utilise for any purposes, it can be utilised for agriculture, silviculture and end numbered uses. There are various scopes to develop park and these may develop into tourist spots. In the current paper an eco restoration plan for the abandoned degraded black stone mined out mines has been proposed.

**Key words: land degradation, black stone mines, eco restoration, Pakur.**

## INTRODUCTION

Mining of black stone in Pakur is going on with only dictum of making profit without any safety measures and conserving environment. This type of haphazard 'slaughter mining' and un-systematic mining over a period of time in the black stone mining areas of Pakur has resulted in relentless land degradation. The landscape resulted into remnants of old abandoned quarries, spoil dumps, subsided depressions, etc. thus resulting into disturbance of biodiversity.

A three tier ecological restoration is needed to restore the degraded and mined out lands by means of afforestation choosing the natural and native species. In fact the Ecological Restoration involves three tier plantations with indigenous plant species with lower level grasses, middle

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level shrubs or bushes trees at the top. The main objective of this to bring back the original normalcy of function, structure, potential, service and process of eco system as existed prior to mining activity by establishing a natural forest eco-system with biodiversity. For this purpose first of identification and selection of the local native species is required there after tie up with forest department to procure and plant the concerned species.

A single layer plantation is also useful in bringing greenery however this provides the green canopy in aerial view only and is not very effective in control erosion, recharging ground water and establishing bio-diversity. For eco restoration purpose the selection of species is not considered to meet socio-economic obligation of the local community. Ecological restoration is the process of short-circuiting the natural recovery of degraded ecosystems through ecological interventions. The ecological restoration is to establish a three-tier vegetation comprising of native species grasses as lower tier, shrubs and bushes as middle tier and trees as upper tier with an objective to establish biodiversity and food chain; to improve the local climate regime and socio-economic conditions. Removal of invasive weeds and addition of biomass to the degraded land creates an opportunity for the native species to germinate and establish biodiversity. Ecological restoration enhances biodiversity at faster rate and over time, end number of species may develop creating natural forest over burden dump. Therefore ecological restoration of mined out areas is the most appropriate ecologically and socio-economically compatible measure. Such restored area can serve as replacement of Reserve forest below which the presence of mineral is found in future. These dumps are profusely invaded by exotic weeds like *Parthenium hysterophorus, Croton bonplandianus, Xanthium Strumarium, Chromolena odorata, Eupatorium odoratum, Lantana camara, Panisettum padicellatum* etc. Due to several years of mining and severe land degradation, there is no soil cover on the dumps and are poor in nutrients.

## MATERIALS & METHOD

The mining sites, currently operational as well as mined out, abandoned, closed mines were visited from the year 2016-2020. The conditions of the closed mines sites were observed closely, monitored and plan for eco restoration is proposed.

There are approximately 775 mines existed in Pakur district from which the black stone extraction took place since 1912 till date. If the average mines area considered as 5.5 acres. The total mining area comes to 5.5ac x 775 = 4262.5ac. Out of that area presently the mining activity going on in 5.5ac x 114 = 627ac. whereas rest of the mines where mining activity need eco restoration for the effective ecosystem services of the mined out area. Same time to bring back conditions prior to the mining.

## REVIEW OF LITERATURE

Very less work on eco restoration has been done in Jharkhand, the eco restoration work has seen initiated and done by BCCl in 1912 at some of its mining site, Hindalco is doing eco restoration at Bagru, Lohardaga in their Bauxite mining site. However some the work of national and international paper has been referred here Amishaebal (2017)[1] worked Impact Assessment of mining achvitied on tree diversity at Limestene & Dolomite mining area-BSLC mines, Biramitrapur, Anderswiderland & Bjornohlander (2014)[2]. Environmental Aspects of mining, Bhargava, D.N. (2001)[3], Strategy for Environmental Management in Mining and Associated Operations, Indian Bureau of Mines, Ministry of mines, Government of India. A detailed aspect of mining has been published in district mining survey of Pakur (2016)[4], Dubey & Dubey (2011)[5] worked on Impact of mining on tree diversity of, the silica mining forest area at Shankargarh, Allahabad, M S Li (2005)[6] worked on particular reference to the metalliferous mine wasteland in China, Pawan Kumar (2013)[7] worked on mining activity and its impact on Plant biodiversity.[8] Mining Plan of Fatehpur Stone Mine of M/s Mishra Stone Works, Mouza-Fatehpur.[9] Published Paper on Ecological Restoration of Coal Mine Degraded Lands, Energy, GnY. Unanaonwi & Amonum (2017)[10] also worked on Effect of mining. Verma *et al.* (1989)[11] worked on Impact of Mining on the flora of Motijharna, Rajmahal Hills.

## RESULT & DISCUSSION

**Proposed Plan**

Restoring a damaged habitat or ecosystem is rarely as straight forward as just planting a few trees. It is often necessary to determine how to propagate and establish key species. Developing appropriate substrates, particularly when there is insufficient soil, is also a challenge. In a troubled nature of mine sites, some time it is important to grow the alien invasive species in the mined out area for the fast cover and to reduce the soil erosion of the degraded lands. Once the soil will be enriched with the nutrients the native species can thrive well.

The Society for Ecological Restoration (SER 2004) defines ecological restoration as 'the process of assisting the recovery of an ecosystem that has been degraded, damaged or destroyed'. The process of natural succession, which happens concurrently with ecosystem development takes time. Several ecosystem functions of mined land require a long time to develop through natural processes.

**Eco restoration planning and implementation:**

Foe eco restoration, re-vegetation planning is required, that requires types of vegetation cover. This depends on characteristics of the plant, nature of mine soils, quality of existing topsoil, closeness to the nearby natural seed banks, climatic conditions like rainfall, temperature and proneness to anthropogenic disturbance. It is necessary that a clear objective of restoration and final land use at the end of mining operation is clearly defined by mine authority along with the consultation of regulators and local bodies. The restoration planning should be well planned. Eco restoration planning involves the following steps-

**Ensuring involvement of local community:**

Mining is a temporary activity and as temporary occupiers of the land, the black stone miners should plan a better post-mining land use by as per the requirement of the local community.

**Transplantation and preservation of flora and development of flora bank:**

This involves the transplantation of important native species and their documentation. The information need to display owing to its conservation. Utmost care has to be taken while transplanting the plant to the new site. Protection of remaining patches of the original vegetation should be encouraged so as to increase seed propagation and regeneration of native species.

**Management of the top soil:**

Accumulation of topsoil plays a vital role in establishing a self-sustaining cover during eco restoration process. This require the preservation wherever possible and its quality should be protected while transferring and storage. The topsoil management include-inventory and removal of topsoil, concurrent reuse and storage of topsoil (if required) including preservation of topsoil fertility by vegetating grass-legume mixture.

**Planning for revegetation**

The establishment of a succession based vegetation cover is the key element of booming restoration. This requires the plantation using grasses and legumes seeds. A list of preferable plant species is given in table -1.

A total of 39 plant species have been proposed for the plantation in lieu of eco restoration. Out of these 39 species 23 species are of tree, 8 are of herb and 8 species are of shrub. These species are selected on the basis of their presence in and around mining sites and their related growth.

Amalgamation of ecological succession: This approach promotes development of native plant species, initially planted species will restructure dump plane and support colonisation of native species. It act as link between primary colonisers and afterward mounting vegetation.

**Table 1. List of Plant species selected for the eco restoration of the degraded mined our areas**

| Sl. No. | Scientific Name | Family | Habit |
|---|---|---|---|
| 1 | *Acacia arabica* | Fabaceae | Tree |
| 2 | *Acacia catechu / Senegalia catechu* | Fabaceae | Tree |
| 3 | *Aegle marmelos* | Rutaceae | Tree |
| 4 | *Agave americana* | Asparagaceae | Shrub |
| 5 | *Alangium salvifolium* | Cornaceae | Small tree |
| 6 | *Albizia lebbeck* | Fabaceae | Tree |
| 7 | *Alstonia scholaris* | Apocyanaceae | Tree |
| 8 | *Anacardium occidentle* | Anacardiaceae | Tree |
| 9 | *Azadirachta indica* | Meliaceae | Tree |
| 10 | *Bambusa arundinacea* | Poaceae | Tree |
| 11 | *Bauhinia purpurea* | Fabaceae | Tree |
| 12 | *Bauhinia variegata* | Fabaceae | Tree |
| 13 | *Bombax ceiba* | Malvaceae | Tree |
| 14 | *Butea monosperma* | Fabaceae | Tree |
| 15 | *Caldenia procumbens* | Boraginaceae | Herb |
| 16 | *Calotropis gigantea* | Apocyanaceae | Shrub |
| 17 | *Calotropis procera* | Apocyanaceae | Shrub |
| 18 | *Clerodendron infortunatum* | Lamiaceae | Shrub |
| 19 | *Chromolena odorata* | Astraceae | Shrub |
| 10 | *Croton bonplandianus* | Euphorbiaceae | Herb |
| 21 | *Cynodon dactylon* | Poaceae | Herb |
| 22 | *Dalbergia latifolia* | Fabaceae | Tree |
| 23 | *Dalbergia sissoo* | Fabaceae | Tree |
| 24 | *Dendrocalamus strictus* | Poaceae | Tree |
| 25 | *Deris pinnata* | Fabaceae | Tree |
| 26 | *Diospyros melanoxylon* | Ebenaceae | Tree |
| 27 | *Ficus religiosa* | Moraceae | Tree |
| 28 | *Flemingia chappar* | Fabaceae | Shrub |
| 29 | *Lagerostroemia parviflora* | Lythraceae | Tree |
| 30 | *Lantana camara* | Verbanaceae | Shrub |
| 31 | *Leucas aspera* | Lamiaceae | Herb |
| 32 | *Madhuca latifolia* | Sapotaceae | Tree |
| 33 | *Opuntia dilleni* | Cactaceae | Shrub |
| 34 | *Saccharum munja* | Poaceae | Herb |
| 35 | *Saccharum officinarum* | Poaceae | Herb |
| 36 | *Solanum xanthocarpum* | Solanaceae | Herb |
| 37 | *Tephrosia purpurea* | Fabaceae | Herb |
| 38 | *Zizyphus jujuba* | Rhymaceae | Tree |
| 39 | *Zizyphus maurtiana* | Rhymaceae | Shrub |

**Plates showing degraded black stone mines to be restored**



**An abandoned mines site near police line, Pakur**



**Mines site at Malpahari filled with water**



**An abandoned mines site at Mali Para, Pakur**



**Mines site at Malpahari filled with water**



**An abandoned mines site at Mal Pahari, Pakur**



**An abandoned mine at Raddipur, with some of aquatic plants adding beauty to it.**



**An abandoned mines filled with water at Mal Pahari, Pakur**



**The mines is being filled with OB at Sidpahari**




**An abandoned mine at Sidpahari**

**Plants species recommended for Eco restoration**



***Acacia nilotica***



***Lantana camara***



***Opuntia dilleni***



***Croton bonplandanus***



***Clerodendron infortunatum***



***Aegle marmelos***



***Azadirachta indica***



***Butea monosperma***



***Solanum xanthocarpum***



***Anacardium occidental***



***Calotropis procera***



***Alstonia scholaris***

## CONCLUSION

There is a close relationship between ecological understanding and successful eco restoration. Technical aspects of restoration are the keys of ecological restoration of mine soils, which should be carried out in an orderly manner. Miners are now realising that reclamation by simple plantation of fast-growing trees are not going to restore a degraded ecosystem. There should be a paradigm shift to restore a degraded ecosystem through an ecological restoration approach. In fact every mine closure plan of a working mining plan an environment management plan is mentioned, that after the complete extraction of the reserve the mined out area will be filled out and plantation will be done, but after the completion of the project , this is hardly followed.

## REFERENCES

1. **Amishaebal. 2017.** Impact Assessment of mining achvitied on tree diversity at Limestone & Dolomite mining area- BSLC mines, Biramitrapur, Odisha. *Journal of Pharmacognosy and Phytochemistry.* **6(5):**1902-1905
2. **Anderswiderland And Bjornohlander. 2014.** Environmental Aspects of mining, Lulea Uni. of Technology Franke Ecke, Swedish univa of agricultural science.
3. **Bhargava, D.N. 2001.** Strategy for Environmental Management in Mining and Associated Operations, Indian Bureau of Mines, Ministry of mines, Government of India.
4. **District Survey Report of Minor Minerals, Pakur. 2006.** Prepared in accordance with Para 7 (iii) of S.O.141 (E) Dated 15th January 2016 of Ministry of Environment, Forest and Climate Change Notification. Government of Jharkhand, District of Pakur.
5. **Dubey K, Dubey K P. 2011.** Impact of mining on tree diversity of, the silica mining forest area at Shankargarh, Allahabad, *India Journal of Forestry Research.* **35:**319-327.
6. **M S Li. 2005.** Particular reference to the metalliferous mine wasteland in China: *A Review of Research and Practice*, PMID: 15992864 DOI: 10.1016/j. scitotenv.
7. **Pawan Kumar. 2013.** "The mining activity and its impact on Plant biodiversity a case study at Bhiwani open cast mining zone- Haryana, India. **3(6):**78-83 Pelagia Research library ISSN: 2249 -7412.
8. **Radha Nand Singh, Reg. No. RQP/ RNC/ 145/ 2011/ A, 2016.** *Mining Plan of Fatehpur Stone Mine of M/s Mishra Stone Works, Mouza-Fatehpur*
9. **Subodh kumar Maiti. 2017.** Ecological Restoration of Coal Mine Degraded Lands, Energy, GnY.
10. **Unanaonwi OE & Amonum JI Jan-fib, 2017.** Effect of mining. Activities On vegetation composition and nutrient Status d Forest Soil benue Cement Company, Benue state, Nigeria. *IJEAB* Vol-**2.1-1** ISSN: 2456-1878
11. **Verma, S.K., R.R. Jha and A.K. Pandey. 1989.** Impact of Mining on the Flora of Motijharna, Rajmahal Hills, Bihar, *The India Forester*, Vol.**115 (12).**

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# Pollution of water in North Koel River basin in Latehar and Palamu districts in the state of Jharkhand

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**Abstract:-** Water is an essential nutrient and plays a key role in the human body. We can survive up to several weeks without food, but only a few days without water. Every system in the body, from cells and tissues, to vital organs requires water to function. Water is uniquely vulnerable to pollution. Known as a "universal solvent" water is able to dissolve more substances than any other liquid on earth. It's also why water is so easily polluted. Toxic substances from farms, towns and factories readily dissolve into and mix with it, causing water pollution. Groundwater gets polluted when contaminants from pesticides and fertilizers to wastes leached from landfills and septic systems make their way into an aquifer, rendering it unsafe for human use. Covering about 70% of the earth, surface water is what fills our oceans, lakes, river etc. Every time it rains, fertilizers, pesticides and animal waste from farms and livestock operations wash the nutrients and pathogens such bacteria and viruses into our water sources. The causes of water pollution include a wide range of chemicals and pathogens as well as physical parameters. Contaminants may include organic and inorganic substances. Water pollution is measured by analysing water samples. The present paper deals in the pollution of water of North Koel river basin in districts of Latehar and Palamu. Samples from various sites were collected and analysed and we could sum up on the preventive measures required in future.

**Key words: Groundwater, pollution, contaminants, pathogens**

## INTRODUCTION

Water is an essential nutrient and plays a key role in the human body. We can survive up to several weeks without food, but only a few days without water. Every system in the body, from cells and tissues, to vital organs requires water to function. Water is uniquely vulnerable to pollution. Known as a "universal solvent," water is able to dissolve more substances than any other liquid on earth. It's also why water is so easily polluted. Toxic substances from farms, towns and factories readily dissolve into and mix with it, causing water pollution.

Water is the main source of livelihood of human being. The water sources are decreasing and life is becoming dependent on the running water available located in the adjacent area of small hamlets. These water resources are also being polluted for various reasons due to which the quality of human life is deteriorating. This is also causing a number of water borne disease in rural areas. The longivity of life is also decreasing due to this problem.

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**Geographical Area**

Geographically the study area is located on western side of the state of Jharkhand covering parts of Latehar and Palamu districts. The area can be approached by road and railways both with a considerable walk to the river side. The study area in the North Koel river basin starts from Kechki to Rehla which passes through Chianki, Sua, Kauria, Medininagar, Kajri, Rajhara and Pandwa areas of both the districts. The nearest railway station is Kechki in Chianki, Daltonganj and Garhwa Road. Roads are close to the river banks.

**Location**

The area falls within the latitudes 23°45' N to 37°30' N and longitudes 83°50' E to 84°23'E covering Latehar and Barwadih blocks of Latehar district and sadar block of Daltonganj and Rehla area of Palamu district. The area is covered in Toposheet No.F45A2, F45A1, G45S4 and G44X16 of Survey of India of India in the scale of 1:50000.




**Geology of the Area**

The study area forms a part of Chotanagpur Gneiss granulites complex, which is characterized by vast expanse of granitic rocks and hence the area carries the name Chotanagpur Gneiss Granulites complex (CGGC).

The area represents a group of meta-sedimentary rocks such as mica-schists, crystalline dolomitic limestones, quartzites and calc-granulites. The granitic rocks and gneisses, which constitute the bulk of the complex, are syn-post tectonic. A swarm of metamorphosed basic intrusion represented by meta-dolerites, meta-gabbros and massive amphibolites are encountered in the meta-sediments. The granites of the area are younger than the meta-sedimentary and basic intrusive and often show intrusive relationship with the country rocks. The pegmatite and quartz veins occur as numerous small bodies throughout the area. It is pertinent.

**North Koel river basin**

The North Koel River originates in the Ranchi plateau (Lohardaga) and flows in Latehar and Palamu districts in Jharkhand state. The length is about 260 km. It is a big source of irrigation and water supply to a large number of populations in both the districts.

**Tributaries of North Koel River basin**

North Koel and its tributaries is the lifeline of Latehar and Palamu district but due to polluted water the quality of life is decreasing and causes several diseases. The Important tributaries of North Koel river basin are:-

**a.** Burha Ghagh
**b.** Auranga
**c.** Amanat
**d.** Maila
**e.** Sapahi



**Rainfall data of the area**

The rainfall data (in mm) of 2018 and 2019 of Palamua are:

| Month | Year (2018) | Year (2019) |
|---|---|---|
| **January** | 000.0 | 003.6 |
| **February** | 00.6 | 009.9 |
| **March** | 000.0 | 008.5 |
| **April** | 005.7 | 008.9 |
| **May** | 009.6 | 001.2 |
| **June** | 078.9 | 077.0 |
| **July** | 206.3 | 263.3 |
| **August** | 236.3 | 311.5 |
| **September** | 144.9 | - |
| **October** | 009.2 | - |
| **November** | 003.0 | - |
| **December** | 012.6 | - |

**Courtesy**- Indian Meteorological Department, Meteorological centre, Birsa Munda Airport, Ranchi, Govt. of India.

**The Main Causes of Pollution:**

**1.** Domestic waste- Sewage discharges, use of water by animals on river banks are the root cause of pollution. Washing of clothes by local people on river banks, solid waste disposal due to wind current from cities is also causing pollution. It consists mostly of grey water, black water, soaps and detergents. The waste water from residences, food wastes, laundry wastes and other waste products are classed as domestic or sanitary sewage.

**2.** Industrial cause- Discharge of toxic chemical waste from industries without any treatment or partially treated is being discharged into the river.

**3.** Agriculture cause- Fertilizers and pesticides are used in cultivation of crop. Rainwater dissolves certain percentage of the fertilizer and pesticides and goes to the river as subsurface and surface flow.

**4.** Leaching of minerals from the rock exposed on the surface by rain water and running water and they finally reach the stream and pollute the river water.

**Extent of pollution**

**The water samples from the various locations were collected during post- monsoon season. The water analysis data of one year for post- monsoon period of various locations are as below-**

| Sl No | Parameter | Units | Results | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | WS1 | WS2 | WS3 | WS4 | WS5 | WS6 | WS7 | WS8 | WS9 |
| 1 | PH value | pH | 7.77 | 8.65 | 8.3 | 8.04 | 7.82 | 8.25 | 8.44 | 8.28 | 8.21 |
| 2 | conductivity | µs/cm | 195.0 | 332.0 | 219.0 | 275.0 | 298.0 | 288.0 | 342.0 | 280.0 | 312.0 |
| 3 | Total dissolved solid | mg/l | 124.0 | 206.0 | 132.0 | 160.0 | 180.0 | 178.0 | 216.0 | 168.0 | 192.0 |
| 4 | Iron( as fe) | mg/l | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) | ND (DL 0.1) |
| 5 | Magnesium | mg/l | 4.37 | 0.48 | 6.8 | 8.74 | 11.7 | 10.20 | 21.38 | 10.20 | 11.7 |
| 6 | Potassium | mg/l | 2.0 | 2.0 | 2.0 | 2.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| 7 | Sodium | mg/l | 17.0 | 27.0 | 18.0 | 25.0 | 26.0 | 25.0 | 30.0 | 25.0 | 28.0 |
| 8 | Phosphate | mg/l | 0.04 | ND (DL 0.003) | 0.01 | 0.03 | 0.01 | 0.01 | 0.04 | 0.01 | 0.05 |
| 9 | Calcium | mg/l | 24.0 | 54.4 | 23.2 | 28.8 | 27.2 | 27.2 | 19.2 | 25.6 | 31.2 |

**Location of Water Sample (WS):-**

WS1 - North Koel near Kechki
WS2 - Auranga River near Kechki
WS3 - North Koel + Auranga River (confluence of both rivers)
WS4 - Pampukal Belwatikar (Koel River enters Daltonganj)
WS5 - Bisfutta Daltonganj (River Koel exists Daltonganj)
WS6 - North Koel at Singra
WS7 - River Amanat at Singra
WS8 - North Koel River + Amanat River (Confluence of both rivers)
WS9 - North Koel River at Rajahra

**Courtesy** - sample analysis have been done from Yugantar Bharti Analytical & Environmental Engineering Laboratory, Namkum Ranchi.

## DISCUSSION

The water analysis of various locations clearly indicates that the extent of pollution before the entry of the river in a human habitat is less but when it crosses the populated zone then due to causes mentioned above the extent of pollution increases. The increase is due to various causes and on the basis of analysis, it is discussed as below:-

The water sample analysis data clearly reflects the decrease and increase of various elements after confluence of the tributaries of the river into the main river. If one point is enriched in one element while the tributary is less concentrated with other elements than after confluence it either increases or decreases.

**The following points clarify the above facts:-**

The water sample WS6 is having 27.2 mg/l Calcium (Ca) in North Koel at Singra. While at same place the water sample WS7 is having 19.2 mg/l Ca in river Amanat. Immediately, after confluence the average comes to 25.6 mg/l. same is the case with magnesium (Mg) too. The water sample WS6 at North Koel near Singra contains 10.20 mg/l Mg. While as the water sample WS7 in Amanat near Singra contains 21.38 mg/l. After confluence it lowers down to 10.20 mg/l.

There is little variation in pH value but there is a marked change in Conductivity and TDS.

The Iron (Fe) has got no variation but potassium is slightly varied at different points. The sodium (Na) contains range from 17-28 mg/l at various locations which also fluctuates due to confluence of tributaries and Main River. The entire system of control can be achieved through awareness to the villagers.

## CONCLUSION

The above discussion clarify the fact that due to disposal of solid waste, discharge of excess dissolve pesticides from agriculture field, washing of cloth by the villagers and towns. Increase the pollution of river water and broad changes in the Ph- value of the water. Running water on the bank of cities is considered as the lifeline of human being. The analysis data of the water collected from various resources clearly reflects the extent of pollution. However, on confluence it decreases certain remedial measures are being suggested to check the pollution at the initial stage.

## REMEDIAL MEASURES

It is suggested that the solid waste disposal should be stopped immediately by creating the collection points in the cities and reducing the volume of solid waste through treatment. The use of small quantity of fertilizer and pesticides will also reduce the agriculture cause of water pollution. Further River banks should not be allowed for washing clothes because it increases the acidic content in water. With this at initial stage the pollution can be controlled.

The process of this study will continue for two years in pre and post- monsoon season and finally after the end of stipulated period of research. The studies likely to come with a definite result and final remedial measures on the basis of old experiment will be suggested to control the pollution.

## REFERENCES

1. **Mishra Shweta, Verma P.K. & Kumar Uday. 2017.** Effects of rain water harvesting induced artificial recharge on the ground water in the areas of Palamu. **Vol.4, No.2(II):** pp.77-82.
2. **Mishra Shweta & Verma P.K. 2010.** Abstract Vol. of Regional workshop of all exploration, development and management of groundwater in hard rock by C.G.W.B. A-16,pp 23-24.
3. **Pandey Vijay, Verma P.K. & Kumar Uday. 2017.** Evaluation of fluoride contamination in ground water in hard rock's terrains in and around chukru- chianki areas, Palamu, Jharkhand, India.
4. **Singh Shyam Lal, Verma P.K. & Mishra Shweta. 2017.** Comparitive study of change in soil composition due to the biomass generation through Chakriya Vikas Pranali Project- A case study of Tandwa, district Palamu.
5. **Verma P.K. & Sinha S.P. 2000.** Artificial recharge of groundwater in hard rock terrain through Chakriya Vikas Pranali – A case study of Palamu district in Bihar.

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# Studies on aquatic macrophytic diversity in ponds of Pakur

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**Abstract:-** The current study deals with the Aquatic Macrophytes of Pakur. The study conducted from the year 2017 to 2020 reveals the presence of total 68 species, these 68 species are distributed over 36 families and 53 genus. Out of that 13 families, 27 genera and 30 species belongs to monocotyledons whereas 20 family, 26 genus and 35 species belongs to dicotyledons. The algae with 1 family, 1 genus and 1 species & Pteridophyte with 2 family, 2 genera and 2 species.

**Key words: Aquatic, Macrophytes, Pakur, Ponds.**

## INTRODUCTION

Pakur literally meaning Pokhar (Pokur in Bangla). Due to the presence of large number of Ponds (Pokhar) the district is called Pakur. However the other historical facts are different. A large number of Ponds are present in this district, being a Bengali dominating area the ponds are used for fishery activities. These ponds harbor a large number of Aquatic Plants (Aquatic macrophytes). Some of the ponds in the area are badly infested with these plants, hampering the fisheries activities as well as other effective uses of water. So far no any scientific studies on Macrophytes of the area has been done, this is the pioneer work on Aquatic Macrophytes.

It is evident that the aquatic plants are different from the terrestrial plant as it is very fragile and soon lose their characters as soon as taken out of the water. These aquatic plants help in the maintenance of the aquatic ecosystem.

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As far as flora of Jharkhand is concerned Botany of Bihar and Orissa (1921-1925) is a well documented flora, supplemented by Mooney (1950)[1], coming to aquatic flora is concerned some important workers are Christopher (2006)[2], Cook (1996)[3], Gupta (1979)[4], Jha (1965)[5], Kachroo (1959)[6], Maheshwari (1960)[7], Majumdar (1965)[8], Mukherjee (2001)[9], Mukherjee & Kumar (2003)[10], Mukherjee & Ghosh (2015)[11], Naskar (1990)[12], Subramanyam (1962)[13], Singh (1998)[14], Singh (1990)[15], Verma and Pandeya (2007)[16], Verma and Pandeya (2008)[17], Mukherjee and Kumar (2019)[18], Mukherjee and Kumar (2020)[19].

## MATERIALS & METHOD

### Study area

The district of Pakur came into existence in the year 1994. It is situated in the north east corner of Jharkhand State. It is located at 23°40' to 25°18' latitude and 86°25' to 87°57'E longitude. Pakur is the administrative headquarters

of this district. It is situated on the north-eastern corner of Jharkhand state. The district is bounded on the north by Sahebganj district, on the south by Dumka district, on the west by Godda district, and on the east by the Murshidabad district of West Bengal. Pakur district consists of 06 Blocks. The following are the list of the Blocks in Pakur district- Pakur, Maheshpur, Hiranpur, Littipara, Amrapara, Pakuria. The district, with a population of 899,200 (census 2011), and covering an area of 686.21 km², this gives it a ranking of 465th in India (out of a total of 640).




**Fig. 1-Map of Jharkhand Showing Pakur District**

## METHODOLOGY

The Ponds of Pakur were visited in regular interval in every season intensively in order to report the number of aquatic Macrophytes. Considering the delicate nature of aquatic plants, extensive care was taken during removing them from water so that they might not lose their significant taxonomic characters. During field work, important characters like habit, colour, height and association of the plants with other plants were noted down. Plants were pressed in the field herbarium press. Some plants were also kept in dilute solution of formaldehyde. Just after returning from the field, the plants were studied in the laboratory on the very same day. The field numbers were verified, flowers were dissected and the plants were identified with available flora.

After proper identification, the plants were dried and pressed under wooden herbarium press. Prior to pressing, the specimen were poisoned in 2% solution of Mercuric Chloride in rectified spirit. The specimens after drying were mounted with the help of synthetic resin adhesive and thread on the herbarium sheets of standard size (41.6 x 29.5). The plants are listed as per the classification system of Bentham and Hooker (1862-1883). The specimens are deposited in the herbarium of Department of Botany, K.K.M. College, Pakur.

## RESULT & DISCUSSION

A survey conducted in Pakur to know the Macrophytes of the area reveals the presence of total of 68 species, these 68 species are distributed over 36 families and 53 genus. Out of that 13 family 27 genera and 30 species belongs to monocotyledons whereas 20 family, 26 genus and 35 species belongs to dicotyledons. The algae with one family, 1 genus and 1 species and Pteridophyte with 2 family, 2 genera and 2 species.

Out of total reported 68 aquatic Macrophytes, the dominating taxa belongs to emergent Macrophytes contributing 57.35%, 39 species, then comes the submerged species, which contributes 16.17% having 11 species, followed by Rooted Macrophytes that contributes 14.70%, 10 species and finally the free floating species contributes 11.76%, 8 species.

The largest family is Cyperaceae having 5 species followed by Poaceae and Hydrocharitaceae with 4 species

each, Amaranthaceae, Lythraceae, Polygonaceae, Pontederiaceae, Polygonaceae, Potamogetonaceae and Araceae with 3 species each, Fabaceae, Aponogetonaceae, Commelinaceae, Convolvulaceae, Onagraceae, Nymphaceae and Menyanthaceae with two species each where as Plantaginaceae, Ceretophyllaceae, Apiaceae, Eriocaulaceae, Molluginaceae, Boraginaceae, Acanthaceae, Lemnaceae, Haloragaceae, Nelumbonaceae, Rubiaceae, Typhaceae, Verbenaceae and Portulacaceae with one species each beside Salvinaceae, characeae and Marsiliaceae have 1 species too. A list of all 68 plants has been given in Table -1.

**Table 1- Aquatic Macrophytes from Pakur**

| S.N. | Name | Family | Dicot / Monocot | Habitat |
|---|---|---|---|---|
| 1 | *Azolla pinnata* R.Br. | Salvinaceae | Cryptogams | Free floating |
| 2 | *Aeschynomene aspera* L. | Fabaceae | D | Emergent |
| 3 | *Aeschynomene indica* L. | Fabaceae | D | Emergent |
| 4 | *Alternanthera philoxeroides* (Mart.) Griseb. | Amaranthaceae | D | Emergent |
| 5 | *Alternanthera sessilis* (L.)R. Br. ex | Amaranthaceae | D | Emergent |
| 6 | *Ammannia baccifera* L. | Lythraceae | D | Emergent |
| 7 | *Aponogeton undulatus* Roxb | Aponogetonaceae | M | Submerged |
| 8 | *Aponogeton natans* (L.) Engl. & K. Krause in Engl., | Aponogetonaceae | M | Submerged |
| 9 | *Bacopa monnieri* (L.) Penell | Plantaginaceae | D | Emergent |
| 10 | *Ceratophyllum demersum* L. | Ceretophyllaceae | D | Free floating |
| 11 | *Chara globularis* J.L.Thuiller | Characeae | Cryptogams | Submerged |
| 12 | *Chenopodium murale* L. | Amaranthaceae | D | Emergent |
| 13 | *Coix lacryma -jobi* | Poaceae | M | Emergent |
| 14 | *Centella asiatica* (L.) Urb. | Apiaceae | D | Emergent |
| 15 | *Commelina benghalensis* L. | Commelinaceae | M | Emergent |
| 16 | *Commelina hasskarlii* | Commelinaceae | M | Emergent |
| 17 | *Cyperus rotundus* L. | Cyperaceae | M | Emergent |
| 18 | *Cyperus corymbosus* Rottb. | Cyperaceae | M | Emergent |
| 19 | *Cyperus iria* L. | Cyperaceae | M | Emergent |
| 20 | *Cyperus difformis* L. | Cyperaceae | M | Emergent |
| 21 | *Cynodon dactylon* | Poaceae | M | Emergent |
| 22 | *Eichhornia crassipes* (Mart.) Solms. | Pontederiaceae | M | Free floating |
| 23 | *Echinochloa colona* | Poaceae | M | Emergent |
| 24 | *Eclipta prostrata* | Asteraceae | D | Emergent |
| 25 | *Eleocharis dulcis* (Burm.F.) | Cyperaceae | M | Emergent |
| 26 | *Enydra fluctuans* DC. | Asteraceae | D | Emergent |
| 27 | *Eriocaulon cinerum* | Eriocaulaceae | M | Emergent |
| 28 | *Glinus lotoides* L. | Molluginaceae | D | Emergent |
| 29 | *Heliotropium indicum* | Boraginaceae | D | Emergent |
| 30 | *Hygrophila auriculata* (Schumach.) Heine | Acanthaceae | D | Emergent |
| 31 | *Hygrorhiza aristata* (Retz.) Nees ex Wight. & Arn. | Poaceae | M | Free floating |
| 32 | *Hydrilla verticillata* (L. f.) Royle | Hydrocharitaceae | M | Submerged |
| 33 | *Ipomoea aquatica* Forssk. | Covolvulaceae | D | Rooted floating |
| 34 | *Ipomoea carnea* Jacq. | Convolvulaceae | D | Emergent |
| 35 | *Ludwigia adscendens* (L.) H.Hara | Onagraceae | D | Rooted floating |
| 36 | *Ludwigia perennis* L. | Onagraceae | D | Rooted floating |
| 37 | *Lemna perpusilla* Torr. | Lemnaceae | M | Fee floating |
| 38 | *Marsilia minuta* | Marsiliaceae | Cryptogams | Emergent |
| 39 | *Myriophyllum tetrandrum* Roxb., | Haloragaceae | D | Rooted floating |
| 40 | *Monochoria hastata* (L.) Solms | Pontederiaceae | M | Rooted floating |
| 41 | *Monochoria vaginalis* (Burm. f.) Presl. | Pontederiaceae | M | Rooted floating |
| 42 | *Nymphaea pubescens* Willd | Nymphaeaceae | D | Rooted floating |
| 43 | *Nymphaea rubra* Roxb. ex Andrews | Nymphaeaceae | D | Rooted floating |
| 44 | *Nelumbo nucifera* Gaertn., | Nelumbonacae | D | Rooted floating |
| 45 | *Nymphoides hydrophylla* (Lour.) Kuntze | Menyanthaceae | D | Rooted floating |
| 46 | *Nymphoides indica* (L.) Kuntze | Menyanthaceae | D | Rooted floating |
| 47 | *Nechamandra alternifolia* Roxb. Ex Wight Thwaites | Hydrocharitaceae | M | Submerged |
| 48 | *Ottelia alismoides* (L.) Pers. | Hydrocharitaceae | M | Submerged |
| 49 | *Oldenlandia corymbosa* L. | Rubiaceae | D | Emergent |
| 50 | *Phyla nodiflora* (L.) Greene | Verbenaceae | D | Emergent |
| 51 | *Perscaria glabra* (Willd.) M.Gomez | Polygonaceae | D | Emergent |
| 52 | *Perscaria barbeta* (L.)H. Hara | Polygonaceae | D | Emergent |

| | | | | |
|---|---|---|---|---|
| 53 | *Potamogeton nodosus* Poir. | Potamogetonaceae | M | Submerged |
| 54 | *Potamogeton crispus* L. | Potamogetonaceae | M | Submerged |
| 55 | *Portulaca oleraceae* L. | Portulacaceae | D | Emergent |
| 56 | *Pistia stratiotes* L. | Araceae | M | Free floating |
| 57 | *Ranunculus sceleratus* L. | Ranunculaceae | D | Emergent |
| 58 | *Rotala indica* Blatt. &Halb. | Lythraceae | D | Emergent |
| 59 | *Rotala rotundifolia* (Roxb.) Koehne | Lythraceae | D | Emergent |
| 60 | *Rumex dentatus* L. | Polygonaceae | D | Emergent |
| 61 | *Sagittaria trifolia* L. | Alismataceae | M | Emergent |
| 62 | *Sphaeranthus indicus* L. | Asteraceae | D | Emergent |
| 63 | *Spirodela polyrrhiza* (L.) Schleid. | Araceae | M | Free floating |
| 64 | *Typha angustata* Bory & Choub. | Typhaceae | M | Emergent |
| 65 | *Utricularia aurea* Lour. | Lentibulareaceae | D | Submerged |
| 66 | *Vallisneria spiralis* L. | Hydrocharitaceae | M | Submerged |
| 67 | *Wolffia globosa* (Roxb.) Hartog & Plas | Araceae | M | Free floating |
| 68 | *Zannichellia palustris* L. | Potamogetonaceae | M | Submerged |



***Ottelia alismatoides***



***Hydrilla verticillata***



***Vallisnaria spiralis***



***Typha angustata***



***Sagittaria sagittifolia***



***Pistia stratiotes***



*Acorus calamus*



*Potamogeton nodosus*



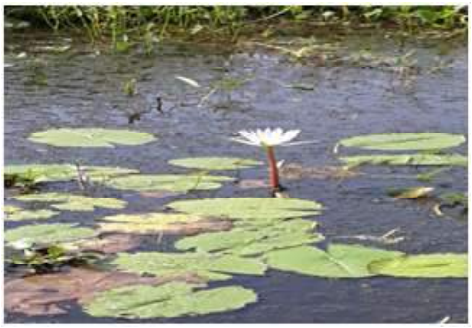
*Hygrorhiza aristata*



*Eriocaulon cinerum*



*Ranunculus sceleratus*



*Nymhaea pubescens*



*Nymphaea nouchali*



*Nelumbo nucifera*



***Ludwigia adscendens***



***Enhydra fluctuans***



***Spaeranthes indicus***



***Nymphoides indicum***

## REFERENCES


1. **Mooney, H.F. 1950.** *A supplement to the Botany of Bihar & Orissa.* Catholic Press, Ranchi.
2. **Christopher, C. 2006.** Study on the diversity of the aquatic vascular plants in Karikkakom, Thiruanant puram. *Ecol. Env. & Conc.* **12(4):** 661- 663.
3. **Cook, C.D.K.1996.** *Aquatic and Wetland plants of India* Oxford University Press, New York.
4. **Gupta, O.P. 1979.** *Aquatic weeds their menace and control. A text book and manual*. Today & tomorrow's Printers & Publisher, New Delhi.
5. **Jha, U.N. 1965.** Hydrophytes of Ranchi. *Trop. Ecol*. **6:**96-105.
6. **Kachroo, P. 1959.** Aquatic vegetation of Damodar Valley. I. Phanerogamic flora of forest water ponds, etc. *J. Asiat. Soc. Bengal.* **1:** 271-289.
7. **Maheshwari, J.K. 1960.** The vegetation of marshes, swamps and riversides in Khandwa District (Madhya Pradesh) J. Bombay Nat. Hist. Soc. **51:** 371-387.
8. **Majumdar N.C. 1965.** Aquatic and semiaquatic flora of Calcutta and adjacent localities. *Bull. Bot. Soc. Bengal*. **19:** 10-17.
9. **Mukherjee, P. 2001.** *The Floristic and Ecological studies of Aquatic Angiosperms of Lohardaga*. Ph.D. Thesis, Ranchi University, Ranchi (Unpublished).
10. **Mukherjee, P. & Kumar, P. 2003.** Studies on the angiospermic flora of ponds of Ranchi (Jharkhand). *Proceeding Nat. Sem.on Biodiversity, Taxonomy and Conservation.* Majalgaon College, Majalgaon.
11. **Mukherjee P. and Ghosh T.K. 2015.** Aquatic and Semi aquatic angiospermic flora of Lohardaga (Jharkhand) *Phytotaxonomy.* **15:** 134-145.
12. **Naskar, K.R. 1990.** *Aquatic & Semi-aquatic plants of the lower Ganga Delta.* Daya Publishing House, Delhi.
13. **Subramanyam, K. 1962.** *Aquatic Angiosperms*. CSIR, New Delhi.
14. **Singh, A.1998.** Floristic composition and vegetation profile of aquatic angiosperms of Hazaribag and adjacent area. Ph.D. Thesis, V.B.University Hazaribagh, (Unpublished).
15. **Singh, M. P. 1990.** Hydrophytes of Ranchi *J. Econ. Taxon. Bot.* **14:3**.
16. **Verma S.K. and Pandeya P.K. 2007.** Ethnomedicinal use of some aquatic plants in Jharkhand. *J. Haemotol & Ecotoxicol*. **2(1):** 34-37.
17. **Verma, S.K. and Pandeya P.K. 2008.** Floristic studies of Aquatic and Semi aquatic Angiosperms of Ratu Maharaja Pond, Ranchi, Jharkhand. *Biospectra*. **2(1):** 139-41.
18. **Mukherjee P. and Kumar J. 2019.** Studies on the aquatic and semi- aquatic Angiosperms of Kanke Dam, Ranchi, Jharkhand, *Phytotaxonomy*, Vol. **18.**
19. **Mukherjee P. and Kumar J. 2020.** Floristic Studies on Aquatic and Semi Aquatic Angiosperms of Major Water Bodies of Jharkhand, *J. Indian bot. Soc*. **100 (3&4):** 119-133.


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# Spermatorrhoea disease and ethnomedicinal plants with reference to Simdega District of Jharkhand, India

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*Received : 8th January, 2021 ; Revised : 9th February, 2021*

**Abstract-** Spermatorrhoea is a disease among males were White mucilage discharge with urine during early morning period, Pain in joints and muscles in lower portion of body, burning sensation in eyes is generally observed. A detailed survey identification work was carried by the author between 2016-2020 in ten blocks of Simdega district and 24 plants were identified along with their local name, Botanical name, Family and Plant parts used against Spermatorrhoea is described in the present paper.

**Key words: Spermatorrhoea, census, ethnomedicinal plants, folk medicine**

## INTRODUCTION

Simdega district ranks twenty-second in terms of total population in the state and twentieth in regards to decadal population growth rate (2001-11) among the twenty-four districts, with a sex ratio of 1000. The district comprises of ten blocks namely, Simdega, Pakar Tanr, Kurdeg, Kersai, Bolba, Thethaitangar, Kolebira, Jaldega, Basjor and Bano. As per Census 2011, the district has 451 villages and 1 town Census 2011 figure indicated that the percentage share of scheduled caste population to total population was 7.45 percent while that of scheduled tribes was 70.78 percent. Total area of Simdega District is 3761.20 sq km. The Major rivers of this area are Sankh, Deo, Girwa, and Palamara. Among these rivers Sankh is the main river of the district. Simdega district has about 32% of forested area. Important forest products are Saal seeds, Cocoon, Lac, Tendu leaves, Karanj, Chiraunji etc.. The major trees are Sal, Bija, Gamhar, Jackfruit, Jamun, Mango, Bamboo, Neem etc. Many folk medicines are also found in Simdega district of Jharkhand as it is a tribal dominated area.[1]

Spermatorrhoea or dhatu rog as it is commonly known is a disease which is prevalent among males of 15-40 yrs age groups, the common symptoms include White mucilage discharge with urine during early morning period, Pain in joints and muscles in lower portion of body, burning sensation in eyes.[2]

## MATERIALS & METHODS

A survey was made by visiting local areas of Simdega district and in different villages the patient suffering from Spermatorrhoea and cured patients were interviewed, the local vaidyas kaviraj, ojhas who could cure Spermatorrhoea were interviewed. The information were gathered regarding plants, plants parts used the way of treatment all were documented and plants were identified and herbaria of plants were made with the help of authentic texts.[3]

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## OBSERVATION

The observations was enlisted in a tabular form, 24 plants were observed after survey and identification, these are enlisted and arranged in alphabetical orders based on their botanical names, family, local names and parts used in Table 1.

**Table 1: List of plants used against Spermatorrhoea by ethnic people of Simdega district of Jharkhand state**

| S.No. | Botanical name | Family | Local name | Parts Used |
|---|---|---|---|---|
| 1. | *Achyranthus aspera* L. | Amaranthaceae | Chitchiri | Root |
| 2. | *Andrographis paniculata* | Acanthaceae | Kalmegh | Leaf, Stem |
| 3. | *Aloe vera* | Liliaceae | Ghrit kumari | Leaf |
| 4. | *Acacia arabica* | Mimosaceae | Babul | Leaf |
| 5. | *Bombax ceiba* | Bombaceae | Simal | Root |
| 6. | *Cajanus cajan* L. | Leguminosae | Arhar | Seed, Leaf |
| 7. | *Curculigo orchioides* | Hyponideaceae | Talmuli | Root |
| 8. | *Costus speciosus* | Costaceae | Kaeon | Root |
| 9. | *Diospyrus embryopteris* | Ebenaceae | Keond | Fruit, Root |
| 10. | *Elephantopus scaber* L. | Asteraceae | Minjur juti | Root |
| 11. | *Ficus glomerata* Roxb. | Moraceae | Gular | Bark |
| 12. | *Ficus benghalensis* | Moraceae | Bargad | Root |
| 13. | *Hemidesmus indicus* Br. | Papilionaceae | Anantmul | Root |
| 14. | *Lamanea coromandelica* M. | Anacardiaceae | Doka | Bark |
| 15. | *Madhuca latifolia* | Sapotaceae | Mahua | Bark |
| 16. | *Momordica dioica* Roxb | Cucurbitaceae | Ban Karela | Bark |
| 17. | *Mucana pruria* Hook | Papilionaceae | Alkusi | Seed |
| 18. | *Pureria tuberosa* D.C. | Fabaceae | Patal Kohra | Tuber |
| 19. | *Phyllanthus emblica* L. | Euphorbiaceae | Amla | Fruit, Leaf |
| 20. | *Scoparia dulcis* L. | Scorphulariaceae | Van dhania | Leaf |
| 21. | *Smilax zeylanica* | Smilaceae | Ramdatun | Root |
| 22. | *Terminalia arjuna* | Combretaceae | Arjun | Bark |
| 23. | *Tinospora cordifolia* | Menispermeaceae | Amrita | Leaf |
| 24. | *Uraria lagopodioides* L. | Papilionaceae | Ishwarjata | Root |

## RESULT & DISCUSSION

The present study is focused on the traditional healing process used and practiced by tribals and local inhabitants of Simdega district of Jharkhand against a male disease Spermatorrhoea commonly known as Dhatu rog. Simdega area is rich in flora diversity and the area is dominated by tribal population as compared to 26% of total Jharkhand population according to Census of 2011, Simdega has 70.78% population of tribals. Similarly Simdega lacks the modern amenities and the entire district has only one town i.e. Simdega, people of Simdega district still lacks the modern infrastructures for normal living and health, for health needs Jungle and the traditional knowledge is a big boon. The traditional knowledge of jari buti as the medicinal plants are described among locals, cures most of the deadly diseases,[4] including the dhatu rog or Spermatorrhoea. The need of the hour is to spread the traditional knowledge and a bridge between the allopathic medicine and herbal or traditional medicine should be established. This work has documented the plants with potential to cure Spermatorrhoea future aspect can be to find out the active principal which are responsible for curing the disease and establishing a link between these plants and allopathic medicines.

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The author is thankful to the local herbal practitioners of Simdega district, who were kind enough to share their ancestral knowledge against Spermatorrhoea, the author is also thankful to local forest department who supported in identifying the local areas and plants, special thanks to the patients of Spermatorrhoea of Simdega district who were ready to be interviewed.

## REFERENCES

1. **Census India.gov.in / 2011** Census / dchb / DCHB- A /20/2022 - PART-A-DCHB-SIMDEGA.pdf.
2. **Rosenman, Ellen Bayuk. 2003.** "Body doubles. The Spermatorrhoea Panic", Journal of the history of Sexuality **12(3):**365-399
3. **Bentham & Hooker. 1833.** Genera plantarum, Ad exemplaria imprimis in herbarlis kewensibusservata definite, III vol.
4. **Hembrom P.P. 1993.** Adivasi Asuadhi (Horopathy) Volume I-VII, Paharia Sewa Samiti, Pakur.
5. **Jain SK, Rao RR. 1997.** A Handbook of Field and Herbarium Methods, Today and Tomorrows' Printers and Publishers, New Delhi.

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# Comparative study of stomatal index of some ethnomedicinal plants of some species of family Malvaceae in Ranchi district of Jharkhand

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*Received : 9th January, 2021 ; Revised : 10th February, 2021*

**Abstract-** Present investigation deals with the study of stomata and stomatal index of two plants of family Malvaceae. Stomata are tiny openings present on the epidermis of leaves. We can see stomata under the microscope. Stomata are cell structures in the epidermis of leaves that involved in the exchange of carbon dioxide and water between plants and the atmosphere. Stomata present in leaves either on dorsal surface or ventral surface. The determination of stomatal index in the ethnomedicinal plants of family Malvaceae viz. *Hibiscus rosa-sinensis* L. and *Hibiscus esculentus* L. has been done for taxonomical purposes.

**Key words: Stomatal index, Malvaceae, *Hibiscus rosa-sinensis* L. and *Hibiscus esculentus* L.**

## INTRODUCTION

Stomata are the tiny microscopic pores present on the leaves. Stomata are the main channels for plants to exchange water and gas with the environment within the main photosynthetic organs. These are correlated with physiological activities such as photosynthesis, respiration and transpiration.[1] Stomata are easily affected by external environmental conditions. The micromorphological characteristics of leaves have been commonly used in plant classification.[2] Stomata are important organs in phylogenetic relationship of plants. It helps in plants origin, evolution and classification.[3]

After counting the stomata and epidermal cells, the stomatal index is determined. Stomatal index is the percentage of total number of stomata with total number of epidermal cells around the stomata in a unit area of leaf.

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## MATERIAL & METHOD

The collected plants were identified with the help of flora Botany of Bihar and Orissa Vol I-VII.[4] The leaves of *Hibiscus rosa-sinensis* L. were collected from Ratu road, Ranchi in January, 2021 and the leaves of *Hibiscus esculentus* L. were collected from Kanke, Ranchi in February 2021. The distribution of stomata on the upper and lower surface of the leaves were studied by removing the peels of the leaf with the help of blade after that it was properly cleaned and stained with the safranin then prepared the slide and observed under the microscope. Leaf of *Hibiscus rosa-sinensis* L. stomata were absent on the upper surface and on the lower surface of the leaf Anomocytic stomata were present. On the both surface of

the leaf of *Hibiscus esculentus* L. two types of stomata's were present that is Anisocytic and Paracytic. The number of stomata and epidermal cells were counted. After that stomatal index of each surface of the leaf were calculated using following formula:-

$$\text{Stomatal index} = \frac{\text{No. of stomata}}{\text{No. of stomata} + \text{No. epidermal cells}} x\ 100$$

S.I = Stomatal index
S = Number of stomata per unit area
E = Number of epidermal cells in the same area

## RESULTS & DISCUSSION

### PLANT - 1

*Hibiscus rosa-sinensis* L.

Stomatal index on upper surface = Nil

**Table 1: Stomatal index on lower surface**

| No. of Observation | No. of stomata per Unit area (s) (40x) | No. of epidermal Cell (E) (40x) | S.I = S / S + E ×100 |
|---|---|---|---|
| 1. | 28 | 56 | |
| 2. | 21 | 58 | |
| 3. | 23 | 52 | |
| 4. | 25 | 46 | S.I = 32.81 % |
| 5. | 32 | 43 | |
| 6. | 24 | 55 | |
| 7. | 26 | 52 | |
| 8. | 25 | 48 | |
| 9. | 21 | 52 | |
| 10. | 27 | 54 | |
| **Total** | **252** | **516** | |

Type of stomata in lower surface = Anomocytic

### PLANT - 2

*Hibiscus esculentus* L.

**Table 2: Stomatal index on upper surface**

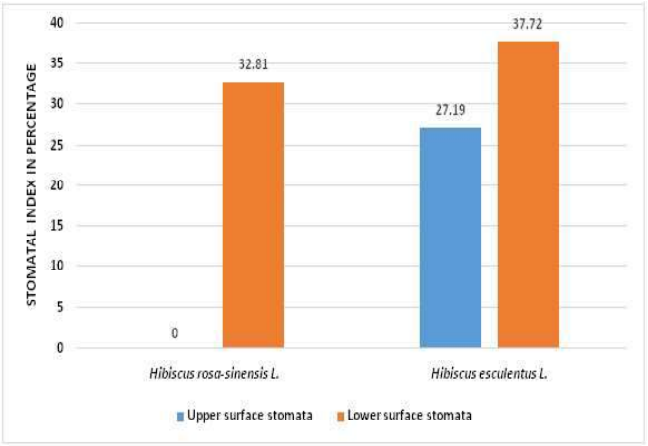
| No. of Observation | No. of stomata per Unit area (s) (40x) | No. of epidermal Cell (E) (40x) | S.I = S / S + E ×100 |
|---|---|---|---|
| 1. | 19 | 45 | |
| 2. | 15 | 49 | |
| 3. | 17 | 38 | |
| 4. | 14 | 41 | 27.19 % |
| 5. | 17 | 43 | |
| 6. | 16 | 46 | |
| 7. | 15 | 48 | |
| 8. | 16 | 36 | |
| 9. | 14 | 47 | |
| 10. | 18 | 38 | |
| **Total** | **161** | **431** | |

Type of stomata in upper surface = Anisocytic and Paracytic

**Table 3 :Stomatal index on lower surface**

| No. of Observation | No. of stomata per Unit area (s) (40x) | No. of epidermal Cell (E) (40x) | S.I = S / S + E ×100 |
|---|---|---|---|
| 1. | 26 | 53 | |
| 2. | 21 | 48 | |
| 3. | 28 | 45 | |
| 4. | 34 | 45 | |
| 5. | 23 | 39 | 37.72 % |
| 6. | 27 | 42 | |
| 7. | 23 | 38 | |
| 8. | 31 | 55 | |
| 9. | 28 | 36 | |
| 10 | 31 | 48 | |
| **Total** | **272** | **449** | |

Type of stomata in lower surface = Anisocytic and Paracytic




**Graph 1. Comparative study of stomatal index on upper and lower surface of the leaf**

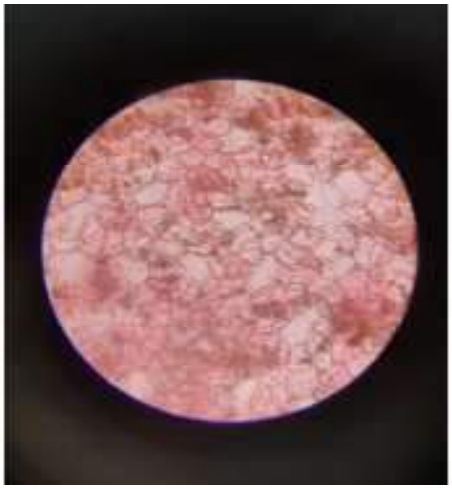


**Figure 1: *Hibiscus rosa-sinensis* L. Upper surface of stomata**



**Figure 2 : *Hibiscus rosa-sinensis* L. Lower surface of stomata**



**Figure 3 : *Hibiscus esculentus* L. Upper surface of stomata**



**Figure 4 : *Hibiscus esculentus* L. Lower surface of stomata**

The present investigation deals with the type of stomata in investigated plant species and their stomatal index were calculated in *Hibiscus rosa-sinensis* L. on upper surface of leaf stomata was absent and in lower surface Anomocytic type of stomata were found the stomatal index calculated was 32.81%. In *Hibiscus esculentus* L. leaves two types of stomata's Anisocytic and Paracytic were present in both upper and lower surface. In upper surface of leaf, stomatal index calculated was 27.19%. In lower surface of leaf, stomatal index calculated was 37.72%. The highest stomatal index was calculated in lower surface of leaf of *Hibiscus esculentus* L. and the lowest stomatal index was calculated in upper surface of leaf of *Hibiscus esculentus* L. The stomatal indexes of some medicinal plants of Malvaceae family were studied.[5] The stomatal study plays an important role in the classification of plants.

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## REFERENCES

1. **Taylor SH, Franks PJ, Hulme SP, Springs E, Christin PA. 2010.** Photosynthetic pathway and ecological adaptation stomatal trait diversity amongst grasses. *New Phytol.* **193(2):**387-396.
2. **Ma D. 1987.** stomates [M]. Beijing: science press:10-30.
3. **Liu J, Wang I, Xing Y. 2014.** Morphology structure of leaf epidermis of Maloideae in Heilongjiang. *J chin Electron Microsesoc.* **3(1):**69-76
4. **Haines, H.H. 1921.** *The Botany of Bihar and Orissa* vol (I-VI). Bishen Singh Mahendra Pal Singh 23-A, New Cannaught place Dehradun.
5. **Chachad D.P. and Meenakshi Vaidya. 2016.** Stomatal studies on some selected plants of Malvaceae. *World Journal of Pharmaceutical Research.* ISSN 2277-710

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# Qualitative phytochemical screening of leafy vegetable consumed by aborigine of Santal Pargana, Jharkhand

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*Received : 09th January, 2021 ; Revised : 10th February, 2021*

**Abstract-** Santal Pargana is a home of tribal groups mainly "Santals". They have wider array of traditional leafy vegetable that are cheap source of nutrition as well as medicinal value. The utilized leafy vegetables are rich source of different minerals as well as very high in phytochemicals such as Flavonoids, Saponin, Tannin, Phenolic acid, Glycosides, Terpenes, Anthocyanin, organic acid, Steroids, oils, Alkaloids and Amino acid. In one hand phytochemicals are used as protective measures of plants while on the other hand they give beneficial effect on human health. The aim of research is to determine the qualitative nature of phytochemicals present in leafy vegetables which are utilized by the tribal of Santal Parganas as well as to make aware about the importance of potency of these phytochemicals which are unknown for the aborigine people of Santal Parganas locality. Further it is also the aim to aware the urban people.

**Key words: Leafy vegetables, Phytochemicals, Aboriginal, Santal Pargana**

## INTRODUCTION

In India there are large no of tribal communities living in forest areas who lead simple primitive life form. They are most dependent upon plants, animals and their products which are present in their surroundings. Out of the whole tribal population in Jharkhand, Santal Parganas has mostly Santal tribals. Santal Parganas comprises 6 districts of Jharkhand which lies between 23°40' N and 25°28'N latitude and between 86°28'E and 87°57'E longitude. The region extends over 5470 square miles. Profile of the people in Santal Parganas as the study area is interesting as this tract of Jharkhand is ethnographically rich with the aboriginal and the migrants.

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Rice is the main cereal food and the communities depends a lot on the wild vegetables, fruits, flowers, tubers and rhizomes that grow profusely on their own without any inputs. The study area is very rich with respect to the diversity of green leafy vegetables that are cultivated and is collected from wild.

The leafy vegetables most often come out from short lived herbaceous plants, whereas leaves of some woody plants are also eaten by local people. They collected these from agricultural field, non-agricultural field as well as from forest land to supplement their staple food. The availability of leafy vegetables totally depends upon seasons. Besides their dietary need they also sale leafy vegetables in nearby markets.

These local leafy vegetables are rich source in nutrient, minerals and phytochemical active molecules.

Leafy vegetables contain number of phytochemicals which helps to protect them from free radicals and reduce oxidative stress. The phytochemicals which are found in these leafy vegetables are mainly Flavonoids, Saponin, Tannin, Phenolic acid, Glycosides, Terpenes, Anthocyanin, Organic acids, Steroids, Oils, Alkaloids, Amino acids etc. The phytochemical analysis thus collected by different researchers help in finding the potential of leafy vegetable which are by aboriginals without knowing their potential.

Several studies have been carried out on ethno-medicine, Ethnobotany and on wild edible of the indigenous and tribal population of this region by different investigators like Verma *et al.* (1986)[1], Sinha and Ranjan (2006)[2], Das (2007)[3], Kumar and Abbas (2012)[4], Das and Mukherjee (2012)[5], Das and Bondya (2016)[6], Kumar (2014)[7], Bondya *et al.* (2015)[8] in the recent past but qualitative phytochemical screening of leafy vegetable studies have been carried out measures of the area.

## MATERIALS AND METHOD

Intensive & were conducted in the study area during June to September 2020 in different region of Santal Pargana such as Dumka, Pakur, Kathikhund, Udhwa, Shikari Pará, Raneshwar, Masaliya, and nearby villages where tribal reside. Voucher specimens and ethnomedicinal information were collected following the methods described by Jain (1965)[9] and Jain & Rao (1976)[10]. The tribal hamlets and forest pockets were first visited to collect a broad range of information about the different types of leafy vegetables used by the tribal community. The information was gathered and views or knowledge about the traditional and medicinal properties of leafy vegetables which are being consumed by them were recorded. Plant's specimens were collected with the informant from home gardens, cultivated land, forests and local markets. Plants were identified with the help of available flora and their vernacular names were verified. Haines (1961)[11], Hembrom (1991)[12], Jain (1991)[13].

## RESULTS

After collecting the data about leafy vegetables which are regularly consumed by aboriginals, we collected the data of their phytochemical nature through different works done by different researchers. The natures of these phytochemicals are:

**Flavonoids:** These are the major groups of compounds that are found in 23 plants out of 32 plants. It acts as antioxidant and has multiple biological activities including Vasodilator, antibacterial, immune stimulating, anti-allergic & antiviral.

**Saponin & Alkaloids:** These are the 2nd major phytochemical constituent among 32 plants. Saponin decreases blood lipids, and lower blood glucose. Alkaloids are used for reducing headache and fever. They are attributed for anti-bacterial and analgesic properties. The plant species use alkaloids to protect themselves against herbivores. They could also be used as a natural source of insecticides and fungicides.

**Tannin and Phenolic acid:** These are 3rd major phytochemicals. Tannin contains anti diabetic properties, astringent in nature and is used for intestinal disorder. The phenolic compounds are most ubiquitous group of plant metabolites appeared to possessed biological properties such as anti-apoptosis, anti-aging, anti-carcinogen, Cardiovascular and protective in nature.

**Terpenes, Glycosides & Steroids:** These are next mostly found phytochemicals in leafy vegetables. Terpenoids molecules have anti-cancer properties. It has also therapeutic properties including anti- parasitic, anti-microbial, anti-allergic, anti-spasmodic, anti-hyperglycemic, anti-inflammatory and immunomodulatory properties. Steroids have been reported to have anti-bacterial properties and they are very important compound especially due to relationship with compounds such as sex hormone. Glycosides are naturally cardio active drugs used in the treatments of congestive heart failure they have potent in curing cardiac in sufficient cough and circulatory problems.

**Amino Acids and Oils:** These phytochemicals are found in few leafy vegetables. It helps in the treatment of imbalance of brain metabolism and neurotransmission. Oils provide flavor and has pharmaceutical and cosmetic properties.

## DISCUSSION

Around 32 leafy vegetable plant species have been collected which are frequently used by tribal communities. The screening and study of 32 different plant specimens belonging to different families for phytochemical constituents such are Flavonoids, Saponins, Tannins,

Phenolic acid, Glycosides, Terpenes, Anthocyanin, Organic acid, Steroids, Oils, Alkaloids and amino acid. The distribution of these constituent in the plant specimens were assessed and compared. A 32 plant specimens Flavonoids, Saponins, Tannins, Phenolic acid and Alkaloids are major phytochemical constituents of these plants. Glycosides, Terpenes, Steroids and Amino acids are found as 2nd major phytochemicals in 15 plants. Anthocyanin have been found in 1 plant while Organic acid and Oils are found in 2 and 7 plants respectively.

The maximum leafy vegetables belong to family Amaranthacea, Lamiaceae, Fabaceae, and Cleomaceae. The 12 types of phytochemicals in these leafy vegetables show the nutritional and medicinal properties.

**Table 1: Phytochemical Analysis of leafy vegetables**

| Sl.No. | Name of leafy vegetables | Santali Name | Family | Flavonoids | Saponin | Tannin | Phenolic Acid | Glycosides | Terpenes | Anthocyanin | Organic acid | Steroids | Oils | Alkaloids | Amino Acid |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *Aerva lanata* (L.) Juss. | Lopong arak | Amaranthaceae | + | + | + | - | - | + | - | - | - | - | + | - |
| 2 | *Alternanthera sessilis* Br. | Garundi arak | Amaranthaceae | + | + | + | + | - | + | - | - | + | - | + | + |
| 3 | *Amaranthus dubius* | Gandhari | Amaranthaceae | + | - | - | + | - | - | + | - | - | - | + | - |
| 4 | *Amaranthus gangeticum* Roxb. | Lal Gandhari | Amaranthaceae | + | + | + | + | + | - | - | - | - | - | + | - |
| 5 | *Amaranthus spinosus* Linn. | KanteliChaulai | Amaranthaceae | - | - | - | - | + | + | - | - | - | - | + | - |
| 6 | *Amaranthus viridis* Linn. | Lotia arak | Amaranthaceae | - | + | + | + | + | - | - | - | - | - | + | + |
| 7 | *Antidesma diandrum* Roxb. | Matha arak | Euphorbiaceae | + | + | + | + | - | - | - | - | - | + | - | + |
| 8 | *Basella alba* L. | Purai arak | Basellaceae | + | + | - | - | - | - | - | + | - | + | - | + |
| 9 | *Bauhinia purpurea* L. | Sing arak | Fabaceae | + | + | + | + | + | - | - | - | + | + | + | + |
| 10 | *Boerhavia diffusa* L. | Khapra arak | Nyctaginaceae | + | + | + | + | + | - | - | - | - | - | + | - |
| 11 | *Celosia argentea* L. | Sirgiti arak | Amaranthaceae | + | + | + | + | + | - | - | - | - | - | + | - |
| 12 | *Centella asiatica* Linn. | Beng arak | Apiaceae | + | + | + | - | + | + | - | - | + | - | + | - |
| 13 | *Chenopodium album* Linn. | Bathua arak | Amaranthaceae | + | + | - | + | - | - | - | - | - | - | + | - |
| 14 | *Cleome gynandra* L. | Hurharia arak | Cleomaceae | + | + | + | - | - | + | - | - | + | - | - | - |
| 15 | *Cleome viscosa* L. | Setakata arak | Cleomaceae | + | + | + | + | + | + | - | + | + | + | - | + |
| 16 | *Colocasia esculenta* L. | Saru arak | Araceae | + | + | + | + | + | + | - | - | + | - | + | - |
| 17 | *Commelina benghalensis* L. | Kana arak | Commelinaceae | + | + | + | - | - | - | - | - | - | + | - | - |
| 18 | *Cordia dichotum* | Buj arak | Boraginaceae | + | + | - | + | + | - | - | - | + | - | + | - |
| 19 | *Enhydra fluctuans* Lour | Hencha arak | Asteraceae | + | + | + | + | - | + | - | - | - | - | - | + |
| 20 | *Ficus geniculata* Kurz | Putkal arak | Moraceae | - | + | + | - | - | - | - | - | - | - | - | - |
| 21 | *Hygrophila auriculata* | Dangrakanta | Acanthaceae | + | - | - | + | + | + | - | - | + | - | - | + |
| 22 | *Ipomoea aquatica* | Kalmi arak | Convolvulaceae | - | + | + | + | + | - | - | - | + | - | + | - |
| 23 | *Leucas aspera* Spreng. | Guma | Lamiaceae | - | - | - | + | + | + | - | - | + | - | + | + |
| 24 | *Leucas cephalotes* Spreng. | Dhurup arak | Lamiaceae | - | - | + | + | - | + | - | - | + | + | + | - |
| 25 | *Marsilea minuta* L. | Chatom arak | Marsileaceae | + | - | + | + | - | + | - | - | - | - | + | + |
| 26 | *Melochia corchorifolia* L. | Thuya arak | Malvaceae | | - | - | - | + | - | - | - | + | - | + | - |
| 27 | *Oxalis corniculata* Linn. | Tandichatom | Oxalidaceae | + | - | + | + | - | - | - | - | + | + | - | - |
| 28 | *Polygonum plebeium* R.Br. | Chimti arak | Polygonaceae | + | - | - | + | - | - | - | - | - | - | - | - |
| 29 | *Portulaca oleraceae* Linn. | Ulinalan arak | Portulacaceae | + | + | - | - | + | + | - | - | + | - | + | - |
| 30 | *Senna tora* L. | Chakaunda arak | Fabaceae | + | + | + | - | - | - | - | - | - | - | + | - |
| 31 | *Talinum paniculatum* | Dakhin arak | Talinaceae | + | + | + | - | - | - | - | - | + | - | + | - |
| 32 | *Vangueira spinosa* Roxb. | Makra arak | Rubiaceae | + | - | - | - | - | + | - | - | - | - | + | - |

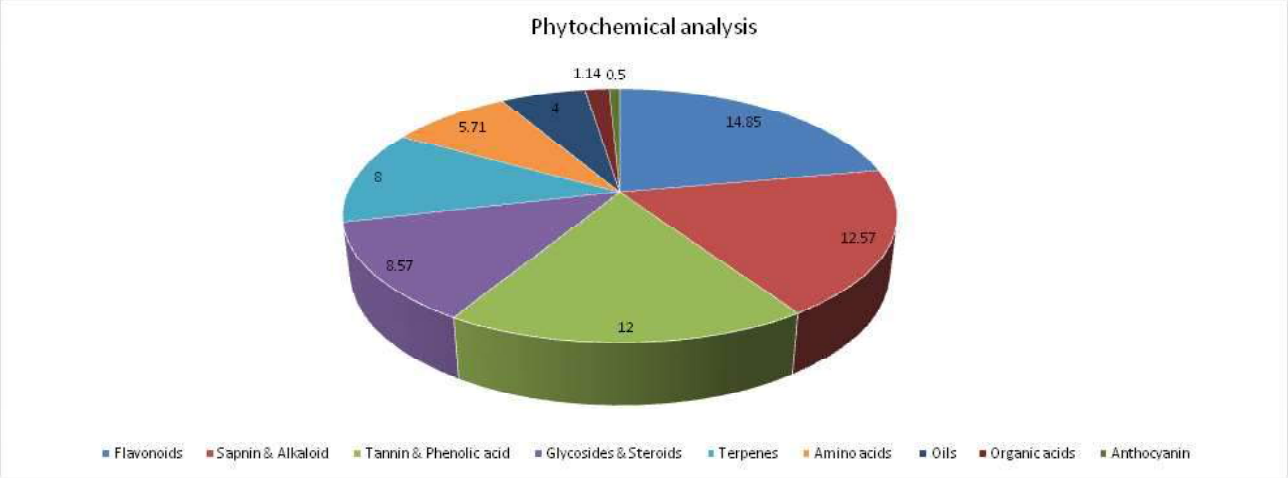



**Fig.1- Percentage analysis of phytochemicals in leafy vegetables**

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## REFERENCES

1. **Verma, S.K., Srivastava, D.K. and Sinha, R.K. 1986.** Tapping some medicinal plants for the tribal development in Santhal Parganas. *Journal of Regional Studies*, pp. 1-2.
2. **Sinha, R.K. and Ranjan, R. 2006.** Phytodiversity of medicinal flora of Santhal Paragana vis-à-vis Human interference, In: *Modern trends of research in ecology and environmental science.* Chapter 4: pp. 43-56. Ed. B.K.Gupta & A.Gupta, Mittal Publications, New Delhi.
3. **Das, A. 2007.** Taxonomic, Anatomical and Ethnobotanical studies of plants used in traditional medicinal system in Jamtara District. Ph.D. Thesis, S.K.M. University, Dumka, Jharkhand.
4. **Kumar, K and Abbas, S.G. 2012.** Ethnomedicinal composition depends on floristic composition: A case studied in Sal forests of Jharkhand. *Int. J. of Pharm. & Life Sci. (IJPLS)*, Vol. **3,** Issue **5**.
5. **Das, A. and Mukherjee, P. 2012.** The conservators, of the natural resources & Traditional farming knowledge, helpful in combating climate change-Tribal communities, of Dumka district (under Santhal Paraganas). *Journal of Recent Advances in Science.* Vol. **II (2):** pp. 17-23.
6. **Das, A and Bondya, S. L. 2016.** Underutilized edibles of Dumka, Santhal Praganas: Scope for future nutrition. *Conservation of medicinal plants conventional and modern approaches.* Ed. H.K. Chourasia, Omega Publications, New Delhi. pp. 264- 269.
7. **Kumar, S. 2014.** *Ethnobotanical studies in India*, Deep publication, New Delhi
8. **Bondya, S. L, Mukherjee, P and Das A 2015.** Traditional Knowledge: Best medicine mines, a way forward. *Recent Advances in Ethnobotany.* Ed. Sanjeev Kumar, Deep Publications, New Delhi. pp. 191-196.
9. **Jain, S.K. 1965.** Medicinal Plantlore of tribals of Bastar. *Econ. Bot*. **19:** 236-250.
10. **Jain, S.K. & Rao, R.R. 1976.** *A Hand Book of Field and Herbarium Methods*. Today and Tomorrow's Publishers, New Delhi.
11. **Hains, HH, 1961.** The Botany of Bihar and Orissa, Vol **3,** Botanical Survey of India Calcutta.
12. **Hembrom,PP, 1991.** Tribal medicine in Chotanagpur and Santhal Pargana of Bihar state, *Ethnobotany*. **3:** 97-99.
13. **Jain, S.K. 1991.** *Dictionary of Indian Folk Medicine and Ethnobotany*. Deep Publications, New Delhi.

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# Comparative study of moisture content of some ethnomedicinal plants of some species of family Malvaceae in Ranchi district Jharkhand

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**Abstract-** The family Malvaceae includes herbs, shrubs and trees. They are widely varied and include several important crops, which are mainly used as source of medicines, fiber, food etc. The importance of the studied species of the family Malvaceae consists in their very rapid emergence and fast growth. The tribal communities are having very deep knowledge about these plants. The tribes are using these plants in their daily life. Water is one of the most vital constituents in plants. The study was conducted to investigate moisture content in leaves. The analysis of the moisture content was done by drying fresh leaves. Some species of family Malvaceae like *Hibiscus rosa-sinensis* L., *Hibiscus esculentus* L. are having medicinal value.

**Key words: Moisture content, Malvaceae, ethnomedicinal plants, tribal communities.**

## INTRODUCTION

Ranchi is the capital of Jharkhand. In Ranchi district different types of tribal communities dwell here. The tribals of Ranchi district use traditional remedies for different diseases are mainly plants. Moisture content is the amount of moisture in the sample given as a percentage of the sample's original weight. Dry content is the amount of solids which are left after drying given as a percentage of the sample's original weight. The comparative studies of moisture content of some species of family Malvaceae are also having medicinal properties. The difference between initial weight of fresh leaves and final weight of same leaves after drying is the moisture content. Percentage of moisture content was calculated which varies from one plant to another.

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## MATERIALS & METHODS

In study of moisture content, plants of family Malvaceae were selected from the different areas of Ranchi district. Plant species were identified with the help of Botany of Bihar and Orissa vol I-VII.[1,2] The fresh leaves of plants were collected then washed. After washing the leaves were kept in a cool place for drying and then weighted in weighing machine. Then these leaves were kept in hot air oven at 55°C temperature for half an hour. Repeated this activity for 2 or 3 times. After that dried leaves were weighted. The difference between initial weight of the green leaves and final weight after drying

was the moisture content. Percentage of moisture content was calculated with the formula:-

$$\text{Moisture content} = \frac{\text{Initial weight of leaves} - \text{Final weight after drying}}{\text{Initial weight of leaves}} x\ 100$$

## RESULT & DISCUSSION

In this paper comparative study of moisture content of two ethnomedicinal plants viz *Hibiscus rosa-sinensis* L. and *Hibiscus esculentus* L. has been done. Several studies in this field have been done in different area of India.[3]

**Table 1 : Moisture content of *Hibiscus rosa-sinensis* L.**

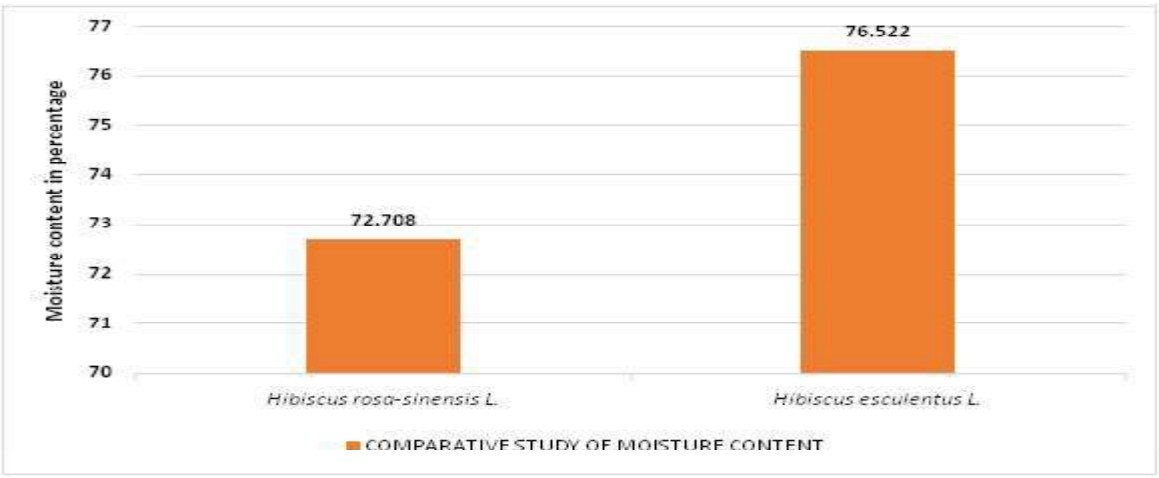
| Material | Number of observation | Weight of fresh leaves | Weight of dried leaves | Difference | % of Moisture content |
|---|---|---|---|---|---|
| Fresh leaves of *Hibiscus rosa-sinensis* L. | 1. | 10 gms | 2,680 gms | 7.320 gms | 73.2 % |
| | 2. | 10 gms | 2.748 gms | 7.252 gms | 72.52 % |
| | 3. | 10 gms | 2.748 gms | 7.252 gms | 72.52 % |
| | 4. | 10 gms | 2.716 gms | 7.284 gms | 72.84 % |
| | 5. | 10 gms | 2.750 gms | 7.250 gms | 72.5 % |
| | 6. | 10 gms | 2.838 gms | 7.162 gms | 71.62 % |
| | 7. | 10 gms | 2.800 gms | 7.200 gms | 72 % |
| | 8. | 10 gms | 2.704 gms | 7.296 gms | 72.96 % |
| | 9. | 10 gms | 2.646 gms | 7.354 gms | 73.54 % |
| | 10. | 10 gms | 2.662 gms | 7.338 gms | 73.38 % |
| **Total** | | | | | **727.08 %** |

Average Moisture content = 72.708

**Table 2 : Moisture content of *Hibiscus esculentus* L.**

| Material | Number of observation | Weight of fresh leaves | Weight of dried leaves | Difference | % of Moisture content |
|---|---|---|---|---|---|
| Fresh leaves of *Hibiscus esculentus* L. | 1. | 10 gms | 2.348 gms | 7.652 gms | 76.52 % |
| | 2. | 10 gms | 2.352 gms | 7.648 gms | 76.48 % |
| | 3. | 10 gms | 2.322 gms | 7.678 gms | 76.78 % |
| | 4. | 10 gms | 2.371 gms | 7.629 gms | 76.29 % |
| | 5. | 10 gms | 2.346 gms | 7.654 gms | 76.54 % |
| | 6. | 10 gms | 2.386 gms | 7.614 gms | 76.14 % |
| | 7. | 10 gms | 2.328 gms | 7.672 gms | 76.72 % |
| | 8. | 10 gms | 2.354 gms | 7.646 gms | 76.46 % |
| | 9. | 10 gms | 2.324 gms | 7.676 gms | 76.76 % |
| | 10. | 10 gms | 2.347 gms | 7.653 gms | 76.53 % |
| **Total** | | | | | **765.22 %** |

Average Moisture content = 76.522 %




**Graph 1- Comparative study of moisture content**

The comparative study of two plant species of family Malvaceae it is found that the moisture content of *Hibiscus rosa-sinensis* L. was 72.708 % having low moisture content and *Hibiscus esculentus* L. was 76.522 % having high moisture content. The investigated plants are highly medicinal. *Hibiscus rosa-sinensis* L. are used to treat hair loss, fever, wounds etc and *Hibiscus esculentus* L. are used in the treatment of gonorrhea, ordorurinae etc. These plants are used in our daily life. Few earlier works have been done in this field in different areas of Jharkhand.[4] It is an urgent need to conserve different types of ethnomedicinal plants which are going to be extinct.

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## REFERENCES

1. **Haines H.H. 1921.** *The Botany of Bihar and Orissa vol* (I-VI). Bishen Singh Mahendra Pal Singh 23-A, New Cannaught place Dehradun.
2. **Jain S.K. and Rao P.P. 1978.** "*A Hand Book of Field and Herbarium Method"* Today and tomorrow Publication, New Delhi.
3. **RMNA Wijewardana, S. B. Nawarathne, I. Wikramasinghe. 2015.** Evaluation of Physicochemical and Antioxidant Property of Dehydrated Hibiscus (*Hibiscus rosa-sinensis*) Flower Petals and Its Stability in Product Preparation. *International Journal of Innovative Research In Technology*. ISSN: 2349-6002
4. **Hembrom P.P. 1995.** *Adivashi-Aushdhi (Horopathy)* I (VII): Paharia Seva Samity. Pakur, Jharkhand.

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# Determinants of antipredator behaviour in birds

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**Abstract-** Predation influences the fitness of prey and leads to antipredator strategies in prey populations. Anthropological changes are also detrimental to birds' survival. Hence, birds have evolved defenses against predation. Prey defenses include a number of components including physical, behavioural and ecological elements that are significant in determining the antipredator response of birds. The anti-predatory response of birds suggests their adjustments towards fear from predation. Study of these factors enables us to understand the selective forces against birds' response to natural predation risk. These components support the animal in immediate survival but they have likely costs on food availability, feeding time, reproductive output and parental care. Identified gaps in the understanding of antipredator behaviour of birds emphasize the need to initiate behavioural studies in different habitats and geographic regions of the country to frame guidelines for conservation of birds.

**Key words: Anthropological, antipredator response, components, predation, survival.**

## INTRODUCTION

Most of the animals in the ecosystem are potential prey, except some upper trophic level carnivores. A behavioural adaptation to minimize the chances of being eaten is called anti-predatory behaviour. Various strategies undertaken by animals and birds as anti-predatory behaviour include individual and social strategies. Birds which play an important role in ecosystem often face various types of disturbance stimuli and show different tactics and anti-predator behaviour against it, like escaping, freezing, deception, mimicry, increased detection of predators, etc. Birds are feathered vertebrates and accord to various ecological functions extending over a broad range of positions in ecosystem. The avian habitat is roughly divided into forest, scrub and wetlands, although many species require a mixed type of habitat.[1]

As the prey becomes aware of the predator's approach, it does not fly away instantly; instead it monitors the predator's approach till flying away becomes useful. The cost of fleeing essentially results into loss of opportunities to feed and engage in social activities like courtship, mating, territorial defence, and performing other activities that increase fitness. Energetic costs of fleeing and risk of injury as a consequence of fleeing are other expenses of fleeing. The opportunity cost of fleeing decreases with the decrease in predator-prey distance. Decrease in the predator-prey distance increases the risk of not fleeing due to increase in predation risk. The distance at which birds start to move or flee against intruder is called

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as Flight initiation distance (FID). Similarly, the term escape distance is also used to indicate minimum distance from the invader. It is also called as flush distance or escape flight distance. The distance at which a predator is detected along with the distance at which the prey flees off the predator is called buffer distance. Buffer zone is the gap within which human action is limited to reduce interruption to wildlife. Birds behave in a different manner in areas with varied levels of human disturbance. The flight initiation distance was found to decrease in areas where more people visited the beach, and specifically in a small area near the pier. North Indian resident birds have been observed to be habituated towards people because of non-violent shielding behaviour of Hindus. Samia *et al.* (2015)[2] investigated 96 bird species and observed morphological, life history and natural history traits to correlate with variation in avian escape strategy. Prey initiate flight soon after detecting a predator so as to minimize attention costs related to ongoing monitoring of the location of the predator. Brain mass, body size, habitat structure and group sizes are the main predictors of escape strategy. FID for 2067 adult barn swallows was registered by Moller (2014)[3] from 1983 to 2012. Of these birds, 18 individuals were captured by predators, domestic cats (*Felis catus*) or sparrow hawks. Significantly shorter FID was observed among depredated individuals than survivors. Moller *et al.* (2013)[4] provided estimates of FID in resident and migratory birds before and after a severely cold winter along a latitudinal gradient in Europe. Decrease in FID was evident before and after the cold winter, but only among resident species, and more so for rural than urban populations of the same species. Fernandez- Juricic *et al.* (2005)[5] reported visibility and detectability as important factors influencing a species response to an approaching threat.

In this review, we focus on a number of biological, ecological, antipredator distances and behavioural responses that influence the antipredator behaviour of birds, eventually leading to bird conservation.

**Biological factors**

**Body size**

Moller *et al.* (2008)[6] studied 63 species of birds in Denmark to analyze susceptibility of predation by sparrow hawks in relation to mean FID, showing a negative relationship accounting for 13% of the variance after adjusting for body size. Preys of intermediate body size are easier to handle and also yield significant resources and hence are preferred by predators. Blumstein (2006a)[7] found that larger flying birds take longer flight before take-off and longer FID is expected in larger species for this reason alone. Approach towards smaller bird species was easier in suburban than in rural areas. Body size can be considered as the best predictor of FID because it is also the best predictor of life history, anatomy, physiology, behaviour, and conservation status. As the prey's size brings it into the range favoured by the predator, predation risk increases.

**Eye size, Lens and tympanic membrane**

In an initial study conducted by Blumstein *et al.* (2004)[8], there was no indication that large eyes had evolved in species living in isolated habitats, or in species eating mobile prey. Contradictorily, Moller and Erritzoe, (2010)[9] analyzed eye size in 97 bird species and reported that species with large eyes and higher ability to resolve visual details will have relatively longer FID as mediated by longer alert distances (ADs). Relatively higher AD in birds with large eyes and increased perch height was reported by Blumstein *et al.* (2004)[8]. They found that birds assessed themselves to be at a lower risk of predation when they were 3 m above the ground compared with being <3 m above ground. Birds of perch height 3m off the ground had greater FIDs in six species, but in three species had the opposite response. A significant positive relationship was found between eye size and alert distance in 23 species. Hearing in birds is related to the tympanic membrane in birds. For a given body size, a larger tympanic membrane implies better hearing ability. FID was reported to be related to the size of the tympanic membrane and the footplate of stapes in 37 species of birds. A significant positive correlation was observed between FID and tympanic membrane. This indicates better hearing ability. It may be deduced that auditory information has a remarkable role in predator-prey interface.[10]

**Brain size**

Analysis of 27 bird species from Australia by Guay *et al.,* (2013)[11] found no significant association between brain size and FID. Moller and Erritzoe, (2014)[12] suggested that a relatively larger brain allows for longer FID because of faster reaction to a likely predator. The intentions of the predator and the chance of attack before fleeing are also assessed by a larger brain. Different results were observed by Cooper and Blumstein (2013)[13] who reported

that in a sample of 107 bird species, FID increased with relative eye size, but decreased with relative brain size. Additionally, FID increased independently with size of the cerebellum, which plays an important role in motor control.

**Wing shape**

Fernandez-Juricic *et al.* (2006)[14] studied the relationship between body mass and size, indices of wingtip shape and hind limb length, flight initiation distance and landing distance of 83 birds of North America, Europe, and Australia and reported that pointed and concave wingtips and shorter hind limbs make birds more vulnerable to predation. Body mass and size, indices of wingtip shape and hind limb length, flight initiation distance and landing distance of 83 birds of North America, Europe, and Australia. Birds with pointed wings fled at longer distances and landed much farther than species having rounded wings, although hind limb length was not associated with flight initiation and landing distance.

**Plumage**

The toll of having conspicuous colours of plumage in birds leads to increased predation risk, affecting their anti-predator behaviour. McQueen *et al.* (2017)[15] employed behavioural observations and alarm call playback in Male superb fairy-wrens (*Malurus cyaneus*) to find out how birds adapt responses to predation risk in accordance with their plumage colour. They observed blue fairy-wrens to flee frequently in reaction to alarm calls, emerged from their cover in a longer time, foraged in cover and were more vigilant. Thus, fairy wrens with pronounced plumage were cognizant of being at a higher predation risk and accordingly change their behaviour. Antipredator behaviours of conspecific males within sexes have been shown to escalate with the potency of colour signals exhibited by them.[16,17]

**Ecological Factors**

**Habitat**

Adaptation to an urban environment in juncos (*Junco hyemalis*) was reported by Atwell *et al.* (2012)[18] as a rapid change in FID between urban and rural populations. Intermediate to large effects of relationship between FID, habitat openness and habitat diversity have been reported in most studies, although Blumstein, (2006a)[7] reported a small effect. Moller and Garamszegi, (2012)[19] reported that the horizontal component of FID was strongly related to habitat openness as compared to the vertical component. However, a strong relationship between FID and habitat was found by Valcarcel and Fernandez-Juricic (2009)[20] while testing the safe habitat hypothesis. There is also indirect evidence of predators affecting FID of prey in urban areas as many birds seek shelter near human dwelling where raptors are rare. Guay *et al.* (2013)[11] reported that FID was longer when individual black swans (*Cygnus atratus*) were farther from water, which acted as a refuge. These findings fit well with the predominant role of predation in predicting urbanization of birds.[3] Carrete and Tella (2017)[21] studied behavioural changes like antipredator response and exploration in burrowing owls (*Athene cunicularia*) living in urban habitats in relation to their fear to humans. Their findings suggested that fear of humans is variable in different bird individuals. This is estimated as FID to the advancing human in the way that urban life would select for fearless individuals. Recent work by Jiang *et al*. (2020)[22] suggests that animals avoid predators by gathering information produced by other species. Mixed species flocks (MSFs) of birds are predominant in forest ecosystems and are partly structured around interspecific information transfer, with "nuclear" species providing information that other species overhear and use against predator encounters. Foraging decisions are mainly governed by the predation risk as observed in granivorous bird species in different habitats and microhabitats. The study done in a Nigerian Forest Reserve suggested that a genuine approach for assessing habitat quality as perceived by animals can be an insight into their foraging decisions.[23]

**Season**

Moller *et al.* (2013)[4] did FID experiments on 207 bird species in 7 pairs of urban and rural areas from northern to southern Europe before and after the extremely cold winter of 2009-2010. FID in different species of birds on an average became shorter after the cold winter, but only in resident urban populations frequently exposed to humans and not in migratory or rural populations of the same species. The difference in FID between urban and rural habitats increased after the cold winter, especially in areas with the coldest winter weather.

**Visibility**

Influence of habitat visibility on time allocation and escape decisions was studied by Boyer *et al.* (2006)[24] in Crimson rosella as (*Playcercus elegans*). It was found that visibility influenced time allocated to both foraging and locomotion and there was no effect of visibility on FID, latency to detect and latency to flee.

**Urbanization**

Diaz *et al.* (2013)[25] studied patterns of FID across Europe and found that urban populations had shorter FID than rural populations of the same species. The difference in FID between urban and rural populations decreased with increasing latitude, paralleling trends in raptor affluence. In analyses of sample genes in 12 paired populations of blackbirds (*Turdus merula*), Moller *et al.* (2013)[4] discovered that a gene involved in harm avoidance differed strongly between urban and rural populations. Carrete and Tella (2011)[26] found that urbanization is the most prevailing cause of habitat transformation worldwide. The relationship between FID and relative brain size (RBS) in urban invasiveness was studied by Moller (2009)[27] who found that RBS indirectly influenced urban invasion through its positive effect on inter-individual variability in FID. FIDs were consistently lower in urban than in rural conspecifics. Short FID allows birds to coexist in the proximity of humans. Initial responses to urbanization are defined by behaviour and provide clues about how certain species deal with urban habitats, human presence, and novel resources of food and nest boxes conditions like noise, artificial light and pollution.[28]

**Food resources**

The link between individual state and responsiveness of turnstones, *Arenaria interpres*, on rocky shores was studied by manipulating condition through the provision of supplementary food by Beale and Monaghan (2004)[29]. They found that birds whose condition had been enhanced showed greater responsiveness to standardized human disturbance, flying away at greater distances from the observer, scanning more frequently for predators and flying further when flushed. An experiment on white-browed sparrow-weavers (*Plocepasser mahali*) was done by Fong *et al.* (2009)[30] to determine the importance of direct and indirect environmental cues in assessing predation risk. Weaver foraging was influenced more by indirect than by direct risk. Distance to cover is much influencing factor rather than availability of food resources. Foraging decisions in weaver are sensitive to variation in risk and indirect cues are relatively more important than direct cues. In a food supplementation experiment conducted on song sparrows, *Melospiza melodia*, Rastogi *et al.* (2006)[31], suggested that daytime nest predation reduced greatly with food supplementation. Fed females exhibited short foraging bouts and time spent in the nest increased. This highlights the fact that food availability guides nest predation through its influence on antipredator behaviour.

**Antipredator distances**

**Effect of Starting Distance (SD) on FID**

In a study of 64 species of Australian birds, significant positive relationship between SD and FID was observed by Blumstein, (2003)[32]. He contributed economic models of escape behaviour and predicted that escape decisions are dynamic and will be influenced by both the costs and benefits of remaining. This demonstrates that species generally assess risk dynamically and flush at a greater distance as starting distance increases. Stankowich and Coss (2006)[33] reported that the relationship between SD and FID might be logarithmic or even quadratic due to suboptimal scanning rates by prey. A strong correlation was found by Cooper and Blumstein (2014)[34] between SD and AD with FID. Approaches with longer SD may have shorter FID than those with a SD at the limits of the zone of awareness. A predator begins its approach far enough from prey and the individual also becomes alert far away from predator, resulting in a longer FID.

**Effect of AD on FID**

Role of physiological cost of monitoring and risk assessment and relation of these factors with AD and FID were observed by Cooper and Blumstein (2014)[34]. With increase in AD, there is an increase in the length of approach and assessed risk at any given predator prey distance. There is a positive correlation between FID, SD and AD. Utilizing data from FID tests, a study was done by Dumont *et al.* (2012)[35] on alpine marmots (*Marmota marmota*) which demonstrated that results on FID are affected by using SD as a proxy for AD. A linear mixed model with covariate AD showed FID to be affected by interaction between previous activity and AD whereas when SD replaced AD as the covariate, no effect was observed in the analysis.

**Population trends and FID**

Relationship between population trends and FID of 193 species of birds in North America, Australia and Europe was studied by Moller *et al.* (2014)[36]. A negative relationship between population trends and FID was observed in Australia and Europe, rather than North American birds. Strong effects of some variables of population trends such as farmland breeding habitat, FID, body size, migratory habit and continent were observed mostly in European birds. Bolnick and Preisser, (2005)[37]

found that costly defensive strategies reduce population density of animals through indirect effects. Declining population trends were observed among 56 species of European birds' species by Moller (2008)[38]. It was concluded that large body sized species with longer FIDs show declining population trends because humans disturb such species more often. Shorter FIDs have been found to scale down the impact of human disturbance and are acquired in birds in countries with higher human population density. However, Reif *et al*. (2011)[39] reported that stronger effects of brain size on population trends in birds is context dependent in North-Western Germany, compared to an area from Eastern Germany to the Czech Republic. A strong negative correlation between population trend and FID was reported by Diaz *et al.* (2013)[25] after analysis of 329 populations with information on both variables in the same country.

**Horizontal and vertical components of FID**

FID can be separated into its horizontal and vertical components as reported by Blumstein *et al.*, (2004)[8]. In their studies it was observed that vertical component becomes zero in organisms moving in two dimensions. Moller (2010)[40] did a comparative study of 69 species of birds and reported four times more variance in FID among species to be due to the horizontal than the vertical dimension of escape. Due to this, large species rely more on the horizontal component of escape as compared to small species which are influenced by cost of flight.

**Predators' eye contact**

Birds respond differently against direct and indirect eye contact of approaching threat. In various studies done by Bateman and Flemming (2011)[41] for *Hadeda ibises* (*Bostrychia hagedash*) and Lee *et al.* (2013)[42] for magpies (*Pica pica*), it was reported that FID or AD significantly increased when approaching humans looked directly at them rather than elsewhere. This was observed to reinforce the hypothesis that many bird species can distinguish between potential predators intently looking at a prey individual and those only looking intermittently.

**Speed of predators' approach**

Different impacts of slow and fast approaches by predators towards prey were observed by Cooper and Avalos (2010)[43]. FID was smaller for slower approach speed of predator as the prey has more time to reach a refuge than if the predator approaches faster. Lank and Ydenberg (2003)[44] found that if two identical predator approach at different speeds, the risk to the prey at any particular distance is greater during the faster approach. Similarly, Bateman & Flemming (2011)[41] reported that *Hadeda ibises* show lower FID when approached quickly rather than slowly by a human.

**Distance from refuge**

Refuge is a safe area for prey to hide from predator. Kramer and Bonenfant (1997)[45] reported AD critical as the distance at which a predator approaching at a continuous speed will reach the refuge at the same time as the fleeing prey. Preys flee before the predator reaches ADcritical to allow them a margin of safety (MOS) to arrive at the refuge before the predator. If the prey is between the predator and the refuge, the predicted FID, d*, is ADcritical + MOS – DP, where DP is the distance between the prey and its refuge.

**Driving speed and direction**

Cars are functional tools for assessing cognitive abilities of birds with respect to risk. Legagneux and Ducatez, (2013)[46] in a study recorded that birds adjusted their FID to road speed limits by increasing FID as speed limit increases, while there was no similar effect of the vehicle speed. This response reduces the risk of collision and decreases mortality. Mukherjee *et al.* (2013)[47] showed that American crows (*Corvus brachyrhynchos*) adjusted their risk-taking behaviour to the driving direction of cars in a particular lane while crows in the opposite lane to that used by an approaching vehicle stayed back.

**Direct and tangential approach**

Attributes of duration and length of the predator's approach are helpful in risk assessment. These factors lead to assessment of greater risk than that attributable to predator-prey distance alone. This causes increase in FID as AD increases. Various studies done separately[5,48,41] show that birds could distinguish direct and tangential approaches by humans at a fixed distance. Moller and Tryjanowski (2014)[49] found that rural birds were better able to assess the direction of approach by humans in rural than in urban habitats. Interestingly it was reported[50] that blackbirds (*Turdus merula*) fled on foot, when exposed to frequent disturbance by humans rather than engaging in potentially expensive flight.

**Behavioural responses**

**Vigilance**

Vigilance is the active behaviour shown by prey against approaching possible threats. In studies by

Stankowich and Coss (2006)[33] it was reported that individuals scan their surroundings for potential threats that might be looming. This vigilance may be shared by members of a group, which would be more beneficial than a solitary animal that must divide its time between scanning and foraging. It is evident that the predation risk is greater to a less vigilant prey. Birds spend much of their time in foraging activities and antipredator vigilance. Proctor *et al.* (2006)[51] explained a vigilance model predicting that birds in the outer region of feeding area are more vigilant as compared to those in the inner region. This testifies to the dominant birds choosing to forage in the centre of the flock. Increased vigilance on exposure to a predator model was reported in Siberian Jay *Perisoreus infaustus*.[52] Jay's breeders retaining offspring were observed to be comparatively more vigilant than breeder groups with no kin. Not allocating time to vigilance can be fatal for a prey, if it fails to detect the predator until captured. Campos *et al.* (2009)[53] studied the relationship between predation risks such as degree of concealment, height in tree/shrub, distance to the forest edge and time allocated to singing and vigilance in a group of 13 passerine species living in an East African Savanna. They found that concealed birds sang more and were less vigilant. Vigilance increased as distance to the forested edge increased. Whether vigilance plays a role in antipredator behaviour and influences escape of animals from predators, a multispecies study was done on wild birds from urban and rural habitats in three European countries.[54] Rural birds were observed to be less vigilant as compared to urban birds. Urban habitat disturbances influence the vigilance behaviour and urban birds exhibit higher perceived predation risk and are therefore better at avoiding predation.

**Detection**

A game-theoretical model was developed by Broom and Ruxton (2005)[55] that applies specifically to the effect of crypsis on FID. Prey is cognizant of the predator before the predator detects the prey, but as the predator approaches closer, its chance of the prey detection increases. In a detailed study done[7], it was observed that at an isolated island the loss of all predators relaxed selection on predator recognition abilities. Mutation rate and the cost of flight had considerable effect on the average and standard deviation of recognition abilities.

**Escape latency**

Starting distance and many of the predation risk factors like intruder speed and directness of approach known to affect FID have influence on escape latency.[56] Longer SD and direct eye contact of predator enhance escape latency. Shorter escape latency is observed in the presence of food and familiar female. Escape latency decreases when opportunities to improve fitness are present. A prey is predicted to emerge when the risk of emerging equals the cost of hiding. A waiting game theory between prey and predator given by Hugie (2003)[57] concluded that large differences in fitness consequences for predator and prey lead to a waiting game where prey stays in refuge until after the predator has left the area. Similarly, Cooper *et al.* (2012)[58] reported that cost of fleeing decreases as escape latency increases. Latency to escape is expected to be greater when predation risk is smaller and cost of fleeing is higher because it prevents the prey from obtaining benefits of feeding, social behavior, or other activities that increase fitness.

**Loss of predator over long period of time**

Escape behaviour and vigilance behaviour reduces when birds are isolated from predators, and insular animals forage more and are less vigilant than mainland ones. In multi predator hypothesis, Blumstein, (2006b)[59] reported that isolation from all predators lead to a rapid loss of antipredator behavior, including loss of the group size effect and breakdown of predator recognition abilities. Studies have shown that local extinction of predators for prolonged periods relaxes natural selection on predator recognition.[60] Over by gone years prey may lose the ability to identify locally extinct predators as dangerous. Experience-dependent behaviour is lost after the first generation in the absence of predators, while more hard-wired antipredator behaviour persists for thousands of years following isolation from predators.

**Prey-predator interaction**

When a prey is threatened by predators, prey adopts defensive tactics, and its cost affects growth, maturation rates, survivorship, fecundity, or population density. This is known as "trait mediated interaction" (TMI) and effects reflect a balance between the conflicting demands of reproduction vs. predator avoidance. An intermediate positive effect on FID was reported in plover species against abundance of mammalian predators.[10] Predator-prey interactions are dependent on predator's perspective, beginning with detection by the predator, identification, approach, subjugation, and consumption. Weak escape

responses are observed in areas lacking predation and having high exposure to humans.

**Habituation**

Runyan and Blumstein (2004)[61] reported that moderate degree of pseudo replication of prey did not significantly affect individual identity because rate of habituation varies for different individuals and is evident only when individual identity is known. A weak evidence of habituation with repeated estimates of FID was reported by Blumstein (2014)[62] in 8 different species of ducks and owls to passerines. Impact of habituation and risk allocation on blackbirds (*Turdus merula*) was studied by Rodriguez-Prieto *et al.* (2009)[63]. They found lower FID with higher pedestrian rates, in response to both usual and novel approaches. Jiang and Moller (2017)[64] estimated the FID of 48 pairs of related bird species with different threat status to human proximity and reported longer FID in more threatened bird species. The study revealed that animals become habituated to human proximity through adaptation or through microevolution.

**Parental investment**

Two parameters, clutch size and fecundity were found to be negatively related to mean FID.[7] Hatching success in curlews (*Numenius arquata*) was maximum at intermediate FID[65], as would be expected if long FID restrained efficient incubation. Seltmann *et al.* (2012)[66] reported longer incubation periods in eiders (*Somateria mollissima*) with longer FIDs. Contradictorily, Moller and Garamszegi (2012)[19] observed that rate of senescence decreased with increasing mean and variance in FID, especially for the vertical component of escape. A positive correlation between FID and corticosterone levels in incubating eiders was found by Seltmann *et al.* (2012)[66]. However, Partecke *et al*. (2006)[67] found lower testosterone and corticosterone levels in urban as compared to rural individuals in blackbirds.

**Nesting and Parental nest visit synchronization**

Predation is the primary cause of nesting mortality in birds. The antipredatory behaviour of *Cyanistes caeruleus*, blue tit parents was studied by Saavedra (2020)[68]. The blue tit parents were laid bare to predators or their stuffed models having chemical cues. Blue tits responded to the behavioural cues of live predators more than to predator models infused with chemical cues. It was suggested that the blue tits are able to gauge the intensity of predation risk and adjust their antipredator behaviour according to the behavioural, visual and chemical cues of the predators. A number of studies suggest that behavioural changes are exhibited by birds in response to the visual cues of the predator.[69,70]

Birds with antiracial offspring need to feed them at regular intervals, but each feeding visit risks drawing attention of the potential predators to the nest and exposing its location. Synchronisation of visits by both parents has been suggested as a behavioural adaptation to reduce the risk of nest predation. Khwaja *et al.* (2019)[71] investigated this prediction over three timescales using nestling provisioning data from 25 passerine species in Tasmania and New Zealand and estimated the level to which parents actively synchronised their nest visits. They concluded that breeding birds may have some capacity to adapt the synchrony with which they provision over short evolutionary timescales. FID may be used as a parameter of measurement of risk-taking behaviour. Risk taking decisions made for offspring care can influence fitness of breeding individuals. The cost of antipredator behaviour leads to parents' trade off between care of offspring and cessation of care for their lives. Therefore, in order to find a way out, predation risk is needed to be assessed exactly.[72,73] Increase in current reproductive value, nest concealment and costs of lost parental care led to decrease in optimal FID and also the parent to stay back at nest.[74] Some instances of costs of fleeing are so much that staying at the nest becomes optimal strategy for nesting species.

## CONCLUSION

The ability of birds to cope with predators is very much dependent on its bodily cues. Studies provide a link between behavioural adaptations and survival success of birds in the wild. These behavioural changes play an effective role in the conservation of birds. The anti-predatory behaviour is modified by the adjustments in the distances at which birds flush after seeing or confronting the predator. Understanding how individuals use biological, ecological, behavioural factors and anti-predatory distances to survive depredation may help us device conservation strategies to the fullest.

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## REFERENCES

1. **Yadav, V. K. and Chauhan, P. S. 2018.** Avifaunal Diversity and Status of Jhalawar Forest Division, South-Eastern Rajasthan, India. *Indian Journal of Ecology.* **45(1):** 107-116.
2. **Samia, D. S. M., Moller, A. P. and Blumstein, D. T. 2015.** Brain size as a driver of avian escape strategy. *Sci. Rep.* 5, 11913.
3. **Moller, A. P. 2014.** Urban birds use cheap escape: Variance in escape direction from predators by prey. Cambridge University Press
4. **Moller, A. P., Vagasi, C. I. and Pap, P. L. 2013.** Risk-taking and the evolution of mechanisms for rapid escape from predators. *Journal of Evolutionary Biology.* **26:** 1143-1150.
5. **Fernandez-Juricic, E., Venier, P., Renison, D. and Blumstein, D. T. 2005.** Sensitivity of wildlife to spatial patterns of recreationist behavior: a critical assessment of minimum approaching distances and buffer areas for grassland birds. *Biological Conservation.* **125:** 225-235.
6. **Moller, A. P., Nielsen, J. T. and Garamszegi, L. Z. 2008.** Risk taking by singing males. *Behavioral Ecology.* **19:** 41-53.
7. **Blumstein, D. T. 2006a.** Developing an evolutionary ecology of fear: How life history and natural history traits affect disturbance tolerance in birds. *Animal Behaviour.* **71:** 389-399.
8. **Blumstein, D. T., Fernandez-Juricic, E., O Le Dee. 2004.** Avian risk assessment: Effects of perching height and detectability. *Ethology.* **110:**273–285.
9. **Moller, A. P. and Erritzoe, J. 2010.** Flight distance and eye size in birds. *Ethology.* **116:** 458-465.
10. **Cooper, W. E. and Blumstein, D. T. 2015.** Escaping from Predators: An Integrative View of Escape Decisions. Cambridge University Press.
11. **Guay, P. J., Weston, M. A., Symonds, M. R. E. and Glover, H. K. 2013.** Brains andbravery: Little evidence of a relationship between brain size and flightiness in shorebirds. *Austral Ecology.* **38:**516-522.
12. **Moller, A. P. and Erritzoe, J. 2014.** Predator–prey interactions, flight initiation distance and brain size. *Journal of Evolutionary Biology.* **27:** 34-42.
13. **Cooper, W. E. and Blumstein, D. T. 2013.** Novel effects of monitoring predators on costs of ûeeing and not ûeeing explain ûushing early in economic escape theory. *Behavioral Ecology.* **25:** 44-52.
14. **Fernandez-Juricic, E., Blumstein, D. T. and Abrica, G. 2006.** Relationships of antipredator escape and post-escape responses with body mass and morphology: A comparative avian study. *Evolutionary Ecology Research.* **8:** 731-752.
15. **McQueen, A., Naimo, A. C., Teunissen, N., Magrath, R. D., Delhey, K. and Peters, A. 2017.** Bright birds are cautious: seasonally conspicuous plumage prompts risk avoidance by male superb fairy-wrens. *Proc. R. Soc. B.* **284:** 20170446.
16. **Iba ìn Þez, A., Lo ìpez, P. and Mart ìýn, J. 2014.** Inter-individual variation in antipredator hiding behavior of Spanish terrapins depends on sex, size, and coloration.*Ethology.* **120:** 742-752
17. **Pascual, J., Senar, J. C. and Dome'nech, J. 2014.** Plumage brightness, vigilance, escape potential and predation risk in male and female *Eurasian siskins* 19. (*Spinus spinus*). *Auk*. **131:** 61-72.
18. **Atwell, J. W., Cardoso, G. C. and Whittaker, D. J. 2012.** Boldness behaviour and stress physiology on a novel urban environment suggest rapid correlated evolutionary adaptation. *Behavioral Ecology.* **23:** 960-969.
19. **Moller, A. P. and Garamszegi, L. Z. 2012.** Between individual variations in risk taking behavior and its life history consequences. *Behavioral Ecology.* **23:**843-853.
20. **Valcarcel, A. and Fernandez-Juricic, E. 2009.** Antipredator strategies of house finches: Are urban habitats safe spots from predators even when humans are around? *Behavioral Ecology &Sociobiology.* **63:** 673-685.
21. **Carrete, M. and Tella, J. L. 2017.** Behavioral Correlations Associated with Fear of Humans Differ between Rural and Urban Burrowing Owls. *Front. Ecol. Evol.* **5:** 54.
22. **Jiang, D., Sieving, K. E., Meaux, E. and Goodale, E. 2020.** Seasonal changes in mixed species bird flocks and antipredator information. *Ecology and Evolution.* **10(2):** 5368-5382**.**
23. **Abdulwahab, U. A., Osinubi, S. T. and Abalaka, J. 2019.** Risk of predation: a critical force driving habitat quality perception and foraging behavior of granivorous birds in a Nigerian forest reserve. *Avian Research* **10:**33.
24. **Boyer, J. S., Hass, L. L., Lurie, M. H. and Blumstein, D. T. 2006.** Effect of visibility on time allocation and

escape decisions in *Crimson rosellas*. *Aust. J. Zool.* **54(5):** 363-367.

25. **Diaz, M., Moller, A. P. and Flensted-Jensen, E. 2013.** The geography of fear: A latitudinal gradient in anti-predator escape distances of birds across Europe. *Public Library of Science One.* 8 e64634.
26. **Carrete, M. and Tella, J. L. 2011.** Inter-individual variability in fear of humans and relative brain size of the species are related to contemporary urban invasion in birds. *PLoS One.* 6(4).
27. **Moller, A. P. 2009.** Basal metabolic rate and risk-taking behavior in birds. *Journal of Evolutionary Biology.* **22:** 2420-2429.
28. **Isaksson, C., Amanda, D. and Rodewald, G. D. 2018.** Behavioural and Ecological Consequences of Urban Life in Birds. *Front. Ecol. Evol.* **6:** 50
29. **Beale, C. M. and Monaghan, P. 2004.** Human disturbance: people as predation free predators? *J. Appli. Ecol.* **41(2):** 335-343.
30. **Fong, T. E., DeLong, T. W., Hogan, S. B. and Blumstein, D. T. 2009.** The importance of indirect cues for white-browed sparrow-weaver (*Plocepasser mahali*) risk assessment. *Acta ethologica.* **12(2):** 79-85.
31. **Rastogi, A. D., Zanette, L. and Clinchy, M. 2006.** Food availability affects diurnal nest predation and adult antipredator behaviour in song sparrows, *Melospiza melodia. Animal Behavior.* **72(4):** *933-940.*
32. **Blumstein, D. T. 2003.** Flight-initiation distance in birds is dependent on intruder starting distance. *J.Wildlife Management.* **67(4):** 852-857.
33. **Stankowich, T. and Coss, R. G. 2006.** Effects of predator behavior and proximity on risk assessment by Columbian black-tailed deer. *Behavioral Ecology.* **17:**246-254.
34. **Cooper, W. E. and Blumstein, D. T. 2014.** Starting distance, alert distance and flushing early challenge economic escape theory: New proposed effects on costs of fleeing and not fleeing. *Behavioral Ecology***25:**44-52.
35. **Dumont, F., Pasquaretta, C., Reale, D., Bogliani, G. and Hardenberg, A. V. 2012.** Flight Initiation Distance and Starting Distance: Biological Effect or Mathematical Artefact? *Ethology.* **118(11):** 1051-1062.
36. **Moller, A. P., Samia, D. S. M., Weston, M. A., Guay, P. J. and Blumstein, D. T. 2014.** American Exceptionalism: Population Trends and Flight Initiation Distances in Birds from Three Continents. *PLoS One.* **9(9):** e107883.
37. **Bolnick, D. I. and Preisser, E. L. 2005.** Resource competition modifies the strength of trait-mediated predator–prey interactions: A meta-analysis. *Ecology.* **86:** 2771-2779.
38. **Moller, A. P. 2008.** Flight distance and population trends in European breeding birds. *Behavioral Ecology.* **19(6):** 1095-1102.
39. **Reif, J., Bohning-Gaese, K., Flade, M., Schwarz, J. and Schwager, M. 2011.** Population trends of birds across the iron curtain: Brain matters. *Biological Conservation.* **144:** 2524-2533.
40. **Moller, A. P. 2010.** Up, up, and away: Relative importance of horizontal and vertical escape from predators for survival and senescence. *Journal of Evolutionary Biology* **23:** 1689-1698.
41. **Bateman, P. W. and Flemming, P. A. 2011.** Who are you looking at? *Hadeda ibises* use direction of gaze, head orientation and approach speed in their risk assessment of a potential predator. *Journal of Zoology.* **285:**316-323.
42. **Lee, S., Hwang, S. and Joe, Y. 2013.** Direct look from a predator shortens the risk- assessment time by prey. *PLoS One.* **8(6):** e64977.
43. **Cooper, W. E. and Avalos, A. 2010.** Escape Decisions by the Syntopic Congeners *Sceloporus jarrovii* and *S. virgatus*: Comparative Effects of Perch Height and of Predator Approach Speed, Persistence, and Direction of Turning. *Journal of Herpetology.* **44(3):** 425-430.
44. **Lank, D. B. and Ydenberg, R. C. 2003.** Death and danger at migratory stopovers: problems with "predation risk". *Journal of Avian Biology.* **34:** 225-228.
45. **Kramer, D. L. and Bonenfant, M. 1997.** Direction of predator approach and the decision to flee to a refuge. *Animal Behaviour.* **54:** 289-295.
46. **Legagneux, P. and Ducatez, S. 2013.** European birds adjust their flight initiation distance to road speed limits. *Biology Letters.* **9(5)**.
47. **Mukherjee, S., Ray-Mukherjee, J. and Sarabia, R. 2013.** Behaviour of American crows (*Corvus brachyrhynchos*) when encountering an oncoming vehicle. *Canadian Field- Naturalist.* **127:** 229-233.
48. **Heil, L., Fernandez-Juricic, E. and Renison, D. 2007.** Avian responses to tourism in the biogeographically isolated high Cordoba Mountains, Argentina. *Biodiversity and Conservation.* **16:** 1009-1026.

**49. Moller, A. P. and Tryjanowski, P. 2014.** Direction of approach by predators and flight initiation distance of urban and rural populations of birds. *Behavioral Ecology.* **25:** 960-966.

**50. Rodriguez-Prieto, I., Fernandez-Juricic, E. and Martin, J. 2008.** To run or to fly: Low cost versus low-riskstrategies in blackbirds. *Behaviour.* **145(8):**1125-1138.

**51. Proctor, C. J., Broom, M. & Ruxton, G. D. 2006.** Antipredator vigilance in birds: modelling the 'edge' effect. *Math Biosci.* **199(1):** 79-96.

**52. Griesser, M. and Nystrand, M. 2009.** Vigilance and predation of a forest-living bird species depend on large-scale habitat structure. *Beh. Ecology.* **20(4):**709-715.

**53. Campos, D. P., Bander, L. A., Raksi, A. and Blumstein, D. T. 2009.** Perch exposure and predation risk: a comparative study in passerines. *Acta Ethol.***12(2):**93-98.

**54. Tatte, K., Diego, J., Ibanez-Alamo, Marko, G., Mand, R. and Moller, A. P. 2019.** Antipredator function of vigilance re-examined: vigilant birds delay escape. *Animal Behaviour.* **156:** 97-110.

**55. Broom, M. and Ruxton, G. D. 2005.** You can run or you can hide: Optimal strategies for cryptic prey. *Behavioral Ecology.* **16:** 534-540.

**56. Martin, J., Luque-Larena, J. J. and Lopez, P. 2009.** When to run from an ambush predator: Balancing crypsis benefits with costs of fleeing in lizards. *Ani.Beh.* **78:**1011-1018.

**57. Hugie, D. M. 2003.** The waiting game: A "battle of waits" between predator and prey. *Beh.Ecology.* **14:** 807-817.

**58. Cooper, W. E., Lopez, P., Martin, J. and Perez-Mellado, V. 2012.** Latency to flee from an immobile predator: Effects of risk and cost of immobility for the prey. *Behavioral Ecology.* **23:** 790-797.

**59. Blumstein, D. T. 2006b.** The multipredator hypothesis and the evolutionary persistence of antipredator behavior. *Ethology.* **112:** 209–217.

**60. Lahti, D., Johnson, N. A. and Ajie, B. C. 2009.** Relaxed selection in the wild. *Tre. in Eco. & Evo.*. **24:** 487-496.

**61. Runyan, A. M., Blumstein, D. T. 2004.** Do individual differences influence initiation distance? *J Wildl Manage.* **68:** 1124-1129.

**62. Blumstein, D. T. 2014.** Attention, habituation and antipredator behaviour: implication for urban birds. *Avian Urban Ecology.* Oxford University Press.

**63. Rodriguez-Prieto, I., Fernandez-Juricic, E., Martin, J. and Regis, Y. 2009.** Antipredator behavior in blackbirds: habituation complements risk allocation. *Behav. Ecol.* **20(2):** 371-377.

**64. Jiang, D. and Moller, A. P. 2017.** Antipredator escape distances of common and threatened birds. *Behavioral Ecology.* **28(6):** 1498–1503.

**65. De Jong, A., Magnhagen, C. and Thulin, C. G. 2013.** Variable flight initiation distance in incubating Eurasian curlew. *Beh. Ecology and Sociobiology.* **67:** 1089-1096.

**66. Seltmann, M. W., M. Ost M. and Jaatinen, K. 2012.** Stress responsiveness, age and body condition interactively affect flight initiation distance in breeding female eiders. *Animal Behaviour.* **84:** 889-896.

**67. Partecke, J., Schwabl, I. and Gwinner, E. 2006.** Stress and the city: urbanization and its effects on the stress physiology in European blackbirds. *Ecology.* **87:** 1945-1952.

**68. Saavedra, I. 2020.** The importance of chemical, visual and behavioral cues of predators on the antipredatory behavior of birds. *Journal of Avian Biology.* **51(7):**

**69. Amo, L., Tomas, G. and Lopez-Garcia, A. 2017.** Role of chemical and visual cues of mammalian predators in nest defence in birds. *Behav. Ecol. Sociobiol.* **71:** 49.

**70. Dure Ruiz, N. M. D., Fasanella, M. and Fernandez, G. J. 2018.** Breeding southern house wrens exhibit a threat-sensitive response when exposed to different predator models. *J. Ethol.* **36:** 43-53.

**71. Khwaja, N., Massaro, M., Martin, T. E. and Briskie, J. V. 2019.** Do Parents Synchronise Nest Visits as an Antipredator Adaptation in Birds of New Zealand and Tasmania? *Front. Ecol. Evol.* **7:**389

**72. Thomson, R. L., Forsman, J. T. and Monkkonen, M. 2011.** Risk taking in natural predation risk gradients: support for risk allocation from breeding pied flycatchers. *Anim. Behav.* **82:** 1443-1447.

**73. Mutzel, A., Blom, M. P. K., Spagopoulou, F., Wright, J., Dingemanse, N. J. and Kempenaers, B. 2013.** Temporal trade-offs between nestling provisioning and defense against nest predators in blue tits. *Anim. Behav.* **85:** 1459-1469.

**74. Dowling, L., and Bonier, F. 2018.** Should I stay, or should I go: Modeling optimal flight initiation distance in nesting birds. *PLoS ONE.* **13:**11

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# Impact of Corona crisis on Environment

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**Abstract-** The novel Corona Virus disease pandemic (Covid-19) has brought the human life to complete standstill as most countries of the World have shut themselves off from the work. Covid-19, just few days back, was foreign to us and today it has indeed routed itself in our lives so much so that we are witnessing lot of things around us which we could not have imagined otherwise or perhaps for the first time in modern industrialized history. To be precise, shut down of all types of industries, drastically reduced plying of vehicle on the road and several activities of people affecting the environment in not so good ways. For a long time, it has been stated that the increased industrialization and anthropogenic activities in the last two decades polluted the atmosphere, hydrosphere, and biosphere. Therefore, the aim of this study is to understand the impact of Covid-19 on our environment viz., Air quality, Water quality, Carbon emission in longer run. Data available through various platforms have helped in this analysis on impact of Covid- 19 on environment. Cleaner atmosphere, hydrosphere and biosphere for a few months may be a tiny silver lining to Covid-19's dark clouds, but will do little in the long run to solve the problem of environmental pollution in all forms viz., air, water etc.

**Key words: Corona Virus, environmental, Air quality, Water quality, Carbon emission**

## INTRODUCTION

The Covid-19 pandemic has impacted every aspect of human life and the global economy. As of May 25, 2020, the coronavirus (Covid-19) has infected over 1.31 lacs people and caused ~3867 deaths in India. (https://economictimes.indiatimes.com/coronavirus). From March 2020, India, the second-most populous country in the world with 1.3 billion people, went into lockdown. Nonetheless, the lockdown has halted quite significantly all forms of transport (flights, trains, automobiles), factories, shops, markets, and other economic and social activities.

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Despite the severe impacts of lockdown on people's social life, global mobility and economy, there are reports that lockdown has temporarily improved environmental conditions. For example, the lockdown improved the air quality of the 103 cities in India- the most polluted country in the world which has 21 out of the 30 world's most polluted cities (Singh and Chakraborty 2020)[1].

**Air pollution** is one of the largest environmental health risks in the world. 7 million people die prematurely every year globally as a result of exposure to air pollution, more than 91% of the world's population lives in places where air quality exceeds the World Health Organization's guidelines (WHO 2020)[2]. PM2.5 is a common cause for adverse health outcomes such as chronic obstructive

pulmonary disease (COPD) and lower respiratory infection (LRI) causing death of nearly three million people globally, according to a study. $NO_2$ is the leading source of childhood asthma in urban areas globally (Achakulwisut *et al.* 2019)[3]. Likewise, acute to chronic exposures to $SO_2$, CO, and $O_3$ can harm the human respiratory system, causing breathing problems including occasional death (Chen *et al.* 2007, Brown 2009)[4,5]. The coronavirus-triggered lockdown has led to a steep fall in global carbon emissions by 17 per cent in early April as compared to 2019 levels with India's emissions dropping by 26 per cent, according to a study.

Therefore, understanding this temporary improvement in air quality at the planetary scale provides a unique opportunity to study its implications in longer run.

## MATERIALS & METHODS

Here, insight on air pollution by utilizing vast networks of ground-based air quality monitoring stations from major Indian cities is being provided. The air quality data is collected from 10 ground-based stations from 1st January 2020 till 25th May 2020 to compare periodical changes pre-covid and post-covid lockdown.

**Selection of the major cities**

10 major cities in India are selected to undergo this study for Air Pollution. Study is based on the following:

(i) The historical level of air pollution in these cities

(ii) Availability of air pollution data.

CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/) is used to download the above discussed periodical data on air pollution indicators. Selection was also based on the fact that these cities are affected heavily by anthropogenic emissions and a visible/ remarkable reduction in air pollution is expected in these cities as a result of lockdown.

**Selection of air pollutants:**

Normally, air quality is evaluated by measuring the atmospheric concentrations of six pollutants:

**i)** Fine particulate matter (PM2.5, mass concentration of particles with diameters d"2.5 um)

**ii)** Coarse particulate matter (PM10, mass concentration of particles with diameters d"10 um)

**iii)** Ground-level ozone ($O_3$)

**iv)** Nitrogen dioxide ($NO_2$)

**v)** Sulfur dioxide ($SO_2$)

**vi)** Carbon monoxide (CO)

All of them except $O_3$ have primary sources, such combustion sources as the automobiles and industries, while $O_3$ is formed in the atmospheric from precursor gaseous species, emitted mostly from primary sources, in presence of sunlight. Therefore, pollutants viz., PM2.5, PM10, $NO_2$, $O_3$, $SO_2$ & CO, to investigate the changes in air quality among cities before and during lockdowns and its future is selected.

**Data and analysis:**

Publicly available air quality data provided by the CPCB - India Central Pollution Control Board which curates air quality data collected from ground-based air quality monitoring stations situated in cities in India has been referred to.

**Analysis Assumptions:**

- It must be noted that the data from CPCB is from the stations managed by government agency and may not be always fully validated.
- Daily concentrations of air pollutants collected from multiple CPCB stations, which were available for further analysis.
- By 'lockdown', it is meant restrictions on vehicles, commercial flights except for cargo and charter, and prohibitions of commercial activities, except for essential services, resulting in a temporary shutdown of air pollution sources from these activities.
- It is assumed that air quality in January, February of 2020 was under the business as usual conditions and hence considered normal. Therefore, daily changes in concentrations of these air pollutants from January 1 to May 25 (146 days) was examined to investigate the effect of lockdown on their daily concentrations of air pollutants using lockdown as dummy variable, i.e., periodical concentration (on a given day) of the six air pollutants of January, February of 2020, (all pre-Covid) with March, April and May of 2020 (post Covid lockdown).

## RESULTS & DISCUSSIONS

After analysing the data, it is seen, that there had been consistent decline in the PM2.5 levels in all the concerned cities throughout. Marginal spikes are visible post 17th May'20 and same is due to the ease in lockdown by the Govt. during Lockdown 4.0. It is worth mentioning here that in absolute term these cities already have cleaner air, i.e., PM2.5 concentrations are very low compared to heavily polluted earlier.

**Table 1: PM2.5 concentration levels analysis: (unit: particle per million (ppm))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 100.74 | 35.45 | 55.92 | 23.57 | 44.21 | 19.69 | 19.8 | 17 |
| Patna | 144.22 | 62.48 | 71.4 | 54.08 | 49.09 | 20.03 | 43.76 | 47.9 |
| Lucknow | 388.95 | 139.37 | 54.43 | 96 | 98.66 | 43.83 | 83.86 | 95.7 |
| Kolkata | 115.14 | 46.13 | 68.19 | 51.52 | 27.67 | 19.58 | 8.27 | 14.25 |
| Amravati | 58.61 | 20.55 | 33.53 | 16.71 | 19.42 | 21.94 | None | 16.82 |
| Delhi | 367.38 | 130.43 | 61.67 | 42.55 | None | None | 96.64 | 69.01 |
| Ahmedabad | 78.79 | 88.63 | 52.56 | 34.65 | 31.44 | 23.85 | 26.89 | 23.59 |
| Mumbai | 54.93 | None | 24.37 | 21.26 | 20.33 | 42.29 | 59.18 | 47.33 |
| Jaipur | 51.64 | 35.2 | 28.47 | 24.36 | 32.08 | 29.65 | 33.14 | 42.31 |
| Chennai | 29.55 | 42.88 | 36.9 | 15.7 | 14.78 | 12.9 | 3.96 | 12.14 |

***Range:** Good (O -30); Satisfactory (31 -60); Moderately Polluted (61 -90); Poor (91 -120); Very Poor (121-250); Severe (250+)*

*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*

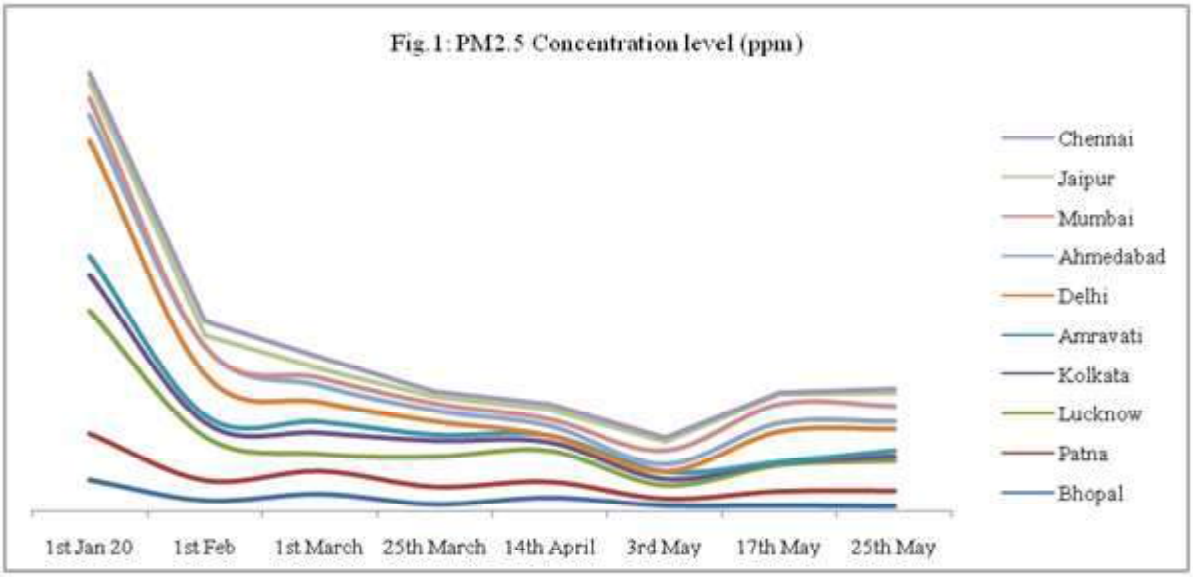



**Table 2: PM10 concentration levels analysis: (unit: particle per million (ppm))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 146.59 | 95.66 | 153.83 | 67.92 | 122.27 | 84.34 | 67.02 | 40.19 |
| Patna | 263.65 | 154.05 | 160.75 | 100.07 | 79.52 | 46.89 | 117.48 | 129.93 |
| Kolkata | 185.33 | 81.35 | 150.87 | 76.69 | 53.19 | 31.59 | 22.44 | 45.89 |
| Amravati | 87.32 | 42.45 | 56.54 | 35.6 | 45.57 | 55.14 | 45.44 | 57.38 |
| Delhi | 460.72 | 269.45 | 190.19 | 72.58 | NA | NA | 165.23 | 104.76 |
| Ahmedabad | 131.51 | 125.72 | 122.71 | 92.68 | 71.5 | 63.16 | 62.76 | 66.46 |
| Mumbai | 129.59 | NA | 71.02 | 71.07 | 78.11 | 61.93 | 100.83 | 123.31 |
| Jaipur | 135.71 | 88.71 | 98.43 | 46.24 | 108.09 | 78.22 | 74.3 | 123.03 |
| *Note: Data for Lucknow and Chennai not available, hence excluded for PM 10 analysis* | | | | | | | | |

***Range:** Good (O -50); Satisfactory (51 -100); Moderately Polluted (101 -250); Poor (251 -350); Very Poor (351-430); Severe (430+)*

*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*

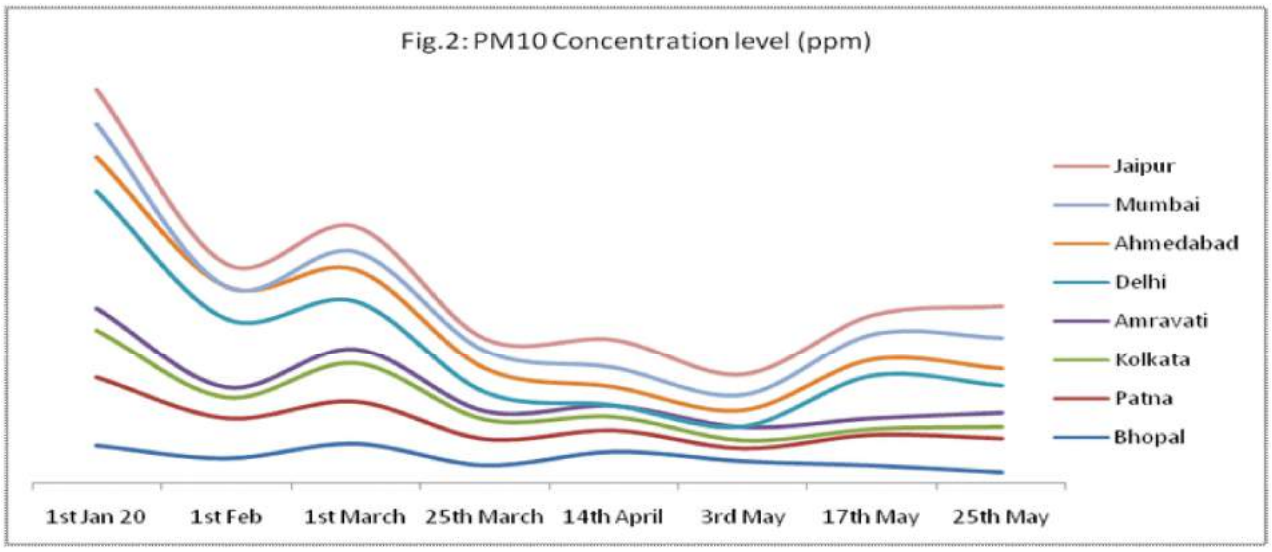



**Table 3: $NO_2$ concentration levels analysis: (unit: particle per billion (ppb))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 42.23 | 36.69 | 34.67 | 9.64 | 10.85 | 6.34 | 9.95 | 4.21 |
| Patna | 77.55 | 80.61 | 84.04 | None | None | None | 79.75 | None |
| Lucknow | 26.99 | 35.76 | 33.66 | 14.34 | 49.62 | 9.89 | 14.3 | 26.99 |
| Kolkata | 49.37 | 34.62 | 39.78 | 9.87 | 4.84 | 5.07 | 4.94 | 4.37 |
| Amravati | 11.95 | 11.54 | 8.52 | 4.32 | 4.94 | 4.36 | 14.39 | 3.55 |
| Delhi | 54.65 | 68.57 | 75.51 | 20.65 | None | None | 28.25 | 27.79 |
| Ahmedabad | 12.56 | 53.05 | 58.4 | 35.7 | 23.07 | 15.77 | 18.69 | 14.13 |
| Mumbai | 33.99 | 3.3 | 33.89 | 23.11 | 4.36 | None | None | 16.96 |
| Jaipur | 23.44 | 42.57 | 20.46 | 10.73 | 15.34 | 7.77 | 21.55 | 14.3 |
| Chennai | 17.41 | 19.65 | 13.79 | 10.35 | 20.85 | 3.85 | 4.78 | 10.32 |

***Range:*** ***Good (O -50); Satisfactory (51 -100); Moderately Polluted (101 -360); Poor (360 -650); Very Poor (650-1250); Severe (1250+)***

*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*

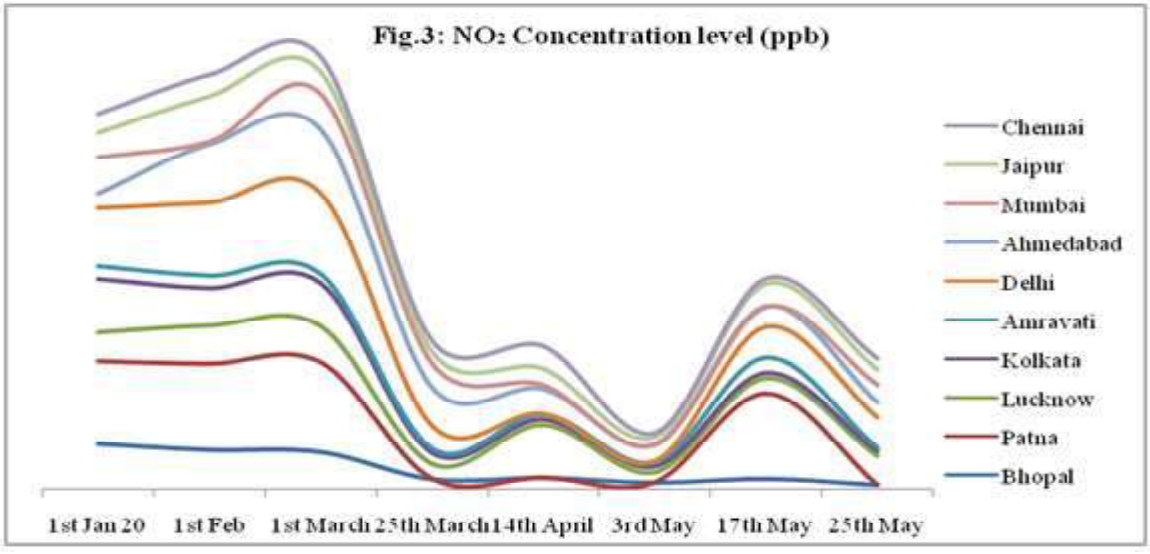


**Table 4: CO concentration levels analysis: (unit: particle per million (ppm))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 1.15 | 0.75 | 0.97 | 0.37 | 0.68 | 0.36 | 0.38 | 0.31 |
| Patna | 2.52 | 1.25 | 1.45 | 0.67 | 0.49 | 0.39 | 0.68 | 0.72 |
| Lucknow | 2.25 | 1.42 | 0.99 | 0.96 | 1.01 | 0.99 | 0.66 | 0.97 |
| Kolkata | 1.3 | 0.55 | 0.86 | 0.45 | 0.4 | 0.4 | 0.3 | 0.31 |
| Amravati | 0.76 | 0.44 | 0.49 | 1.51 | 0 | None | None | 0.45 |
| Delhi | 3.67 | 2.1 | 2.18 | 1.24 | None | None | 2.94 | 1.04 |
| Ahmedabad | 0.88 | 0.99 | 1.23 | 1.24 | 0.37 | 0.65 | 0.67 | 0.51 |
| Mumbai | 2.54 | 3.41 | 1.9 | 1.05 | 0.93 | 0.96 | 0.62 | 0.53 |
| Jaipur | 0.92 | 0.86 | 0.57 | 0.58 | 0.91 | 0.82 | 0.67 | 0.74 |
| Chennai | 0.52 | 0.42 | 0.7 | 0.29 | 0.36 | 0.47 | 0.39 | 0.51 |

***Range:** Good (O -5); Satisfactory (5 -10); Moderately Polluted (10 -12); Poor (13 -16); Very Poor (17-30); Severe (30+)*

*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*

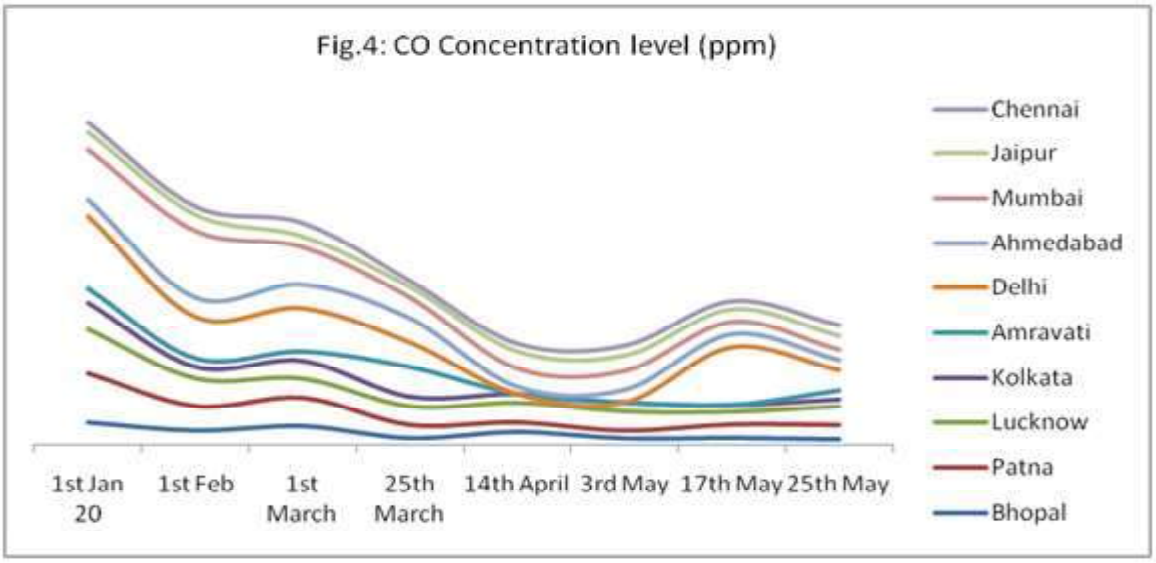



**Table 5: $O_3$ concentration levels analysis: (unit: particle per billion (ppb))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 38.6 | 62.72 | 68.87 | 64.27 | 47.6 | 68.24 | 88.75 | 37.9 |
| Patna | 3.51 | 2.22 | 5.92 | 21.1 | 56.15 | 87.58 | 95.04 | 43.39 |
| Lucknow | 54.74 | 8.25 | 3.07 | 3.11 | 3.53 | 4.18 | 3.25 | 2.37 |
| Kolkata | 34 | 76.69 | 67.22 | 54.88 | 26.12 | 46 | 20.43 | 29.35 |
| Amravati | 56.9 | 32.13 | 51.55 | 27.85 | 27.18 | 31.91 | 30.3 | 42.57 |
| Delhi | 60.06 | 20.55 | 42.59 | 45.52 | None | None | 106.38 | 130.09 |
| Ahmedabad | 23.67 | 45.37 | 40.87 | 38.72 | 43.88 | 54.43 | 55.41 | 59.66 |
| Mumbai | 35.1 | 25.25 | 14.57 | 34.5 | 0.28 | None | 10.51 | 10.1 |
| Jaipur | 71.54 | 51.98 | 68.97 | 51.25 | 47.83 | 55.24 | 72.35 | 57.47 |
| Chennai | 19.19 | 34.04 | 50.34 | 11.52 | 16.34 | 20.28 | 19.86 | 21.81 |

***Range:** Good (O -50); Satisfactory (51 -70); Moderately Polluted (71 -85); Poor (86 -105); Very Poor (106-200); Severe (200+)*

*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*

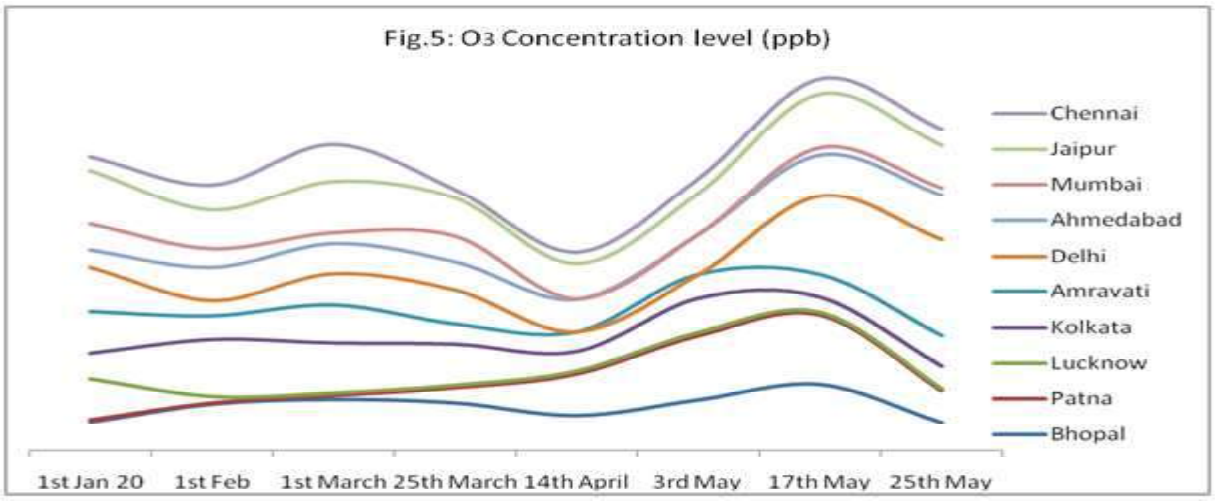
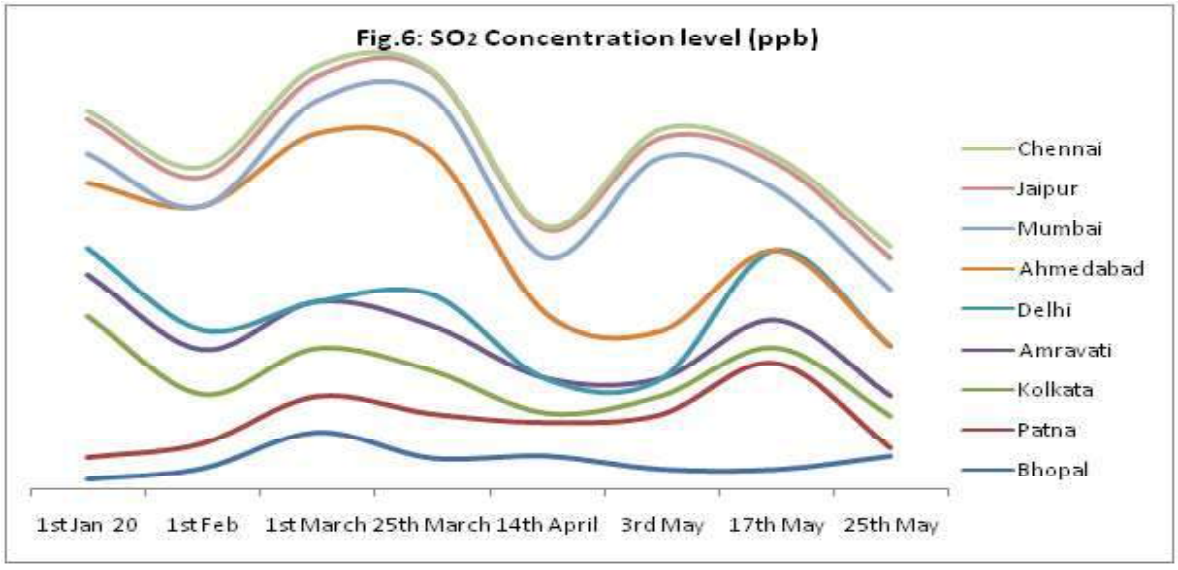



**Table 6: $SO_2$ concentration levels analysis: (unit: particle per billion (ppb))**

| City | 1st Jan 20 | 1st Feb | 1st March | 25th March | 14th April | 3rd May | 17th May | 25th May |
|---|---|---|---|---|---|---|---|---|
| Bhopal | 3.82 | 8.22 | 22.91 | 12.48 | 13.26 | 7.9 | 7.77 | 13.22 |
| Patna | 9.35 | 10.57 | 15.06 | 18.36 | 14.09 | 22.66 | 43.75 | 3.87 |
| Kolkata | 57.78 | 19.78 | 19.67 | 17.56 | 3.43 | 7.33 | 6.33 | 12.52 |
| Amravati | 17.62 | 18.82 | 19.71 | 19.08 | 14.6 | 7.45 | 11.83 | 8.49 |
| Delhi | 10.66 | 8.28 | None | 12.83 | None | None | 28.71 | 20.89 |
| Ahmedabad | 27.07 | 51.16 | 69.35 | 58.95 | 26.97 | 19.89 | None | None |
| Mumbai | 12.32 | 0.04 | 13.91 | 23.15 | 23.48 | 71.79 | 25.4 | 23.45 |
| Jaipur | 13.89 | 11.78 | 9.46 | 9.4 | 11.32 | 7.84 | 10.87 | 13.16 |
| Chennai | 3.74 | 4.33 | 4.56 | 1.29 | 1.36 | 3.77 | 2.79 | 4.7 |
| Lucknow | NA | NA | NA | NA | NA | NA | NA | NA |

***Range:** Good (O -35); Satisfactory (36 -75); Moderately Polluted (76 -185); Poor (186 -300); Very Poor (300-600); Severe (600+)*
*Data Source: CPCB - India Central Pollution Control Board portal (https://cpcb.nic.in/)*



## RESULTS

On the basis Time series of daily concentrations from January 1 to May 25, 2020 in 10 cities across India, following outcomes are noted:

**i.** PM2.5 concentration showed that after the lockdown, a statistically significant decrease in PM2.5 concentration was observed in almost all the cities. Delhi and Lucknow have witnessed highest reduction in PM2.5 concentration levels. Though the concentration in Mumbai & Jaipur is back to pre-Covid levels. For other cities, concentration of PM2.5 has commenced peaking towards pre-Covid levels gradually. It is construed that, concentration levels have begun peaking due ease down in lockdown and commencement of anthropogenic activities.

**ii.** PM10 concentration is available for 8 cities out of the 10 considered. Among 8 cities, despite reduction in PM10 concentration, 4 cities are still under moderately polluted range. Kolkata & Bhopal are under Good range. Delhi and Patna have witnessed considerable reduction. Mumbai is showing PM10 concentration around pre-Covid levels. It is evidenced by the data that PM 10 levels are gradually peaking towards pre-Covid levels.

**iii.** $NO_2$ concentration shows significant reduction in Kolkata and Bhopal post lockdown. Patna is showing consistency higher levels. Ahmedabad & Lucknow are back to pre-Covid levels. Others have seen slight improvement initially post lockdown, however, trends have gradually commenced to move upward.

**iv.** CO concentration shows that all the cities under study are within good range. Delhi, Patna, Lucknow and Mumbai have seen significant decline in carbon emission levels. Chennai though is back to pre-Covid levels with ease in lockdown.

**v.** Time series of daily $O_3$ concentrations shows mixed trends. Patna & Bhopal have seen increase in $O_3$ concentration post lockdown. Lucknow and Mumbai have seen drastic reductions, and other cities have shown stagnancy.

As $O_3$ is a secondary air pollutant, the local availability of precursor gases (nitrogen oxides and volatile organic compounds) and local meteorology (sunlight and temperature) influences the production and destruction of $O_3$ in air. These two factors, particularly more incoming sunlight in March than February, are expected to vary widely across the cities, and hence the $O_3$ concentrations too.

**vi.** $SO_2$ concentration has shown mixed trends. Kolkata having significant decline. Other cities have shown stagnancy and increasing trend.

## CONCLUSION

Evidence based on ground level stations shows significant reduction in concentrations of four major air pollutants PM2.5, PM10, $NO_2$ and CO in majority of the cities. Lockdown introduced to contain the Covid-19 has led to a near-simultaneous shutdown of many sources of air pollutants at a global scale and has set a natural condition to examine the planetary scale reduction in air pollution albeit temporarily. Though the reduction is expected to be short-lived, it provides us encouraging evidence that a swift and widespread implementation of air pollution measures can result in immediate air quality benefits. Yet, there is nothing to celebrate as the Covid-19 pandemic has taken an enormous toll on humanity and nobody wanted air pollution reduction this way.

## REFERENCES


1. **Singh, R. K. & Chakraborty, D. 2020.** Singh World's Dirtiest Air Gets Cleaner After India's Lockdown https://www.bloomberg.com/news/articles/2020-04-07/world-s-dirtiest-air-gets-cleaner-after-india-s-lockdown (Date accessed: May 25, 2020)
2. **WHO. 2020.** Air Pollution http://www.who.int/airpollution/en/ (Date accessed: May 25, 2020)
3. **Achakulwisut, P., Brauer, M., Hystad, P., & Anenberg, S. C. 2019.** Global, national, and urban burdens of paediatric asthma incidence attributable to ambient $NO_2$ pollution: estimates from global datasets. *The Lancet Planetary Health*, **3(4):** e166-e178.
4. **Chen, T. M., Kuschner, W. G., Gokhale, J., & Shofer, S. 2007.** Outdoor air pollution: nitrogen dioxide, sulfur dioxide, and carbon monoxide health effects. *The Ame. Jour. of the Medical Sciences*. **333(4):** 249-256.
5. **Brown, J. S. 2009.** Acute effects of exposure to ozone in humans: how low can levels be and still produce effects? *Ame. Jour. of Res. & Crit. Care Med.* **180(3):** 200-201.

## ADDITIONAL REFERENCES

1. **ET. 2020.** The Corona virus Outbreak, News Release, Economic Times (ET), https://economictimes.indiatimes.com/coronavirus (Date accessed: May 25, 2020)
2. **Cadotte, M. 2020, March 30.** Early evidence that Covid-19 government policies reduce urban air pollution.
3. **Chandrashekhar, V. 2020.** 1.3 billion people. A 21-day lockdown. Can India curb the coronavirus? Science
4. **CPCB. 2019.** Air Quality Monitoring: Continuous Station Status, CPCB – Central Pollution Control Board, India, https://app.cpcbccr.com/ccr/#/caaqm-dashboard-all/caaqm-landing (Date accessed: May 25, 2020)

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# The status of *Lytocestus laturensis* Kale & Kalshetty, 2020- a critical study and its placement under INCERTAE SEDIS

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**Abstract-**Kale and Kalshetty (2020)[1] while describing a new species of a Caryophyllaeid cestode *Lytocestus laturensis* (Lytocestidae: Caryophyllidae) from *Clarias batrachus* at Latur, Maharashtra inadvertently committed certain mistakes. The present authors have critically examined the details, pointed out the mistakes and have suggested re-examination of the species, undertake molecular characterization till then the species in question has been kept under "INCERTAE SEDIS".

**Key words: *Lytocestus laturensis*, status, lacunae, 'incertae sedis'**

## INTRODUCTION

Caryophyllaeids are unique in having a single set of reproductive organs within a non segmented body utilizing oligochaetes as their intermediate host. These are said to have originated from acoelomate turbellarian larvae and branched out at the beginning of Paleozoic era as parasites of aquatic vertebrates Kulakovaskaja and Demshin (1978)[2]. These are having a unique morphology, evolutionary status, genetic stability and show high degree of endemism.

The cestodes are interesting because they show low fecundity yet are extremely successful showing abundance in fishes, which is on account of 'vitelline cells' which can synthesize & store glycogen in the nuclei, a normal function- a unique phenomena in the animal kingdom Agarwal (1985)[3].

The increase in glycogen without concomitant increase in the number of vitelline cells or in egg size could lead to prolonged period of infectivity & utilization of glycogen and not lipid as energy source. Glycogen being heavier than lipid, can be utilised under anerobic condition are consistent with life cycle that has benthic intermediate host Mackiewicz (1981a)[4]. Sahay *et al*. (2007)[5] opined that higher level of glycogen in Caryophyllaeids is indicative of lower caloric value than lipid, which are the result of lower oxygen concentration in the enteric environment but low fecundity and relative abundance is due to K-strategies as an environmental condition.

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Out of the four families falling under Caryophyllidea, one of them is Lytocestidae Wardle & McLeod (1952)[6]. One of the genus belonging to Lytocestidae is *Lytocestus* Cohn (1908)[7] to which 58 species have been reported. These are:-

1. *L.adhaerens* Cohn (1908)[7]
2. *L.filiformis* Woodland (1923)[8]
3. *L.indicus* Moghe (1925)[9]
4. *L.cunnigtoni* Fuhrmann *et al*. (1925)[10]
5. *L.chalmerisius* Woodland (1926)[11]
6. *L.javanicus* Bovien (1926)[12]
7. *L.birmanicus* Lynsdale (1956)[13]
8. *L.alestesi* Lynsdale (1956)[13]
9. *L.parvulus* Furtado (1963)[14]
10. *L.moghei* Murhar (1963)[15]
11. *L.longicollis* Ramadevi (1973)[16]
12. *L.lativitellarium* Furtado & Kim Low (1973)[17]
13. *L.fossilis* Singh (1975)[18]
14. *L.marathawadensis* Shinde & Phad (1988)[19]
15. *L.alii* Jadhav and Gavahne (1991)[20]
16. *L.clariasae* Jadhav and Gavahne (1991)[20]
17. *L.naldurgensis* Kadam, Hiware & Jadhav (1998)[21]
18. *L.chalisgaonensis* Khalse & Shinde (1999)[22]
19. *L.kopardaensis* Shinde & Borde (1999)[23]
20. *L.teranaensis* Kolpuke, Shinde and Begum (1999)[24]
21. *L.batrachusae* Pawar & Shinde (2002)[25]
22. *L.clariasae* (minor) Pawar & Shinde (2002)[25]
23. *L.govindae* Patil & Jadhav (2002)[26]
24. *L.vishnupurensis* Shomendra *et al.* (2003)[27]
25. *L.nagapurensis* Lakhe, Pawar & Shinde (2004)[28]
26. *L.shindei* Khadap *et al.* (2004)[29]
27. *L.assamensis* Tandon, Chakravorty & Das (2005)[30]
28. *L.attenuatus* Tandon, Chakravorty & Das (2005)[30]
29. *L.clariae* Tandon, Chakravorty & Das (2005)[30]
30. *L.heteropneusti* Tandon, Chakravorty & Das (2005)[30]
31. *L.bokaroensis* Poonam (2007)[31]
32. *L.majumdari* Poonam (2007)[32]
33. *L.paithanensis* Shelke (2007)[33]
34. *L.jagtai* Tripathi *et al*. (2007)[34]
35. *L.punensis* Jadhav, Bhure & Padwal (2008)[35]
36. *L.subhapradhi* Jawlikar, Pawar and Shinde (2008)[36]
37. *L.moghei* Sharma (2009)[37]
38. *L.murhari* Kaul & Suryavanshi (2010)[38]
39. *L.follicullariae* Bhure *et al.* (2010)[39]
40. *L.osmanabadensis* Bhure *et al*. (2010)[39]
41. *L.shindei* Suryavanshi *et al.* (2010)[40]
42. *L.vyasaei* Pawar & Hiware (2011)[41]
43. *L.purnensis* Pawar & Hiware (2011)[41]
44. *L.geriapinusae* Kadam *et al.* (2011)[42]
45. *L.khami* Jawle *et al.* (2011)[43]
46. *L.thapari* Sawarkar (2012)[44]
47. *L.alii* (minor) Sawarkar (2012)[44]
48. *L.manjaraensis* Solunki *et al.* (2012)[45]
49. *L.rekhaensis* Nimbalkar *et al.* (2012)[46]
50. *L.indica* Deshmukh (2015)[47]
51. *L.godavariensis* Pawar (2016)[48]
52. *L.mastacembellusi* Pardeshi (2016)[49]
53. *L.ambae* Kankale (2017)[50]
54. *L.paithanensis* Kale (2017)[51]
55. *L.bhartae* Patil (2018)[52]
56. *L.elongatus* Barshe *et al.* (2018)[53]
57. *L.laturensis* Kale & Kalshetty (2020)[1]
58. *L.sahayi* Bhavsar (2020)[54]

Hunter (1927)[55] transferred 4,5 into the genus *Monobothrioides* Gupta (1961)[56] accorded support.

Serial 8 was considered synonym of *L.birmanicus* (serial 7). Mackiewicz (1962)[57] however, considered it to be synonym of *L.filiformis* (serial 2). In *L.birmanicus* Lynsdale (1956)[13] Mackiewicz observed 85-95 testicular follicles whereas Chakravorty & Tandon (1989)[58] recorded 170-384.

Serial nos 7,9,10,11,31 & 32 were considered by Ash (2012)[59] to be synonym of *Pseudocaryophyllaeus tenuicollis* Bovien (1926)[12]. He also synonimised serial no. 12 & 27 with *Lucknowia microcephla* Bovien (1926)[12]. Ash (2011a)[60] considered serial no. 14 synonym of *Pseudocaryophyllaeus ritai* Gupta & Singh (1983)[61].

Most of the species are reported from the state of Maharashtra.

Ash (2011a, b, 2012)[60,59] synomised *L. heteropneustii* and *L. jagtai* (serial no. 30& 34) with *Lucknowia fossilis* Gupta (1961)[56]. *L.lativitellarium* Furtado & Kim Low (1973)[17] & *L.assamensis* Tandon *et al.* (2005)[30] Serial no 12 & 27 were synonimised with *Lucknowia microcephala* Bovien (1926)[12]

*Lytocestus alii* Jadhav *et al.* (1991)[20], *Lytocestus clariasae* Jadhav *et al*. (1991)[20], *Lytocestus chalisgaonensis* Shinde *et al.* (1999)[23], *Lytocestus kopardaensis* Shinde *et al*. (1999)[23], *Lytocestus naldurgensis* Kadam *et al.* (1998)[21], *Lytocestus teranaensis* Kolpuke *et al.* (1999)[24], *Lytocestus batrachusae* Pawar & Shinde (2002)[25], *Lytocestus clariasae* (minor) Pawar *et al.* (2002)[25] (homonym), *Lytocestus govindae* Patil *et al.* (2002)[26], *Lytocestus nagapurensis* Lakhe *et al.* (2004)[28],

*Lytocestus shindei* Khadap *et al.* (2004)[29], *Lytocestus paithanensis* Shelke (2007)[33], *Lytocestus punensis* Jadhav *et al.* (2008)[35], *Lytocestus subhapradhi* Jawlikar *et al.* (2008)[36], *Lytocestus murhari* Kaul *et al.* (2010)[38], *Lytocestus shindei* Suryavanshi *et al.* (2010)[40] were considered synonyms of *Lytocestus indicus* Moghe (1925)[9] by Ash (2012)[59]

Sahay & Khalkho (2017)[62] kept serial no. 49 [*L.rekhaensis* Nimbalkar *et al.* (2012)[46]] under enquiry. Sahay *et al.* (2017)[63] revalidated *L.heteropneusti* Tandon *et al.* (2005)[30]. Singh *et al.* (2018)[64] held *L.bishnupurensis* (serial 24) synonym of *L.indicus* Moghe (1925)[9]. Sahay *et al.* (2019)[65] kept *L.mastacembellusi* Pardeshi (2016)[49] a species under enquiry. Sahay & Ekka (2019)[66] considered *L.jagtai* Tripathi *et al.* (2007)[34] & *Lytocestus fossilis* Singh (1975)[18] to be synonym of *Lucknowia fossilisi* Gupta (1961)[56].

The aim of the present study is to assess the status of *Lytocestus laturensis* Kale & Kalshetty (2020)[1] as the authors of the said species seem to have inadvertently committed mistakes while describing the species.

## MATERIALS & METHODS

Several research papers have been consulted along with the literature of *L.laturensis* Kale *et al.* (2020)[1].

## OBSERVATION & DISCUSSION

**"***Lytocestus laturensis* Kale & Kalshetty (2020)[1] is supposed to show, following characters:- Head spatulate narrow than body, broad at the base narrow at the apex, neck medium in length wide narrow anteriorly, broad posteriorly with constrictions on the lateral side, testicular follicles 850-900 small oval, preovarian scattered in medullary region unevenly in 4-7 rows, not found in posterior 1/3rd region, vas deferens short, thin curved coiled. Cirrus pouch large, oval anteroposteriorly situated on left or right side & open through a genital pore tapering at anterior end curved. Cirrus thin curved slightly coiled within the cirrus pouch. Ovary bilobed butterfly shaped, medium H-shaped, lobes anteroposteriorly elongated with irregular margins, lobes spindle shaped broader in middle, tapering at both ends, uneven in size, poral lobe short, aporal lobe long, isthmus short, bag-pipe like wide, connecting lobes at 1/3rd from posterior end. Vagina thin, starts from genital pore open into ootype. Ootype large oval, antero-posteriorly elongated, **1-2 rows on each lateral side** (?). uterus tubular, 1-2 turns, wide loop shaped with 4-5 folds, narrow, open by uterine pore (oval, large double walled in middle of the worm)**"**.

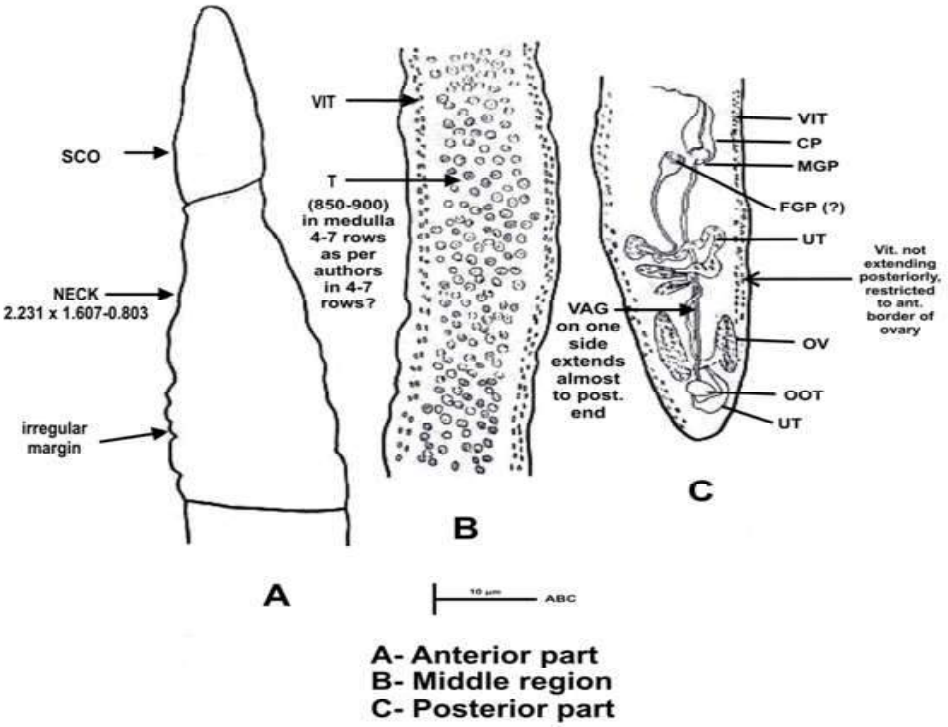



**Fig.1-** ***Lytocestus laturensis*** **(Kale & Kalshetty, 2020)[1] from** ***International Journal of Entomology Research.*** **Volume 5 Issue 5 on page 63 (labeling done for convenience)**

**Lacunae in the paper:**

There are a number of spelling mistakes in the said paper. In the abstract the authors wrote "fifteen species of cestode parasites were collected from the intestine of *Clarias batrachus*, the word "species" should have been replaced by 'specimens'.

In case the authors were in possession of 15 specimens why there should be single measurement in the followings:

a. Length & breadth- **4.973** x 0.446 – 0.603
b. Head – spatulate **2.803** in length x 0.982 – 0.446
c. Neck – **2.231** x 1.607 – 0.803
d. Cirrus pouch – **0.893** x 0.338 – 0.178 in breadth
e. Cirrus – **0.767** x 0.053
f. Vas deferens – **0.535** x 0.035
g. Ovary – **0.446** x 0.893
h. Isthmus – at 1/3rd from post. end **0.357** x 0.089
i. Uterus – **0.446** x 0.893

It seems that the authors studied only one specimen and the range in width in a,b,c etc in all probability was their convenience. If they had fifteen specimens they should have given range in the lengths & breadth of all the parameters, taken out the mean value, SD and CV. Because these could be used to define statistically the species. Ratios such as:

1. Commencement of vitellarial follicles and testicular follicles/ body length.
2. Position of ovary from posterior end/ body length
3. Position of cirrus sac from posterior end/ body length
4. Distance of genital pore from posterior end/ body length
5. Distance of ootype from posterior end/ body length

Of the above 5 parameters would have characterized *Lytocestus laturensis* (?) better than linear measurements.

The present authors are of the opinion that if only whole samples (of different populations of same species) are studied and quantitatively analysed, species determination would not remain entirely subjective.

The authors of the said species state "vagina is a thin tube, starts from the genital pore, runs medially and posteriorly, slightly curved, reaches and opens into the ootype and measures 0.357 in length and 0.213 in breadth."

Mackiewicz (1972)[67] opines that- "vagina communicates between oviduct and the ventral surface (fig. 6) Caryophyllidea- a review. At its proximal end it may join the uterus to form utero-vaginal duct as in *C.laticeps* or *P.differtus*) opening posterior to the male gonopore or it

- may form the uterovaginal duct that also receives the ejaculatory duct thus forming a short distinct canal (hermaphroditic duct) that- terminates at a small, common gonopore (as in *Caryophyllaeids, Isoglaridacris* and *Biacetabulum*).
- A third condition (as in *Atractolytocestus* occurs when the ejaculatory duct and utero-vaginal duct open together or very close to the surface forming a shallow atrium with a single large gonopore through which one can see the male & female gonopore.)

Vagina never opens directly in ootype rather "posteriorly it drains into oviduct but it does not open independently on the ventral surface of body, usually it joins with the terminal part of the uterus to form an uterovaginal duct which communicates to the ventral surface of body as female pore"- Hafezullah (1993)[68].

Kale and Kalshetty (2020)[1] have not mentioned anything about vagina joining the uterus and forming the utero-vaginal duct which opens through uterovaginal pore is a serious lapse.

Developmental studies may reveal, which one develops first, the genital pore or the vagina. But this much is sure that vaginal's posterior end does not open directly into the ootype rather swells to form receptaculum seminis which continues posteriorly & joins the oviduct, before joining the ootype. In *Lytocestus* species male and female genital openings opens separately.

**Vitellaria:** The distribution of vitellaria in *Lytocestus laturensis* (vide fig. 1c) has been shown to be unequal upto almost posterior end on one side whereas, the extension is upto anterior border of ovary. This needs clarification as regards its commencement (anteriorly) and termination (posteriorly).

**Testes:** The number of testicular follicles in *Lytocestus laturensis* has been shown to be 850-900 preovarian scattered in medullary region, unevenly distributed in 4-7 rows from the base of neck to the gonads, not distributed in posterior 1/3rd region of the worm.

The authors of the said species *Lytocestus laturensis* seems to have given conigence to the number of testicular follicles while separating it from other species of

*Lytocestus* [*L.filiformis* Woodland (1923)[8], *L.indicus* Moghe (1925)[9], *L.alestesi* Lynsdale (1956)[13], *L.birmanicus* Lynsdale (1956)[13], *L.longicollis* Ramadevi (1973)[16], *L.marathawadensis* Shinde & Phad (1988)[19], *L.alii* Jadhav and Gavahne (1991)[20], *L.naldurgensis* Kadam *et al.* (1998)[21], *L.teranaensis* Kolpuke *et al.* (1999)[24], *L.nagapurensis* Lakhe *et al.* (2004)[28], *L.assamensis* Tandon *et al.* (2005)[30], *L.kopardaensis* Shinde & Borde (1999)[23], *L.govindae* Patil & Jadhav (2002)[26], *L.murhari* Kaul & Suryavanshi (2010)[38], *L.geriapinusae* Kadam *et al.* (2011)[42], *L.khami* Jawle *et al.* (2011)[43], *L.vyasaei* Pawar & Hiware (2011)[41] a total of 17 species.

Although it has already been pointed out by Sahay *et al.* 2019 that species identification on the basis of number of testicular follicles is questionable because the range depicts that the worm in question were not of the same age and that the worm were more than one when studied. If the worms are of the same age, the number of testicular follicles should normally remain more or less constant for a species.

It is surprising that the authors compared their specimens only with above 17 species & did not compare it with rest of the species. Not only that, out of the above 17 species

1. *Lytocestus* (*alii, naldurgensis, teranaensis, kopardaensis, murhari, govindae, assamensis*) have already been synonymized.
2. *L.marathawadensis* Shinde & Phad (1988)[19], synonimized with *Pseudocaryophyllaeus ritai* Gupta & Singh (1983)[61] by
3. *L.longicollis* Ramadevi (1973)[16] was synonimized with *Pseudocaryophyllaeus tenuicollis* Bovein (1926)[12] by Ash (2012)[59]
4. *L.khami* Jawle *et al.* (2011)[43], was kept by Sahay, *et al.* (2021)[69] under INSERTAE SEDIS

On the above grounds the present authors are left with the following alternatives:-

1. Invalidate *Lytocestus laturensis* as a new species.
2. Keep *Lytocestus laturensis*- a species under INCERTAE SEDIS
3. Request the authors of the paper to restudy the slides in their possession clarify the above points, provide photo-micrograph of the slides and provide cross sections. Take up proper identification of the cestode in question upto molecular level to ascertain their taxonomic status. The molecular identification has been used for delineating species specially between cryptic and sister species- Nadler & Perez-Ponae de Leon (2011) otherwise traditional taxonomy based on phenotypes will fall into distress.

The present authors though support the 3rd option yet advocate till then that the said species be kept under "INCERTAE SEDIS".

## REFERENCES


1. **Sanjay Kale & SG Kalshetty. 2020.** A study of a new cestode *Lytocestus laturensis n. sp.* (Lytocestdae, Hunter 1927) from *Clarias batrachus. International Journal of Entomology Research.* **5(5):** 62-65.
2. **Kulakovskaja, O.P. & Demshin N.I. 1978.** Origin and phylogenetic relationships of Caryophyllideans in Problemy Gidro Parasitologii (ed) *A.P.Markevic,Kiez. Riev. Naukova* Dumka) pp. 95-104.
3. **Agarwal, S.M. 1985.** *Caryophyllaeids* and *Caryophyllidiosis* in India. *Indian Rev. Life Sci.,* **5:**139-161.
4. **Mackiewicz, J.S. 1981.** Synoptic review of the Caryophyllidea (Cestoidea) of India, Pakistan and Nepal. *Himalayan. Journal of Science.* **1:** 1-14.
5. **Umapati Sahay, H.C.P. Sah & Sarojini Sahay. 2007.** Diseases caused by caryophyllaeids (caryophyllidea: cestoidea) in fishes-a critical review. *Biodiversity* (Edit. B.N.Pandey, Sadhana Deshpande & B.D.Joshi) APH Publishing Corporation, New Delhi. Chap.**16:**175-201.
6. **Wardle, R.A & J.A McLeod. 1952.** The Zoology of Tapeworms, University of Minnesota Press, Minneapolis p.780.
7. **Cohn, J.A. 1908.** Die Anatomie eines neuen Fishcestoden. *Centralbl. Bakt. Parasitenk*, infection skrankheiten und Hygiene. *Abetelung I. originale*. **46:**134-139.
8. **Woodland, W.N.F. 1923.** On some remarkable new forms of *Caryophyllaeidae* from Anglo Egyptian Sudan and a revision of the families of Cestodaria. *Q.J.Microsc. Sci.* **67:** 435-472
9. **Moghe, M.A., 1925.** *Caryophyllaeus indicus* n.sp. (Trematoda) from the catfish *Clarias batrachus*. (L). *Parasitology*. **17:** 232-235

10. **Fuhrmann, O and J.G.Baer. 1925.** Zoological results of third Tanganyika expedition conducted by Dr. W.A Cunnington 1904-1905. Report on the Cestoda. *Proc. Zool. Soc. Lond.* 79-100

11. **Woodland, W.N.F. 1926.** On the genera and possible affinities of the Caryophyllaeidae: a reply to Dr. S. O. Fuhrmann and J.G Baer. *Proc Zool Soc Lond*. 49-69.

12. **Bovien, P. 1926.** Caryophyllaeidae from Java. *Videnskabelige Meddeleleser fra Dansk. Naturhistorisk Forening Kobenhavn*. **82:**157-181.

13. **Lynsdale, J.A. 1956.** On two new species of *Lytocestus* from Burma and Sudan respectively. *J. Helm*. **30(2-3):** 87-96.

14. **Furtado. J.I. 1963.** A new Caryophyllaeid Cestode *Lytocestus parvulus* sp.nov. from a Malayan catfish. *Ann. Mag. Nat.Hist* (Ser. XIII) **6:** 97-106.

15. **Murhar, B.M. 1963.** *Crecentovitus biloculus* gen.nov; sp.nov a fish (cestode: *Caryophyllaeidae*) from Nagpur. India. *Parasitology*. **53:** 413-418.

16. **Ramadevi, P. 1973.** *Lytocestus longicollis* sp. nov. (Cestoda:Caryophyllidea) from the catfish, *Clarias batrachus* in India. *J.Helm*. **47(4):** 415-420.

17. **Furtado, J.I and Jan Kim-Low. 1973.** Incidence of some helminth parasites in the Malaysian Catfish *Clarias batrachus* (L) *Verhandlungen Internationale fur Theoritische and Angewandte Limnologie.* **18(3):** 1674-1685.

18. **Singh, S. S. 1975.** On *Lytocestus fossilis* n.sp. (Cestodea: Lytocestidae) from *Heteropneustes fossilis* from Nepal. In Dr. B.S. Chauhan commemoration volume K.K.Tiwari and CB Srivastava (eds) *Zool. Soc. of India*. Utkal Univ. Vani Vihar Orrisa. 79-82.

19. **Shinde, G.B. and A.N. Phad. 1988.** On a new Cestode *Lytocestus marathawadensis* from freshwater fish. *Riv. Di Parasit*. **47(2):** 295-298.

20. **Jadhav, B.V. and A.V. Ghavane. 1991.** Two new Cestodes from Caryophyllaeidae at Aurangabad. *Ind.J.Inv.Zool and Aq. Biol*. **3(1):** 28-31.

21. **Kadam, M.N., C.J. Hiware and B.V. Jadhav. 1998.** On a new Caryophyllaeid Cestode of the Genus *Lytocestus* Cohn, 1908 from *Clarias batrachus* Dr. B.A.M. Univ. Aurangabad *J.of.Sci.* **29(6):**143-148.

22. **Khalse, A.T and G. B. Shinde. 1999.** *Lytocestus chalisgaonensis* n. sp. (Cestoidea: Caryophyllidea) from catfish *Clarias batrachus* at Chalisgaon M.S. India. *Riv.Di.Parasit.* **XVI(LX)N-1:** 39-42.

23. **Shinde, G.B. and Sunita Borde. 1999.** On *Lytocestus kopardaensis* n.sp. Cestoode (Lytocesidae:Hunter) from a fish in Maharashtra State India. *Utt.Pra.Jour.Zool.* **19(3):** 211-213.

24. **Kolpuke, M.N., G.B.Shinde and I.J.Begum. 1999.** On a new species of the genus *Lytocestus* Cohn, 1908 (Cestode:Caryophyllidea) from *Wallago attu* from Terna river at Aurangabad, India. *Uttar Pradesh.J.Zool.* **19(1):** 93-95.

25. **Pawar, S.B. and G.B.Shinde. 2002.** A new species *Lytocestus clariasae* n.sp. (Cotyloda: Lytocestidae) from *Clarias batrachus* at Kallam, India. *Riv.Di.Parasit.* **XIX(LXIII) 2:** 157-160.

26. **Patil, D.N and B.V.Yadhav. 2002.** On a new Caryophyllaeid Cestode of the genus *Lytocestus* Cohn, 1908 from *Clarias batrachus*. *Ind.J.Helm*. (N.S). **20:**45-48.

27. **Shomendra M., A.N. Jha & Pankaj Kumar. 2003.** A new cestode *Lytocestus bishnupurensis* from a fresh water fish *Mystus senghala* (Sykes). *J.Freshwater.Biol.* **15(1-4):** 43-45.

28. **Lakhe,A.D; S.B.Pawar and G.B.Shinde. 2004.** A new Cestode *Lytocestus nagapurensis* n.sp. (Cotyloda:Lytocestidae). *Riv.Di.Parasit*. **XXI(LXV) 2:** 95-98.

29. **Khadap, R.M; B.V.Jadhav and N.V. Suryavanshi. 2004.** A new species of the genus *Lytocestus* (Cohn, 1908) from *Clarias batrachus* at Aurangabad. *Nat.J.Life.Sciences*. **1(2):** 413-416.

30. **Tandon, V.; R. Chakravarty & B. Das. 2005.** Four new species of the genus *Lytocestus* (Caryophyllidea: Lytocestidae) from edible catfishes in Assam and Meghalaya, India. *Jour. Parasitic Diseases*. **29(2):** 131-142.

31. **Poonam. 2007.** On *Lytocestus bokaroensis* n.sp. (Caryophyllidea:Lytocestidae) from *Clarias batrachus. Proc.Zool.Soc.India*. **6(2):** 73-78.

32. **Poonam. 2007.** On a new species of the genus *Lytocestus* (Caryophyllidea: Lytocestidae) from *Clarias batrachus*. *Proc.Zool.Soc.India*. **6(1):** 77-81.

33. **Shelke, V.P. 2007.** *Lytocestus paithanensis* n.sp. from *Clarias batrachus. Nat.J.Life. Sciences.* **4(3):** 151-152.

34. **Tripathi, N.P., S.P.Singh and A.K. Mishra. 2007.** A new species of the genus *Lytocestus* (Cestoda:Lytocestidae) from *Heteropneustes fossilis* at Rewa, M.P., *Nat.J.Life.Sciences*. **4(3):** 111-114.

35. **Jadhav,B.V; D.B.Bhure and Nitin Padwal. 2008.** Caryophyllidean review from catfishes of Maharashtra (India). *Flora and Fauna*. **14(1):** 3-22.

36. **Jawalikar, J.D; S.B.Pawar and G.B.Shinde. 2008.** A new species *Lytocestus subhapradhi* n.sp. (Eucsetoda:Lytocestidae) from *Clarias batrachus. Uttar Pradesh. J. Zool*. **28(3):** 354-369.

37. **Sharma Bindu. 2008-2009.** Morphology, population dynamics, biochemistry and physiopathology of caryophyllaeids of stiging catfish, *Heteropneustes fossilis* (Bloch) Ph,D thesis. Dept. of zoology ch. C.S. University Meerut pp.1-94

38. **Kaul, S.S., A.T.Kalse and R.B.Suryavanshi. 2010.** *Lytocestus murhari* sp.nov. (Cestoda:Caryophyllidea) from the catfish *Clarias batrachus* (L) at Chalisgaon. *Decc.Curr.Sci*. **3(1):** 73-84.

39. **Bhure, D.B., S.B.Waghmare, C.R., Kasar & K.M.Shaikh. 2010.** Taxonomic observation of the Caryophyllidean tapeworm *Lytocestus* Cohn, 1908 from *Clarias batrachus* (Linneus, 1758). *J.Eco.Environ.Sci*. **1(1):** 01-06.

40. **Suryavanshi S.G., D.K. Maske., G.B.Shinde and H.K.Bhagwan. 2010.** A new tapeworm *Lytocestus shindei* n.sp. (Cestoda:Lytocestidae) from *Clarias batrachus* at Rahuri district Ahmednagar (M.S) *Life.Sc.Bull*. **1:** 148-150.

41. **Pawar, R.T and C.J.Hiware. 2011.** Two new species of the genus Lytocestus (Caryophyllidea: Lytocestidae) from freshwater catfish *Clarias batrachus* (Linnaeus, 1758). *Recent Research in Sciences & Technology.* **3(12):** 25-28.

42. **Kadam, K.N and Jaswant S. Dhole. 2011.** New tapeworm *Lytocestus gariepinusae* n.sp. from a freshwater fish *gariepinus* at Makani Dam, Dist. Osmanabad M.S. India. *Recent.Res.Sc.Tech*. **3(8):** 19-23.

43. **Jawale, Sushil and Sunita Borde. 2011.** New species of the genus *Lytocestus* (Caryophyllidea: Lytocestidae) from catfish at Aurangabad district (M.S) India. *Int.Multidisciplinary Res.J.,* **1(8):** 27-30.

44. **Sawarkar, B.W. 2012.** Record of new tapeworm *Lytocestus alii* n.sp. from freshwater fish *Clarias batrachus* (Bleeker, 1862) at Amravati, Maharashtra, India,. *Jour.of Biology & Life Sciences*. **3(1):** 281-287.

45. **Solunke Ravi, Swati Fadke, Sunita Borde and Sushil Jawale. 2012.** New species of the genus *Lytocestus* (Caryophyllidea: Lytocestidae) from cat fish in Latur district M.S. India, *Trends in Parasitology Research*. **1(2):** 25-30

46. **Nimbalkar R.K., R.V.Deolalikar & S.P.Muley. 2012.** Study on a new species of *Lytocestus* (Cohn, 1908) *Heteropneustes fossilis* (Bloch) at Jaiwadi Dam of Aurangabad district (M.S). *Life Science Bulletin*. **9(2):** 239-242.

47. **Deshmukh,V.S., Nanware & D.B. Bhure. 2015.** Biosystematic studies on Caryophyllidean cestode genus *Lytocestus* from freshwater catfish *Clarias batrachus* with description of new species. *Flora & Fauna.* **21(2):** 179-190.

48. **Pawar, R. G.; R. R. Dandawate and R. S. Lonkar. 2016.** *Lytocestus godavariensis* n.sp. from freshwater fish *Clarias batrachus* from Godavari river at Pravarasangam, Taluka Newasa, District Ahmednagar, M.S. *International Journal of Researches in Biosciences, Agriculture and Technology*. **4 (1):** 142-145

49. **Pardeshi, K.S. 2016.** Cestode; *Lytocestus mastacembellusi*. *Intl.Jour.Sci. Res & Education*. **4(4):** 5140-5143.

50. **Kankale, N.M. 2017.** A new species of the genus *Lytocestus ambae* from a freshwater fish *Clarias batrachus. Int. J. of Research in Bioscience, Agri. & Tech special issue*. **5(2):** 242-243.

**51. Kale, S.S. 2017.** A new A new species of cestode *Lytocestus paithanensis* (*Lytocestus* Cohn, 1908) from *Clarias batrachus* at Paithan, MS, India. *Int. J. of Life Sciences*. **5 (3):** 455-458

**52. Patil, D.P. 2018.** A new species of the genus *Lytocestus* (Caryophyllidea:Lytocestidae) from *Clarias batrachus. Review Research*. **7(6):**1-4

**53. Barshe, M.U.; D.B. Bhure and S.Nanware. 2018.** Morphotaxonomic studies on Caryophyllidean cestode genus *Lytocestus* Cohn, 1908 from freshwater catfish *Clarias batrachus* with description of new species. *Annals of Nat. Sciences*. **4(3):** 7-19.

**54. Bhavsar, Khushal; Avinash Bhangale and Ajit Kalse. 2020.** Reporting a new Caryophyllidean worm from a fresh water fish *Clarias batrachus. JETIR*. **7(10):**284-292

**55. Hunter, G.W. III. 1927.** Notes on the *Caryophyllaeidae* of North America. *Journal of parasitology* **14:** 16-26.

**56. Gupta, S.P. 1961**. *Caryophyllaeids* (Cestoda) from freshwater fishes of India. *Proc. Helm. Soc. Wash*. **28(1):** 38-50.

**57. Mackiewicz. J.S. 1962.** Systematic position of *Lytocestus* from Burma & Sudan respectively *J.Helm.* **30(2-3):** 87-96.

**58. Chakravarty, R and Veena Tandon. 1989.** On the present status of Caryophyllidea with a report on some Caryophyllaeid infections in the freshwater catfishes *Clarias batrachus* (L) in Northeast India and record of an anamolous form, *Indian.J.Helminth* (n.s). **5(1):**37-54.

**59. Ash, Anirban. 2012.** Diversity of tape worms (Cestoda) in freshwater fishes of India. Ph.D thesis. Institute of Parasitology, Biology Centre, Academy of Sciences of the Czech Republic & Faculty of Science, University of South Bohemia in Ceske Budejevice. 1-135.

**60. Ash A., Scholz T., Oros M., Kar P. K. 2011,a.** Tapeworms (Cestoda: Caryophyllidea), parasites of *Clarias batrachus* (Pisces: Siluriformes) in the Indomalayan Region. *Journal of Parasitology*. **97(3):** 435–459.

**61. Gupta, V. and Singh, S.R. 1983**. On a new species *Pseudocaryophyllaeus ritai* sp.nov. (Family Caryophyllaeidae) from the intestine of a freshwater fish, *Rita rita*, from river Gomati at Lucknow, *U.P. Ind. Jour. Helminth*. **35 (1):** 11-14.

**62. Sahay, Umapati & A.P.V.Khalkho. 2017.** A discussion on the status of *Lytocestus rekhaensis* Nimbalkar *et al.* 2012. *Biospectra*. **12(1):** 1-8.

**63. Sahay, Umapati; Dimple Mandal; Nayni Saxena & Ravi Rahul Singh. 2017.** On the validity of *Lytocestus heteropneustii* (cestode) Tandon, Chakravarty & Das 2005- a critical review. *Biospectra* **12(2):**115-120.

**64. Singh, Ravi Rahul, Umapati Sahay & Fauzia Sadaf. 2018.** On the synonymy of *Lytocestus bishnupurensis* Shomendra *et al.* (2003) with *L.indicus* Moghe (1925). *Jour. of Exp. Zool*. **21(2):** 893-896

**65. Sahay, Umapati; A.P.V. Khalkho; Ravi Rahul Singh & Dimple Mandal. 2019.** On the status of *Lytocestus mastacembellusi* (Caryophyllidea: Lytocestidae) Pardeshi 2016- a critical study. *Asian Jour of Agriculture & Life Sciences.* **4(1):**13-21

**66. Sahay, Umapati & Pranaati Prabha Ekka. 2019.** On the status of *Lytocestus jagtai* (Caryophyllidea: Lytocestidae) Tripathi, Singh & Mishra. 2007- a critical study. *Trends in Fisheries Research.* **8(2):**78-85.

**67. Mackiewicz, J.S. 1972.** Caryophyllaeidae (Cestoidea): A review. *Experimental Parasitology*. **31:** 417-512.

**68. Hafeezullah, M. 1993.** *Caryophyllidean* Cestode fauna of India, *Rec. Zool. Sur. India*. Occasional Paper No. **157**; pp.1-107.

**69. Sahay, Umapati; Dimple Mandal; Kunjlata Lal; Ravi Rahul Singh and Shalini Kamal. 2021.** A critical study of *Lytocestus khami* Jawle & Borde (2011) and a discussion on its placement under incertae sedis. *J. Experimental Zool. India*. **24(1):**119-124.

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# Review study on altitudinal zonation of floral and faunal species, its richness and distribution

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**Abstract-**Many floral and faunal species are restricted in a particular narrow zone of habitat characteristically in mountain ecosystem. The mountain ecosystem acts as an isolated and unique habitat, and sometimes functions as an isolated island like wetland natural ecosystem. Isolated or higher altitudinal ranges create unique habitat for a range of floral and faunal species and increases the degree of endemism as compared to other habitat which can be more vulnerable to local extinction and very sensitive towards topographical changes in a particular habitat. In this regard a collective effort was needed to review the altitudinal zonation in terms of floral and faunal species richness and distribution. During present review studies it has been found that biotic and abiotic factors influences distribution and diversity pattern of floral and faunal species from low elevation level to higher elevation level. Higher altitudinal survival species are very sensitive towards micro changes in their habitat which can affect them adversely. In present time many anthropogenic activities (like mining in hill areas, deforestation, poaching etc.) and natural disasters (like forest fires on mountains, landslides etc.) leads to unfavorable and irreversible alteration in floral and faunal species mountain ecosystem and at certain time vanish them at all. It has been reviewed that variation in altitudinal ranges also affect breeding biology, morphological and physiological adaption in floral and faunal communities and create wide range of diverse habitat and ecosystems.

**Key words: Mountain ecosystem, altitude, endemism, adaption, habitat**

## INTRODUCTION

Biodiversity encompasses the wide variety and variability of living life forms in an ecosystem and ecological processes, at all levels of biological organization and its foundation for human survival and economic well being.[1] Biodiversity closely linked with human demands for food, fuel, fodder, fiber, medicine, oil, timbers and resins[2] and are also helpful and essential for maintaining climatic stability, pollution control, water and soil conservation, nutrient and mineral recycling, ground water recharging, pollination and recreation values of that particular region.[3-5] Biodiversity and ecosystem stability and sustainability mainly depends upon productivity and utilization rate of resources by human beings in recent and past years.[6-8] Low productivity and higher exploitation rate of resources increases risk of loss of biodiversity and shrinkage of natural ecosystem which creates imbalance amongst various ecosystems and their food chains and food web. Variation in abiotic environmental factors like-

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temperature, rainfall, humidity and wind velocity largely influence the distribution and type of species and vegetation composition. Floral and faunal distribution and its density is also affected by spatial, temporal, latitudinal and altitudinal variations at different geographic regions.[9-11] Altitudinal variation show four types of species distribution and richness patterns namely decreasing at low plateau, low plateau with mid elevation peak; with increase in elevation species richness, pattern and number decreases and generally species richness trends increases with higher elevation levels.[12-14]

In the present review study, a compressive overview that effects altitudinal variation on species distribution, density and diversity in relation to various altitudinal and latitudinal ranges was done which can be enumerated as :-

1. **EFFECT OF ELEVATION RANGES ON VEGETATION COMPOSITION:-** The evolution and composition of vegetation in a particular region is usually decided by several factors like- time, slope, aspect of elevation level, humidity and seasonal or annual precipitation. All these factors also changes with altitudinal ranges, which lead to climatic and spatial changes among animal and plant community at various elevation levels.[15,16] Altitudinal ranges greatly influence diversity and abundance of plant community at various geographical ranges[8], with strong impact of elevation on vegetation structure of most of the mountain ecosystems of the world, as reported in temperate forest of China, Mexico and central Europe.[9,17] At present time various anthropogenic and natural activities like- inefficient erosion of glaciers, soil and various environmental activities are responsible for reduction in mountain ranges height, which are responsible for modification and alternation of previous ecosystems. Trees play efficient and crucial role in defining and sustaining the structural and functional environment complexity and habitat heterogeneity in forest ecosystems of the Earth.[18] Global warming, invasion of exotic species, anthropogenic activities and climatic changes are also responsible for alteration in vegetation composition in native ecosystem at various altitudinal and latitudinal ranges.
2. **EFFECT OF ALTITUDINAL RANGES ON AVI-FAUNAL SPECIES DISTRIBUTION:-** Avi-faunal species composition and distribution with altitudinal range is determined by biotic and abiotic environmental factors such as elevation or height, prevailing wind intensity and direction, slope, temperature, humidity, rainfall and vegetation composition.[19] Abiotic factors cause major effect on species distribution pattern at higher altitudinal ranges and latitudinal levels which also greatly influences diversity and distribution pattern at ecosystem and community levels. Different elevation gradient leads to changes in composition of species distribution and its pattern. Nöske *et al.* (2008)[20] studied different factors such as canopy area of trees and; altitude may influence changes of species composition among flora and fauna. Rahbek (1995)[21] observed a monotypic decline in species number when altitude level increases. Brown and Gibson (1983)[22] observed that the association between plants and animals and extrapolated the same obtaining a dome shaped curve relationship. When defined in relationship with increase in elevation with mid elevation peak of avi-faunal species richness. Species richness of birds decrease enormously at altitudes above 1000 meter; largely affected ones were insectivore species with sustainable declined in all feeding guild of birds.[23,24] It was also observed that bird activities like feeding, foraging, breeding and singing patterns were also influenced by higher elevation levels. Grytnes and Ventaas (2002)[25] observed interpolated species occurrence between the highest and lowest altitude and concluded that it can cause artificial lump in species elevation curve. Slope shape and size also affected the number and variety of birds with increased altitude. It was observed that changes in the number and variety of birds with increased altitude can be co-related to uniqueness in eco-tone of different ecosystems. Elevation pattern among avian communities identified by Mc Cain (2007)[26] includes mid altitude peak, low altitude that has a mid altitude peak and low plateau common at the low altitude and decreasing pattern exhibited takes form of uni-modal peak at the center than at the base and top on quantifiable definition which can be defined as mid altitude peak pattern. High species richness occurred at low elevation ranges while high diversity was observed at the base of low plateau and mid altitude. Mc Cain and Grythes (2010)[27] Observed a specific pattern followed by species shown along different elevations as a reflection of the relationship between different flora and fauna in the ecosystem.

Higher mountains generally have smaller land area and are more isolated and have simple vegetation composition as compared to the open or forest lands.

3. **EFFECT OF ALTITUDINAL VARIATION ON DISTRIBUTION OF INVERTEBRATE AND VERTEBRATE PHYLUM:-** Variation in invertebrate and invertebrate species richness along altitudinal gradients[21] and across taxonomic groups[28] and continents[29] reveals striking patterns of biodiversity distribution[30]. Distribution patterns along elevation gradients have been noted in invertebrates[31-34] mammals[23,35-39] birds[40-45] and herpato-fauna[46-48]. In general it was observed that species richness along altitudinal gradients followed one of the three patterns namely- decreasing with altitude (monotonic decline), mid domain (hump-shaped pattern or curved at middle altitudinal ranges) and increasing with elevation,[28,21] amongst which most widely observed pattern were mid domain peak and monotonic declination.[21, 43] Observing and analyzing the association between species richness and elevation gradient is necessary and important as it provides insights into the observed patterns and processes responsible for maintaining the relationship in between different species which is turn supports conservation efforts.[28,49,50]
4. **EXTENSION OF MOUNTAIN ECOSYSTEM AND SPEICES DISPERSAL:-**Mountain ecosystems comprises of approximately one fifth of the earths' surface out of which tropical mountain forests represents about 11% of tropical forests of the world.[51-53]. Species richness and biodiversity was observed to be highly diverse in mountain ecosystem as compared to other ecosystems. Maximum biodiversity hot spots were also located at higher altitudinal and mountain ranges which consists of majority endemic floral and faunal species.[53-55] Higher altitudinal ranges and mountain ecosystems were observed to have largely influenced distribution pattern and biodiversity curve.[56-60]. In our review studies it was found that altitude comprises of different multi abiotic factors that controls and regulates floral and faunal species distribution and its diversity pattern and ecology in the surrounding ecosystem.
5. **EFFECT OF ALTITUDE VARIATION ON CLIMATIC CONDITIONS:-** Four main trends were observed at high elevation levels in relation to species richness and environmental factors. Several factors predictably with increasing elevation were responsible for native climatic conditions out of which the most obvious was linear decrease in temperature at higher altitudinal ranges. Temperature decreases by an average 0.6°C at every 100 meter elevation level.[61] Climatic variables such as temperature, rainfall, humidity and atmospheric pressure were also observed to be decreasing with gradual increase in elevation peak. Emission of solar radiations and fraction of UV radiations especially UV-B was seen to increase as altitude increases.[24, 60] Other abiotic climatic factors such as light intensity, cloudiness, soil texture and depth (organic matter, humus content, available free nitrogen, carbon-nitrogen ratio) increases with altitude, whereas pH, mineralization and nitrification rates were observed to decrease when altitude range increases at higher elevation levels.[9,24,60,62-65] All environmental factors were observed to be responsible for habitat alteration at various elevation levels which largely influences the distribution, diversity, dispersal patterns of various species and animal taxa.[66-69] Relationship between elevation and species richness may also vary and majority depends upon species specific availability of suitable habitat like- herbaceous life forms which grows highly and diverse in more light intensity tolerant and shade intolerant species.[67,70-74] Other climatic conditions and abiotic factors vary along mountain gradient but observed to have more complex relationship with altitude. Altitudinal variation largely influences annual precipitation at various ranges and create wider variety of micro environments.[61] Precipitation can be in the form of rainfall, condensation from clouds, snowfall (e.g. horizontal rainfall pattern in clouded forests) its pattern slightly increases when elevations gradient increases.
6. **EFFECT OF ALTITUDE VARIATION ON TETRAPOD LIFE:-** In present review studies it was observed that terrestrial tetra pods are predominantly distributed in a particular altitudinal range. Non flying small mammals (e.g. rodents, shrews etc.) were found to be abundant at mid elevation ranges[23] where as Pteropus and other flying mammals were evenly distributed at low elevation levels and sharply declined at mid elevation ranges.[75] Avian taxa follow all four common patterns of distribution and abundance of

species at higher altitudinal ranges[43] and reptiles predominantly declined at higher elevation levels.[76] Preliminary analysis of amphibians like Salamanders preferred mid elevation peaks where as other amphibians follow all four common distribution patterns evenly and with similar frequencies as it was the case with other tetrapods.

7. **EFFECT OF ALTITUDINAL VARIATION ON EGG SIZE AND CLUTCH SIZE:-** One of the most frequently studied trait has been variation in egg clutch size with variation in avi-faunal species along latitudinal gradient[77]; typically tropical species lay smaller egg clutch size as compared to northern temperate climatic conditions both intra and inter specifically.[78] Altitudinal gradient ranging from low to high altitude produces variety of species distribution patterns namely tropical-temperate gradient, lower temperatures, greater seasonality and high environmental variability for bird's behavior like-higher altitudinal resident birds having smaller breeding season as compared to low land birds.[79] Also it was reviewed that bird's egg clutch size variation increase elevation levels increasing from lowland to mountain ecosystem.
8. **EFFECT OF HIGHER ELEVATION LEVEL ON ANIMAL PHYSIOLOGY:-** It was observed in our present review studies that lower temperature and oxygen availability at higher elevation levels can increase metabolic rates and thermoregulation in an organism while in contrast slow metabolic rates and high thermoregulation reduces reproductive effort through reduction in egg clutch size its mass and periods of development and times of hatchling.[79-80] Lower atmospheric pressure at high elevation level leads to water loss from eggs and create hypoxic condition and increase water loss content from eggs during early developmental stages.[81] Increased seasonality in temperature and precipitation at higher elevation level implies that favorable for breeding may be constrained to a shorter period at high elevations, thus limiting components of fecundity such as number of broods.[79,82] Food availability play critical role for egg size, clutch size, incubation periods and nesting success as well as juvenile survivorship.[83] Food limiting factors and scarcity also reduces breeding strength among birds at higher elevation level due to colder environment and hypoxic conditions.[84] Pattern of variation in nest predation risk across elevation gradients are unclear with studies reporting decreasing at higher elevations[85-86] and increasing at higher elevations.[87] It was found in review studies that lower to higher altitudinal level highly influences animal physiology and behavioral biology.

## CONCLUSION

Different altitudinal and latitudinal ranges highly influence the distribution and diversity patterns of floral and faunal composition. Altitudinal gradient also determines the distribution, density and abundance pattern of various invertebrate and vertebrate animals and as well as plant communities. Higher altitudinal ranges in habitat wide variety of floral and faunal species particularly of native nature due to which mountain ecosystems are occupied with maximum number of biodiversity hotspot areas. Our review studies conclude that high species richness and biodiversity plays essential and curial role for human existence and survival and are necessary for factors for ecosystem balance and stability. At present time over exploitation, habitat alteration, deforestation and natural disasters are becoming limiting factor for biodiversity decline at both regional and global levels.

## REFERENCES

1. **McNeely, G., Mille, K.R., Reid, W.V., Mittermeier, R.A., Werner, T.R. 1990.** Conserving the World's Biological Diversity, IUCN. Gland.
2. **Gaur,R.D., 1999.** Flora of the District Garhwal Northwest Himalaya: with Ethanobotanical Notes. Trans Media,Srinagar, U.P., India.
3. **Singh,J.S. 2002.** The biodiversity crisis: a multifaceted review. *Current Science*. **82:**638-647.
4. **Kumar, A., Sharma, M.P., Yang, T. 2018.** Estimation of carbon stock for green house gas emissions from hydropower reservoirs. *Stoch. Environ. Res. Risk.* **10:** 3183-3319.
5. **Kumar, A., Yang, T., Sharma, M.P. 2019.** Long term prediction of green house gas risk to the Chinese hydro power reservoirs. *Sci. Total Environ*. **646(1):**300-308.
6. **Ayensu, E.,Van, R.C., Laasen, D., Collins, M., Dearing, A., Fresco, L., Gadgil, M., Gitay, H., Glaser,**

**G., Juma, C., Krebs, J., Lenton, R., Lubchenco, J., McNeely, J.A., Mooney, H.A., Pinstrup Andersen, P., Ramos, M., Raven, P., Reid, W.V., Samper, C., Sarukha´n, J., Schei, P., Tundisi, J.G., Watson, R.T., Guanhua, X., Zakri, A.H. 1999.** *International ecosystem assessment. Science*. **286:**685-686.

7. **Malik, Z. A. 2014.** Phyto sociological Behavoiur, Anthropogenic Disturbances and Regeneration Status along an Altitudinal Gradient in Kedarnath Wildlife Sanctuary (KWLS) and its Adjoining Areas. Ph.D. Thesis. HNB Garhwal University, Srinagar Uttarakhand.

8. **Malik, Z.A., Hussain, A., Iqbal, K., Bhatt, A.B. 2014a.**Species richness and diversity along the disturbance gradient in Kedarnath Wildlife Sanctuary and its adjoining are as in Garhwal Himalaya, India.*Int J. Curr. Res.* **6:**10918-10926.

9. **Zhang, J.T., Xu, B., Li, M. 2013.** Vegetation patterns and species diversity along elevational and disturbance gradients in the baihua mountain Reserve, Beijing, China. *Mt.Res.Dev.* **33(2):**170-178.

10. **Kumar,A., Sharma, M.P. 2016.** Carbon stock estimation in the catchment of kotlibhel1 Ahydro electric reservoir, Uttarakhand, India. *Ecotoxicol. Environ. Saf.* **134:**365-369.

11. **Kumar,A.,Sharma, M.P. 2017.** Estimation of green house gas emissions from Koteshwar hydropower reservoir, India. *Environ. Monit. Assess.* **189(5):**240.

12. **Martin, P.S. 1958.** A biogeography of reptiles and amphibians in the Gomez Farias region. Tamaulipas, Mexico. *Miscellaneous Publications of the Museum of Zoology, University of Michiga*n. **101:** 1-102.

13. **Wake, D.B., Papenfuss, T.J., and Lynch, J.F. 1992.** Distribution of salamanders along elevational transects in Mexico and Guatamala. *Tulane Studies in Zoology & Botany.* **1(suppl):** 303-319.

14. **Grytnes, J.A., Heegaard, E., & Ihlen, P. G. 2006.** Species richness of vascular plants, bryophytes, and lichens along an altitudinal gradient in western Norway. *Acta Oecologica*. **29:** 241-246.

15. **Gauthier, S., Grandpre, L.D., Bergeron, Y. 2000.** Differences in forest composition in two boreal forest eco regions of Quebec. *Journal of Vegetation Science.* **11:**781-790.

16. **Kharkwal, G.. 2009.** Qualitative analysis of tree species in evergreen forests of Kumaun Himalaya, Uttarakhand, India. *African Journal of Plant Science,* **3(3):**049-052.

17. **Sabatini, F. M., Jime´nez Alfaro, B., Burrascano, S., Lora, A., Chytrý, M. 2018.** Beta diversity of Central European forests decreases along an elevational gradient due to the variation in local community assembly processes. *Ecography*. **41:**1038-1048.

18. **Jones, C. G., Lawton, J.H., Shachak, M., 1997.** Positive and negative effects of organisms as physical ecosystem engineers: positive interactions in communities. *Ecology*. **78(7):**1946e1957.

19. **Poulsen, B. O. 2002.** Avian richness and abundance in temperate Danish forest: tree variables important to birds and their conservation. *Biodiversity and Conservation.* **11:**1551-1566.

20. **Noske, N. M., Hilt, N., Werner, F. A., Brehm, G., Fiedler, K., Sipman, H. J. M., and Gradstein, S. R., 2008.** Disturbance effects on diversity in montane forest of Ecuador: sessile epiphytes versus mobile moths. *Basic and Applied Ecology.* **9:**4-12.

21. **Rahbek, C. 1995.** The elevational gradient of species richness: a uniform pattern? *Ecography* **18(2):** 200-205.

22. **Brown, J. H., & Gibson, A. C. 1983.** Biogeography. Sunderland. Sinauer Associates.

23. **Mc Cain, C.M. 2005.** Elevational gradients in diversity of small mammals. *Ecology*. **86:** 366-372.

24. **Körner, C. 2007.** The use of 'altitude' in ecological research. *Trends in Ecology and Evolution*. **22:**569-574.

25. **Grytnes, J.A.,Vetaas, O.R. 2002.** Species richness and altitude: a comparison between simulation models and interpolated plant species richness along the Himalayan altitude gradient, Nepal. *American Nature.***159:** 294-304.

26. **McCann, K. 2007.** Protecting bio structure. *Nature* **446(7131):** 29-29.

27. **McCain C.M., and Grytnes, J.A., 2010.** Elevational gradients in species richness. In: Encyclopedia of Life Sciences (ELS). Chic ester: John Wiley & Sons, Ltd. 1- 10.

28. **Stevens, G.C. 1992.** The elevational gradient in altitudinal range: an extension of Rapoport's latitudinal rule to altitude. *The American Naturalist.* **140(6):** 893-911.

29. **Cavarzere, V., and Silveira, L.F. 2012.** Bird species diversity in the Atlantic forest of Brazil is not explained by the mid-domain effect. *Zoologia*. **29(4):** 285-292.

30. **Sanders, N.J., and Rahbek, C. 2012.** The patterns and causes of elevational diversity gradients. *Ecography.* **35:** 1-3.

31. **Sanders, N.J. 2002.** Elevational gradients in ant species richness: area, geometry, and Rapoport's rule. *Ecography.* **25(1):** 25-32.

32. **Khan, Z. H., Raina, R.H., Dar, M.A., & Ramamurthy, V.V. 2011.** Diversity and distribution of butterflies from Kashmir Himalayas. *Journal of Insect Science.* **24(1):** 45-55.

33. **Levanoni, O., Levin, N., Pe'er, G., Turbe, A., and Kark, S. 2011.** Can we predict butterfly diversity along an elevation gradient from space? *Ecography.* **34(3):** 372-383.

34. **Carneiro, E., Mielke, O.H.H., Casagrande, M.M., and Fiedler, K. 2014.** Community structure of Skipper butterflies (Lepidoptera, Hesperiidae) along elevational gradients in Brazilian Atlantic forest reflects vegetation type rather than altitude. *PLoS One.* **9(10):** 108-207.

35. **Patterson, B.D., Pacheco, V. and Solari , S. 1996.** Distribution pattern of bats along an elevational gradient in the Andes of southeastern Peru. *Journal of Zoology.* **240:** 637-658.

36. **Patterson, B.D., Stotz, D.F., Solari, S., Fitzpatrick, J.W. and Pacheco, V. 1998.** Contrasting patterns of elevational zonation for birds and mammals in Andes of Southern Peru. *Journal of Biogeography.* **25(3):** 593-607.

37. **Brown, J.H. 2001.** Mammals on mountainsides: elevational patterns of diversity. *Global Ecology and Biogeography*. **10:** 101-109.

38. **McCain, C.M. 2004b.** The mid-domain effect applied to elevational gradients: species richness of small mammals in Costa Rica. *Journal of Biogeography.* **31:** 19-31.

39. **McCain, C.M. 2006.** Do elevational range size, abundance and body size patterns mirror those documented for geographic ranges? A case study using Costa Rican rodents. *Evolutionary Ecology Research* **8:**435-454.

40. **Terborgh, J. 1977.** Bird species diversity on an Andean elevational gradient. *Ecology.* **58(5):** 1007-1019.

41. **Blake, J.G., and Loiselle, B.A. 2000.** Diversity of birds along an elevational gradient in the Cordillera Central, Costa Rica. *Auk*. **117:** 663-686.

42. **Lee, P.F., Ding, T.S., Hsu, F.H., & Geng, S. 2004.** Breeding bird species richness in Taiwan: distribution on gradients of elevation, primary productivity and urbanization. *Journal of Biogeography.* **31:** 307- 314.

43. **McCain, C.M. 2009.** Global analysis of bird elevational diversity. *Global Ecology and Biogeography.* **18:** 346-360.

44. **Joshi, K., and Bhatt, D. 2015.** Avian species distribution along elevation at Doon valley (foothills of western Himalayas), Uttarakhand, and its association with vegetation structure. *Journal of Asia-Pacific Biodiversity.* **8(2):** 158-167.

45. **Montano-Centellas, F.A., and Garitano-Zavala, A., 2015.** Andean bird responses to human disturbances along an elevational gradient. *Acta Oecologica.* **65(66):** 51-60.

46. **Hofer, U., Bersier, L.F., Borcard, D. 1999.** Spatial organization of a herpetofauna on an elevational gradient revealed by null model tests. *Ecology*. **80(3):** 976-988.

47. **Chettri, B., Bhupathy, S., and Acharya, B.K. 2010.** Distribution pattern of reptiles along an eastern Himalayan elevation gradient, India. *Acta Oecologica.* **36(1):** 16-22.

48. **Srinivas, G.. 2011.** Distribution pattern of amphibians in Megamalai landscape, Western Ghats, Tamil Nadu. PhD Thesis. Bharathiar University, Coimbatore, India, 144pp.

49. **Raman, T.R.S., Joshi, N.V., and Sukumar, R. 2005.** Tropical rainforest bird community structure in relation to altitude, tree species composition and null models in the Western Ghats, India. *Journal of the Bombay Natural History Society.* **102(2):** 145-157.

50. **Acharya, B., Sanders, N.J., Vijayan, L., and Chettri, B. 2011.** Elevational gradients in bird diversity in the eastern Himalaya: an evaluation of distribution patterns and their underlying mechanisms. *PLoS One* **6(12):**290-297

51. **Doumenge, C., Gilmour, D., Ruiz Perez, M., Blockhus, J. 1995.** Tropical montane cloud forests: conservation status and management issues. 24-37. In: Hamilton LS, Juvik JO, Scatena FN (eds) Tropical montane cloud forests. *Ecolological Studies*. 110. Springer, New-York.

52. **Tse-ring, K., Sharma, E., Chettri, N., Shrestha, A.B. 2010.** Climate change vulnerability of mountain ecosystems in the Eastern Himalayas. Synthesis report. Kathmandu: International centre for integrated mountain development (ICIMOD).

53. **Price, M.F., Gratzer, G., Duguma, L.A., Kohler, T., Maselli, D., Romeo, R., (eds) 2011.** Mountain forests in a changing world-realizing values, addressing challenges. International year of forests 2011. FAO, Rome.

54. **Bussmann, R.W. 2001.** The montane forests of Reserva Biológica San Francisco (Zamora Chinchipe, Ecuador)-vegetation zonation and natural regeneration. *Erde*. **132:**11-24

55. **Admassu, A., Teshome, S., Ensermu, K., Abyot. D., Alemayhu, K. 2016.** Floristic composition and plant community types of Agama Forest, an Afromontane Forest in Southwest Ethiopia. *Journal of Ecology and Natural Environment.* **8:**55-69.

56. **Doumenge, C. 1998.** Forest diversity, distribution, and dynamique in the Itombwe Mountains, South-Kivu, Congo Democratic Republic. *Mount Res Dev.* **18:**249-264.

57. **Eilu, G., Hafashimana, D. L. N., Kasenene, J. M. 2004.** Density and species diversity of trees in four tropical forests of the Albertine rift, western Uganda. *Diversity and Distribution.* **10:**303-312.

58. **Hemp, A. 2009.** Climate change and its impact on the forests of Kilimanjaro. *African Journal of Ecology*. **47:**3-10

59. **Pickering, C.M., Green, K. 2009.** Vascular plant distribution in relation to topography, soils and micro-climate at five GLORIA sites in the Snowy Mountains, Australia. *Aust J Bot.* **57:**189-199.

60. **Chain-Guadarrama, A., Finegan, B., Vilchez, S., Casanoves, F. 2012.** Determinants of rain-forest floristic variation on an altitudinal gradient in southern Costa Rica. *Journal of Tropical Ecology.* **28:**463-481.

61. **Barry, R. G.. 2008.** Mountain Weather and Climate 3rd edn (Cambridge Univ. Press).

62. **Lebrun, J., Gilbert, G. 1954.** Une classification écologique des forêts du Congo.-Publications INEAC; Série Sciences. N° 63.

63. **Doležal, J., and Šr?tek, M. 2002.** Altitudinal changes in composition and structure of mountain-temperate vegetation: a case study from the Western Carpathians. *Plant Ecology*. **158:**201-221.

64. **Unger, M., Homeier, J., Leuschner, C. 2013.** Relationships among leaf area index, below-canopy light availability and tree diversity along a transect from tropical lowland to montane forests in NE Ecuador. *Tropical Ecology.* **54:**33-45.

65. **Zhang, W., Huang, D., Wang, R., Liu, J., Du, N., 2016.** Altitudinal patterns of species diversity and phylogenetic diversity across temperate mountain forests of Northern China. *PLoS One.* **11:**0159995.

66. **Grime, J.P. 1977.** Evidence for the existence of three primary strategies in plants and its relevance to ecological and evolutionary theory. *American Nature*, **111:**1169-1194.

67. **Kessler, M. 2000.** Elevational gradients in species richness and endemism of selected plant groups in the central Bolivian Andes. *Plant Ecology.* **149:**181-193.

68. **Pausas, J.G., & Austin, M.P. 2001.** Patterns of plant species richness in relation to different environments: an appraisal. *Journal Vegetation Science.* **12:**153-166.

69. **Chawla, A., Rajkumar, S., Singh, K. N., Lal, B., Singh, R.D., Thukral, A.K. 2008.** Plant species diversity along an altitudinal gradient of Bhabha Valley

in western Himalaya. *Journal of Mount Science*. **5:**157-177.

70. **Pierlot, R. 1966.** Structure et composition de forêts denses d'Afrique Centrale, spécialement celles du Kivu. 1966. Belgique, Académie royale de Bruxelles.

71. **Grytnes, J. A. 2000.** Fine-scale vascular species richness in different alpine vegetation types: relationships with biomass and cover. *Journal of Vegetation Science*. **11:**87-92

72. **Jiang Z, Ma K, & Anand, M. 2016.** Can the physiological tolerance hypothesis explain herb richness patterns along an elevation gradient? A trait-based analysis. *Community Ecology*. **17:**17-23.

73. **Lomolino, M.V. 2001.** Elevation gradients of species-density: historical and prospective views. *Global Ecology and Biogeography*. **10:**3-13

74. **Salas-Morales, S.H. and Meave, J. A. 2012.** Elevational patterns in the vascular flora of a highly diverse region in southern Mexico. *Plant Ecol*. **213:**1209-1220

75. **Mc Cain, C.M. 2007b.** Could temperature and water availability drive elevational species richness? A global case study for bats. *Global Ecology and Biogeography*. **16:**1-13.

76. **McCain, C.M. 2010.** Global analysis of reptile elevational diversity. *Global Ecology and Biogeography*. **19:** 541-553.

77. **Lack, D. 1947.** Darwin finches: An Essay on the General Biological Theory of evolution: Cambridge: Cambridge University Press.

78. **Böhning-Gaese, K., Halbe, B., Lemoine, N., & Oberrath, R. 2000.** Factors influencing the clutch size, number of broods and annual fecundity of North American and European land birds. *Evol. Ecol. Res.* **2:**823-839.

79. **Martin, K. 2001.** Wildlife in alpine and sub-alpine habitats. Wildlife Habitats and Relationships in Oregon and Washington (eds D.H. Johnson & T.A. O'Neil), pp. 239-260. Oregon State University Press, Corvallis, OR, USA.

80. **Conway, C.J., & Martin, T.E. 2000.** Evolution of passerine incubation behavior: influence of food, temperature, and nest predation. *Evolution*. **54:** 670-685.

81. **Rahn, H., & Ar.,A. 1974.** The avian egg: incubation time and water loss. *Condor*. **76:** 147-152.

82. **Hardesty, J.L. 2008.** Seasonality in Equatorial Cloud Forest Birds. ProQuest.

83. **Martin, T.E. 1987.** Food as a limit on breeding birds: a life-history perspective. *Annual Review of Ecology and Systematics*. **18:**453-487.

84. **Chalfoun, A.D., &Martin, T.E. 2007.** Latitudinal variation in avian incubation attentiveness and a test of the food limitation hypothesis. *Animal Behaviour*. **73:**579-585.

85. **Badyaev, A.V. 1997.** Avian life history variation along altitudinal gradients: an example with cardueline inches. *Oecologia*. **111:**365-374.

86. **Martin, T.E., &Briskie, J.V. 2009.** Predation on dependent offspring: a review of the consequences for mean expression and phenotypic plasticity in avian life history traits. *Annals of the New York Academy of Sciences*. **1168:**201-217.

87. **Sandercock, B.K., Martin, K., & Hannon, S.J. 2005.** Life history strategies in extreme environments: comparative demography of arctic and alpine ptarmigan. *Ecology*. **86:** 2176-2186.

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# Black Carbon - A dangerous climate forcerer mitigating strategies

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**Abstract-**"Human activities are changing the earth's climate in two ways:
1) By way of greenhouse gas emission -$CO_2$etc.
2) By changing the atmospheric aerosol load (BC etc.) -Richard Black
Black Carbon is released during incomplete combustion of fossil fuels or biomass, is a component of soot (from inefficient cooking stoves, diesel emissions from vehicles and from generators etc.), ultrafine particles or PM 0.1 posing health risks, strongly absorb light and convert the energy to heat. It has a brief atmospheric life and its climatic effects seem to be regional yet due to strong wind may be driven to far flung areas and unfortunately it covers the ice or snow, it would darken them thereby increasing the absorption of sunlight. This would in turn increase heat absorption and melt the ice or snow. It is therefore, a significant contributor of recent warming in changing Arctic which has accelerated sea ice loss and melting Himalayan Glaciers. Contrarily $CO_2$, a greenhouse gas has much longer lifetime (over 100 years) and has significant influence on the global temperature. The present authors have dealt with the black carbon emission in rural and urban area of chotanagpur area, how black carbon impacts cloud formation and rainfall etc. and remedial strategy in reducing Black Carbon emission

**Key words: Black Carbon, Soot, Remedial measures**

## INTRODUCTION

Black Carbon (BC) a short lived climate forcer (SLCF) is a charcoal like residue left by burning fuels that are rich in carbon often seen in fires/ fire places, due to incomplete combustion of fossil fuels, biomass and biofuels. It is an ultrafine fraction of the particulate matter emitted by diesel driven vehicles, ships on seas, coal or wood burning, stoves used in residential premises from forest fires or burning of agricultural wastes (open biomass burning).

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There are two terms often used while reading black carbon. These are:

**A. Black Carbon** - It is dark, light absorbing components of aerosol that contain two forms of elemental carbon.

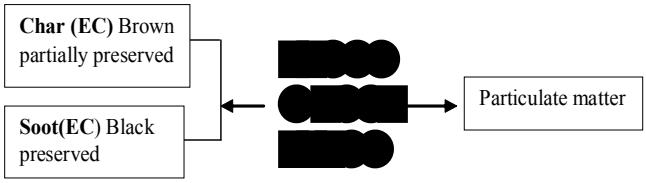
**1. Char-** It is the carbonaceous material obtained by partially heating / burning organic substances with limited access of air. This results into an impure form of graphite carbon. [Exp. Burning Vegetation or wood in residential houses]

**2. Soot-** It is carbon particles that form at high temperature via gas phase process. [Exp. From diesel engines]

**B. Elemental Carbon (EC)-** It is atmospheric particulate matter (PM) derived from varieties of combustion sources and contains:-

**1. Char-** The graphite like structure of natural carbon having brown colour (partially preserved)

**2. Soot-** This is original structure of carbon not preserved which is black in colour.




The above two types possess different chemical and physical properties and different light absorbing properties.[1,2]

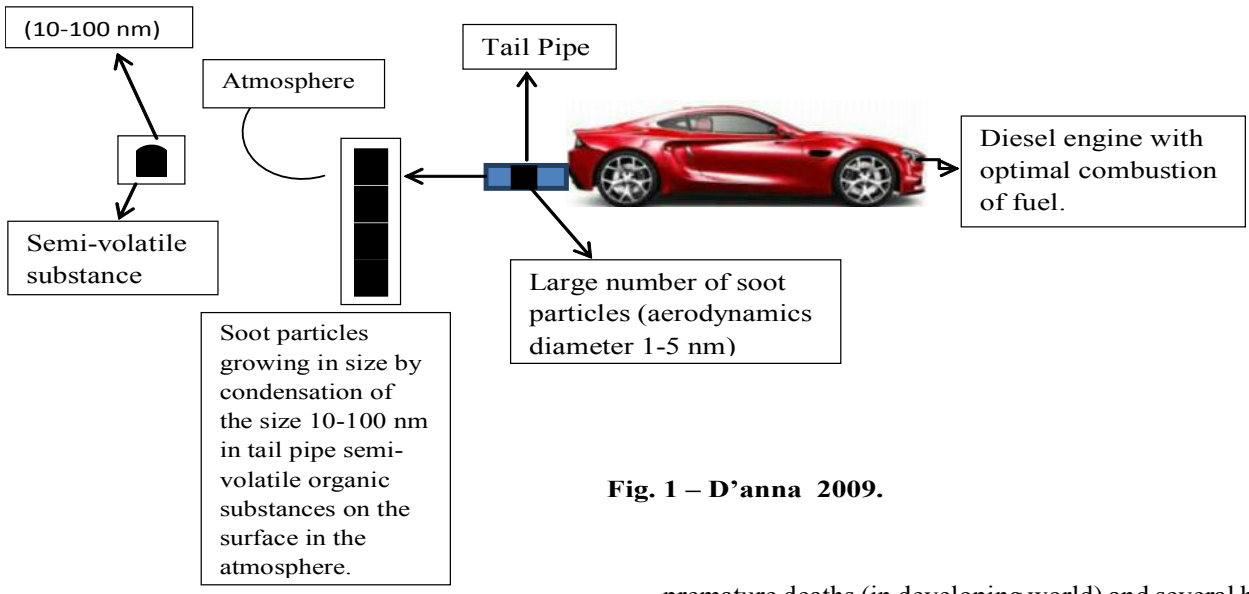



**Fig. 1 – D'anna 2009.**

**A)** When fuel undergoes optimal combustion at high temperature (as in diesel engine), emits large number of small soot particles (aerodynamic diameter being 1-5 nm). These soot particles grow in size (10-100 nm) in tail pipe and undergoes simultaneous condensation of semi-volatile organic substances in the atmosphere.[3] (Fig. 1) He further opines that the speed of growth is dependent upon air temperature, sunlight & the concomitant oxidants etc.

**B)** Contrarily due to incomplete combustion of solid fuel (wood, coal), the emitted particulate matters have aerodynamic diameter 150-16- nm which continue to grow taking semi-volatile organics on their surface (wood and coal produce large amounts of semi-volatile organics).

Such particles, in due course of time (short period to several days) become upto 1µm as inorganic salts emerging from $NO_2$ and $SO_2$ and atmospheric water attach to the surface of hydroscopic carbonaceous particles.

Black Carbon (a charcoal like residue) emitted by burning fuels are rich in carbon (seen in fires or fire places etc) and stays in the atmosphere for a short time (several hours or days). Particulate Matter (PM) of which Black Carbon (BC) is a main constituent responsible for premature deaths (in developing world) and several health complications.

Black Carbon has been also called Elemental Carbon (EC), its effects should not be underestimated as it can remain in the air for months and travel hundreds of kilometres. Its coarse particles though settle on the ground and do not get easily transported to large area or beyond as courser particles fail to be driven by air streams and thus get easily removed on contact with surfaces.

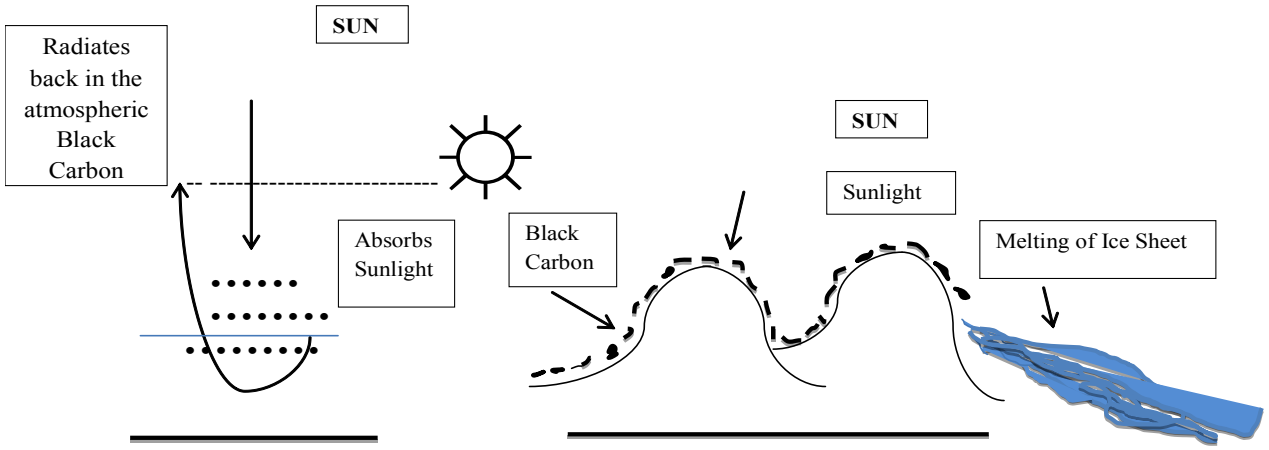
**Black Carbon vis a vis– climate change / global warming:**

Some scientists believe that it does have major role to play in the climate change yet others believe it does.

Jacobson (2007)[4] estimated that "Black Carbon accounts for about 40% of net global warming or 16% of gross global warming" whereas $CO_2$ is the major player. Therefore Black Carbon is placed 2nd as 'Warming agent'.

Chow *et al.*(2009)[5] however, had left technical problems in the instruments used in the calculation and intercomparision studies.

Black Carbon absorbs sunlight and radiates it back in the atmosphere as heat, it also affects cloud and rainfall patterns.




In colder regions, Black Carbon settles on ice and snow as well as on other surface of the earth. If increases the melting rate of ice and snow.

D'Anna, (2009)[3] reported that half or more of the warming measured in Aretic from 1976-2007 is due to Black Carbon plus Organic Carbon. Black Carbon coming from fuel combustion has been implicated in reducing reflectivity rate and increasing the rate of melting of ice.

Arctic and Himalayan region is covered by ice/ snow, when Black Carbon deposits, it darkens the surface and increases the absorption of solar radiation. This Black Carbon ice Albedo* effect leads to warming of the lower atmosphere and melting of snow /ice sheets.

Ice and snow covered areas showing light coloured surfaces show higher albedo reflecting much of the sun's rays (radiations) which get absorbed by the oceans resulting into heating up the earth surface.

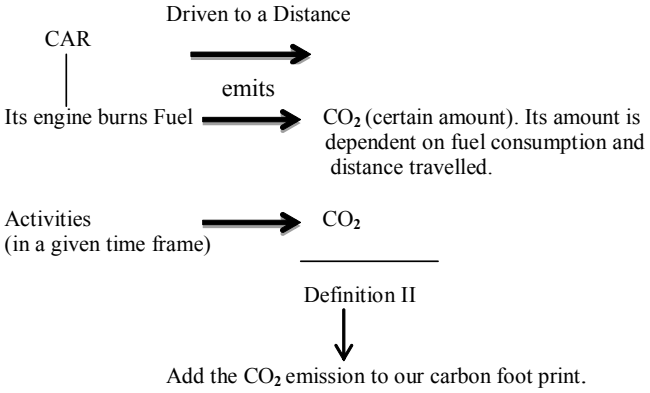
Contrarily dark surface (showing low albedo) absorbs higher energy from the sun and has thus warming effect.

**Carbon Foot Print:**

Carbon Foot Print is defined as the total amount of greenhouse gases produced, to directly and indirectly support human activities, usually expressed in equivalent tons of $CO_2$. Or

It is the sum of all emissions of $CO_2$ which are induced by activities performed in a given time frame (usually calculated for the time period of a year)

*Albedo effect is the ability of the surface to reflect sunlight (heat from the sun). Light coloured surface return a large part of the sun's rays back in the atmosphere (HIGH ALBEDO) whereas dark surface absorbs higher energy from the sun's rays (LOW ALBEDO)




Example:

| Fuel | Amount of $CO_2$ |
|---|---|
| UK 1 gallon Petrol | 10.4 kg $CO_2$ |
| US 1 gasoline | 8.7 kg $CO_2$ |
| A car with diesel (7.5 L) | travel 100 Km |
| ↓ Multiply by a factor 0.27 | Then for 300 Km it will consume 3 x 7.5= 22.5 L Diesel i.e, 22.5 x 2.7 Kg = 60.75 Kg $CO_2$ (Personal Carbon Foot Print) = 16.40 Carbon |
| (Note: 1000 kg $CO_2$= 270 Kg Carbon) | |

If a car is driven, its engine burns fuel and emit certain amount of $CO_2$. Its amount is dependent on the fuel consumption and distance travelled.

Similarly, in a given time frame certain activities (burning) have been performed. Therefore its Carbon Foot Print will be equal to sum of all emissions of $CO_2$ which have been induced by the activities.

**CALCULATION OF $CO_2$ EMISSION:**

Some examples,

| Fuel | Emitted Amount of $CO_2$ |
|---|---|
| UK 1 gallon Petrol consumed | 10.4 kg $CO_2$ |
| US 1 gasoline consumed | 8.7 kg $CO_2$ |
| Car in India | |

Diesel car supposing it consumes 7.5 litres of diesel for travelling 100 km.

- A drive of 300 will consume
  7.5 x 3 = 22.5 litre diesel

This is multiplied by 2.7 Kg

22.5 x 2.7 Kg = 60.75 Kg $CO_2$

Becomes the personal Carbon Foot Print

By multiplying 60.75 Kg $CO_2$ with a factor 0.27
(1000 Kg $CO_2$ equals to 270 Kg Carbon)

- 60.75 x 0.27 = 16.4 Kg Carbon

Unfortunately, conventional Carbon Foot Print account for only homogenous greenhouse gases (GHGs) specified by the IPCC (Intergovernmental Panel on Climate Change) as global warmers: $CO_2$, $CO_4$, $N_2O$ & halocarbons. Foot prints generally do not include gases or particles not specified by the IPCC such as Black Carbon and other heterogenous species such as Carbon-mono-oxide (CO), Hydrogen ($H_2$), Organic Carbon (OC), Ozone ($O_3$), Nitrogen Oxide (NOx) or Sulphates (SOx)- vide www. WORLDLPGAS.Com.

**METHODS GENERALLY USED**

1) Light transmission through integrated Particulate Matter samples collected at 24 hours intervals on thin cellulose fibre filter and subsequent convention of the optical measurement units to man based unit.

2) Photometer absorption of Particulate Matter sample spots (Black Carbon) at 1-5 minute intervals and automatic readings in mass based unit.

3) Chemical determination Elemental Carbon and Organic Carbon using thermal optical analysis either semi-continuously with mass based reading given every 30 minutes to 3 hours, or from integrated Particulate Matter samples collected at 24 hours intervals on quartz filter.[4]

**REMEDIAL STRETEGIES:**

Since Black Carbon is a component of particulate matter, hence particulate matter can be minimised. Jacobson (2002,2005)[6,7] suggested converting vehicles from fossil fuel to electric, plug in hybrid or hydrogen fuel cell vehicles. He opines that such conversion would eliminate 160 G g/yr (24% of United States or 1.5% of world) fossil fuel soot and 26% of us (or 5% of world $CO_2$ emission by 1.63 Gt $CO_2$ eq per year). He felt that "the elimination of hydrocarbons and nitrogen oxides would also eliminate some cooling particles, reducing the net benefit by at most, half but improving human health."

**DIESEL PARTICULATE FILTER:**

Technologies are available for replacing the diesel vehicles where diesel particulate filter can be used which in turn eliminate 90% of Black Carbon.[8] Ultra Low Sulphur Diesel (ULSD) is a form of diesel fuel with exceedingly lower sulphur content which is comparatively better compared to petroleum based diesel. Enactment of Acts/ Laws: Strict Act has to brought for implementing such fuel.

**REFERENCES**

1. **Han, Yongming, Junji Cao, Judith C. Chow, John G. Watson, Zhisheng An, Zhangdong Jin, Kochy Fung & Suixin Liu. 2007.** Evaluation of the thermal/ optical reflectance method for discrimination between char-and soot-EC. *Chemosphere*. **69(4):** 569-574.
2. **Han, Y. M., Cao, J. J., Lee, S. C., Ho, K. F., & An, Z. S. 2010.** Different characteristics of char and soot in the atmosphere and their ratio as an indicator for source identification in Xi'an China *Atmospheric Chemistry & Physics*. **10:**595-607.
3. **D'Anna A. 2009.** Combustion formed nanoparticles. Proceedings of the combustion institute **32:** 593-613.
4. **Jacobson.M. Z. 2007.** Black Carbon and Global Warming. W.H.O Representative. Washington DC.
5. **Chow, J.C., Watson, J.G., Doraiswamy, P., Chen, L.W.A., Sodeman, D.A., Lowenthal, D.H., Park, K., Arnott, W.P. and Motallebi, N., 2009.** Aerosol light absorption black carbon & elemental carbon at the Fresno Supersite, California. *Atmospheric Research.* **93:**874-887.
6. **Jacobson, M.Z. 2002.** Control of fossil fuel particulate black carbon & organic matter, possibly the most effective method of slowing global warming. *Journal of Geophysical Research.* **107(D19).**
7. **Jacobson, M.Z. 2005.** Correction to 'Control of fossil fuel particulate black carbon & organic matter, possibly the most effective method of slowing global warming. *Journal of Geophysical Research.***110 D14 105**.
8. **Chow JC, Watson JG, Chen LW, Paredes-Miranda G, Chang MC, Trimble D, Fung KK, Zhang H, Zhen Yu J. 2005.** Refining temperature measures in thermal/ optical carbon analysis. *Atmospheric Chemistry & Physics.* **5(4):** 2961-2972.

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# Resveratrol's myriad biological activities and interest: A brief review

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**Abstract-** Organic legacy of testing, evaluation and exploration is an influential tool for discovering bioactive with distinct structure and action mechanism. Resveratrol is one amongst the foremost investigated bioactive compounds.Findings have provided mounting proof that resveratrol, a dietary polyphenol, might confer health benefits and may defend against a variety of medical conditions and age-related complications. Studies have shown that resveratrol possesses several bioactivities and health advantages like anti-inflammatory and immunomodulatory effects, in the treatment of cancer, liver diseases, diabetes, obesity, Alzheimer's sickness, and Parkinson's sickness. Throughout this review, efforts have been made to survey the newest advances about bioavailability, toxicity, health benefits, and resveratrol-drug interactions studies. Additional bioactivities and health advantages of resveratrol ought to be evaluated, and more clarification of the underlying mechanisms of action is needed. In this perspective, this report focuses on the actions of resveratrol treatment in combatting diseases, like cancer, diabetes, neurodegeneration, and various age-related ailments, with the aim to develop resveratrol into functional dietary supplement and drug.

**Key words: Resveratrol, antioxidant, polyphenols, bioactive, stilbene synthase**

## INTRODUCTION

Michio Takaoka a Japanese scientist was first to isolate resveratrol as a new phenolic compound in 1939 from *Veratrum grandiflorum*[1] followed by in 1964 in the roots of *Polygonum cuspidatium*[2]. In 1974 the resveratrol was described as a phytoalexin found in the leaves of Vitaceae[3]. Ancient Asian medicine identifies resveratrol as Ko-Jo-Kon used as an anti-inflammatory obtained from the Polygonum (Japanese knot weed) root in powdered form. Recently the discovery of resveratrol in grape wine brings great attention towards the natural antioxidants. In recent years the studies on stilbenoids have become popular among researchers due to its myriad biological activities in human.[3]

Recent studies describe ninety two new resveratrol compounds from nine families of the plant kingdom (Dipterocarpaceae, Paeoniaceae, Vitaceae, Leguminosae, Gnetaceae, Cyperaceae, Polygonaceae, Gramineae, and Poaceae) identifying these as 6 monomers, 39 dimers, 23 trimers, 13 tetramers, 4 pentamers, 6 hexamers, and one octamer. Fifty resveratrol compounds are only identified from seven genera of (*Vatica, Vateria, Shorea, Hopea, Neobalanocarpus, Dipterocarpus* and *Dryobalanops*).

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**Resveratrol sources from plants**

| S.No. | Family | Species |
|---|---|---|
| 1. | Cyperaceae | *Carex miyabei* |
| 2. | Dipterocapaceae | *Vatica pauciflora* |
| 3. | Gramineae | *Hordeum vulgare , Poa annua , Festuca ovina , Stipa tianschania* |
| 4. | Gnetaceae | *Gnetum parvifolium, Gnetum montanum, Gnetum hainanense* |
| 5. | Leguminosae | *Cassia quinquangulata, Cassia tora, Cercis chinensis,Glycine max, Bauhinia racemosa,Arachis hypogaea, Lysidice rhodostegia,Caesalpinia millettii, Maackia amurensis, Caragana sinica, Caragana jubata , Caragana stenophylla, Medicago sativa, Ammopiptanthus mongolicus, Vigna umbellata, Albizia kalkora* |
| 6. | Liliaceae | *Aloe vera ,Veratrum grandiflorum,Veratrum nigrum, Veratrum taliense, Veratrum maackii* |
| 7. | Pinaceae | *Pinus palustris , Pinus taeda, Pinus echinata, Pinus elliotti , Pinus sibirica , Pinus sylvestris , Pinus koraiensis , Picea abies* |
| 8. | Polygonaceae | *Rheum nanum. Reynoutria japonica . Fallopia multiflora* |
| 9. | Vitaceae | *Vitis vinifera , Vitis amurensis , Ampelopsis cantoniensis, Ampelopsis japonica, Tetrastigma hypoglaucum, Cissus quadrangularis,Cissus sicyoides, Parthenocissus tricuspidata , Parthenocissus quinquefolia* |

**Figure: 1 Different source of Resveratrol [2-7]**

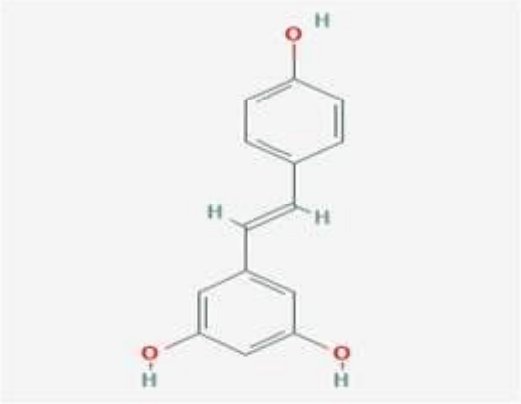


**Figure: 2- Chemical structure of Resveratrol (source: NCBI)[8]**

**Resveratrol's Chemical Properties**

Resveratrol (3,5,4'-trihydroxy-trans-stilbene) belongs to polyphenols' stilbenoids group, having two phenol rings attached to each other by an ethylene bridge.[9] Chemical name or IUPAC name of resveratrol is E-5-(4-hydroxystyryl) benzene-1,3-diol. Normally resveratrol exists in plants as *cis*- and *trans*-isomers. The *trans*- isomer has more importance due to its higher activity and stability than *cis*- isomer. C-C double bond existence in resveratrol helps in remodelling of *cis*- and *trans*- isomer configuration into each other under certain conditions.[10,11]

Modification of hydroxyl group, benzene ring and double bond in resveratrol shows improvement in the initial stages of oxidation and the resultant by products have better biochemical activities.[5,6] It is established that pulverized trans- resveratrol doesn't undergo any chemical decomposition when kept at 40°C and 75% humidity. Restricted medicinal use of resveratrol is due to its low bioavailability. Hence, it accrued exceptional heed from the researchers all over the world and resulted in synthesis of a good number of new resveratrol derivatives.[7,12] Halogenated, methoxylated, glycosylated, hydroxylated derivatives of resveratrol are widely accepted for its medicinal properties and applications.[13-15]

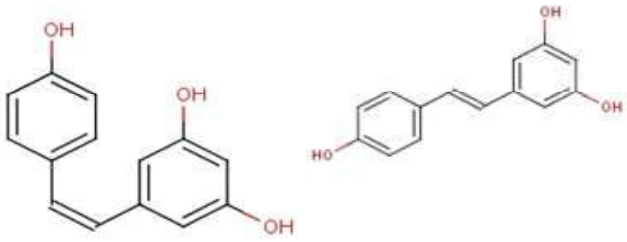


**Figure: 3- Structure of *Cis*- and *Trans*- Resveratrol (source: NCBI)[16-17]**

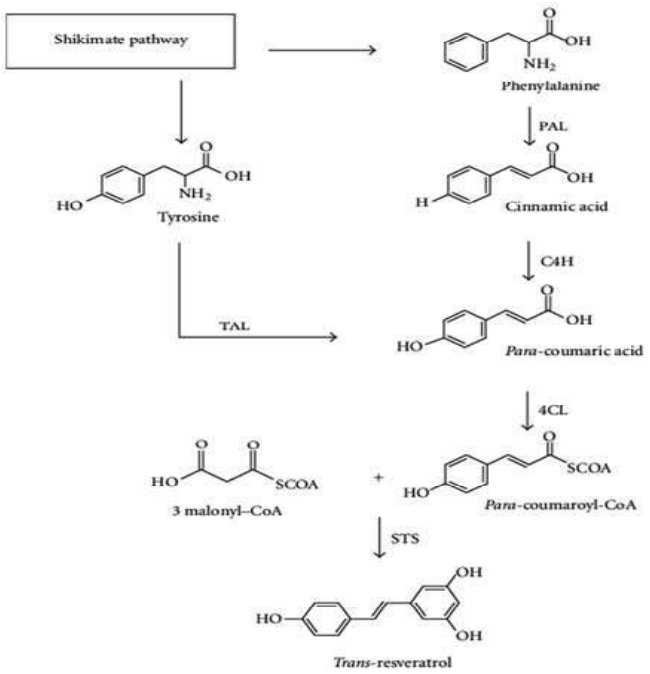



**Figure: 4 - Biosynthesis of Resveratrol [18]**

**Biosynthesis of Resveratrol**

An immense number of valuable natural products are synthesized in plants by Phenylpapanoid pathway (secondary metabolisms pathways).[19] The biosynthesis of plants secondary metabolites are generally regulated and controlled by plant's own genotype and ontogeny.[20] phenylalanine ammonia-lyase.Pinosylvinmethoxy transferase, 4-coumarate-CoA ligase, Cinnamic acid 4-hydroxylase stilbene synthase are the key enzymes involved in the biosynthesis of stilbene.

**Resveratrol as a bio active**

Resveratrol proves to be very effective as an anti-inflammatory[21] and anti- microbial natural compound. Resveratrol's engrossing biological activities and molecular structure have coruscated several novel technique, principles or means towards its biosynthesis. However, poor bioavailability of resveratrol raised concerns about its efficacy in its oral administration. Recent studies suggest that 'Biological Activities' of resveratrol derivatives may have similar or even high potent structure and activity.[22,23]

Cellular redox balance of cell is also sustained by resveratrol. Resveratrol potential for cell defence may be elucidated by its capabilities to counterpoise the oxidation and triggering the cellular shielding gene. Unmediated activities of resveratrol forages different oxidative reactions and is a detriment to safeguard biomolecules from damage. Resveratrol has hardly any reported cytotoxic reactions and is effective to infiltrate blood-brain blockades and has its future as the restorative medicine in the control, the prevention and the treatment of the brain malignancies those befall at early ages.[24-26]

Recent studies suggest that the physical activity and longevity in mice shows remarkable results. There has been identification of phosphoesterase enzyme (PDE) as recipient of resveratrol and by its implications sirtuin pathway is triggered.[27-32]

Resveratrol assimilation by the cells accompanies both the facilitated transport and diffusion process.[33] Hepatic cells metabolize resveratrol as *sulfo-* and *tauto-*conjugate, glucuronide and three Hydrophilic resveratrol finishing off by MRP- I.[12,34] Resveratrol tie up to albumin and lipoprotein for their transportation through blood. 20 to 50 micromoles usually brings the required potential for the expression of resveratrol *in-vitro* analysis.[35]

**Resveratrol and Cancer**

Introduction of Resveratrol to the area of cancer research gained evidences suggesting the hypothetical effects of it. Resveratrol due to its least reported toxicological effects and because of extensively studied polyphenols, is set to be solitary in domain of polyphenols or flavonoids based chemotherapy and chemopreventive.[36-38]

| Effect | Pathway |
|---|---|
| Anti oxidant | Inhibition QR2 |
| | Up- regulation SOD |
| | Copper chelant |
| | Activation p38 MAPK |
| | |
| Vasolidation | Activation K+ channel $Ca^{2+}$ →↑eNOS,↑eNOS |
| | |
| Metabolic Disorder ( CVD / T2DM / Obesity) | Activation Sirt -I → Activation PGC -I alpha → plasma levels glucose & glycated |
| | |
| Cancer | Downregulation HIF I alpha |
| | Up regulation p21 / p53 |
| | Inhibition AMPK / YAP |
| | Inhibition NF – k B / STAT3 |
| | Inhibition Sirt – I / PTEN/ P13K / AKT |
| | |
| Anti inflammatory | Blocked MAPK |
| | Inhibition COX1 |
| | Blocked xA2 |
| | Inhibition COX2 |

**Figure: 5- Main effects and pathways for action of Resveratrol[39-68]**

Resveratrol is a powerful antioxidant possessing antitumor activities.[69] It have been considered as a potential candidate for prevention, control and treatment of several types of disease including cancer as confirmed by many *in-vitro* and *in-vivo* studies.[6,70-71] Resveratrol has been shown to inhibit all cancinogenesis stages (initiation, promotion and progression). Resveratrol also influences metabolism and increases life span by calorie restriction mechanism.[72,73] Resveratrol is seen as a substitute for the chemopreventive activity due to its higher viral efficacy increased absorption and minimal phase 2 metabolism.[74,75]

**Resveratrol an antiviral**

Resveratrol shows inhibitory effects on the Epstein Barr virus lytic cycle.[76] Resveratrol is giving a very strong conformation to SARS *Cov*- 2 Protein : Human angiotensin Coy enzyme 2 in in-silco study models. The high affinity for spike protein S1 : ACE 2 was confirmed in molecular dynamics simulations and the free energies study stating that the resveratrol inhibits ACE 2 receptor to form S1: ACE 3 complex and restricting entry of the virus into the host cells.[77]

**Resveratrol as anti-diabetic**

Resveratrol has beneficial effects toward reduction in blood sugar, preservation of pancreatic β cells and improvement in insulin action.[78-79] Resveratrol molecule might be the Scaffold structure for the development of different synthetic compounds for the treatment and management of Diabetes. It was seen that the effectiveness of Resveratrol treatment depends on the metabolic status and the improvement in hormone sensitivity in response to resveratrol was tissue-specific and occurred solely underneath insulin-resistant conditions.[79] Resveratrol conjointly exerts glucose-lowering effects in general, the management of diabetes involves in three main aspects: reduction of blood Glucose[80], preservation of cells particularly in the case of type 2 diabetes mellitus (T2DM)[81], and improvement in insulin secretion from Pancreatic β cells.[77] Resveratrol Improves glucoregulation and insulin sensitivity in diabetic patients.[82,84]

Moreover, Resveratrol supplementation significantly decreased fasting and postprandial serum glucose level and insulin concentration.[85] Insulin resistance was

| The effects of Resveratrol on different biological entities | |
|---|---|
| Coronary Artery Disease | |
| | LIPIDS |
| | Plasma levels descrease of LDL – C , total cholesterol and triglycerides , HDL -C |
| | There is blunted reduction in the total cholesterol and LDL – C plasma levels following exercise. |
| | Total cholesterol and LDL- C plasma levels increases . |
| | Plasma levels of LDL – *ox* and ApoB reduces |
| | INFLAMMATION |
| | Inflammatory pathway and cytokine signaling downregulation |
| | Plasma levels of IL – 6 , TNFa, hCRP , IFN-γ and IL – 8 , leptin and leukocytes reduced |
| | Leukocytes adhesion molecules downregulation |
| | microRNA's involved in inflammatory modulation modifies |
| | Rise in plasma levels of IL- 10 |
| | HYPERTENSION |
| | Decrease in systolic blood pressure ( pose dependent) |
| | Dull Decrease in mean arterial blood pressure following exercise |
| | ENDOTHELIAL |
| | Reduction in arterial stiffness |
| | Improvement in flow mediated dilation |
| | |
| Heart Failure/ left ventricular functions | |
| | LEFT VENTRICLE FUNCTION |
| | Left diastolic function improves |
| | Brain natriuretic peptide reduction |
| | |
| | |
| DIABETES | Protection against insulin resistance, hyperinsulinemia , and hypoglycemia |
| | Decrease in fasting glucose, insulin , glycated haemoglobin with short term consumption. |
| | |

**Figure 6 : Effects of Resveratrol on different biological entities[95-100]**

improved in diabetic receiving a low Resveratrol dose via activation of the Akt signalling pathway and decrease in the levels of oxidative stress markers.[86] Similarly, diet supplementation with Resveratrol not solely complemented standard antidiabetic medication, but it also provided more protection in T2DM patients already on antidiabetic therapies.[87,88] This was evident by the decrease in systolic blood pressure and the average HbA1c levels, lower fasting blood glucose and plasma insulin concentrations, and improved insulin resistance index (HOMA-IR), all of which reaching levels attained after metformin treatment.[89] Literature highlights the beneficial effects of Resveratrol in polygenic disease, and its aspects.[90,91] However, limited clinical information is available on the potential effects of Resveratrol.

**Resveratrol in Neuro disorders**

Resveratrol have the capability to ameliorate the neurodegenerative disorders. Resveratrol is shown to slowdown the process of aggregation of amyloid beta by attuning particular proteins responsible for proteostasis. Memory loss in Alzheimer may be stopped or suspended by resveratrol as it gave a rise to high amount of mitochondrial complex IV protein through the SiTR I and AMPK pathways.[92,93] Caspase 3 activity is slowed and improvements in oxidative damage through restoration of Xanthine oxidase and protein carbonyl genesis along with initiation of glutathione peroxidase Nrf2 pathways reduces the Parkinson's side effects by resveratrol. Resveratrol attenuated Parkinson via prompting MiR 214 expression and decreasing m RNA expression of synuclein.[94] Resveratrol help in the improvement of Alzheimer and Parkinson by triggering the SiTRI, AMPK and Nrf2 pathways regulating the linked gene expression.

## CONCLUSION

Resveratrol is one amongst the foremost investigated bioactive compounds.Findings have provided mounting proof that the resveratrol, a dietary polyphenol, might confer health blessings and defend against a variety of medical conditions and age-related complications. Studies have shown that resveratrol possesses several bioactivities and health advantages like Anti-inflammatory and immunomodulatory effects, and in liver diseases, diabetes, obesity, Alzheimer's sickness, Parkinson's sickness etc. However, there isn't any consistent proof against metabolic disorders and various ailments on studies in laboratory animals and humans. Throughout this review, efforts have been made to survey the newest advances about bioavailability, toxicity, health benefits, and resveratrol-drug interactions studies. Additional bioactivities and health advantages of resveratrol ought to be evaluated, and more clarification of the underlying mechanisms of action is needed.This report focuses on the actions of resveratrol treatment in combatting diseases, like cancer, diabetes, neurodegeneration, upset, and various age-related ailments, so as to develop Resveratrol into functional dietary supplements and drug.

## REFERENCES


1. **Takaoka M., 1939.** Resveratrol a new phenolic compound, from *Veratrum grandiflorum*. *Nippon Kagaku kaishi.* **60:** 1090-1100.
2. **Nonomura, S.; Kanagawa, H.; Makimoto, A., 1963.** Chemical Constituents of Polygonaceous Plants. I.Studies on The Components of Ko-jo-kon. (Polygonum Cuspidatum Sieb. Et Zucc.). *Yakugaku* Zasshi *J. Pharm. Soc*. Japan. **83:**988–990
3. **Langcake P and Pryce R J., 1976.** The production of resveratrol by *Vitis vinifera* and other members of the *Vitaceae* as a response to infection or injury. *Physiological Plant Pathology*. **9**(**9**). DOI:10.1016/0048-4059(76)90077-1.
4. **Weiskirchen S and weistein R., 2016.** Resveratrol: How much Wine do you have to drink to stay Healthy? *Advances in Nutrition.* **7(4):**706-718.
5. **Bingren Tian and Jiayue. 2020.** Resveratrol: A review of plant sources, synthesis, stability, modification and food application. DOI:10.1002/jsfa.10152
6. **Burns, J.; Yokota, T.; Ashihara, H.; Lean, M.E.J.; Crozier, A., 2002.** Plant foods and herbal sources of resveratrol. *J. Agric. Food Chem*. **50:**3337–3340.
7. **Keylor, M.H.; Matsuura, B.S.; Stephenson, C.R.J., 2015.** Chemistry and biology of resveratrol-derived natural Products. *Chem. Rev*. **115:**8976–9027
8. **w.w.w .ncbi.nlm.nih.gov/compound/445154#section = 2D-Structure**
9. **Rivera, L.; Morón, R.; Zarzuelo, A.; Galisteo, M., 2009.** Long-term resveratrol administration reduces metabolic Disturbances and lowers blood pressure in

obese Zucker rats. *Biochem. Pharmacol.***77:**1053–1063.

10. **Chen Y.H., Chung Y.L. and Lin C.H., 2002.** Low temperature non aqueous capillary electrophoretic separation- 77 K fluorescence spectroscopic detection for the online identification of photo converted analysis of trans-resveratrol. *J.Chromatogr* **A943:** 287-294.

11. **Lin C.H. and Chen Y.H., 2001.** Online identification of trans- and cis- resveratrol by non aqueous capillary electrophoresis/fluorescence spectroscopy at 77k**.** *Electrophoresis* **22:** 2574-2579.

12. **Carrizzo, A.; Puca, A.; Damato, A.; Marino, M.; Franco, E.; Pompeo, F.; Traficante, A.; Civitillo, F.; Santini, L.; Trimarco, V. 2013**. Resveratrol improves vascular function in patients with hypertension and dyslipidemia by modulating no metabolism. *Hypertension*. **62:**359–366.

13. **Jung, K.-H.; Lee, J.H.; Thien Quach, C.H.; Paik, J.-Y.; Oh, H.; Park, J.W.; Lee, E.J.; Moon, S.-H.; Lee, K.-H., 2013**. Resveratrol suppresses cancer cell glucose uptake by targeting reactive oxygen species-mediated Hypoxia-inducible factor-1á activation. *J. Nucl. Med*. **54:**2161–2167.

14. **Kala, R.; Shah, H.N.; Martin, S.L.; Tollefsbol, T. 2015.** Epigenetic-based combinatorial resveratrol and pterostilbene Alters DNA damage response by affecting SIRT1 and DNMT enzyme expression, including SIRT1-dependent Ã-H2AX and telomerase regulation in triple-negative breast cancer. *BMC Cance.* **15:**672.

15. **Corpas, R.; Grinan-Ferre, C.; Rodriguez-Farre, E.; Pallas, M.; Sanfeliu, C, 2019.** Resveratrol induces brain resilience Against Alzheimer neurodegeneration through proteostasis enhancement. *Mol. Neurobio*. **56:**1502–1516.

16. **Maugeri, A.; Barchitta, M.; Mazzone, M.G.; Giuliano, F.; Basile, G.; Agodi, A., 2018.** Resveratrol Modulates SIRT1 and DNMT Functions and Restores LINE-1 Methylation Levels in ARPE-19 Cells under Oxidative Stress and Inflammation. *Int. J. Mol. Sci.* **19:**2118**.**

17. **Salehi, B., Mishra, A.P., Nigam,M. Et Al. 2018.** Resveratrol: A Double-Edged Sword in Health Benefits. *Biomedicine.* **6(3):** 91.

18. **Park, S.-J., Ahmad, F., Philp, A., Baar, K., Williams,T., Luo, H., Ke, H., Rehmann, H., Taussig, R.,Brown, A.L. 2012.** *Cell.* **148:** 487–501.

19. **w.w.w .ncbi.nlm.nih.gov/compound/47425#section =2D-Structure**

20. **W.w.w .ncbi.nlm.nih.gov/compound/47426#section =2D-Structure**

21. **Adrian M, Jeandet P. 2012.** Effects of resveratrol on the ultra structure of Botrytis *Cinerea conidia* and biological significance in plant/pathogen Interactions. *Fitoterapia.* **83:**1345–1350.

22. **Zhang, H.C, Liu, J.M., Chen,H.M., Gao,C.C., Lu, H.Y. 2011.** Up Regulation of *Licochalcone* A Biosynthesis and Secretion Tween 80 in Hairy Root Culture of *GlycyrrhizauralenisisFisch Mol. Biotechnol.* **47:**50-56.

23. **Klienbenstein D.J. 2004.** Secondary metabolites and plant/environment interaction: a view through Arabidposis Thailand tinged glasses. *Plant Cell Environment.* **27:** 675-684.

24. **Tili E, Michaille J J, Adair B, Hansjuerg A, Emetic Ki, Cristina T, Carlo M C., 2010.** Resveratrol decreases the level of miR -155 by upregulating miR-663, a microRNA targeting Junb and JunD. *Carcinogenesis*, **31:**1561. DOI: 10.193/carcin/bgq143.

25. **Wang ,M., Jin,Y., C.T., 1999.** Evaluation of resveratrol derivatives as potential antioxidant and identification of a reaction product of resveratrol and 2,2-diphenyl 1-1-picryhydrazyl 1 radical. *J. Agric. Food. Chem.* **47:**3974-3977.

26. **Fauconneau, B., Waffo-Teguo P., Higher, F., Barrier, L., Decendit, A., Merillon, J.M. 1997.** Comparative study of radical scavengers and antioxidant properties of phenolic compounds from *Vitis vinifera* cell culture using in vitro test. *Life Science*. **61:**2103-2110.

27. **Stivala, L.A.; Savio, M.; Carafoli, F.; Perucca, P.; Bianchi, L.; Maga, G.; Forti, L.; Pagnoni, U.M.; Albini, A.; Prosperi, E. 2001.** Specific structural determinants are responsible for the antioxidant activity and the cell Cycle effects of resveratrol. *J. Biol. Chem.* **276:**22586–22594.

28. **St John, S.E.; Jensen, K.C.; Kang, S.; Chen, Y.; Calamini, B.; Mesecar, A.D.; Lipton, M.A., 2013.** Design, synthesis, Biological and structural evaluation of functionalized resveratrol analogues as inhibitors of quinone Reductase 2. *Bioorg. Med. Chem.* **21:**6022–6037.

29. **Wang Q, Xu J, Rottinghaus GE, Simonyi A, Lubahn D, Sun GY. 2002.** Resveratrol Protects against global cerebral ischemic injury in gerbils. *Brain*. **958:**439–47.

30. **Park, E.-J.; Pezzuto, J.M., 2015.** The pharmacology of resveratrol in animals and humans. *Biochim. Biophys. Acta*. **1852:** 1071-1113.

31. **Delmas, D.; Jannin, B.; Latruffe, N., 2005.** Resveratrol: Preventing properties against vascular alterations and ageing. *Mol. Nutr. Food Res.* **49:**377–395.

32. **Huang, H.; Lin, H.; Zhang, X.; Li, J., 2012.** Resveratrol reverses temozolomide resistance by down regulation of MGMT in T98G glioblastoma cells by the NF-êB-dependent pathway. *Oncol. Rep*. **27:**2050–2056.

33. **Magyar, K.; Halmosi, R.; Palfi, A.; Feher, G.; Czopf, L.; Fulop, A.; Battyany, I.; Sumegi, B.; Toth, K.; Szabados, E., 2012.** Cardioprotection by resveratrol: A human clinical trial in patients with stable coronary artery disease. *Clin. Hemorheol. Microcirc*. **50:**179–187.

34. **Tomé-Carneiro, J.; Gonzálvez, M.; Larrosa, M.; García-Almagro, F.J.; Avilés-Plaza, F.; Parra, S.; Yáñez-Gascón, M.J.; Ruiz-Ros, J.A.; García-Conesa, M.T.; Tomás-Barberán, F.A. 2012.** Consumption of a grape Extract supplement containing resveratrol decreases oxidized LDL and ApoB in patients undergoing primary Prevention of cardiovascular disease: A triple-blind, 6-month follow up, placebo-controlled, randomized Trial. *Mol. Nutr. Food Res.***56:** 810–821.

35. **Aires, V.; Colin, D.J.; Doreau, A.; Di Pietro, A.; Heydel, J.M.; Artur, Y.; Latruffe, N.; Delmas, D., 2019.** P-glycoprotein 1 affects chemo activities of resveratrol against human colorectal cancer cells. *Nutrients*. **11:** 2098.

36. **Lançon, A.; Hanet, N.; Jannin, B.; Delmas, D.; Heydel, J.M.; Lizard, G.; Chagnon, M.C.; Artur, Y.; Latruffe, N., 2007.** Resveratrol in human hepatoma HepG2 cells: Metabolism and inducibility of detoxifying enzymes. *Drug Metab. Dispos.* **35:** 699–703.

37. **Langcake P and Pryce R J., 1976.** The production of resveratrol by *Vitis vinifera* and other members of the Vitaceae as a response to infection or injury. *Physiological Plant Pathology.* **9(9).** DOI:10.1016/0048-4059(76)90077-1.

38. **Savouret J.F., M. Quesne, 2002.** Resveratrol and cancer: a review. *Biomed.* **56:** 84-87

39. **Kanemaki, T.; Kitade, H.; Kaibori, M.; Sakitani, K.; Hiramatsu, Y.; Kamiyama, Y.; Ito, S.; Okumura, T. 1998.** Interleukin 1beta and interleukin 6, but not tumor necrosis factor alpha, inhibits insulin-stimulated glycogen Synthesis in rat hepatocytes. *Hepatol. Baltim*. **27:**1296–1303**.**

40. **Belguendouz, L.; Frémont, L.; Gozzelino, M.T, 1998.** Interaction of transresveratrol with plasma lipoproteins. *Biochem. Pharmacol.,* **55:**811–816.

41. **Bhatt, J.K.; Thomas, S.; Nanjan, M.J., 2012.** Resveratrol supplementation improves glycemic control in type 2 Diabetes mellitus. *Nutr. Res.* **32:**537–541.

42. **Brasnyó, P.; Molnár, G.A.; Mohás, M.; Markó, L.; Laczy, B.; Cseh, J.; Mikolás, E.; Szijártó, I.A.; Mérei, A.;Halmai, R. 2011.** Resveratrol improves insulin sensitivity, reduces oxidative stress and activates the Akt Pathway in type 2 diabetic patients. *Br. J. Nutr.* **106:** 383–389.

43. **Chatterjee, B.; Ghosh, K.; Kanade, S.R., 2019.** Resveratrol modulates epigenetic regulators of promoter histone Methylation and acetylation that restores BRCA1, p53, p21CIP1 in human breast cancer cell lines. *Bio Factors Oxf. Engl.,* **45:**818–829

44. **Della-Morte, D.; Dave, K.R.; DeFazio, R.A.; Bao, Y.C.; Raval, A.P.; Perez-Pinzon, M.A., 2009.** Resveratrol pre-treatment Protects rat brain from cerebral ischemic damage via a sirtuin 1-uncoupling protein 2 pathway. *Neuroscience*. **159:**993–1002.

45. **Fiori JL, Shin YK, Kim W, Krzysik-Walker SM, Gonzalez-Mariscal I, Carlson OD, Sanghvi M, Moaddel R, Farhang K, Gadkaree SK, Doyle ME, Pearson KJ, Mattison JA, de Cabo R, Egan JM., 2013.** Resveratrol prevents beta-cell dedifferentiation in nonhuman primates given a high-fat/high-sugar diet. *Diabetes*. **62:**3500–3513.

46. **Kim, C.-W.; Hwang, K.-A.; Choi, K.-C., 2016.** Anti-metastatic potential of resveratrol and its metabolites by the inhibition of epithelial-mesenchymal transition, migration, and invasion of malignant cancer cells Phytomed**.** *Int. J. Phytother. Phytopharm.* **23:**1787–1796.

47. **Kong, Q.; Ren, X.; Hu, R.; Yin, X.; Jiang, G.; Pan, Y., 2016.** Isolation and purification of two antioxidant isomers of Resveratrol dimer from the wine grape by counter-current chromatography. *J. Sep. Sci.* **39:**2374–2379.

48. **Kutil, Z.; Temml, V.; Maghradze, D.; Pribylova, M.; Dvorakova, M.; Schuster, D.; Vanek, T.; Landa, P. 2014.** Impact of wines and wine constituents on cyclooxygenase-1, cyclooxygenase-2, and 5-lipoxygenase catalytic activity. *Mediat. Inflamm.*178931.

49. **Mo, W.; Xu, X.; Xu, L.; Wang, F.; Ke, A.; Wang, X.; Guo, C., 2011.** Resveratrol inhibits proliferation and induces Apoptosis through the hedgehog signalling pathway in pancreatic cancer cell. *Pancreatology*. **11:**601–609.

50. **Miyagi, Y.; Miwa, K.; Inoue, H., 1997.** Inhibition of human low-density lipoprotein oxidation by flavonoids in red Wine and grape juice. *Am. J. Cardiol*. **80:** 1627–1631.

51. **Li, W.; Cao, L.; Chen, X.; Lei, J.; Ma, Q., 2016.** Resveratrol inhibits hypoxia-driven ROS-induced invasive and Migratory ability of pancreatic cancer cells via suppression of the Hedgehog signalling pathway. *Oncol. Rep.* **35:** 1718–1726.

52. **Singh, S.K.; Banerjee, S.; Acosta, E.P.; Lillard, J.W.; Singh, R., 2017.** Resveratrol induces cell cycle arrest and apoptosis with docetaxel in prostate cancer cells via a p53/p21WAF1/CIP1 and p27KIP1 pathway. *Oncotarget*. **8:**17216–17228.

53. **Shen, M.Y.; Hsiao, G.; Liu, C.L.; Fong, T.H.; Lin, K.H.; Chou, D.S.; Sheu, J.R., 2007.** Inhibitory mechanisms of Resveratrol in platelet activation: Pivotal roles of p38 MAPK and NO/cyclic GMP. *Br. J. Haematol.* **139:**475–485.

54. **Thirunavukkarasu, M.; Penumathsa, S.V.; Koneru, S.; Juhasz, B.; Zhan, L.; Otani, H.; Bagchi, D.; Das, D.K.;Maulik, N., 2007.** Resveratrol alleviates cardiac dysfunction in streptozotocin-induced diabetes: Role of nitric oxide, Thioredoxin, and hemeoxygenase. *Free Radic. Biol. Med*. **43:**720–729.

55. **Wici ´nski, M.; Malinowski, B.; W²eclewicz, M.M.; Grze´sk, E.; Grze´sk, G., 2017.** Resveratrol Increases Serum BDNF Concentrations and Reduces Vascular Smooth Muscle Cells Contractility via a NOS-3-Independent Mechanism. *BioMed. Res. Int.* 9202954.

56. **Yang, J.; Wang, N.; Li, J.; Zhang, J.; Feng, P., 2010.** Effects of resveratrol on NO secretion stimulated by insulin and its dependence on SIRT1 in high glucose cultured endothelial cells. *Endocrine*. **37:** 365–372.

57. **Yar, A.S.; Menevse, S.; Alp, E.; Helvacioglu, F.; Take, G., 2010.** The effects of resveratrol on cyclooxygenase-1 and Cyclooxygenase-2 mRNA and protein levels in diabetic rat kidneys. *Mol. Biol. Rep*. **37:**2323–2331.

58. **Kimura, Y.; Okuda, H.; Arichi, S., 1985.** Effects of stilbenes on arachidonate metabolism in leukocytes. *Biochim. Biophys. Acta.* **834:**275–278.

59. **Khan O S, Bhar A A, Krishna kitty R, Mohammed R M and Uddin S.2016 .** Therapeutic Potential of Resveratrol in Lymphoid Malignancies. *Nutrition and Cancer*. **68**:365-373.

60. **Jang, M.; Cai, L.; Udeani, G.O.; Slowing, K.V.; Thomas, C.F.; Beecher, C.W.; Fong, H.H.; Farnsworth, N.R.; Kinghorn, A.D.; Mehta, R.G. 1997.** Cancer chemopreventive activity of resveratrol, a natural product derived from grapes. *Science*. **275:** 218–220.

61. **Asadi, S.; Rahimi, Z.; Saidijam, M.; Shabab, N.; Goodarzi, M.T. 2018.** Effects of Resveratrol on FOXO1 and FOXO3a Genes Expression in Adipose Tissue, Serum Insulin, Insulin Resistance and Serum

SOD Activity in Type 2 Diabetic Rats. *Int. J. Mol. Cell. Med.***7:** 176–184.

62. **Cao, L.; Chen, X.; Xiao, X.; Ma, Q.; Li, W. 2016.** Resveratrol inhibits hyperglycemia-driven ROS-induced invasion and migration of pancreatic cancer cells via suppression of the ERK and p38 MAPK signalling pathways. *Int. J. Oncol*. **49:**735–743.

63. **Cilibrasi, C.; Riva, G.; Romano, G.; Cadamuro, M.; Bazzoni, R.; Butta, V.; Paoletta, L.; Dalprà, L.;Strazzabosco, M.; Lavitrano, M. 2017.** Resveratrol Impairs Glioma Stem Cells Proliferation and Motility by Modulating the WntSignalling Pathway. *PLoS ONE*. **12:**e0169854.

64. **Estrov, Z.; Shishodia, S.; Faderl, S.; Harris, D.; Van, Q.; Kantarjian, H.M.; Talpaz, M.; Aggarwal, B.B. 2003.** Resveratrol blocks interleukin-1beta-induced activation of the nuclear transcription factor NF-kappaB, Inhibits proliferation, causes S-phase arrest, and induces apoptosis of acute myeloid leukaemia cells. *Blood*. **102:**987–995.

65. **Manna, S.K.; Mukhopadhyay, A.; Aggarwal, B.B., 2000.** Resveratrol suppresses TNF-induced activation of nuclear Transcription factors NF-kappa B, activator protein-1, and apoptosis: Potential role of reactive oxygen Intermediates and lipid peroxidation. *J. Immunol. Baltim*. **164:** 6509–6519.

66. **Giménez-Bastida, J.A.; Ávila-Gálvez, M.Á.; Espín, J.C.; González-Sarrías, A., 2019.** Conjugated Physiological Resveratrol Metabolites Induce Senescence in Breast Cancer Cells: Role of p53/p21 and p16/Rb Pathways and ABC Transporters. *Mol. Nutr. Food Res*. **63:**e1900629.

67. **Haslbeck, K.-M.; Schleicher, E.; Bierhaus, A.; Nawroth, P.; Haslbeck, M.; Neundörfer, B.; Heuss, D., 2005.** The AGE/RAGE/NF-(kappa)B pathway may contribute to the pathogenesis of polyneuropathy in impaired Glucose tolerance (IGT). *Exp. Clin. Endocrinol. Diabetes.* **113:**288–291.

68. **Iuga, C.; Alvarez-Idaboy, J.R.; Russo, N., 2012.** Antioxidant activity of trans-resveratrol toward hydroxyl and Hydroperoxyl radicals: A quantum chemical and computational kinetics study. *J. Org. Chem*. **77:** 3868–3877.

69. **Jiang, Z.; Chen, X.; Chen, K.; Sun, L.; Gao, L.; Zhou, C.; Lei, M.; Duan, W.; Wang, Z.; Ma, Q. 2016.** YAP Inhibition by Resveratrol via Activation of AMPK Enhances the Sensitivity of Pancreatic Cancer Cells to Gemcitabine. *Nutrients*. **8:** 546.

70. **Aires, V.; Limagne, E.; Cotte, A.K.; Latruffe, N.; Ghiringhelli, F.; Delmas, D., 2013.** Resveratrol metabolites inhibit Human metastatic colon cancer cells progression and synergize with chemotherapeutic drugs to induce cell Death. *Mol. Nutr. Food Res*. **57:** 1170–1181.

71. **Ducible, R.; Shah, H.N.; Martin, S.L.; Tollefsbol, T., 2015.** Epigenetic-based combinatorial resveratrol and pterostilbene Alters DNA damage response by affecting SIRT1 and DNMT enzyme expression, including SIRT1-dependent Ã-H2AX and telomerase regulation in triple-negative breast cancer**.** *BMC Cancer*. **15:** 672.

72. **Kapetaonvic I.M, Muzzio M., Huang Z., Thompson T.N. and McCormick D.L. 2001.** Pharmacokinetics, oral bioavailability, and metabolic profile of resveratrol and it's dimethyl ether analog, petrostilbene, in rats. *Cancer Chemother. Pharmacol.* **68:**593-601.

73. **Bonkowski M.S. and Sinclair D.A., 2016.** Slowing ageing by design: the rise of NAD+ and sirtuin-activating compounds. *Nature Reviews Molecular Cell Biology.* **17:** 679-690.

74. **Low, I.C.C.; Chen, Z.X.; Pervaiz, S., 2010.** Bcl-2 modulates resveratrol-induced ROS production by regulating Mitochondrial respiration in tumor cells. *Antioxid. Redox Signal.* **13:**807–819.

75. **Fulda S., 2010.** Resveratrol and derivatives for the prevention and treatment of Cancer. *Drug Discovery Today*. **15:**757-765.

76. **Kang W, Hong HJ, Guan J, Kim DG, Yang EJ, Koh G, Park D, Han CH, Lee YJ, Lee DH. 2012.** Resveratrol improves insulin signalling in a tissue-specific manner under insulin-resistant conditions only: in vitro and in vivo experiments in rodents. *Metabolism*. **61:**424–433.

77. **Yiu, C.Y., Chen, S.Y., Chang,L.K., Chin.,Y.F., and Lin, T.P., 2010. Inhibitory** effects of Resveratrol on

the Epstein Barr virus lytic cycle. *Molecule*, **15:** DOI:10.3390/molecule 15107115.

78. **Hussain, M.W, Sajjad, A. And Sumra, W. A, 2020.** Stilbene based Natural Compound as promising Drug Candidate against COVID 19, *Journal of Biomolecular structure and Dynamics*. DOI:10.1080/07391102. 2020.1762743.

79. **Jimenez-Gomez Y, Mattison JA, Pearson KJ, Martin-Montalvo A, Palacios HH, Sossong AM, Ward TM, Younts CM, Lewis K, Allard JS, Longo DL, Belman JP, Malagon MM, Navas P, Sanghvi M, Moaddel R, Tilmont EM, Herbert RL, Morrell CH, Egan JM, Baur JA, Ferrucci L, Bogan JS, Bernier M, de Cabo R. 2013.** Resveratrol improves adipose insulin signalling and reduces the inflammatory response in adipose tissue of rhesus monkeys on high-fat, high-sugar diet. *Cell Metab.* **18:**533–545.

80. **H.C. Su, L.M. Hung, J.K. Chen. 2006.** Resveratrol, a red wine antioxidant, possesses an Insulin-like effect in streptozotocin-induced diabetic rats, *Am. J. Physiol.Endocrinol. Metab.* **290(6):** E1339–E1346.

81. **P. Palsamy, S. Subramanian. 2009.** Modulatory effects of resveratrol on attenuating the Key enzymes activities of carbohydrate metabolism in streptozotocinnicotinamide induced diabetic rats, *Chem. Biol. Interact.* **179(2):**356–362.

82. **T. Szkudelski. 2006.** Resveratrol inhibits insulin secretion from rat pancreatic islets, *Eur.J. Pharmacol.* **552(1):** 176–181.

83. **T. Szkudelski, K. Szkudelska, 2011**. Anti-diabetic effects of resveratrol. *Ann. N. Y. Acad. Sci.* **1215 (1):**34-39.

84. **Latruffe, N.; Menzel, M.; Delmas, D.; Buchet, R.; Lançon, A. 2014.** Compared binding properties between resveratrol and other polyphenols to plasmatic albumin: Consequences for the health protecting effect of dietary plant Micro components. *Molecules*. **19:**17066–17077.

85. **Hausenblas HA, Schoulda JA, Smoliga JM. 2014.** Resveratrol treatment as an adjunct to pharmacological Management in type 2 diabetes mellitus-systematic review and meta-analysis. *Mol Nutr Food Res*.doi: 10.1002/mnfr.201400173

86. **Elliot PJ, Walpole S, Morelli L, Lambert PD, Lunsmann W, Westphal CH, Lavu S. 2009.** Resveratrol/SRT-501**.** *Drugs of the future.* **34:**291-295.

87. **Li, W.; Ma, J.; Ma, Q.; Li, B.; Han, L.; Liu, J.; Xu, Q.; Duan, W.; Yu, S.; Wang, F. 2013.** Resveratrol inhibits the epithelial-mesenchymal transition of pancreatic cancer cells via suppression of the PI-3K/ Akt/NF-êB Pathway. *Curr. Med. Chem.***20:** 4185–4194.

88. **Movahed A, Nabipour I, Lieben LX, Thandapilly SJ, Yu L, Kalantarhormozi M, RekabpourSJ ,Netticadan T. 2013.** Antihyperglycemic effects of short term resveratrol supplementation in type 2 Diabetic patients. Evid Based Complement *Alternat Med.* **2013:**851267.

89. **Xu, J.; Liu, D.; Niu, H.; Zhu, G.; Xu, Y.; Ye, D.; Li, J.; Zhang, Q., 2017.** Resveratrol reverses Doxorubicin resistance by inhibiting epithelial-mesenchymal transition (EMT) through modulating PTEN/ Aktsignaling pathway in Gastric cancer. *J. Exp. Clin. Cancer Res*. **36:** 19.

90. **Moyano-Mendez JR, Fabbrocini G, De SD, Mazzella C, Mayol L, Scognamiglio I, Carnuccio R, Ayala F, La Rotonda MI, De Rosa G, 2014.** Enhanced antioxidant effect of trans-resveratrol: potential of Binary systems with polyethylene glycol and cyclodextrin. *Drug Dev Ind Pharm*. **40:**1300–1307.

91. **F.K. Knop, E. Konings, S. Timmers, P. Schrauwen, J.J. Holst, E.E. Blaak. 2013.** Days resveratrol supplementation does not affect postprandial incretin hormone responses, but suppresses postprandial glucagon in obese subjects. *Diabet. Med.* **30(10)**: 1214–1218.

92. **Porquet, D.; Grinan-Ferre, C.; Ferrer, I.; Camins, A.; Sanfeliu, C.; Del Valle, J.; Pallas, M., 2014.** Neuroprotective role of *trans*-resveratrol in a murine model of familial Alzheimer's Disease. *J. Alzheimers Dis*. **42:**1209–1220.

93. **Baur, J.A.; Pearson, K.J.; Price, N.L.; Jamieson, H.A.; Lerin, C.; Kalra, A.; Prabhu, V.V.; Allard, J.S.;Lopez-Lluch, G.; Lewis, K. 2006.** Resveratrol improves health and survival of mice on a high-calorie diet. *Nature*. **444:**337–342.

94. **Wang, Z.; Zhang, J.; Duan, Y.; Zhang, Q.; Li, G.; Zheng, D. 2015.** MicroRNA-214 participates in the neuroprotective Effect of resveratrol via inhibiting alpha-synuclein expression in MPTP-induced Parkinson's disease mouse. *Biomed. Pharmacother.* **74:** 252–256.

95. **Baur J.A. and Sinclair D.A., 2006.** Therapeutic Potential of Resveratrol: the in vivo evidence. *Nat. Rev. Drug. Didcov.* **5:**493-506.

96. **Beaudoin, M.-S.; Snook, L.A.; Arkell, A.M.; Simpson, J.A.; Holloway, G.P.; Wright, D.C. 2013.** Resveratrol Supplementation improves white adipose tissue function in a depot-specific manner in Zucker diabetic fatty Rats. *Am. J. Physiol. Regul. Integr. Comp. Physio*l. **305:**R542–R551.

97. **Belguendouz, L.; Fremont, L.; Linard, A. ,1997**. Resveratrol inhibits metal ion-dependent and independent Peroxidation of porcine low-density lipoproteins. *Biochem. Pharmacol.* **53:**1347–1355.

98. **Huang, J.-P.; Hsu, S.-C.; Li, D.-E.; Chen, K.-H.; Kuo, C.-Y.; Hung, L.-M. 2018**. Resveratrol Mitigates High-Fat Diet-Induced Vascular Dysfunction by Activating the Akt/eNOS/NO and Sirt1/ER Pathway. *J. Cardiovasc. Pharmacol.* **72:**231–241.

99. **K.-H.; Lee, J.H.; Thien Quach, C.H.; Paik, J.-Y.; Oh, H.; Park, J.W.; Lee, E.J.; Moon, S.-H.; Lee, K.-H. 2013.** Resveratrol suppresses cancer cell glucose uptake by targeting reactive oxygen species-mediated Hypoxia-inducible factor-1á activation. *J. Nucl. Med*. **54:**2161–2167.

100. **Liu K, Zhou R, Wang B, Mi MT. 2014**. Effect of resveratrol on glucose control and insulin sensitivity: a meta-analysis of 11 randomized controlled trials. *Am J. ClinNutr*. **99:**1510–1519.

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# Indigenous isolate *Bacillus subtilis* strain mp1 mediated degradation of di -(2-ethyl hexyl) phthalate and analysis of the degradation intermediates by lc-ms.

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**Abstract-** DEHP (di-ethyl hexyl phthalate) is an endocrine disrupting chemical that is commonly used as a plasticizer worldwide and is reported to adversely affect humans as well as wildlife. In the present study, strain MP1, an indigenous isolate from landfill soil has been selected which was able to grow and metabolize DEHP as sole carbon and energy sources in the MSM liquid media and showed DEHP degrading potential . It was capable of sustaining a wide range of environmental conditions with the optimum pH for bacterial growth 7.0, temperature 28°C and tolerance upto salinity 5%. The analysis of morphological and biochemical characteristics along with gram staining technique followed by 16s rRNA sequencing, identified it as *Bacillus subtilis* (Genbank Accession no. KP241787.1). LC-MS) analysis was done for the determination of metabolic intermediates. There are four main peaks identified having different RT values and mass. These are 301.15 at RT value 9.02, 413.29 at RT value 10.14, 441.32 at RT value 10.33 and 757.68 at RT value 11.27. These peaks are degradation intermediates identified as Sodium isooctyl phthalate (mass 300.13), 2-(Dodec-1-yn-1-yl)-1,1,1,3,3,3-hexamethyl-2-(trimethylsilyl)trisilane (mass 412.28), bis[4-(oxolan-2-yl) butan-2-yl] nonanedioate (mass440.31),n-(2-hydroxyethyl)-n,n-dimethyl-2,3-bis(octadecanoyloxy)propan-1-aminium) acetate (mass 755.66). These findings can provide some information in the bioremediation of DEHP from contaminated soil.

**Key words: DEHP, Bacteria, Degradation, *Bacillus subtilis*, 16sr RNA, LC-MS**

## INTRODUCTION

Phthalates are a group of compounds employed in the fabrication of plastics and diverse other industrial applications.[1] Di-ethyl hexyl phthalate (DEHP) is a kind of phthalic acid esters (PAEs), widely used as a plasticizer. It is also known as environmental secretion disruptor or environmental hormone. It is used in the manufacture of plastics, rubber, medicine, cosmetics, toys and other industries as a plasticizer.[2] As DEHP physically adhere to the polymeric matrix it can easily leak from the polymers into the environment through manufacturing, storage, use and disposal.[3] The women would have been related with endometriosis and gestational duration with precocious breast development due to increase in the level of phthalate.[4] Due to its significant lethal or mutagenic effects DEHP could cause central nervous system inhibition and

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renal injury in mice.[5] DEHP is considered as one of the most resistant phthalate esters because of its long hydrocarbon chain.[6] Many researchers reported that the humans are affected by DEHP due to its exposure in daily lives which may result in adversely affecting the functioning of endocrine, reproductive and nervous systems.[7,8] Biological methods of degradation have been paid more attention due to their environmental friendly nature, low cost, and contaminant mineralization. DEHP is liable to the microbial degradation and studies had shown this phthalate could be degraded by microorganisms because the sole carbon and energy sources.[9-11]

This study focused on degradation capability of *Bacillus subtilis* towards DEHP, isolated from the landfill soil that could utilize DEHP as a sole source of carbon and energy. Batch experiments were performed at different temperatures, pH and salt concentration for optimizing the conditions for better degradation of DEHP. LC-MS analysis was done to detect metabolic intermediates of degradation.

## MATERIALS & METHODS

**Chemicals and media-** DEHP was acquired from Accu Standard Inc. USA. The stock solution of DEHP (20 mg/ml) was prepared in corn oil and kept at 4°C until use. Corn oil used as a vehicle was purchased from Sigma Pvt. Ltd and Nieshiel Chemical Pvt. Ltd. which were of analytical grade. The minimal salt medium (MSM) consisted of the following chemicals (mg/l): $(NH_4)2SO_4$, 1,000; $KH_2PO_4$, 800; $K_2HPO_4$, 200; $MgSO_4 \cdot 7H_2O$, 500; $FeSO_4$, 10; $CaCl_2$, 50 and the pH was maintained to 7.0±0.1 with HCl or NaOH.[12]

**Isolation of DEHP degrading bacterial strain and enrichment-**

An aliquot of 1g of fresh soil was added into 100 ml minimal salt medium (MSM) supplemented with 100 mg/l of DEHP to enrich the culture and the culture was incubated in a shaker (30°C, 125rpm).[13,14] After 7 days of incubation, 2 ml of the culture was incubated further for another 7 days. There was repetition of process by increasing the DEHP concentration. Finally, the samples were streaked onto nutrient agar plates supplemented with 100 mg/l DEHP as sole carbon sources, and plates were incubated under 37°C in the dark. After incubation at 37°C, few well-separated individual colonies of different morphological types appeared and it pure culture was obtained by further repeated streaking on agar plates. Among the isolates, strain MP1 was able to significantly degrade DEHP.

**Microbial Identification and Phylogenetic Analysis**

The genomic DNA of the bacterial isolate was extracted using Bacterial Genomic DNA Extraction kit (Yaazh xenomics, Coimbatore, India) and subjected to PCR amplification of 16S rRNAgene sequence using 8F and 1541R. The PCR products were sequenced and deposited into Gen Bank, and related sequences were obtained by BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi). The multiple sequence alignments were performed by program MUSCLE 3.7. The resulting aligned sequences were cured by employing the program G block 0.91b. This G blocks eliminates poorly aligned positions and alignment noise was removed by divergent regions. The phylogeny analysis and Program PhyML 3.0 was performed. The program Tree Dyn 198.3 was employed for the rendering of tree.

**Biodegradation tests:**

To determine the effect of environmental factors,optimization experiments were performed, as pH (5.5, 7.0 and 10.5), temperature (28, 37 and 50°C)and salinity (5%, 10% and 15%).After 48 hrs of incubation interval, up to the period of 8 days, the response was recorded in the terms of absorbance at 600nm. Experiments were performed in triplicate.[15]

**Analysis of metabolic intermediates**

DEHP degradation intermediateswere analysed using LC-MS. The mass spectrometry was operated with ESI ionization source, positive mode, mass scanner range of 100~300 amu, fragment at 100 V; drying gas flow was 13.0 L- min; nebulizer pressure was 60 Psig; drying gas temperature was 350ºC and capillary voltage was 4000 V.[16]

## RESULT & DISCUSSION

**1. Microbial Isolation, Identification and Phylogenetic Analysis-**

The most potent bacterial isolate named MP1 was selected after several screening. The strain was gram-positive, rod-shaped and flagellated with non-spore forming. After cultured for 18-24 h at 37°C, this strain MP1 formed off white, opaque and irregular flat colonies with smooth margin on nutrient agar. The strain MP1 identified as *Bacillus subtilis* through its morphological

characteristics, biochemical characteristics and phylogenetic analysis. According to the NCBI BLAST analysis of its 16S rRNA gene sequence, strain MP1 belonged to the *Bacillus subtilis* (GenBank accession no. KPO241787.1). Fig. 1 illustrates the phylogenetic relationship of MP1 with its close relatives.[17]

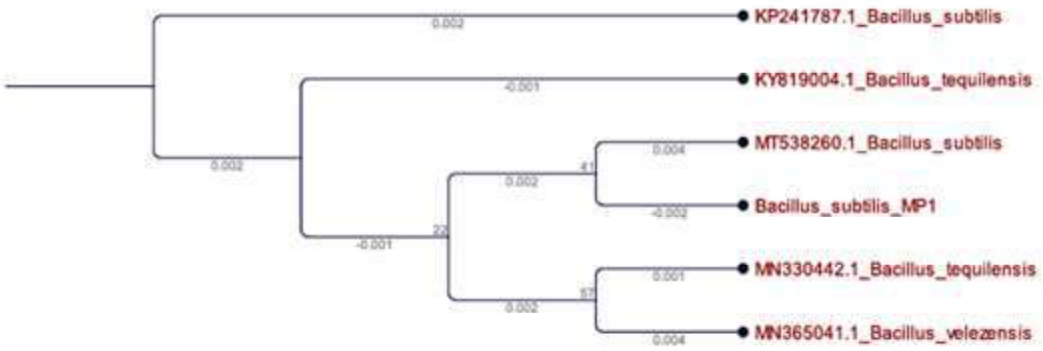



**Fig. 1.Phylogenetic tree of the isolated strain MP1.**

## 2. Biodegradation Experiments

Environmental factors such as pH, temperature, salinity and pollutant affect biodegradation of phthalates.[18] In this study, strain MP1 was able to grow and metabolize DEHP as sole carbon and energy sources in the MSM liquid media. It was capable of sustaining in a wide range of environmental conditions.

### 2.1 Effect of pH on DEHP degradation

The effect of pH on DEHP degradation by strain is shown in Fig. 2a. The degradation rate increased rapidly when pH was increased from 5.5 to 10.5. The highest DEHP degradation for MP1 was achieved at pH 10.5. Some other microorganisms such as *Rhodococcus* sp.HS-D2[19],*Gordonia alkanivorans* YC-RL2[19], *Acinetobacter* sp. SN13[14] and *Pseudomonas fluorescens* FS1[13] have been reported a wide pH range at 5-10, 6-11, 3-9, and 4-9 respectively.

### 2.2 Effect of temperature on DEHP Degradation

Strain MP1 was able to grow and degrade DEHP in temperatures of 28-50$^{o}$C with the optimum being 28$^{o}$C as shown in Fig. 2b. At 37$^{o}$C, there was little or minimal growth and degradation. As the temperature was increased from 37$^{o}$C to 50$^{o}$C, the rate of degradation increased drastically. The maximum rate of degradation was at 50$^{o}$C. The capability of MP1 to degrade DEHP at higher temperatures in accordance with the ability of *B. subtilis* 3C3.[20] There are few other strains reported which are capable of degrading DEHP at a wide range of temperatures.[21]

### 2.3 Effect of NaCl concentration on DEHP degradation

Strain MP1 was able to grow and degrade DEHP in salt concentrations ranging from 0 to 5%. There was a relatively slow growth in higher salt concentrations of 5-15%. At 5% salt concentration the microbial growth was highest as in Fig.2c. Salinity is an important factor ofwaste water treatment as many industries such as the leather industry and landfill leachate releases waste water containing high salt concentrations.[22] A study has identified halo tolerant isolate *Sphingobium* sp. which degraded DBP at the salinity ranging from 0 to 4%.[23] In biological waste treatment, salinity content of wastes is a major factor affecting bacterial growth and hence enzymatic activity.[24]

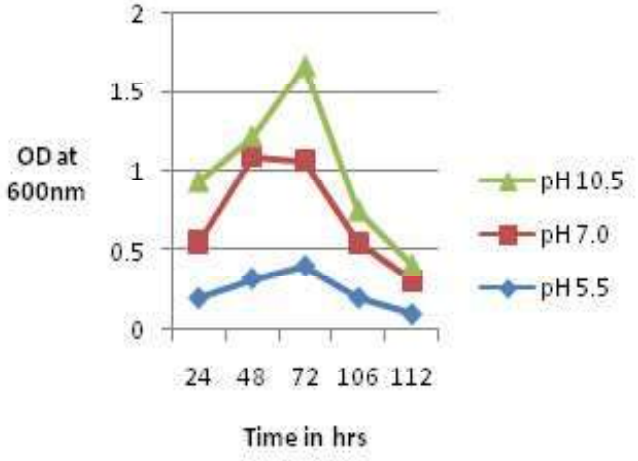



**Fig.2a: Effect of pH on DEHP degradation**

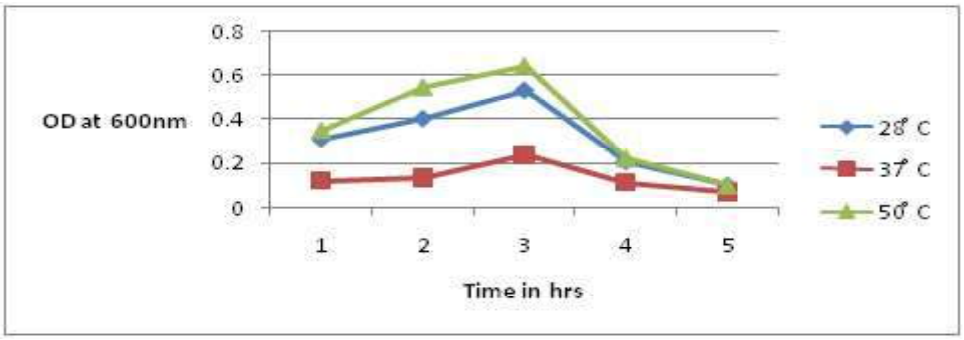



**Fig. 2 b: Effect of temperature on DEHP degradation**

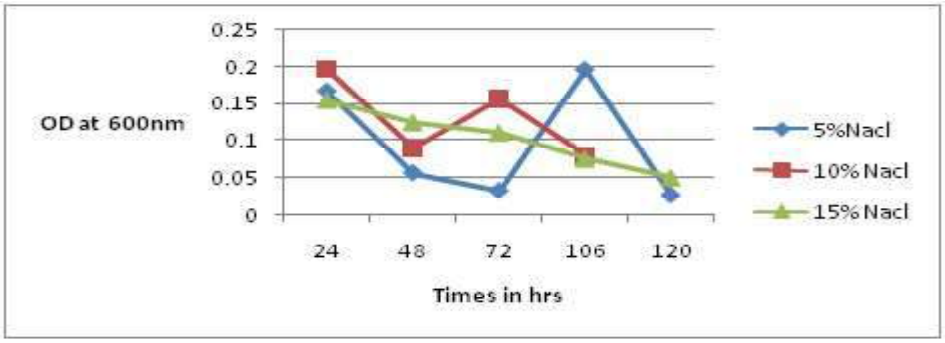



**Fig 2 c: Effect of salinity on DEHP degradation**

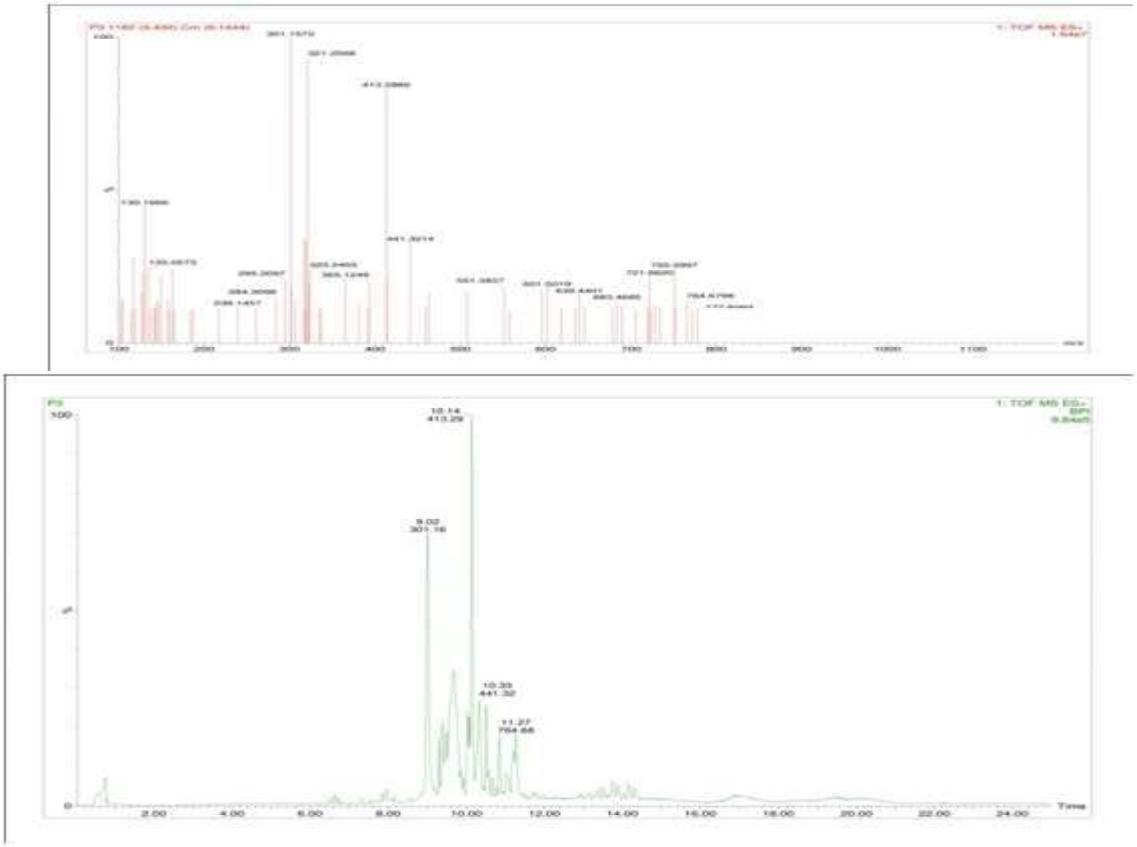


**Fig.3 - The mass spectrum of DEHP degraded intermediate compounds, X-Axis contains mass in relative to charge ratio (m/z) while y-axis contains a relative abundance**

**Table 1- Biochemical characterization result**

| | Biochemical tests | Bacterial strain (T20) |
|---|---|---|
| 1 | Amylase | - |
| 2 | Casein Hydrolysis | - |
| 3 | Catalase | + |
| 4 | Gelatin Hydrolysis | - |
| 5 | Nitrate Reduction | + |
| 6 | Citrate Utilization | - |
| 7 | Indole Production | + |
| 8 | Methyl Red | + |
| 9 | VogesProskuer | - |
| 10 | Lactose | - |
| 11 | Dextrose | + |
| 12 | Sucrose | + |
| 13 | $H_2S$ production | - |

**Identification of DEHP degradation intermediates**

The mass spectral analysis of the compound showed the parent ion peak at m/z 321.25 as shown in Fig.3. The fragment peaks patterns showed at m/z ion peaks at m/z 301.15 base peaks Fig.3 presents the metabolic intermediated identified in DEHP degradation by *Bacillus subtilis*.

In the present study, peak was obtained at 10.14 min. in the chromatogram which exhibited mass spectrum depicted in figure 3. Increase in the other peaks within chromatogram revealed the degradation products of DEHP by *Bacillus subtilis* MP1. The metabolites were identified by comparing the mass spectrum at a particular retention time (RT) with published mass spectra from the database. There are four main peaks identified having different RT values and mass. These are 301.15 at RT value 9.02, 413.29 at RT value 10.14, 441.32 at RT value 10.33 and 757.68 at RT value 11.27. These peaks are degradation intermediates identified asSodium isooctyl phthalate (mass 300.13), 2-(Dodec-1-yn-1-yl)-1,1,1,3,3,3-hexamethyl-2-(trime thylsilyl) trisilane (mass 412.28), bis[4-(oxolan-2-yl) butan-2-yl] nonanedioate (mass440.31), n-(2-hydro xyethyl) -n, n-dimethyl-2,3-bis (octadecanoyloxy) propan-1-aminium) acetate (mass 755.66).

Strain MP1 showed an outstanding ability to degrade DEHP in a wide range of temperatures (20-50$^{o}$C), wide range of pH (5.5-10.5) and superior salinity tolerance (0-5% NaCl) and therefore it could be used for remediating various phthalate contaminated environment.

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## REFERENCES

1. **Lin Y., Wang L., Li R., Hu S., Wang Y., Xue Y., Y. H., Jiao Y., Wang Y., and Zhang Y. 2018.** How do root exudates of bokchoy promote dibutyl phthalate adsorption on mollisol? *Ecotoxicol. Environ. Saf.* **161:**129–136.
2. **Nahurira R., Ren L., Song J.L., Jia Y., Wang J.H., Fan S.H., Wang H.S., Yan Y.C. 2017.** Degradation of Di(2-Ethylhexyl) Phthalate by a Novel *Gordonia alkanivorans* Strain YC-RL2, *Curr Microbiol*. **74:**309-319.
3. **Zhang L., Liu J., Liu H., Wang and Zhang S. 2015.** The occurrence and ecological risk assessment of phthalate esters (PAEs) in urban aquatic environments of China. *Ecotoxicol*. **24(5):**967–984.
4. **Toft G., BO A., Jonsson G., and J., P., B. 2012.** Association between pregnancy loss and urinary phthalate levels around the time of conception, *Environ. Health Perspec*. **120(3):** 458–463.
5. **Gao H. T., Xu R., Cao W. X., QianL. L., Wang M., Lu L., Xu Q., and Yu S. Q. 2017.** Effects of six priority controlled phthalate esters with long-term low dose integrated exposure on male reproductive toxicity in rats, *Food Chem. Toxicol.* **101:** 94–104.
6. **Chang B. V., Yang C. M., Cheng C. H., and Yuan S. Y. 2004.** Biodegradation of phthalate esters by two bacteria strains. *Chemosph*. **55:** 533e538.
7. **Net S., Semp R., Delmon T. A., Paluselli A., and Ouddane B. 2015.** Occurrence, fate, behaviour and ecotoxicological state of phthalates in different environmental matrices. *Environ. Sci. Technol*. **49:** 4019e4035
8. **Chen H., Zhang W., Rui B. B., Yang S. M, Xu W. P., and Wei W. 2016.** Di (2-ethylhexyl) phthalate exacerbates non-alcoholic fatty liver in rats and its potential mechanisms. *Environ. Toxicol. Pharmacol.* **42:** 38-44

9. **Chen J. A., Li X., Li J., Cao J., QiuZ., Qing Z., and Shu W. 2007.** Degradation of environmental endocrine disruptor di-2-ethylhexyl phthalate by a newly discovered bacterium, *Microbacterium sp. strain CQ0110Y. Appl. Microbiol.* **74(3):** 676-82.

10. **Meng X., Niu G., Yang W., and Cao X. 2015.** Di (2-ethylhexyl) phthalate biodegradation and denitrification by *Pseudo xanthomonas sp*. Strain. *Bioresour. Technol*. **180:** 356e359.

11. **Pradeep S., Josh M. S., Binod P., Devi R. S., Balachandran S., Anderson R. C., Benjamin S. 2015.** *Achromobacter denitrificans strain* SP1 efficiently remediates di (2-ethylhexyl) phthalate. *Ecotoxicol. Environ. Saf*. **112:** 114e121.

12. **Fang C. R., Yao J., Zheng Y. G., Jiang C., J., Hu L. F., Wu Y. Y., and Shen D. S. 2010.** Dibutylphthalate degradation by *Enterobacter sp.* T5 isolated from municipal solid waste in landfill bioreactor. *Int. Biodeter. Biodeg*. 64: 442e446.

13. **Zhao H., Du H., Lin J., Chen X., Li Y., Li H., Cai Q, Mo C, Qin H and Wong M. 2016.** Complete degradation of the endocrine disruptor di-(2-ethylhexyl) phthalate by a novel *Agromyces sp.* MT-O strain and its application to bioremediation of contaminated soil. *Sci. Tot. Environ*. **562:**170e178

14. **Li C., Chian X. L., Chen Z., X., Deng Z. Y., and Xu H., 2012.** Biodegradation of an endocrine-disrupting chemical di n-butyl phthalate by *Serratia marcescens* C9 isolated from activated sludge. *Afr. J. Microbiol. Res*. **6(11):** 2686–2693, 702e710.

15. **Ting Y., Lei R., Yang J., Shuanghu F., Junhuan W., Jiayi W. R., Nahurira , H., Wang and Yanchun Y. 2018.** Biodegradation of Di-(2-ethylhexyl) Phthalate by *Rhodococcus ruber YC-YT1* in Contaminated Water and Soil. *Int. Journal of Environ. Res. Public Health*. **1(5):** 15-3

16. **Wu N. and Thompson R. 2006.** Fast and efficient separations using reversed phase liquid chromatography. *J. Liq. Chrom. Related Tech*. **29(7-8):** 949-988. doi:10.1080/10826070600574903

17. **Quan C. S., Liu Q., Tian W. J., Kikuchi J., and Fan S. D. 2005.** Biodegradation of an endocrine-disrupting chemical ethylhexyl phthalate, by *Bacillus subtilis* No. 66. *Appl. Microbiol. Biotechnol*. **66(6):**702-10

18. **Thomas Johnson B., Heitkamp M. A., Jones J. R. 1984**. Environmental and chemical factors influencing the biodegradation of phthalic acid esters in freshwater sediments. *Environ Pollut Ser B. Chem Phys*. **8:**101–118. doi:10.1016/0143-148X(84)90021-1

19. **Zhang Y., Chen H., Liu J., Geng G., Liu D., Geng H. and Xiong L. 2016.** Genome sequencing and biodegradation characteristics of the *n*- butyl benzyl phthalate degrading bacterium-*Rhodococcus* sp. HS-D2. *Inter. Biodetr. & Biodegr.* Doi:10.1016/j.ibiod.2016.08.024

20. **Navacharoen A., Vangnai A.S. 2011.** Biodegradation of diethyl phthalate by an organic-solvent-tolerant *Bacillus subtilis* strain 3C3 and effect of phthalate ester coexistence. *Int. Biodeterior. Biodegrad.* **65:**818–826. doi:10.1016/j.ibiod.2011.05.005

21. **Ren L., Jia Y., Ruth N.., Qiao C., Wang J., Zhao B., Yan Y. 2016.** Biodegradation of phthalic acid esters by a newly isolated Mycobacterium sp YC-RL4 and the bioprocess with environmental samples. *Environ. Sci. Pollut. Res*. **23:**1–11. doi:10.1007/s11356- 016-6829-4

22. **Kartal B., Koleva M., Arsov R., vander Star W., Jetten MSM, Strous M. 2006.** Adaptation of a freshwater anammox population to high salinity wastewater. *J Biotechnol.* **126:**546–553. doi:10. 1016/j.jbiotec.2006.05.012

23. **Jin D., Kong X., Cui Z., Bai H. and Zhang. 2013.** Biodegradation of Di-*n*-Butyl Phthalate by a Newly Isolated Halotolerant *Sphingobium* sp. *Inter. J. Mol. Sci. 14***:** 24046-24054.

24. **Cortes-Lorenzo C., Corte´s-Lorenzo C., Rodrý´guez-Dý´az M., Lo´pez- Lopez C., Sa´nchez-Peinado M., Rodelas B., Gonza´lez-Lo´pez J.2012.** Effect of salinity on enzymatic activities in a submerged fixed bed biofilm reactor for municipal sewage treatment. *Bioresour. Technol*. **121:**312–319. doi:10.1016/j.biortech.2012.06

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# Seasonal variation in physico-chemical parameters of the water of River Karamnasa at Buxar, Bihar

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**Abstract-** The quality of river body is determined by its physico-chemical parameters. The physico-chemical parameters refer as Temperature, Transparency, pH, DO, Free $CO_2$, Total Alkalinity, Nitrate, Phosphate, BOD and COD. These parameters provide all necessary informations regarding the suitability of water for its use as well as for improving its quality. The present investigation is done to find out the impact of season on physico-chemical parameters of the water of River Karamnasa at Buxar, Bihar. For this purpose, various physico-chemical parameters were studied during the whole period of investigation from March 2019-Feburary 2020 and noticed the changes on the parameters due to season. The result of present finding showed that the physico-chemical parameters of the water of River Karamnasa Varied seasonally. Temperature and Free $CO_2$ were maximum in summer while these were minimum in winter. pH, Transparency and total alkalinity were recorded maximum in summer while minimum in monsoon season. The DO was maximum in winter while it was minimum in monsoon. Nitrate, Phosphate, BOD and COD were found maximum in monsoon while these were minimum in winter.

**Key words: Physico-chemical parameters, Season change, Karamnasa River, Buxar, Bihar**

## INTRODUCTION

Riverine system is one of the important freshwater resources on the earth. River bodies are the most abundant and familiar liquid that closely related to social, economical and political development of the country. Because, river serves as a source of water supply to meet domestic, industrial, agricultural, fisheries and power generation needs. But, increasing population, rapid growth of industrialization and urbanization deteriorate the quality of river body. The quality of a river body depends upon the different physico-chemical parameters. Physico-chemical parameters also provide all necessary information regarding the suitability of water for its use as well as for improving its quality. It is essential to know the physico-chemical parameters of river body for assessment of its quality. The physico-chemical parameters refer as temperature, Transparency, pH, DO, Free $CO_2$, Total Alkalinity, Nitrate, Phosphate, BOD and COD. These parameters are directly influenced by season.

Various works contributed towards the better understanding of seasonal variation in physico-chemical parameters of aquatic body. Some of them are cited here such as Odum (1971)[1], Das *et al.*(1992)[2], Khaiwal *et*

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*al.*(2003)[3], Srivastava *et al.* (2003)[4], Singh *et al.* (2004)[5], Deshmukh *et al.*(2006)[6],Chaurasia *et al.* (2007)[7], Bhandari *et al*. (2008)[8], Prasad *et al*. (2008)[9], Ahmed *et al.* (2010)[10], Bhatt *et al.*(2010)[11], Pathak *et al.*(2011)[12], Sahni *et al.* (2012)[13], Singh P. (2014)[14], Singh *et al.* (2016)[15], Balkrishna *et al.* (2017)[16], Ramanathan and Asmath (2018) [17].

From a review of literature cited above, it appears that a lot of works on seasonal variation in physico-chemical parameters of water body has already been done. But no work was done to evaluate the seasonal impact on physico-chemical parameters of the water of river Karamnasa at Buxar, Bihar. Thus the present project deals with the seasonal variation in physico-chemical parameters of the water of river Karamnasa.

## MATERIAL & METHOD

River Karamnasa is one of the largest and Perinnial Rivers of Bihar. It originates at a height of 350 meters on the Northern face of Kaimur range near Sarodag in Kaimur district of Bihar. It flows in North-westernly direction through the plain of Mirzapur, then forms the boundary between U.P. and Bihar and finally joins with River Ganga at Chausa (Buxar). River Karamnasa is a tributary of River Ganga and hence shows various similarities in biological features with River Ganga. The total drainage area of river Karamnasa is approx. 4251 sq mile. Along its course, the river carries the discharge of many canals, household and agricultural effluents containing organic and inorganic wastes from adjoining areas.

**Area of study:** To analyse the different parameters, three sampling sites were selected that stretches to a distance of 5 km covering the entire length of river passing through district Buxar upto the confluence with River Ganga. Site I was situated near the village Banarpur, were river carries domestic sewage, agricultural and ricemill effluents. Site II was placed near the village Narbatpur. At this site, the river carries household, washing, bathing, cattle wallowing and agricultural effluents. This site was 2 km downstream freom site I. Site III was about 1 km downstream from site II, near the confluence of River Karamnasa and River Ganga. All the above sites showed a varied ecological conditions and hence helpful to analyse the seasonal variation in physico-chemical parameters.

**Collection and preservation:** The samples were collected in different season i.e. summer (March to June), monsoon (July to October) and winter (November to February) by using plastic canes of size 1 litre. The collected samples were transferred to laboratory with taking precautions. The water samples were preserved at $4^0$C temperature and different physico-chemical parameters were mostly tested within one day of collection. The parameters were analysed by standard method of APHA (2005)[18]. Temperature was calculated by simple thermometer. pH metre was used to find the pH of water. Transparency was noticed by using Secchi-disc method. Wrinkler Idometric method was used to find out DO content. Free $CO^2$, total alkalinity, nitrate, phosphate, BOD and COD were calculated by Titration method.

## RESULT & DISCUSSION

During the whole period of investigation, the result of findings of different physico-chemical parameters of river Karamnasa are shown in table (1). Temperature of river body is directly influenced by solar radiation (Yadav *et al.*, 2008)[19] and photoperiod (Odum,1971)[1]. Both solar radiation and photoperiod were higher in summer than in winter. In present investigation, the temperature was ranged from 21.0°C to 32.6°C. The minimum temperature was 21.0°C at site I in winter season while maximum temperature was 32.6°C at site III in summer. The result was supported by the findings of Khanna *et al.* (2016)[20].

Transparency refers the suspended dirt and other particles in water. It restricts the penetration of light. In the present investigation, maximum transparency was 79.0 at site I in summer season while minimum transparency was 15.9 at site II during monsoon. Transparency showed an inverse relationship with light penetration. In present study, light penetration was high in monsoon while it was low during summer. The similar result was found by Srivastava *et al.*(2003)[4] and Ahmed *et al.*(2010)[10].

pH plays an important role in biological activities. Such activities occur only within a narrow range of pH. The pH of river Karamnasa was maximum as 8.2 at site I in summer season while it was minimum as 7.1 at site II in monsoon. Hence, the pH of present finding was ranged from 7.1 to 8.2. the observed result was supported by the finding of Das *et al.*(1992)[2].

Dissolved Oxygen (DO) of water body directly influenced the survival and distribution of fauna and flora in it. DO of present finding was ranged from 6.6 mg/l to 10.1 mg/l. The maximum DO was 10.1 mg/l at site I in winter and minimum DO was 6.6 mg/l in monsoon at site III. Similar finding was observed by Sahni *et al.* (2012)[13] and Singh (2014)[14].

Free $CO_2$ of a river body was inversely related with concentration of DO (Odum,1971)[1]. In present study maximum Free $CO_2$ was 10.4 mg/l at site II in summer while it was minimum at as 1.0 mg/l at site I in winter season. The maximum level of Free $CO_2$ was due to increase population density of aerobic organisms. Bhandari *et al.* (2008)[8] also found similar result.

Total alkalinity is one of the important parameter which decides the quality of water for the use of irrigation and drinking. It neutralizes acid and hence it assess the presence of pH. In present finding, the total alkalinity was maximum as 195.8 mg/l at site II in summer and it was minimum as 136.5 mg/l at site III in monsoon season. the findings was also supported by Chaurasia *et al.* (2007)[7].

Nitrate concentration in river body is associated with discharge of $N_2$ wastes as well as sewage water. The nitrate content of river Karamnasa was maximum as 0.6 mg/l at site II in monsoon. The minimum value of nitrate was 0.1 mg/l at site I in winter season. The similar result was also found by Deshmukh *et al.*(2006)[6].

Phosphate value of present finding was maximum in monsoon at site II as 4.7 mg/l and value was minimum in winter season at site I as 0.4 mg/l. the above finding was supported by Venkatesharaju *et al.* (2010)[21].

BOD reduces the DO concentration in river body. Excess BOD load damage the quality of river body. Highest BOD of river Karamnasa was recorded 6.2 mg/l at site II in monsoon while minimum BOD was 3.2 mg/l at site II in winter. High value of BOD indicates higher $O_2$ consumption and heavy pollution load. The findings was also supported by finding of Bhatt *et al.*(2010)[11] and Deshmukh *et al.*(2006)[6].

COD value was maximum as 10.9 mg/l at site II in monsoon while it was minimum as 7.1 mg/l at site I in winter season. Singh *et al.* (2014)[14] also found the same result.

**Table 1- Seasonal variation in different physico-chemical parameters from March 2019-Feburary 2020**

| Parameters | Season | Site I | Site II | Site III | Mean |
|---|---|---|---|---|---|
| Temperature($^oC$) | Summer | 30.8 | 31.5 | 32.6 | 31.6 |
| | Monsoon | 29.6 | 28.8 | 27.9 | 28.7 |
| | Winter | 21.0 | 21.2 | 21.1 | 21.1 |
| Transparency(NTU) | Summer | 79.0 | 77.5 | 75.7 | 77.4 |
| | Monsoon | 16.4 | 15.9 | 18.3 | 16.9 |
| | Winter | 70.1 | 72.0 | 70.6 | 70.9 |
| pH | Summer | 8.2 | 7.8 | 8.0 | 8.0 |
| | Monsoon | 7.6 | 7.1 | 7.2 | 7.3 |
| | Winter | 7.3 | 7.7 | 7.9 | 7.6 |
| Dissolved Oxygen (mg/l) | Summer | 8.3 | 7.8 | 7.1 | 7.7 |
| | Monsoon | 6.7 | 7.6 | 6.6 | 7.0 |
| | Winter | 10.1 | 8.5 | 8.3 | 8.9 |
| Free $CO_2$(mg/l) | Summer | 5.0 | 10.4 | 6.9 | 7.4 |
| | Monsoon | 1.3 | 8.7 | 4.9 | 4.9 |
| | Winter | 1.0 | 4.7 | 2.5 | 2.7 |
| Total Alkalinity (mg/l) | Summer | 157.3 | 195.8 | 160.3 | 171.1 |
| | Monsoon | 137.8 | 155.5 | 136.5 | 143.3 |
| | Winter | 136.8 | 156.6 | 142.6 | 145.3 |
| Nitrate(mg/l) | Summer | 0.3 | 0.3 | 0.4 | 0.3 |
| | Monsoon | 0.4 | 0.6 | 0.2 | 0.4 |
| | Winter | 0.1 | 0.3 | 0.2 | 0.2 |
| Phosphate(mg/l) | Summer | 0.8 | 2.3 | 0.9 | 1.3 |
| | Monsoon | 0.8 | 4.7 | 0.8 | 2.1 |
| | Winter | 0.4 | 3.3 | 0.6 | 0.7 |
| BOD(mg/l) | Summer | 4.1 | 4.4 | 5.2 | 4.6 |
| | Monsoon | 5.8 | 6.2 | 6.0 | 6.1 |
| | Winter | 3.5 | 3.2 | 3.8 | 3.5 |
| COD(mg/l) | Summer | 9.3 | 9.4 | 9.5 | 9.4 |
| | Monsoon | 10.3 | 10.9 | 10.6 | 10.6 |
| | Winter | 7.1 | 7.2 | 7.4 | 7.2 |

## CONCLUSION

In the present investigation, it was concluded that seasonal changes are the major factor in variation of the physico-chemical parameters of the water of river Karamnasa. During the whole study period, it was also found that different parameters of this river were within the permissible limits and with the aggrement with Prasad *et al.* (2008)[9], Singh *et al.* (2016)[15] and Ramanathan S. (2018)[17]. The water quality of river Karamnasa is also suitable for human use as well as appropriate for fish culture.

## REFERENCES


1. **Odum E.P. 1971.** Fundamentals of ecology.3rd edition W.B., Philadelphia,pp-554.
2. **Das H.B., Kalita H. & Kannur S.B. 1992.** Physico-chemical characteristics of Brahmaputra water Tezpur. *Pollution Research*.**11(3):**169-172.
3. **Khaiwal, Ravindra & Anubha Kaushik. 2003.** Seasonal variations in physico-chemical characteristics of river Yamuna in Haryana. *Journal of Environmental Monitoring.* **(5):**419-426.
4. **Srivastava, N., Agrawal, M., & Tyagi, A. 2003.** A study of physico-chemical characteristics of water bodies around Jaipur. *Journal of Environmental Monitoring.* **24:**177-180.
5. **Singh M. & Gupta K.C. 2004.** Physico-chemical studies of river Yamuna of Mathura. *Eco. Env. and Conv.* **10(2):**193-196.
6. **Deshmukh, J. U., & Ambore, N. E. 2006.** Seasonal variation in physico-chemical aspects of pollution in Godavari River at Nanded, Maharashtra. India. *J.Aqua.Biol.* **21(2):** 93-96.
7. **Chaurasia, M., & Pandey, G. C. 2007.** Study of physico-chemical characteristics of some water ponds of Ayodhya- Faizabad. *JEP*. **27(1):**1019-1023.
8. **Bhandari N.S. & Nayal K. 2008.** Correlative study on physico-chemical parameters and quality assessment of Kosi river water, Uttarakand. *Electronic Jour. of Chem*. **5(2):**342-348.
9. **Prasad, N. R., & Patil, J. M. 2008.** A study of physico-chemical parameters of Krishna river water particularly in western Maharashtra. *Rasayan J. Chem*. **1(4):**943-958.
10. **Ahmed. 2010.** Water quality assessment of river Gomati at Lucknow (U.P.), India. *Aquaculture*. **11(1):**39-46.
11. **Bhatt M. & Yunus M. 2010.** Seasonal variation of physico-chemical characteristics in several ponds of Lucknow city affected by Urban drainage. *Advance Environmental Biology*. **6(10):** 2654-2663.
12. **Pathak P., Sinha U.S. & Srivastava M.L. 2011.** Physico-chemical quality of river Thora at Buxar. *Modern J. Life Science.* **10(1-2):**55-58.
13. **Sahni K. & Yadav S. 2012.** Seasonal variations in physico-chemical characteristics of Bharawas pond, Haryana. *Asian J. Exp. Sci.* **26(1):**61-61.
14. **Singh P. 2014.** Studies on seasonal variations in physico-chemical parameters of the river Gomati (U.P.), India. *International Journal of Advanced Research.* **2(2):**82-86.
15. **Singh, P. K., Shrivastava, P., & Borana, K. 2016.** Seasonal variations in physico-chemical parameters of river Narmada (M.P.), India. *IJSR*. **5(1):**367-371.
16. **Balakrishnan, S., Chelladurai, G., Mohanraj, J. and J. Poongodi. 2017.** Seasonal variations in physico-chemical characteristics of Tuticorin coastal waters southeast coast of India. *Applied Water Science*. **7(4):**1881-1886.
17. **Ramanathan S. & Asmath A. 2018.** Seasonal variation in physico-chemical parameters of Puthukulam pond, Tamil Nadu. *Res. J. of Life Science.* **(10):**181-187.
18. **APHA. 2005.** Standard method for examination of water and waste water. 21st Edn,
19. **Yadav S.K.,Sharma S.P. and Sahu B. 2008.** Physico-chemical characteristic and phytoplankton of Betwa river, Vidisa district (M.P.), *Indian. J. Environ & Ecoplan.* **15(3):**557-580.
20. **Khanna, D. R., Bhutiani, R., Matta, G., Singh, V., & Bhadauriya, G. 2011.** Physico-chemical property of river Ganga at foothills of Garhwal Himalayas. *Environment Conservation Journal.* **12(3):**163-168.
21. **Venkatesharaju, K., Somashekar, R. K., & Prakash, K. L. 2010.** Study of seasonal variation in surface water quality of Cauvery river stretch in Karnataka. *Journal of Ecology and Natural Environment*. **2(1):**001-009.


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# A study on analysis of gut content in certain freshwater food fishes found in River Kosi

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**Abstract-** Gut content analysis is widely used to find out or to ascertain the food & feeding habit of fish species. To have knowledge about the inter - specific relation between the aquatic fauna & and the productivity of the water body the study of food and feeding habit of the fish is very important. Behavior, condition, energy intake, etc are the ecological components that can be only represented when the diet of fishes are known. In the present study the gut content analysis was done in four fish species *Labeo rohita, Channa punctata, Cirrhinus mrigala* & *Cyprinus carpio*. Volumetric methods & preponderance index method was used for the estimation of diet in various fishes. The result thus obtained shows that dead and decayed organic matter, rotifers, sand and mud etc.

**Key words: Gut content, food fishes, Madhepura, Bihar**

## INTRODUCTION

Fish is regarded as one of the important part of human food since the time immemorial. Harvesting of fish by the fisherman is passed from one generation to other.[1] Survivability of the fish usually depends on the aquatic availability of the organic matter and optimal sized prey in appropriate concentration, lack of which influences the larval survival. Heavy mortality rate has been seen in fishes usually during the time when these fishes transit from endogenous to exogenous mode of feeding. It is regarded as one of the most critical phase of their life because it is the time when they require right type of food for their growth & nourishment. Hence low survival rates are regarded as one of the most important drawback in the successful culture of many marine and freshwater fishes.[2] In India major carps, rohu (*Labeo rohita*) is considered as one of the most intensively cultured species.[3] It account for about 15% of worlds freshwater aquaculture production. *Cirrhinus mrigala* is one of the very fast growing fish which is generally large in size and commonly it is known as mrigala.[4] Knowledge about its feeding habit is very important to evaluate the ecological role and position of fish in the food web of ecosystem.[5] Thus it is very important to understand the ecology as well as ethology of feeding.

The annual fish production of River Kosi in Madhepura region was estimated as 2.9 ton in 2015. The major species of fishes found in this region were generally *Catla catla, Cirrhinus mrigala, Labeo calbasu, Labeo rohita, Labeo bata, Labeo dyocheilus, Puntius sarana,*

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*Cyprinus carpio, Notopterus notopterus* and *Ompok bimaculatus*. However there has been decline in the fish production for last couple of years rated the recruitment of catla as very feeble.[6] Mrigal showed a steep decline in the reservoir and its recruitment was described as moderate to poor. Volumetric methods & preponderance index method was used for the estimation of diet in various fishes.[6,7]

## MATERIALS & METHODS

**Study area**

Different specimens of fish having different age groups were collected from the River Kosi in Madhepura district of Bihar. The specimen of four predatory fish species *Labeo rohita, Channa punctata, Cirrhinus mrigala* & *Cyprinus carpio* were collected by random sampling method from different localities in the River Kosi. Samples of fish species were collected during the period of January 2016 to December 2016 in fresh condition by using the planktonic net.

**Species collection**

Total of 200 specimens of fishes were collected and separated according to different species. Out of the 200 species 80 of the species were found to be *Labeo rohita,* 70 of the species belonged to *Channa punctata*, 30 belonged to *Cirrhinus mrigala* & 20 of it belonged to *Cyprinus carpio*.

For the further study the collected species were then taken to the laboratory in the water medium. In the laboratory the species were identified. Later on the species were dissected under the supervision of the professor in charge. The content in the stomach and intestine which was very partially or totally in undigested state were removed by using forceps and soft brush. The gut content was then kept in petridish for the further analysis. The gut content was then examine and counted under the microscope and the percentage composition was worked out. Index of preponderance, a method was adopted for grading the food element in the stomach during analysis of fishes along with volumetric method.[8]

## OBSERVATION

From the above study it was found that the gut content of 80 fishes belonging to *L. rohita* mainly contained the foods items *closterium* sp., *Pediastrum simplex*, and *spirogyra* sp. among the chlorophyceae, *Nostoc* sp., *Oscillatoria* sp., *Anabaena* sp., *Microcyctis aeruginosa* among the Cyanophyceae; *Synedra ulna* among the Bacillarophyceae were observed. Organic matter along with mud content was also found in significant amount. Since the food of these fishes comprised of micro organisms, individual separation of food items was not possible therefore percentage was used to indicate its value.

In *Channa punctatus* the gut content showed zooplanktons and rotifers in dominance amount. In *Cirrhinus mrigala* the gut content analysis showed large amount dead and decaying organic matter, mud along with other green algae consisting of *Spirogyra* sp., *Pediastrum simplex*, *Closterium* sp & *Diatoms*. In *Cyprinus carpio* the gut content showed large amount of *Daphnia rosea*, *Cyclops insignis*, *Brachionus tridens* tender roots and leaves of higher aquatic plants. Thus percentage of food content found in the four different species of the fishes have been given in the table 1 and in fig; 1.

## RESULT & DISCUSSION

Thus the result obtained from the above research on gut content of the different fish shows that *Labeo rohita, Channa punctata, Cirrhinus mrigala* & *Cyprinus carpio* feed on wide variety of food items. The ability of these fishes to digest some algae and not others largely depends on the nature of the algal cell wall. Gupta (1996)[9] reported that food of *Labeo rohita* mainly composed of algal and crustacean but in the present investigation the food of *Labeo rohita* in River Kosi mainly composed of algae and rotifers., George(1964)[10] observed large quantities of mud

**Table 1- Percentage composition of food items in different fishes as revealed from the gut content**

| Name of the fish | Sand & mud | Green algae | Bluegreen algae | Diatoms | Rotifers | Copepods | Decaying matter | Miscellaneous items |
|---|---|---|---|---|---|---|---|---|
| *L.rohita* | 18 | 23 | 14 | 3 | 12 | 1 | 22 | 6 |
| *C. punctata* | 8 | 7 | 4 | 8 | 26 | 18 | 6 | 2 |
| *C.mrigala* | 18 | 8 | 6 | 7 | 11 | 5 | 33 | 4 |
| *C.carpio* | 8 | 5 | 6 | 3 | 16 | 12 | 40 | 3 |

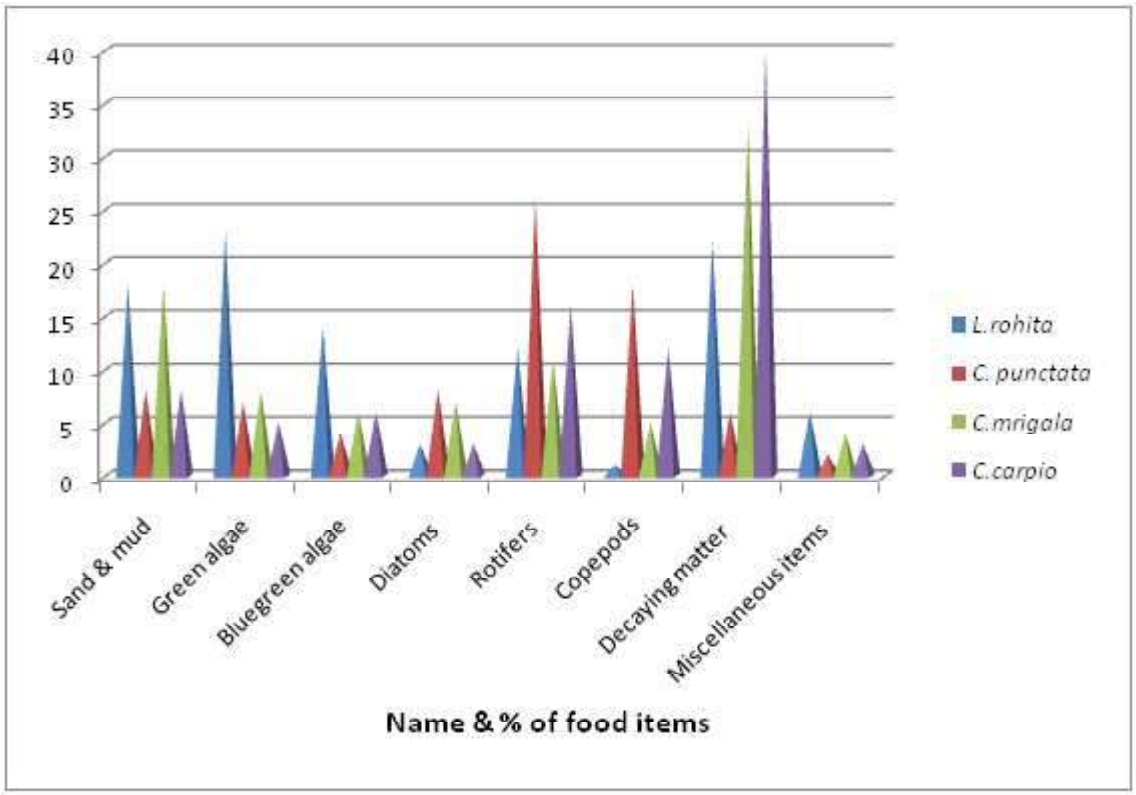



**Fig. 1- Percentage composition of different food items of fresh water fish of River Kosi**

sand , and fragments of decaying plants in addition to the planktonic algal food. According to the above findings also confirms that blue green algae were not utilized as food for many fishes which is also mentioned by Gupta (1996).[9]

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## REFERENCES

1. **Tynsong, H. & B. K. Tiwari 2008**. Traditional knowledge associated with fish harvesting practices of war Khasi community of Meghalaya. Indian *Journal of Traditional Knowledge*. **7(4):** 618-623.
2. **Rao, T.R. 2003.** Ecological and Ethological Perspectives in Larval Fish Feeding. *Journal of Applied Aquaculture.* **13:** 145-178.
3. **Jhingran, V. G. and Pullin, R. S. 1985.** A hatchery manual for the common, Chinese & Indian major carps. *Worlds fish*.
4. **Dwivedi, A.C. Mayank, P. and Pathak, R. K. 2016**. Size composition and exploitation structure of Indian major carp, *Cirrhinus mrigala* (Hamilton, 1822) from the Ganga River, India, *Journal of Fisheries and Life Sciences*. **1(1):** 30-32
5. **Pradhan, S. and Patra, A. 2015.** Seasonal climate change of water quality indices and impact on feeding habits and bio indices of Cirrhinus mrigala. *International Journal of Bioassays*. **4(9):** 4254-4261.
6. **Natarajan A.V., Jhingram A.G. 1961.** Index of preponderance a method of grading the food in the stomach analysis of food. *Indian J. fish*. **8(1):** 54-59.

7. **Pillay TVR. 1952.** A critique of the methods of study of fishes. *Journal of Zoological Society.* **4(2):** 185-200.

8. **Rao.P Yedukondala & Sirisha, I.R. 2005.** Food and feeding habits of Pufferfish *Lagocephalus* & *padiceus* of Visakhapatnam, east coast of India. *Journal of Aqua. Biol.* **20(2):** 150-154.

9. **Gupta R.K. 1996.** Ecology of fresh water pond in Dhanbad with special reference to plankton and weed fauna. Ph.D Thesis (V.B Univ. Hazaribagh).

10. **George, M.R. 1964.** Limnological investigations on ponds plankton, macrofauna and chemical constituents of water and their bearing on fish production. Unpublished Ph. D Thesis, Univ of Calcutta.

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# Significance of aquatic/gill/branchial respiration in *Colisa fasciata* with concurrent exchange mechanism and role of some important factors

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**Abstract-** Gills are the prime respiratory organs in fishes which enable them to remove dissolved oxygen from water. The gills of fishes receive venous blood or de-oxygenated blood through afferent blood vessels and get oxygenated by the absorption of oxygen from water through thin and much vascularised epithelial layers of secondary lamellae of gill filaments. Thus gill is the centre for oxygenation. The gill filaments along with its primary and secondary lamellae provide greater surface area for the exchange of gases. The water with dissolved oxygen passes over the surface of gill filaments consisting of primary and secondary lamellae, oxygen is absorbed through thin wall and carbon dioxide is released into water. The absorbed oxygen comes into blood and then to the different cells for the purpose of intracellular oxidation and liberation of energy for the maintenance of life process. In addition to the gill or branchial respiration, most of the tropical fishes including *Colisa fasciata* show aerial respiration supplementing the gill respiration and involves accessory respiratory organs developed in response to the exceptional environmental conditions which include life in polluted water or life out of water for short period or while taking excursion on land and enable the fish to tolerate oxygen depletion in water during summer season.

**Key words: Gills, *Colisa fasciata,* respiratory organ, epithelial layers, secondary lamellae**

## INTRODUCTION

The gills of *Colisa fasciata* are located in gill/ branchial chamber covered by bony operculum. The respiratory water current enters through the mouth, flows over the surface of gill filaments and then goes out to the outside through the opercular slit or external branchial aperture. The dissolved oxygen of passing water current is absorbed by much vascularised gill filaments and carbon dioxide diffuses out of them into water.

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The aquatic/gill respiration and air breathing in fishes both have to carry the same function, the differences being that the former get the dissolved oxygen of water and later get oxygen of air and both are interdependent supplementing each other. Earlier published research report states that gill area in air breathing fishes is found to be half of the area of the fishes which have only aquatic respiration and has also shown that the gills area in two air breathing fishes *Heteropneustes fossilis* (Singhi) and *Clarias batrachus* (Mangur/Mangoor) is extremely low in comparison to the water breathing fishes. The

physiology of respiration in fishes is similar to that of other vertebrates but water consists of 1/17th amount of oxygen present in the air, so a aquatic breathers require a volume of oxygen containing medium 17 times greater than that required by air breathers. Notable contribution are those of Carter G.S. (1957)[1], Fry,F.E.J,(1957)[2], Beamish F W H (1964)[3], Day F. (1958)[4], Fry F E J and Hart J S (1948)[5], Weber M and De Beaufort L F (1911)[6], Mitra *et al.* (2007)[7], Talwar PK and Jhingran AG.(1991)[8].

## MATERIALS & METHODS

The Banded gourami,*Colisa fasciata* was collected as samples from the different collecting sites of Hardia Chaur (Wetland) situated in Saran district Bihar in neat and clean plastic jars/polythenes filled with water taken from the sites. The collected samples fishes were brought to the laboratory of P.R. College, Sonpur for further study under certain conditions. In the present investigation the sample fishes were collected in summer seasons of a year of this wetland. The sample fishes were dissected with the help of required instruments like forceps, seasors and sharp knife in wax coated tray for anatomical details of gills of *Colisa fasciata*. Following anatomical details correlated with aquatic respiration were displayed:-

**(a) Basic structural design of gills of *Colisa fasciata* as a bony fish:-**

For efficient respiratory function structural design of primary respiratory organ (Gills) of *Colisa fasciata* is well suited according to its habitat and characterised by having

(i) A large surface area for diffusion
(ii) Short diffusion distance and low diffusion barriers i.e., provided with thin epithelium over the gill surface.
(iii) Effective structural support for renewal of the water.
(iv) Much vascularisation i.e., supplied with net work of capillaries.
(v) Effective structural support for continued passage of blood passing beneath the exchange surface.
(vi) A spatial arrangement of the water current and blood current in a counter streaming fashion.

**(b) Typical structure of gill of *Colisa fasciata*:-**

There are four pairs of gills situated in branchial chamber/gill chamber on either side of pharynx and covered with a bony covering , the operculum. Each gill is supported with gill arch bearing two rows of primary gill filaments. Each primary gill filament bears a pair of primary lamellae and are separated & supported by independent gill rays. Each primary lamella is provided with numerous flat leaf like secondary lamellae. The secondary lamellae are the actual site for the exchange of gases and provides ten times more surface area then a fish itself. The secondary lamella is made up of two epithelia which are continuous along with the distal margin of the lamella. The epithelia are held apart by pillar cells and the space between the two epithelia takes the form of a lacunar compartment, the blood spaces.

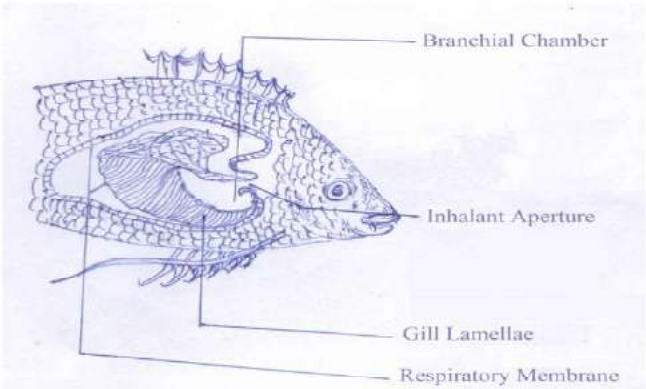



Fig. Location of gills in Colisa fasciata.

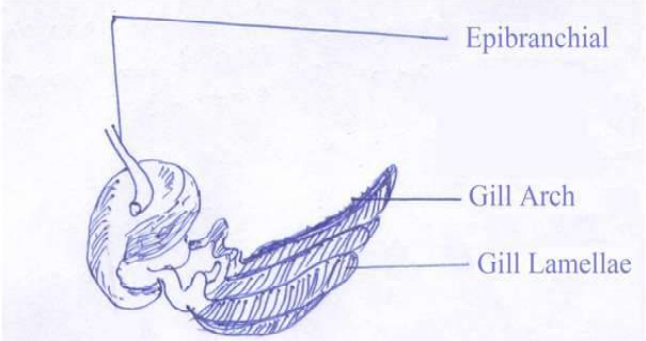



Fig. Anatomical details of gills of Colisa fasciata

**(c) Blood supply to the gills:-**

The gills of *Colisa fasciata* are much vascularised structure. It has got one afferent and two efferent brachial vessels in each gill arch. Each afferent branchial vessel gives off a series of primary afferent branches to the primary gill lamellae. Each primary afferent vessel divides laterally into a no. of secondary vessels which run across gill rays dividing into two-four tertiary branches to supply the secondary lamellae. In secondary lamellae each afferent vessel breaks up into a no. of minute capillaries which

finally join to form a short vessel carrying blood to the primary efferent vessel, thus it collects blood from the secondary lamellae of both sides and finally carries the oxygenated blood to the main efferent branchial vessel of the gill arch for the supply to the whole body directly without going to the heart.

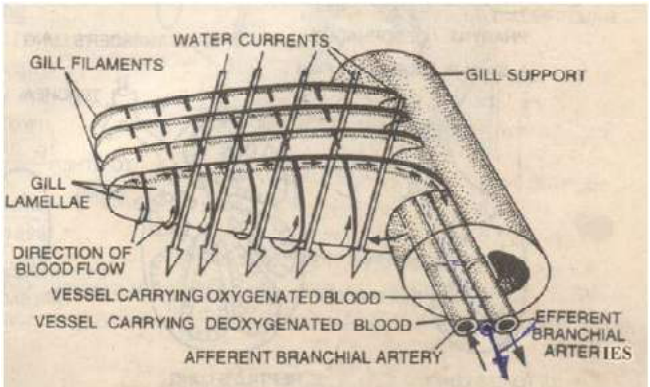


Fig. Direction of blood flow and water flow through the gill filaments in colisa.

**(d) Mechanism of aquatic respiration:-**

The *Colisa fasciata* pumps water over their gills by expanding and contracting the buccopharyngeal cavities in rhythmical manner like other teleosts with the help of muscle activities. During expansion mouth is opened & water enters into branchial chamber and the oxygenated water bathes the highly vascular gill lamellae for exchange of gases. The rhythmical water flow is created by suction of water into the buccopharyngeal cavity and its subsequent expulsion through the gill slits. The buccopharyngeal cavity, therefore, applies both suction and pressure to propel water through the gills. The blood becomes oxygenated by regular intake of oxygen from water by the process of diffusion.

## RESULTS & DISCUSSION

Respiration is most important biological /vital/ metabolic activity of the living beings including *Colisa fasciata* in which biological energy is obtained by the breakdown of respiratory substrate for carrying out all metabolic activities of the body.

**Counter Current Arrangement/Concurrent Exchange Mechanism**.

In fishes including *Colisa fasciata* this mechanism greatly enhances the efficiency of gills as water flows across the gills, in the opposite direction of blood flow. In other way water has its unidirectional path and blood flows opposite to it as a maximum diffusion of gases occurs. About 80-90% of the oxygen is extracted from water by fishes due to this counter current arrangement. Fishes exchange gases by pulling the oxygen rich water content through their mouth and by pumping it across the gills. Once the oxygen has been absorbed by the gills the remaining water is thrown out through the gill slits or openings. The rate of diffusion of gases depends upon the distance (sometimes less than 1 micron) between water flowing outside the gills and blood flowing inside the gills. Less the distance rapid the exchange of gases.

The relative number and size of the gill lamellae determine the respiratory area and varies with the habit of

**Table 1. Showing relative respiratory area of gills**

| Sl.No. | Name of fishes | Gill lamellae/mm of filament | Gill area/Gram of body weight(sq.mm) |
|---|---|---|---|
| 1 | *Colisa fasciata* | 20 | 375 |
| 2 | *Colisa lalia* | 18 | 298 |
| 3 | *Colisa sota* | 15 | 312 |
| 4 | *Heteropneustes fossilis* | 21 | 359 |
| 5 | *Clarias batrachus* | 19 | 295 |
| 6 | *Channa striatus* | 15 | 318 |

**Table 2. Showing effects of temperature, Dissolved Oxygen and pH of water on the opercular beats per minute in *Colisa fasciata***

| Species | Normal condition | Increased temperature | Increased D.O. | pH | Average Range | |
|---|---|---|---|---|---|---|
| | | | | | Low | High |
| *Colisa fasciata* | 80±5 | 6'7 | 13'5 | 10'0 | 80 | 120 |
| *Colisa lalia* | 70±5 | 6'0 | 11'6 | 8'6 | 70 | 105 |
| *Colisa sota* | 60±5 | 6'6 | 13'2 | 10'0 | 60 | 100 |

fishes. Generally fast swimming fishes have more gill area and a large number of gill lamellae per mm of gill filament than the sedentary species. The branchial movements are found to be dependent upon various physic chemical factors of water like concentration of oxygen & carbon dioxide, temperature and pH of water.

## REFERENCES

1. **Carter G.S. 1957.** In 'Physiology of Fishes' ed Brown.**1:**65-69.
2. **Fry, F.E.J. 1957.** 'Physiology of Fishes' ed Brown.**1:**1-63.
3. **Beamish F. W. H. 1964.** Respiration of fishes with special emphasis on standard oxygen consumption. II. Influence of weight and temperature on respiration of several species. *Can. J. Zool*. **42:**177-178
4. **Day F. 1958.** The Fishes of India 1 (London: William Dowson).
5. **Fry F. E. J. and Hart J. S. 1948.** The relation of temperature to oxygen consumption in gold fish. *Biol. Bull.* **94:** 66-77
6. **Weber M. and De. Beaufort L. F. 1911.** The fishes of Indo-Australian Archipelago. ed. E J Brill1 (Holland: Leyden)
7. **Mitra K., Suresh V.R.,Vinci G.K., Mazumdar N.N., Biswas B.K. 2007.** Biology and fishery of banded gourami, *Colisa fasciata* (Bloch and Schneider 1801) in a flood plain wetland of Ganga river basin. *Asian Fisheries Science* **20:**409-423.
8. **Talwar PK & Jhingran AG. 1991.** Inland fishes of India and adjacent countries. *Zoological Society of Bangladesh*. **2:**364-592.

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# Assessment of duration dependent toxic responses induced by dietary administration of phthalate on liver indices, hormonal parameters and associated histopathological changes in reproductive organs of female swiss albino mice

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**Abstract-** Phthalate esters are reported to exert harmful effects on mammalian reproduction and fertility and to elucidate in detail the duration dependant toxic responses of phthalate DEHP, it was orally administered by gavage at the dose of 15 mg /kg b.wt to female swiss albino mice for 4, 6 and 12 weeks period. The observed changes in body weight were statistically not significant ($P>0.05$) after four and six weeks while it was statistically very significant ($P<0.01$) after twelve weeks of DEHP administration in female mice. Progressive decrease in the level of the hormones estrogens and progesterone were recorded and the values were found statistically very significant ($P<0.001$).The increase in serum levels of SGPT and ALP in DEHP treated group of mice was observed from fourth to the twelfth week of the experimental period and the observed values were statistically significant. The present study also showed that the serum levels of albumin and total protein decreased progressively from sixth week to twelfth week and were found to be statistically significant. The biochemical findings were associated with histological changes in the tissues of ovary and uterus of treated group III of mice. The ovarian tissues showed degenerations in the graaffian follicles, stroma and in corpus luteum after 6 and 12 weeks as compared to normal group I and control group II mice. In uterine tissue, lumen with degenerative changes after 6 weeks as well as marked cellular degeneration in the gland & endometrial lining changes after 12 weeks were observed. Duration of exposure to phthalate is a matter of concern apart from dose and the future course of study should focus on examining the effects of phthalates at doses that mimic human exposure.

**Key words: Phthalates, female swiss albino mice, DEHP, reproductive toxicity, liver indices, hormone,histopathology**

## INTRODUCTION

Plasticizers are ubiquitous and many studies have confirmed the existence of plasticizers and their metabolites in air, soil, water and animal species.[1] There are more than 180 types of plasticizers but the most common plasticizer is phthalate. Numerous studies have shown that these chemicals cause endocrine toxicity in mammals and thus make them an emerging concern about

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public health, as they show potential effects on production, development, obesity and other public health problems.[2]

Plasticizers are used in cosmetics, household materials, paints etc. Human population is exposed to these plasticizers in day to day life but women are often exposed to plasticizers than men because of their frequent use of personal care and cosmetic products. Studies have suggested that exposure to phthalates affects human reproductive system, neural growth and cause endocrine disorders, asthma, obesity, autism, diabetes, respiratory effects, and toxicity in many organs including the liver, thyroid, kidneys. and lungs.[3] However, their toxic effects on reproduction and development in human are of major concern at the moment.[4] Numerous animal studies indicate that plasticizers mainly phthalate is an endocrine-disrupting chemical and reproductive toxicant.[5] Evidence has confirmed that endocrine disrupting chemicals (EDCs) have potentially adverse effects on development, growth, metabolism and reproduction, as they may interfere with the production, release, transport, metabolism, binding action or elimination of natural hormones in the body.[6] Developmental toxicity of plasticizersinclude reduced implantations, increased resorptions, decreased fetal body weight, and increased malformations.[7]

Exposure of plasticizers to females especially phthalates can develop many reproductive problems where ovary is the most affected organ.[8] Normal ovarian function is important for reproductive, cardiovascular, neurological, brain and skeletal health.[9] Phthalates have the ability to target the ovary at all stages of growth and development. These toxic effects can lead to premature ovarian failure, anovulation, infertility, and decreased steroidogenesis.[10] High doses of common phthalate, have been shown to suppress serum estradiol levels and rat maturity in adult females.[11] Chronic exposure to phthalate from gestation through pregnancy reduces the fertility of rats.[12] In humans, a report of female workers in a phthalate processing plant found that women exhibited anovulatory cycles in response to high levels of exposure to phthalates.[13]

In addition, high levels of urinary phthalate are associated with pregnancy complications such as anemia, toxemia, and preeclampsia in women.[14] The mechanisms that use phthalates to suppress these endocrine and reproductive factors remain unknown. Interestingly, the ovary is a critical regulator of these processes, and the effects of phthalates on ovarian function remain unclear.[15] Understanding the impact on ovarian function from the exposure to phthalates is very important, especially since the number of women is often exposed to phthalates.[16]

When the ovary is the target of these plasticizers, it will eventually affect the uterus in bringing some abnormal changes[17] and low dose exposure to DEHP promotes histological changes. However, there are few reports of DEHP affecting the uterus of adult female rats during cycling conditions. The effect of DEHP on the uterus at low concentration level is less clear, although it can cause uterine-related defects. The exposure to an environmentally relevant phthalate mixture during pregnancy has caused a wide range of multigenerational alterations in uterine histology.[18] However, Li *et al.* (2020)[18] reported that phthalate compounds did not affect uterine weight, endometrium size, number of glands, or intrauterine or external myometrium thickness in any generation.

Plasticizers can also affect the liver, kidney, and cardiovascular systems, alongwith reproductive organs.[19] Out of which liver is the most targeted organs for phthalate toxicity. In a study it has been concluded that the dietary administration of DEHP (2%) to mice for 10 days resulted in liver dysfunction as evidenced from histological observations and serum analysis.[20] Increased level of glutamic pyruvic transaminase (SGPT) and alkaline phosphatase in serum indicates liver disorders as these enzymes are usually present inside the liver. Albumin and total protein levels are known to be useful in assessing liver function integrity. Therefore, in order to evaluate the liver damage, estimation of these enzymes in serum is essential.[21]

Liver function tests can be performed by measuring the activity of marking parameters such as SGOT, SGPT and ALP, which are initially present with high concentration in the cytoplasm. In the case of hepatocellular damage these enzymes leak into the bloodstream.[22] Elevated levels of these enzyme markers in serum phthalate-treated rats were associated with significant liver damage.[23] Several workers have studied the potency of hepatotoxic phthalates in various animal species. Phthalate is usually removed from the body most of the time and removed within 24 to 72 hours after absorption as reported by Kluwe (1982)[24] but chronic low dose feeding studies in rhesus monkeys has shown long-term retention of this chemical, especially in the liver. Oral administration of phthalate (DEHP) has caused adverse

effects on the liver depending on the type of animal.[25] SGPT and ALP are considered bio-markers of liver function.[26] Serum Glutamic Pyruvate Transminase (SGPT) is specifically localized in hepatocyte cytosol[27] and increased and decreased serum level of it is the specific indicators of hepatocellular necrosis.

In the present study, we investigated the duration dependent toxic responses of phthalate after dietary administration by gavage on liver indices , hormonal parameters and histopathological changes in reproductive organs of female swiss albino mice .

## MATERIALS & METHODS

**Chemical:** Phthalate DEHP was purchased from AccuStandard USA (Marketed by Rankem) and corn oil that was used as vehicle, was procured from Sigma Pvt. Ltd. and Nieschel Chemical Pvt. Ltd. All chemicals used were of Analytical grade.

**Experimental Animal:** Swiss albino mice *Mus musculus* that were purchased from CDRI, Lucknow were reared in the animal house facility of S. S. Hospital & Research Institute, Patna. Adult female mice weighing 30 - 32 grams were selected for study and were allowed to acclimate for at least two weeks. The mice were housed in the stainless steel wire cages at a temperature of (25±1ºC), humidity (56 ±5%) and lighting (12h light / dark cycle). Food and tap water were provided ad libitum. All animal experiments were carried out as per CPCSEA guidelines( Approval No. 1840/PO/ReBi/S/15/CPSCEA).

**In vivo Dosing Regimen :** Mice were divided into three groups of six mice each. After acclimatization, mice of group I (normal) received only food and water *ad libitum*, group II mice (control) were administered corn oil at a dose of 15 ml/kg/b.wt once a day with regular food and water, while mice of group III (treated) were orally administered Phthalate- (Di (2-ethyl hexyl) phthalate), using corn oil as vehicle, at a dose of 15 mg/kg/b.wt by gavage inaddition to food and water ad libitum

**Experimental Design:** Assessment of liver function test parameters and hormonal alterations were conducted after the time intervals of 4,6 and 12 weeks in all the experimental groups of mice. Histological changes in reproductive organs, ovary and uterus were observed after 6 and 12 weeks in all the groups of mice.

**Hormonal Study:** It was done by the ELISA (Elisa plate reader - Tulip diagnostic model - lisaquant elisa plate reader) method. After the time period of 4, 6 and 12weeks, blood sample was collected by ocular puncture from normal, control and treated mice groups. Serum was separated from blood by centrifugation at 3000 rpm for 15 minutes for further biochemical analysis and hormonal assay. At the time of blood collection, body weight was recorded and then blood was collected for serum for the estimation of various biochemical and hormonal parameters.

**Biochemical Analysis:** Biochemical assays related to liver function tests were done by kit method. Estimation of SGPT/ALT/ALAT: By Reitman & Frankel's method, Estimation of Alkaline Phosphatase: By Mod. Kind & King's method, Estimation of total Protein: By Biuret Method, Estimation of Albumin in serum: By BCG method.

**Histopathology:** Tissue samples of ovary and uterus of experimental groups of mice were fixed in 10% neutral buffered formalin, were washed, dehydrated withethanol, and embedded in paraffin. Tissue sections of 4to 5µm thickness were cut and stained with hematoxylinand eosin (H&E) for light microscopic examination.

**Statistical Analysis:** Data were analysed digitally using the SPSS version software 11.0 of statistical software package. All values are expressed in mean ± SEM. The effects of treatment over time were compared between the control and management groups by covariance analysis. P values of less than 0.05 were considered statistically significant.

## RESULTS & DISCUSSION

In the present study,experimental animal female Swiss albino mice was used to assess the toxic potential of orally administered phthalate DEHP,orally administered at the dose of 15mg/kg/b.wt. on hormonal indices, liver function test parameters and histological alterations in its reproductive organs. About 40-52% mortality was recorded among the DEHP treated mice during the experimental period of twelve weeks. Negligible deaths were recorded during the experimental period of twelve weeks in the control groups, which indicated that corn oil at the dose of 15mg/kg/ body weight can be used as a vehicle for toxicological studies. In the experimental period, a progressive decrease in the body weight was observed after four weeks of oral administration of DEHP at a dose of 15mg/kg/b. wt/day.The changes in body weight were

statistically not significant (P>0.05) in four and six weeks while it was statistically very significant (P<0.01) in twelve weeks DEHP treated female mice (Table 1). Body weight is an important indirect indicator that fully reflects the toxicity of substances and can be used to assess the effect of phthalate in the rat growth environment. It has been reported that phthalate can limit body weight gain by disrupting fat metabolism and synthesis.[28] Previous reports have shown that high levels of DEHP (> 1000 mg / kg / day) adversely affect body weight and body weight in adult female rats.[29]

The current study showed an increase in serum SGPT levels in phthalate-treated groups from week six but compared to week six there was not much change that could be seen in the twelve weeks treated group (Table 2). The result suggests that increased SGPT levels may be due to hepatocellular necrosis which causes an increase in the permeability of the cell membrane leading to the release of transaminase into the bloodstream and possibly an enzyme leak in the damaged plasma membrane or an increase formation of enzyme in the liver. In our study the serum ALP level also showed a similar increase in DEHP treated groups of female mice (Table-2).

ALP (Alkaline Phosphatase) is histochemically found in the microvilli of the bile canaliculi and in the sinusoidal portion of hepatocytes[30] and separates various phosphate esters at an alkaline pH and mediates membrane transport[31] and its inactivity leads to damage to the hepatic cell membrane.[32]

High levels of alkaline phosphate occur in cholestatic disorders and that elevation occurs as a result of both intra hepatic and extra hepatic obstruction to bile flow as reported by Friedman *et al.*, (2003)[33]. These enzymes are associated with transmembrane transport, ion transport, ionic energy retention and cell growth in the organ. Significant effects on the activity of acid and alkaline phosphatases can be allocated to the destruction of all membranes and lysosomes which can also cause tissue damage.[34]

Present research has shown an increase of serum ALP levels in groups of phthalate-treated mice exhibiting chronic phthalate toxicity at selected doses as described in Table 2. It was also reported by Jain *et al.* (2009)[23] that parameters of liver function tests such as SGPT and ALP increased significantly after administration of phthalate (DEHP) (1000mg/kg b.wt./day) in the rat. Therefore, it can be assumed that phthalate (DEHP) toxicity can cause intrahepatic and extrahepatic obstruction in the bile duct.

The present study showed that the serum level of albumin decreased progressively from sixth week to twelfth week. The total protein level also decreases in DEHP treated mice. The decreased value of total protein serum level in treated mice was found to be statistically significant in six week and twelve week of treated mice (Table-2).

Phthalate is also a known hepatotoxin in animals and belongs to a class of chemicals called peroxisome proliferators (PPs)[35] as it produces liver hypertrophy.[36] Albumin is a plasma protein, made only by the liver[30] and its serum level is affected by liver disease, nutritional status and hormonal imbalances. Albumin maintains the fluid volume within the vascular space and its low value is a sign of poor health. Recent studies have shown that serum albumin levels and total protein levels decreased by the sixth and twelfth week in mice treated with phthalate at a selected dose as shown in Table 2.

Jain *et al.* (2009)[23] also observed a decrease in total protein levels in mice treated with phthalate (DEHP) (1000mg / kg / b.wt. / day). Total proteins represent the total amount of albumin and globulin. Plasma proteins are the function of a healthy diet which is one of the factors affecting the state of animal health.[37]

The above mentioned results may be due to hepatic tissue damage leading to an increase in cell membrane permeability leading to excessive release of these enzymes in the blood circulation. Since the liver is the centre of the toxin in the body a lot of foreign matter enters the body. The phthalate (DEHP) (15mg / kg b.wt./day) given to Swiss albino mice probably caused toxic effects on the liver.It is therefore, suggested that phthalate (DEHP) containing plastic consumer products should be used with caution.

The biochemical findings were associated with histopathological changes in the ovarian and uterus tissues stained by hematoxylin and eosin stains and observed under light microscope. The results of the present study also showed histopathological changes in the ovarian and uterine specimen of female mice treated with phthalate DEHP in the form of degenerated graaffian follicles, stroma and corpus luteum. These results were consistent with those of Takai *et al.* (2009)[29] who observed vacuolations of stromal cells in female rats receiving 300

mg / kg or more of DEHP and an increase in large atretic follicles in groups 1,000 and 3000 mg / kg doses. Female mice exposed to DEHP have led to a decrease in the number of growing follicles (primary, secondary and antral follicles), and an abnormal increase in estrous cycles and granulosa cell apoptosis.[38] Somasundaram *et al.* (2016)[17] suggested that DEHP treatment reduced the number of endometrial glands and disrupted their formation which is in support of our results. Similar to our work, Liu *et al.*, (2017)[39] have reported that plasticizers are considered a toxic hazard and can produce several endocrine and reproductive disorders in mice, in addition to disrupting the structural and functional aspects of female reproduction, it alters cellular, endocrinological, cytological and biochemical factors and, in turn, causes abnormalities in the ovarian cycle and reproduction .

In the present study, female mice treated with DEHP at the dose of 15mg/kg/b.wt showed progressive decrease in the level of the hormone estrogen and progesterone respectively (Fig.1 and 2). Sex steroids contribute to the growth, differentiation, and functioning of the female reproductive system. During the estrous cycle, variable levels of estradiol and progesterone produce adverse effects on epithelial proliferation and cytodifferentiation of these organs.[40] Estrogen is responsible for the development and control of female fertility and secondary sex characteristics. With regard to the reproductive role of estrogens, they stimulate follicular growth and maturation.

According to Holesh *et al.* (2020)[41], progesterone helps in maintaining pregnancy and implants the egg in the uterus.In the present study, mice treated with phthalate showed decrease in the levels of the hormones estradiol and progesterone which may have increased uterine cell proliferation, which is a major target for ovarian hormones as shown in Fig 1 and 2. Some reports also shown a decrease in E2 levels of proestrus from 50 mg of DEHP / kg / day and are particularly significant at 300 mg of DEHP / kg / day.[42]

The amount of phthalates can also be a source of controversy over the results. Phthalates, like hormones, act on their physiological effects on low rather than in high levels. This phenomenon is called non-monotonic toxicity.[43] This type of toxicity has been demonstrated in various studies.[44] For example, Hannon *et al.* (2015)[15], has shown that exposure to phthalate (DEHP) at 10 µg / mL has increased levels of Cyp19a1, Hsd17b1 and exposure to phthalate (DEHP) to 100 µg / mL has reduced levels of those enzymes in cultured mouse antral follicles. In contrast, some studies indicate that phthalates cause specific toxicity.[45] For example, with an increased dose of phthalate (DEHP) in Sprague-Dawley mice, testosterone levels, FSH and LH levels decreased. In some animal species, they appear to be less sensitive to phthalate-induced toxins, and part of this variation may be allocated to differences in phthalate biotransformation.[46] It was noted that in female Sprague-Dawley mice, prenatal exposure of phthalate (DEHP) at 300 mg / kg / day caused a decrease in estradiol levels.[47]

On the other hand, Brehm *et al.* (2018)[48], observed elevated estradiol levels in female CD mice after prenatal exposure to phthalate (DEHP) at 20, 500, 750 mg / kg / day. Exposure to phthalates can cause a variety of toxic effects depending on the sex of the animal, as there are intersexual differences in the individual isoforms of biotransformation enzymes. There are specific intersexual differences in the activities of those enzymes as well.[49]

The timing of exposure to phthalates can be a source of variability. Prenatal exposure to phthalates can have very serious side effects because pregnancy is a sensitive window of toxic exposure due to the development of the baby.[50] According to Meltzer *et al.* (2015)[42], prenatal exposure to phthalate lowers estradiol levels in female Sprague-Dawley rats. In contrast, in a study by Brehm *et al.* (2018)[48], increased estradiol levels in female CD mice have been reported after prenatal exposure to DEHP. From the present study we observe that the duration of exposure to phthalate is a matter of concern apart from dose and future course of study should focus on examining the effects of phthalates at doses that mimic human exposure.

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**Table 1: Changes in body weight in normal, control and treated groups of mice**

| SL No. | Parameter in (g) | Gr-I (Normal female mice) | Gr-II (Corn oil treated female mice) | | | Gr-III (DEHP treated female mice) | | |
|---|---|---|---|---|---|---|---|---|
| | | | 4 weeks | 6 weeks | 12 weeks | 4 weeks | 6 weeks | 12 weeks |
| 1. | Body weight | 30.02± 0.22 | 30.68± 0.23 | 27.16 ± 0.98 | 25 ± 0.63 | 29.02 ± 0.26 | 24.16$^{**}$ ± 0.75 | 22.3$^{***}$ ± 1.03 |

Values expressed as mean ±SEM. P>0.05 was considered statistically significant; P>0.01$^{*}$, P>0.001$^{**}$, P>0.0001$^{***}$

**Table 2: Liver Function test parameters in normal, control and treated groups of mice**

| SL No. | Liver Function test Parameters | Gr-I (Normal female mice) | Gr-II (Corn oil treated female mice) | | | Gr-III (DEHP treated female mice) | | |
|---|---|---|---|---|---|---|---|---|
| | | | 4 weeks | 6 weeks | 12 weeks | 4 weeks | 6 weeks | 12 weeks |
| 1. | SGPT (IU/L) | 30.5 ± 1.081 | 29.05+1.095 | 28.80±1.34 | 26.87±1.330 | 30.99± 0.89 | 33.6$^{*}$ ±1.23 | 38.6$^{**}$ ±0.61 |
| 2. | ALP (IU/L) | 218.9 ± 33.300 | 217.80+20.071 | 200 ± 20.33 | 218.91±12.567 | 224± 13.22 | 230.2$^{*}$ ± | 275.8$^{***}$ ± |
| 3. | TP (gm/dl) | 6.09 ± 0.016 | 6.02 + 0.456 | 6.11±0.232 | 6.08±0.454 | 5.81± 0.534 | 5.99$^{*}$ ±0.77 | 5.10$^{**}$ ± 0.55 |
| 4. | Alb (gm/dl) | 4.02 ± 0.123 | 3.94± 0.340 | 3.90± 0.476 | 3.41±1.232 | 3.76 ±1.45 | 3.21$^{**}$ ±1.34 | 2.90$^{***}$ ±0.52 |

Values are given as mean ± SEM for 6 animals each. Values expressed as mean ±SEM. P>0.05 was considered statistically significant; P>0.01$^{*}$, P>0.001$^{**}$, P>0.0001$^{***}$

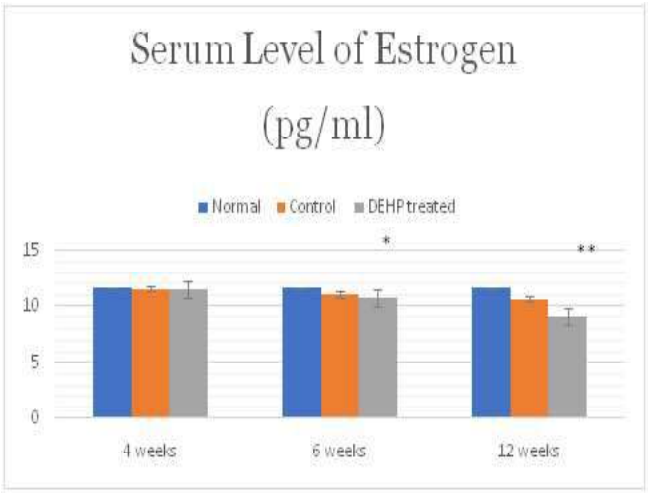



**Fig 1: Showing serum level of hormone estrogen after 4, 6 and 12 weeks in normal, control and treated groups of mice (*considered significant and ** considered very significant)**

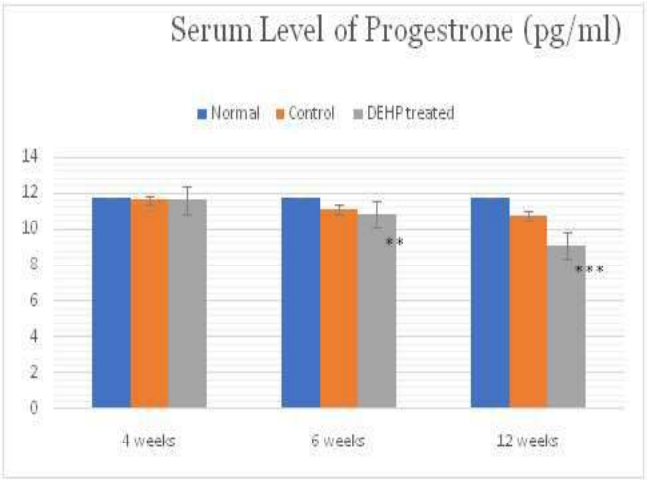



**Fig 2: Showing serum level of progestrone after 4, 6 and 12 weeks in normal, control and treated groups of mice (**considered very significant and *** considered very significant)**

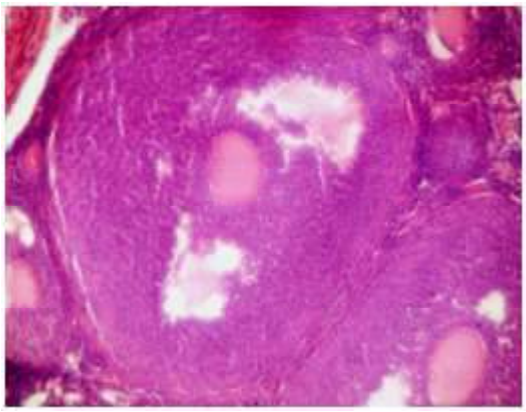


**Fig 3a: Photomicrograph of C.S of ovary of normal mice showing normaldevelopment of graafian follicle (H&E)×400**

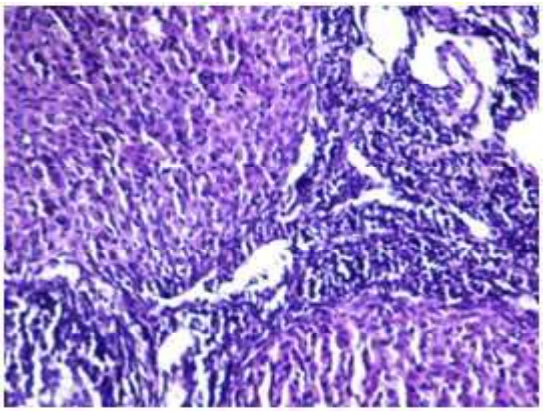


**Fig-3d: Photomicrograph of C.S. of ovary of 12 week DEHP treated mice showing degeneration in graafian follicles ,stroma and in corpusluteum (H&E)×400**

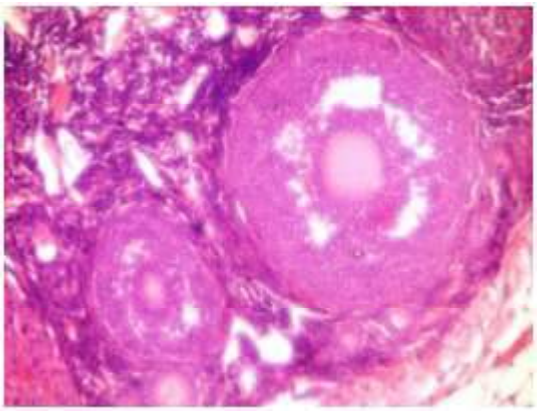


**Fig-3b: Photomicrograph of C.S. of ovary of Control mice of 6 week corn oil treated mice showing very mild degeneration in graafian follicle . (H&E)×400**

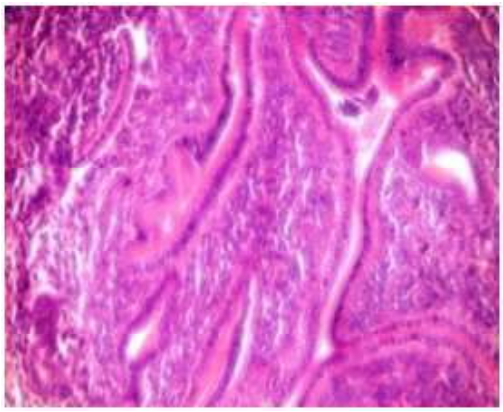


**Fig-4a: Photomicrograph of C.S. of uterus of normal mice showing uterine lining with glandular part (H&E)×400**

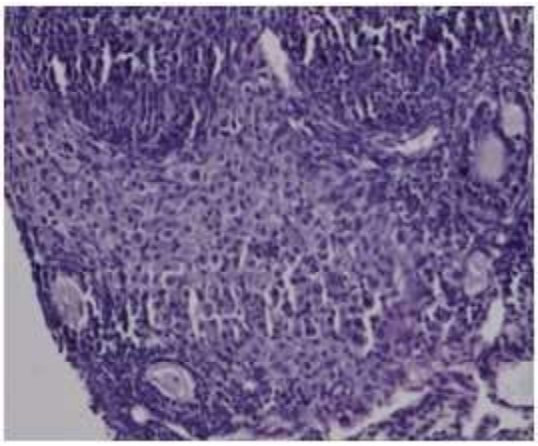


**Fig-3c: Photomicrograph of C.S. of ovary of 6 week DEHP treated mice showing degeneration in graafian follicle and in corpusluteum (H&E)×400**

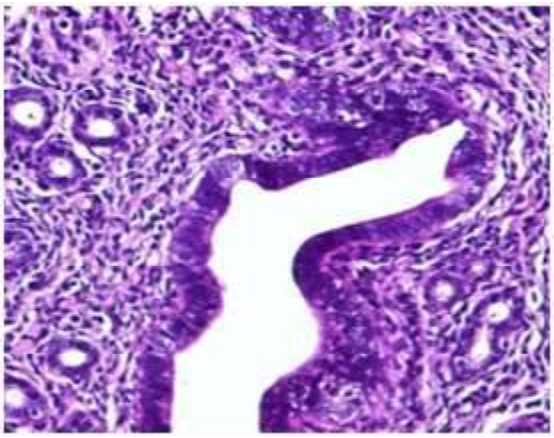


**Fig-4b: Photomicrograph of C.S. of uterus of 6 weeks corn oil treated mice showing uterine lining with glandular part (H&E)×400**

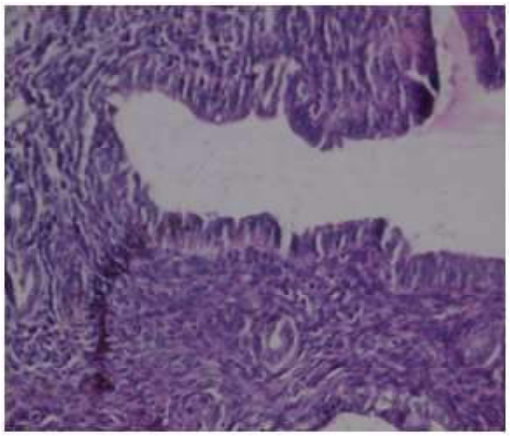


**Fig-4c: Photomicrograph of C.S. of uterus of 6 weeks DEHP treated mice showing uterine lumen with degenerative changes (H&E) × 400**

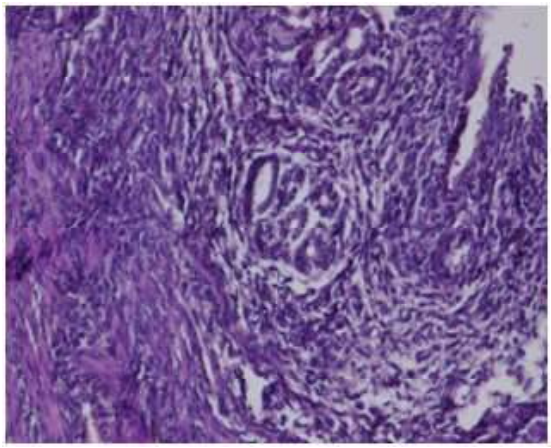


**Fig-4d:Photomicrograph of C.S. of uterus of 12 weeks DEHP treated mice showing marked cellular degeneration in the gland &endometrial lining changes (H&E) × 400**

## REFERENCES


1. **Horn, O., Nalli, S., Cooper, D., Nicell, J. 2004.** Plasticizer metabolites in the environment. *Water Res*. **38:**3693–3698.
2. **Dutta S, Haggerty DK, Rappolee DA and Ruden DM. 2020.** Phthalate Exposure and Long-Term Epigenomic Consequences: A Review. *Front. Genet*. **11:**405.
3. **Chen CC, Wang YH, Wang SL, Huang PC, Chuang SC, Chen MH, Chen BH, Sun CW, Fu HC, Lee CC, Wu MT, Chen ML, Hsiung CA. 2017.** Exposure sources and their relative contributions to urinary phthalate metabolites among children in Taiwan. *Int. J. Hyg. Environ. Health*. **220:**869–879.
4. **Cheon YP. 2020.** Di-(2-ethylhexyl) Phthalate (DEHP) and Uterine Histological Characteristics. *Dev Reprod.* **24(1):**1-17.
5. **Wang W, Craig ZR, Basavarajappa MS, Hafner KS, Flaws JA. 2012.** Mono-(2-ethylhexyl) phthalate induces oxidative stress and inhibits growth of mouse ovarian antral follicles. *Biol Reprod*. **87(6):**152.
6. **Nassouri AS, Archambeaud F and Desailloud R. 2012.** Endocrine disruptors: Echoes of congress of endocrinology in 2012. *Ann Endocrinol* (Paris). **73:** S36-S44.
7. **Lovekamp-Swan T, Davis BJ. 2003.** Mechanisms of phthalate ester toxicity in the female reproductive system. *Environ Health Perspect*. **111(2):**139-45.
8. **Martino-Andrade, A.J. and Chahoud, I. 2010.** Reproductive toxicity of phthalate esters. *Mol. Nutr. Food Res*. **54:** 148-157.
9. **Couse JF, Yates MM, Walker VR, Korach KS. 2003.** Characterization of the hypothalamic-pituitary-gonadal axis in estrogen receptor (ER) Null mice reveals hypergonadism and endocrine sex reversal in females lacking ERalpha but not ERbeta. *Mol Endocrinol*. **17(6):**1039–53.
10. **Bhattacharya P, Keating AF. 2012.** Impact of environmental exposures on ovarian function and role of xenobiotic metabolism during ovotoxicity. *Toxicol Appl Pharmacol*. **261(3):**227–35.
11. **Moyer B, Hixon ML. 2012.** Reproductive effects in F1 adult females exposed in utero to moderate to high doses of mono-2-ethylhexylphthalate (MEHP). *Reprod Toxicol*. **34(1):**43-50.
12. **Gray Jr LE, Laskey J, Ostby J. 2006.** Chronic di-n-butyl phthalate exposure in rats reduces fertility and alters ovarian function during pregnancy in female Long Evans hooded rats. *Toxicol Sci.* **93(1):**189–95.

13. **Aldyreva MV, Klimova TS, Iziumova AS, Timofeevskaia LA. 1975.** The effect of phthalate plasticizers on the generative function. *Gig Tr Prof Zabol*. **12:**25-9.

14. **Tabacova SLR, Balabaeva L. 1999.** Maternal exposure to phthalates and complications of pregnancy. *Epidemiology*.**10.**

15. **Hannon P.R., Brannick K.E., Wang W., Gupta R.K., Flaws J.A. 2015.** Di(2-ethylhexyl) phthalate inhibits antral follicle growth, induces atresia, and inhibits steroid hormone production in cultured mouse antral follicles. *Toxicol. Appl. Pharmacol*. **284:**42–53.

16. **Hogberg J, Hanberg A, Berglund M, Skerfving S, Remberger M, Calafat AM. 2008.** Phthalate diesters and their metabolites in human breast milk, blood or serum, and urine as biomarkers of exposure in vulnerable populations. *Environ Health Perspect*. **116(3):**334–9.

17. **Somasundaram DB, Manokaran K, Selvanesan BC, Bhaskaran RS. 2017.** Impact of di-(2-ethylhexyl) phthalate on the uterus of adult Wistar rats. *Hum Exp Toxicol*. **36(6):**565-572.

18. **Li K, Liszka M, Zhou C, Brehm E, Flaws JA, Nowak RA. 2020.** Prenatal exposure to a phthalate mixture leads to multigenerational and transgenerational effects on uterine morphology and function in mice. *Reprod Toxil.* **93:**178-190.

19. **Li N, Liu K, Yuan H. 2015.** The effect of mono-(2-ethylhexyl)phthalate on apoptosis of rat ovarian granulosa cells in vitro. *Environmental Toxicology and Pharmacology*. **39:** 643–650.

20. **Adibi JJ, Perera FP, Jedrychowski W, Camann DE, Barr D. 2003.** Prenatal exposures to phthalates among women in New York City and Krakow, Poland. *Environ Health Perspect.* **111:** 1719–1722.

21. **Singh A., Singh J.K., Kumar R and Tanuja. 2012.** Chronic Effects of Di (2-Ethylhexyl) Pthalate on Biochemical Indices of Liver in Female Swiss Albino Mice *Mus musculus*. *Asian Journal of Sci. and Tech*. 0976-3376.

22. **Sallie R, Tredger JM, William R. 1991.** Drugs and the liver. *Biopharm. Drug Dispos*. **12:** 251-255.

23. **Jain G. C., Pareek H, Khajja B.S., Jain K., Jhalani S., Agarwal S. and Sharma S. 2009.** Modulation of di- (2- ethylhexyl) phthalate induced hepatic toxicity by *Apium graveolens* L. seeds extract in rats. *African Journal of Biochemistry Research*. **3(5):**222-225.

24. **Kluwe, W.M., Hasemean, J.K., Douglas, J.F. and Huff, J.E. 1982.** The carcinogenicity of dietary di (2 ethylhexyl) phthalate (DEHP) in Fischer 344 rats and B6C3F1 mice. *J. Toxicol Environ Health*. **10:**797-815.

25. **Carpenter, D.,Weil,C.S.and Smyth,S.F. 1953.** Chronic oral toxicity of di(2-ethylhexyl) phthalate for rats, guinea pigs and dogs. *Arch Ind Hyg*. **8:** 219-226.

26. **Mazumder, U.K., Gupta, N., Chakrabarti, S. and Pal, D. 1999.** Evaluation of hematological and hepato renal functions of methanolic extract of *Moringa oleifera* (Lam) root treated mice. *Ind. J. Expl. Biol*. **37:** 612-614.

27. **Rosen, H.R. and Keefe, E.B. 2000.** Evaluation of abnormal liver enzymes, use of liver test and the serology of viral hepatitis: Liver disease diagnosis and management. 1st ed New York Churchill living stone publisher. 24-35.

28. **Itsuki-Yoneda A, Kimoto M, Tsuji H. 2007.** Effect of a hypolipidemic drug, Di (2-ethylhexyl) phthalate, on mRNA-expression associated fatty acid and acetate metabolism in rat tissues. *Biosci Biotechnol Biochem* **71:** 414–420.

29. **Takai R, Hayashi S, Kiyokawa J. 2009.** Collaborative work on evaluation of ovarian toxicity. Two or four-week repeated dose studies and fertility study of di-(2-ethylhexyl) phthalate (DEHP) in female rats. *J Toxicol Sci.* **34(1):**111–119.

30. **Rosalki, S.B., Mcintyre, N. 1999.** Biochemical investigations in the management of liver disease. Oxford textbook of clinical hepatology. 503-521.

31. **Smith M, Weiss MJ, Griffin CA. 1988.** Regional assignment of the gene for human liver/bone/kidney alkaline phosphatase to chromosome 1 p36.1–p34. *Genomics*. **2:**139–143.

32. **Sharma U, Pal D, Prasad R. 2014.** Alkaline phosphatase: an overview. *Indian J. Clin Biochem*. **29(3):**269-278.

**33. Friedman, S.F., Martin, P. and Munoz, J.S. 2003.** Laboratory evaluation of the patient with liver disease. Hepatology, a text book of liver disease. *Philadelphia Saunders publication*. **1:**661-709.

**34. Bull H, Murray PG, Thomas D, Fraser AM, Nelson PN. 2002.** Acid phosphatases. *Mol Pathol*. **55(2):**65-72.

**35. Moody, D.E. and Reddy, J.K. 1978.** Phthalate ester as Peroxisome Proliferator Carcinogens. *Toxicol Appl Pharmacol.* **45:**497-504.

**36. Rusyn, I., Peter, J.M. and Cunnigham, M.L. 2006.** Modes of action and species – specific effects of Di (2 ethylhexyl) phthalate in the liver. *Crit Rev Toxicol*. **36:** 459-479.

**37. Igwebuike, J.U., Anugwa, F.O.I., Raji, A.O., Ehioba, N.G. and Ikuriiz, S.A. 2008.** Nutrient digestibility, hematological and serum biochemical indices of rabbits fed graded levels of *Acacia albida* pods. *J Agricul Biol Sci.* **3(4):** 33-40.

**38. Xu C, Chen JA, Qiu Z, Zhao Q, Luo J, Yang L, Zeng H, Huang Y, Zhang L, Cao J, Shu W. 2010.** Ovotoxicity and PPAR-mediated aromatase down regulation in female Sprague-Dawley rats following combined oral exposure to benzo[a]pyrene and di-(2-ethylhexyl) phthalate. *Toxicol Lett.* **199(3):**323-32.

**39. Liu, J.C.; Lai, F.N.; Li, L.; Sun, X.F.; Cheng, S.F.; Ge, W.; Wang, Y.F.; Li, L.; Zhang, X.F.; De Felici, M.; Dyce, P.W. and Shen, W. 2017.** Di (2- ethylhexyl) phthalate exposure impairs meiotic progression and DNA damage repair in fetal mouse oocytes in vitro. *Cell Death Dis*. **8(8):**e2966.

**40. Boutin EL and Cunha GR. 1997.** Estrogen-induced epithelial proliferation and cornification are uncoupled in sinus vaginal epithelium associated with uterine stroma. *Differentiation*. **62:** 171–178.

**41. Holesh JE, Bass AN, Lord M. Physiology. January, 2021.** Ovulation. [Updated 2020 Aug 22]. In: StatPearls [Internet]. Treasure Island (FL): *Stat Pearls Publishing*.

**42. Meltzer, D., Martinez–Arguelles, D. B., Campioli, E., Lee, S., & Papadopoulos, V. 2015.** In utero exposure to the endocrine disruptor di(2-ethylhexyl) phthalate targets ovarian theca cells and steroidogenesis in the adult female rat. Reproductive *Toxicology*. **51:** 47–56.

**43. Vandenberg L.N., Colborn T., Hayes T.B., Heindel J.J., Jacobs D.R., Lee D.H. 2012.** Hormones and Endocrine Disrupting Chemicals: Low-Dose Effects and Nonmonotonic Dose Responses. *Endocr. Rev.* **33:**378–455.

**44. Lv Y., Dong Y., Wang Y., Zhu Q., Li L., Li X. 2019.** Benzyl butyl phthalate non-linearly affects rat Leydig cell development during puberty. *Toxicol. Lett.* **314:**53–62.

**45. Liu T., Jia Y., Zhou L., Wang Q., Sun D., Xu J. 2016.** Effects of Di-(2-ethylhexyl) Phthalate on the Hypothalamus–Uterus in Pubertal Female Rats. Int. J. Environ. Res. Public Health. **13:**1130.

**46. Ambe K., Sakakibara Y., Sakabe A., Makino H., Ochibe T., Tohkin M. 2019.** Comparison of the developmental/reproductive toxicity and hepatotoxicity of phthalate esters in rats using an open toxicity data source. *J. Toxicol. Sci.* **44:**245–255.

**47. Martinez-Arguelles D.B., Guichard T., Culty M., Zirkin B.R., Papadopoulos V. 2011.** In Utero Exposure to the Antiandrogen Di-(2-Ethylhexyl) Phthalate Decreases Adrenal Aldosterone Production in the Adult Rat. *Biol. Reprod.* **85:**51–61.

**48. Brehm E., Rattan S., Gao L., Flaws J.A. 2018.** Prenatal exposure to Di (2-ethylhexyl) phthalate causes long-term transgenerational effects on female reproduction in mice. *Endocrinology*. **159:**795–809.

**49. Johnson K.J., Heger N.E., Boekelheide K. 2012**. Of mice and men (and rats): Phthalate-induced fetal testis endocrine disruption is species-dependent. *Toxicol. Sci.* **129:**235-248.

**50. Kelley A.S., Banker M., Goodrich J.M., Dolinoy D.C., Burant C., Domino S.E. 2019.** Early pregnancy exposures to endocrine disrupting chemical mixtures are associated with inflammatory changes in maternal and neonatal circulation. *Sci. Rep.* **9:**1–14.

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# Hormonal effects on oxygen consumption in an air breathing snakehead fish, *Channa punctata*

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**Abstract-** An attempt has been made to study the effects of hormones on dual mode of oxygen consumption in an air breathing snakehead fish, *Channa punctata* in this present investigation. The treatment of Hydrocortisone and Thyroxine in *Channa punctata* has showed increase in aquatic, aerial and total oxygen uptake. Adrenaline on the other hand showed a slight decrease in aquatic but an abrupt and higher decrease is seen in case of aerial and total oxygen uptake. Progesterone caused significant increase in aquatic but decrease in aerial and total oxygen uptake. Testosterone increased the aquatic and total oxygen uptake but decreased aerial oxygen uptake. The complete details are provided in this paper.

**Key words: Hormones, Endocrine activity, snakehead fish, *Channa punctata,* oxygen**

## INTRODUCTION

Hormones are chemical agents which are released from a group of cells ad its secretions travel via the bloodstream to affect a different group of cell, or the target organ.[1] Ectothermic animals adapt to their range of external environment via the help of endocrine glands which have effect the action of development, growth and reproduction and change to adapt successfully along with their surroundings. Fishes with bimodal respiration differ in the extent of their reliance on air breathing to support aerobic metabolism, which is reflected in their lifestyles and ecologies. Many freshwater species undertake seasonal and reproductive migrations that presumably involve sustained aerobic exercise. The metabolic activities of these fishes depend upon a variety of factors, hormones being the ones with top priority.[2] Endocrine activity of fishes was correlated with the metabolic activity like respiratory metabolism by some important workers like Matty (1957)[3], Punt and Jungbeod (1963)[4] Kaya (1973)[5]. Exact information about air breathing fishes physiological mechanisms and how hormones affect the oxygen consumption is not yet known. Our present study is an attempt to find the degree of relationship among the different hormones on oxygen consumption in the fish *Channa punctata*.

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## MATERIALS & METHODS

Fresh and live fish were bought from the local fish market of Madhepura, Bihar (25° 92' N, 86° 79'E). A glass aquarium of dimensions 90x60x60 cm was used to keep the fish. As food the fish were fed with chopped goat liver

on a daily basis and were acclimatized for 15 days in the laboratory. Oxygen consumption from air and still water was measured routinely with the help of glass respire meter after. The aquarium originally contained 3 litres of water and 0.5 ml of air. Wrinklers volumetric method was used to estimate the concentration of dissolved oxygen in the water. The experimental fishes were divided into 6 groups including the controls. Each group has 10 fishes. Different doses of Hydrocortisone, Adrenaline, testosterone, Progesterone, and thyroxin were injected intra peritoneally on the abdominal side of the fishes slightly anterior to the pelvic fin to study the effects of these hormones on the oxygen consumption of fishes. Different doses of hormonal injections in number of days were recorded.

The hormones were bought locally. The effective doses of different hormones were standardized. Oxygen consumption of animals treated with different hormones was measured in every 24 hours. While in case of Adrenaline the oxygen consumption was measured after 48 hours.

## RESULTS & DISCUSSION

A summary of the effects of certain hormones on oxygen consumption, the aquatic, aerial and total oxygen consumption after different doses is given in table 1. The mean values or averages of aquatic, aerial and total oxygen consumption were 52.44, 61.56 and 114.0 (ml/kg/hr) respectively.

The fishes were given single dose of hydrocortisone (i.e. 1.0 mg/100gm) for 2 days. An increase in aquatic oxygen consumption was seen in control fishes from 52.44 to 60.76 (ml/kg/hr). The aerial oxygen consumption also increased from 61.56 to 63.24 (ml/kg/hr), while the total oxygen consumption increased upto 124.00 from 114.00 (ml/kg/hr).

Single doses of Adrenaline (0.000006) were also injected for 2 days. However, in this case the aerial oxygen consumption decreased upto 6.96 against 61.56 (ml/kg/hr) in control fishes. The aquatic oxygen consumption slightly decreased to 51.04 (ml/kg/hr) in control fishes. This decrease shows that adrenaline makes the fish sluggish, compelling it to respire via its gills under water. This also decreased the total oxygen consumption from 114 to 58 (ml/kg/hr) in the control fishes.

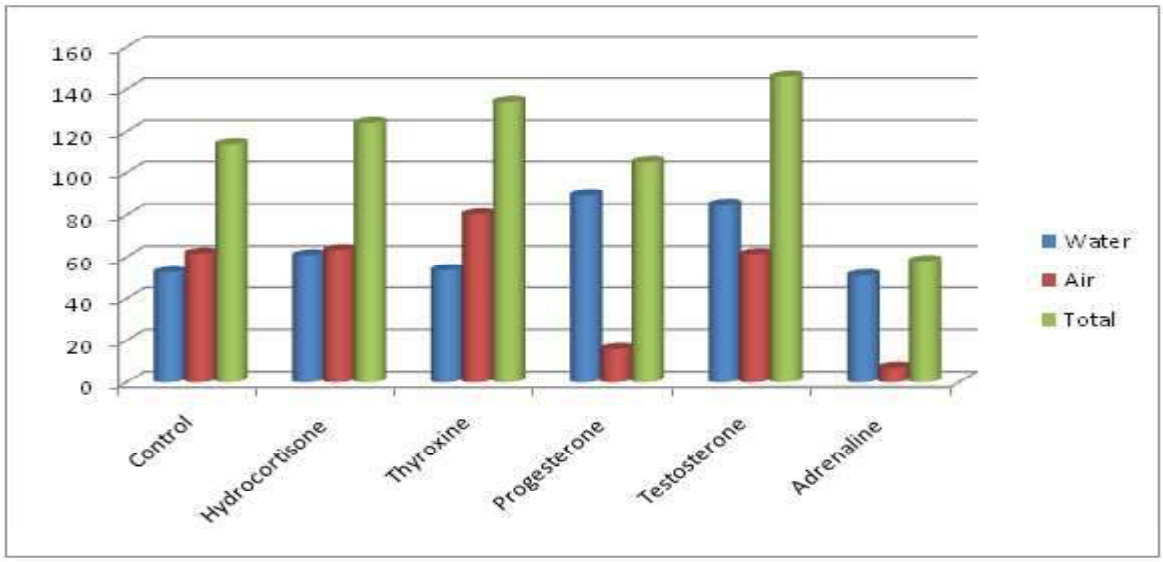
On injecting single dose of testosterone (2mg) for 2 days, an increase in aquatic oxygen consumption was observed upto 84.68(ml/kg/hr). The aerial oxygen uptake remained mainly unchanged but it was slightly lower than that of control ones. The aerial consumption was 61.32(ml/kg/hr) while the total consumption increased upto 146.00(ml/kg/hr). We can conclude that testosterone does not have any effect on the aerial oxygen consumption but it does enhance the aquatic oxygen consumption.

When a dose of 1mg of progesterone is given to the fish for 2 days, we observe an increase in the aquatic oxygen consumption significantly upto 89.25 (ml/kg/hr) as compared to controls 52.04(ml/kg/hr). However the aerial oxygen consumption was reduced to 15.75(ml/kg/hr), decreasing the total oxygen consumption to 105.00(ml/kg/hr). It can be said that progesterone decreased the aerial oxygen consumption but enhanced the aquatic oxygen consumption.

Single dose of Thyroxine (0.5mg) was injected to the fish for 2 days. It showed slight increase in the aquatic oxygen consumption, from 52.04 (ml/kg/hr) to 53.60 (ml/kg/hr). The aerial oxygen consumption also increased upto 80.40, therefore increasing the total oxygen consumption upto 134.0 (ml/kg/hr). Thyroxine does not affect the aquatic oxygen consumption but it elevates the aerial oxygen consumption.

**Table 1- The oxygen uptake in *Channa punctata* at different dose of hormones and body weight 40.0±2.5 gm**

| Sl. no. | Condition | Injection in no. of days | Total dose (mg/100g) | $O_2$ (ml/kg/hr) | | |
|---|---|---|---|---|---|---|
| | | | | Water | Air | Total |
| **1** | Control | -- | -- | 52.44 | 61.56 | 114.00 |
| **2** | Hydrocortisone | 2 | 1.0 | 60.76 | 63.24 | 124.00 |
| **3** | Thyroxine | 2 | 0.5 | 53.60 | 80.40 | 134.00 |
| **4** | Progesterone | 2 | 1.0 | 89.25 | 15.75 | 105.00 |
| **5** | Testosterone | 2 | 2.0 | 84.68 | 61.32 | 146.00 |
| **6** | Adrenaline | 2 | 0.000006 | 51.04 | 06.96 | 58.00 |




**Graph 1- Graphical representation of oxygen uptake in *Channa punctata* at different dose of hormones and body weight 40.0±2.5 gm**

A very little information is there about the effect of adrenocortical hormones on the homeostasis of oxidative metabolism in fishes specially *Channa punctata.* In *Gobius melastomus* the treatment of adrenaline induced depression of metabolism, which was reported by Marinescu (1971)[6]. In *Macrognathus aculeatum* it was found that the treatment of adrenaline causes sharp decrease in oxygen consumption,[7] which were similar to the findings of Marinescu (1971)[6]. Decrease in oxygen consumption may be firstly due to carotid sinus response to increase the blood pressure. Secondly, it may be due to relaxation of respiratory muscles and slow heart beat evoked by epinephrine contrary to adrenaline.[8] In *Macrognathus aculeatum* treatment with hydrocortisone shows rise in oxygen consumption.[7] Oxygen consumption may be due to adreno-cortical hormones, however exact reason is not understood that cause the rise in oxygen consumption.[9]

A view consistent with the findings of Marinescu (1971)[6] and Pandey *et al.* (1985)[7] was found in the present investigation in *Channa punctata*. According to it, the treatment of adrenaline caused a decrease in total and aerial oxygen uptake, whereas treatment with hydrocortisone resulted in increase in total oxygen consumption similar to findings.[9]

Gonadal steroids fail to provide any conclusion and are contradictory when attempted to evaluate their effects on the respiratory metabolism in fishes. The effects seen to depend on the concentration of gonadal steroids, as in some cases gonadal steroids have shown stimulating effects while on some they have shown depressing effects on cellular metabolism.

An increase in oxygen consumption, when treated with testosterone have been reported in gold fish by Hoar (1958)[10] and in *H.fossilis* by Pandey (1976)[11]. Significant decrease in oxygen uptake in fishes may be due to fall in concentrations of sex hormones in the circulating blood and it failed to bring any change after treatment with progesterone, was reported by Pandey (1976)[11] in ovary sectomised *H.fossilis*. A significant correlation was found in *Anabas testudineus* in between the cyclic changes in gonadal activity and seasonal variation in oxygen consumption in both sexes.[12] This concludes that gonadal steroids are one of the many factors that affect the oxidative metabolism directly or indirectly through increased locomotion activities. Two possibilities were suggested by Hoar (1958)[10] : **(a)** demand of oxygen is increased by the steroid by increasing the reactivity of neuromuscular mechanism thus promoting locomotion activity. **(b)** steroid can act throughout the body stimulating oxidative metabolism in the tissue.

Male and female hormones do not have same effect on the oxidative metabolism. Males being metabolically more active than females the oxygen uptake is higher in them.[12] Male hormones directly and indirectly shown an increase in oxygen consumption in *Channa*, when treated with testosterone, views consistent with the finding of Pandey (1976)[11] and Hoar (1958)[12]. Female hormones show contradictory effect on the respiratory metabolism.

All female hormones (progesterone, ovacyclin and duogynon forte) lowered the oxygen consumption. Progesterone decreases the aerial and total oxygen consumption.[11-13] It shows a catabolic effect.[14]

With the increase in cell height of thyroid follicles, an increase in oxygen consumption was found in *H.fossilis*.[15] It seems that thyroid hormones are important regulators of oxygen uptake.[16,17] Injection of thyroxine produced an increase in metabolic rate. Injection of iodothyro acetic acid also elevated oxygen uptake after radio thyroidectomy.[16] These findings make thyroid gland an important regulator of oxygen consumption. In cold blooded or poikilothermal animals thyroid glands effect is more metamorphic than metabolic.

## CONCLUSION

In the present investigation *Channa punctata* was found to be responding to the treatments of thyroxine by increasing total as well as aerial oxygen uptake. A comprehensive idea is given about the effects of hormones on bimodal oxygen uptake in snakehead *Channa punctata*. A few precautions were taken during the study such as

**(a)** Same weight fishes must be bought for experimental and control methods as oxygen consumption tends to differ with body weight.

**(b)** Acclimatization of fishes to the respirometer must be ensured before taking measurements of oxygen consumption

**(c)** Various dose levels of hormones should be used for experimentation.

**(d)** Simultaneously one must do the oxygen consumption for both experimental and control fishes.

**(e)** Minimization of the locomotor activity of the fish should be done by the respirometer.

## REFERENCES

1. **Bayliss W M and Starling E H 1904**. The mechanism of pancreatic secretion *J. physiol.* **28:**325
2. **Raza Bushra, Luxmi Sah and G Lakra. 2000.** Effect of adrenaline in the cortical secrtion in an air breathing teleost *Clarias batrachus* proc. *Nat. Conf. On Endocrinology held at Ranchi Abst.* No. **35:** PP 35
3. **Matty AJ. 1957.** Thyroidectomy and its effect upon oxygen consumption of telost fish *Pseudoscarus gaucamia. J.Endocr.* **15:** 1-8
4. **Punt and Jungbloed, 1963.** Quoted by A Gorbman (Thyroid Hormones) in comparative endocrinology. **1:** 219-324.
5. **Kaya M K. 1973.** Effects of temperature on response of the gonads of green sunfish *Lepomia cynellus* to treatment with carp pituitaries and testosterone propionate. *Fish.Res. Bd.Can*. **30:** 905-912
6. **Marinescu AG. 1971.** The influence of adrenaline on oxygen consumption in the pertilor Gobius melanostomus in high temperature condition stud. *Cerlet. Biol.Ser.Zool.* **23:** 536-546
7. **Pandey B N, S Prasad and Sinha D P. 1985.** Cyclic variations in the activity of internal tissue and oxygen consumption in the fresh water Mud eel *Macrognathus aculeattim* Ad. *Bios*. **411:** 87-91
8. **Krantz J C Jr and Carr. 1958.** Pharmacological principles of medical practice. William & Wilkins Co. Baltimore 1260 PP
9. **Chatterjee C C. 1975.** Human physiology, Vol 1 medical allied agency Calcutta, PP 1-764
10. **Hoar W 5. 1958.** Effect of synthetic thyroxine and gonadal steroids on the metabolism of gold fish. *Can. J. Zool.* **36:** 113-121
11. **Pandey B N. 1976.** Effects of gonadal steroids on oxygen consumption in *Heteropneustes fossilis* (Bloch) *Pol.Arch.Hydrobiol*. **23:** 327-332
12. **Chanchal A K, Pandey B N, Singh S B And Prasad S. 1979.** Cylic Changes in haematalogical values of *A. testudineus. Annals of Zoology*. **15(3):** 111-124
13. **Prasad S, 1979**. Ph.D. Thesis Entitled "Cyclic activity of EndoCrine glands" Magadh University, Bodh Gaya, Bihar.
14. **Chatterjee C C. 1973.** Human physiology, Vol II 8"' Edn. New central book agency Calcutta, India
15. **Pandey B N and Munshi JSD, 1976.** Role of the thyroid gland in the regulation of metabolic rate in an air breathing siluroid fish *Heteropneustes fossilis* (Bloch) *J. Endocrino.* **69:** 421-425
16. **Prichard A W and Gorbman A, 1960.** Thyroid hormone treatment and oxygen consumption in embryos of the spiny dog fish. *Biol. Marine. Biol.lab.Wood.Hole Mass.* **119:**109-119
17. **Smith D C and Mathews S A. 1948.** Parrot fish Thyroid Extract and its effect upon the oxygen consumption in the fish *Bathy Stoma Am. J. physiology.* **153:** 215-221

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# Greening textile industry reduces environmental pollution- A research on eco-friendly garments

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**Abstract-** Consumers are becoming more conscious of the importance of environmental protection, &businesses use these words to advertise their products or services with eco-labels. Environmentally sustainable (also eco-friendly, nature friendly & green) refers to products and services, regulations, rules & strategies that claim to do little or no damage to the ecosystem. Clothing is an important part of our lives, & green or eco-friendly issues have begun to gain momentum in the textile and clothing industries. This paper examined how greening the textile &clothing industry can help in reducing environmental pollution with respect to the product lifecycle, From raw materials to design, manufacturing, & logistics, as well as disposal, key points & criteria for greening the industry are highlighted.

**Key words: Textile, Eco-friendly, Pollution, Industry, Green Industry.**

## INTRODUCTION

Since time immemorial, the textile industry has been well known for its superior quality and design. India has long been a key manufacturer of textiles to countries on all continents. It is one of the country's largest and oldest industries and one of the most significant in terms of efficiency, development & employment. The textile industry directly or indirectly employs approximately 35 million people. Besides this, it makes major financial investments. It accounts for 4% of the GDP, 9% of excise collection, 14% of total industrial output, 16% of total export earnings & 18% of industrial employment. One out of every six households in the country is estimated to rely on this sector for their livelihood, either directly or indirectly. Global production of processed fibers & massive synthetic materials increased by 3.3% per year from 52.6 million tons in 2000 to 70.5 million tons in 2008. During that time, the percentage of imported fibers rose from 59 to 63 per cent. In 2011, global textile production in 2011 reached over 51 million metric tons of processed fibers & 30 million metric tons of synthetic materials, total record reflecting 1 per cent rise from fiber production in 2010 & per capita production of 12 kg. Manufactured fiber consumption has gradually increased over the last 3-4 years, synthetic fibers, on the other hand, consumption has remained constant & has started to decrease.

The textile industry is the largest relevant industrial sector including fiber to ginning, weaving, cloth, dyeing & printing, made ups & apparel. It has the largest distribution network, & intrinsic value-added capacity from each production point. The industry generates approximately one-fourth of all industry real worth & employs 40 per cent of all workers in the industry. Textiles have accounted for around 60% of national exports on average, excluding cyclical & seasonal variations. Textile

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& garment exports now account for approximately 6% of overall global exports. Many LDCs and small developing countries have formed a heavy dependence on the industry, which accounts for more than 90% of industrial exports and more than half of total jobs in many of them. Many sectors of the textile industry are rising production efficiency as a result of intensified global competition.

Environmental pollution is a serious problem that humanity is currently dealing with. Clients were becoming more environmentally conscious as a result of global warming and increased awareness. 'Green' principles and techniques have become essential for organisations as widespread awareness of their environmental impacts has grown. Green textiles offer consumers more eco-friendly options. Climate-related words are used here to refer to legislation, products & services, strategies & guiding principles that do the least amount of harm to the environment.

It is also essential in green clothing to recognize the item's existence and the extent to which it affects the environment during its life expectancy. Furthermore, to obtain green clothing, effective management is needed at all levels, starting with environmental planning, purchasing raw materials, manufacturing garments, disseminating them to channels and shops, and, finally, considering their reverse logistics. Essentially, the goal of green clothing is to transform the manufacturing system to become more energy-efficient while avoiding environmental damage. Green manufacturing plants have developed suppliers or subcontractors' networks to purchase environmentally friendly products and then enforce basic practices to minimise waste and increase operating efficiencies. The purpose is to use environmentally friendly fabrics to mitigate environmental risks and obtain sustainable clothing while also opening new markets and rising competitive advantage in the textile industry.

The word "eco-friendly" refers to products and services, regulations, rules, & regulations that are believed to cause little or no damage to the environment. Green is a highly subjective concept. Green or environmentally products, facilities, & activities ensure using eco-friendly products, free from hazardous materials, chemicals & wastage that don't drain the eco-system while production & manufacturing. Environmentally garments are made from eco-friendly & natural resources, & effective management of acquiring sustainable garments necessitates considering all steps, beginning with constructing for the climate, sourcing raw materials, manufacturing clothes, selling items through networks & retailers, and considering reverse coordination and waste disposal.

Clothing industry is the world's largest & oldest manufacturing industry. According to the United Nations Environment Program, the fashion industry, which includes textile and apparel development and processing, is the world's second-largest trade activity. The global industry is worth $1.44 trillion (about $4,400 per person in the US). In 2010, Indian market produced $63 billion (about $190 per person in the US) accounted to 14 per cent of manufacturing output & 4.5 per cent of Gross domestic product. India is world's biggest garments & textile industries, recruits 35 million people (about twice the population of New York), second only to farming. In India, there are 30,000 garment manufacturing companies that only manufacture for export. 5,000 of these organizations are well known in terms of efficiency and sustainability.

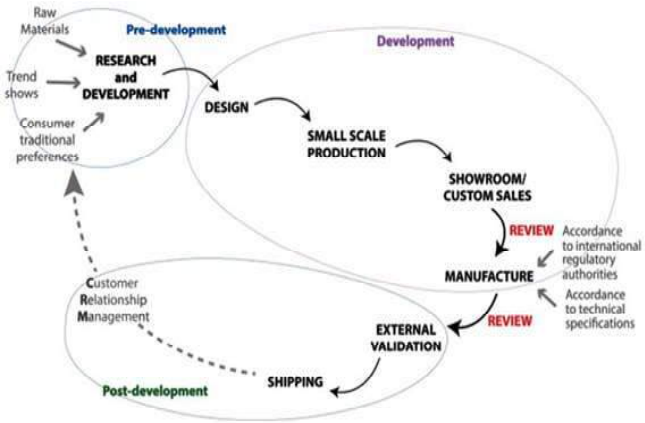



**Fig. 1 Current Framework of Textile development process in the industry**

## LITERATURE REVIEW

In his paper, Eryuruk, S. H. (2012)[1] examined "how green the fabric& garment business is concerning product lifecycle, from natural resources via manufacturing, design & logistics up to disposal to highlight key points & criteria for greening the industry." Clothing is an essential part of our lives, & environmental or green concerns have begun to gain traction in the textile and clothing industries.

According to Clarke *et al.* (2013)[2], "pollution reduction programmes in Bangladesh are significantly underutilized as compared to the most successful companies in developing countries."All of the industries engaged in some form of pollution control, though each had room for development. For instance, neither of the

selection function was involved in emission reduction by developing eco-friendly goods.

In his article, Rao (2004)[3] investigates the significance of "greening of production" in the South-East Asian context. It studies the numerous initiatives undertaken by the companies in this region and, as a result, identifies the factors crucial in the phase of greening development in this area. The factors' importance is determined & confirmed via an experimental study conducted across five countries in the area using the structural equation modelling method.

In their report, Caniato *et al.* (2012)[4] present "the findings of an explorative scenario study that sought to identify 3 elements: the forces driving businesses to implement "Green" practices, various practices which can be used to enhance sustainable development, & the ecological KPIs assessed by fashion companies." The study's results compare two approaches taken by established multinational firms with green-positioned brands with the initiatives of small businesses who have embraced alternative supply chain models.

In his paper, Dawson (2012)[5] stated that "many concepts from the comparatively recent study of green technology, that seeks to use energy effectively & reduce wastage, are relevant in the field of textiles." It is a practical aim to increase higher quantities while minimizing environmental effect in manufacture & eventual coloration of synthetic fibers. While the demand for organic fibers is still limited, public's interest in sustainably grown synthetic materials have mirrored that in sustainably food grown. With special reference to these newer fibers, the methods used to reduce the environmental impact of fiber, pigment manufacturing and corresponding coloration methods are listed.

In their article, Hole (2019)[6] address eco-friendly achievements in the textile industry's growth, focusing on three key participants: Recycling, Consumption & Production. Which demonstrates, despite substantial eco-friendly technology advances in textile industry, reuse & market recognition has a low level of sustainability in the manufacturing industry. Textile industry has compared to others (paper, aluminum, glass) and a better recycling rate. Textile waste disposal legislation and the lack of international textile waste disposal standardization is also addressed with an increasing demand & a lower recycling rate in fabric industry. It isn't easy to achieve required stability between environmental & industrial processes to extend the life span of textile industry.

Mukherjee (2015)[7] wrote in his paper, "The fashion industry is complex & diverse, with four or more levels, including layout, yarn production, raw material gathering, dyeing, weaving, stitching, & fabric creation". Quick apparel refers to clothing with a limited project life cycle have arisen a formidable competitor. Quick fashion has resulted in an increase in consumption as well as increased waste. Quick fashion produces waste at any point in the textile life cycle, posing occupational & environmental risks. Because of the vast number of goods made, used, and discarded, & the clothing business retains the major eco-friendly effect at every single point of produce life cycle. While speak just about the world, we refer to both the social & natural environments, and the development process includes the manipulation of both social & natural environments. Sustainable & ethical design is a reaction to the social & environmental destruction caused via traditional manufacturing methods. The informative, analytical report seeks to raise customer awareness of the physiological needs which apparel serves & know the complete life cycle effects on clothes, from raw sources to waste.

According to Mia *et al.* (2019)[8], this study aims to focus on the current pollution situation in Bangladesh's printing & dyeing industry as a result of various textile pollutants while our country's economy is growing as a result of small & medium-sized manufacturing processes. The dyeing and printing industry's hazardous waste disposal is heavily polluting the atmosphere. Since the country's textile industries' growth, liquid and solid effluents from various industries have been wreaking havoc on the atmosphere, biodiversity, farming, aquaculture, and public health. It's necessary to take appropriate measures to reduce emissions in every industrial firm, especially in printing & color which discharge significant amounts of liquid waste into waterways daily.

Choi *et al.* (2012)[9] write, "This article aims to illustrate why garment industry must "go green," as well as analyze business strategies & sustainable development." The authors use Five R method to analyze the launch, execution & institutionalization voyage of a regional style business, yielding valuable observations & results. This is qualitative experimental research. The Five R abstract structures is tested, suggested, & used to true-world study.

According to this literature review that have been congregate & evaluated thus far, fashion industries gain a contrasting edge via strategic control of natural difficulties. Apparel industries closely recognize the product development cycle & expand governance through several product lifecycles in their greening initiatives. The Five R platform, along with its potential expansions, will provide an opportunity to specifically illustrate what the company has accomplished to date and concisely illustrate where the company is missing or where there is space for more beneficial growth. Desore *et al.* (2018)[10] review the existing literature on numerous environmental problems affecting the global textile industry in their paper. The researchers categorize the research to address the factors, obstacles, & reactions of textile industry regarding sustainability. Even though there is an increasing lot of literature in the field, they find substantial differences in this paper, particularly regarding managerial approaches to integrating sustainability. While it is critical to understand why push firms toward sustainability, and the barriers to adoption, research on managerial perceptions of current methods & techniques is relevant. Future development possibilities are also provided in the form of questions in the paper, which reinforce the current literature in this area. Shen *et al.* (2017)[11] seek to address resource group along with low carbon footprint activities in the textile industry of China in their paper. To ensure a low-carbon economy's viability, the Chinese government has implemented a restrictive regulatory framework and placed a power consumption limit in the Textile Company. The authors want to see how energy consumption constraints influence optimal decisions of distribution network members & solve the distribution network management problem.

The analysts investigate the effect of energy usage restrictions on Chinese textile companies' development & business operations using two methodologies. The analysts create straight forward analysis method for low carbon distribution chain comprised of a single retailer & a single producer based on actual world business practices, the producer prefers renewable sources improved efficiency & to reduce pollution. The authors also discover that fabric business implement renewable technology to minimize emissions during the manufacturing process to comply with energy usage regulations.

In the empirical method, the analysts extract the appropriate decision of distribution chain representatives and demonstrate that supply chain cooperation accomplished if the producer accurately fixed the selling price. However, manufacturing capacity will satisfy a portion of market conditions under a cost-sharing arrangement & the cost-sharing agreement can encourage supplier to expand clean technology investing while decreasing the optimal WS.

## ANALYSIS & DISCUSSION

A substantial extent of study has been done on environmentally customer habits that have contributed to the increase of procedures like ecologically marketing (Thgersen, 1999)[12] & eco-labelled items (D'Souza, 2004)[13]. For starters, results provide clear empirical evidence to the role of societal standards in a democratic society, the perspective of consumption. Previous research has often changed TRA & TPB to examine the effect of arbitrary rules on eco-friendly activities.[14] However, there has been no research on the efficacy of descriptive norms in apparel shopping, considering the difference between injunctive & descriptive norms. Existing research demonstrates that both of the explanatory standard & injunctive. Even though environmental considerations are significant, rules have a larger impact on purchasing intentions. The last is a big factor of environmentally responsive behaviour.[15]

Thus, the present results verify formed concept that studying the behaviour of others immense significance in the decision-making in eco logical usage as behaviours frequently take position in a societal perspective.[16-18] As a result, current results support long-held belief that studying the actions of others.

Next, results revealed that manufacturer's eco-friendly advertising argument mitigates environmental issue & injunctive norms on customer purchasing intentions. In particular, when paired with an extrinsic argument, descriptive norms have a stronger impact on purchase intention than when paired with an inherent claim. In contrast, concern for the environment has a stronger impact on buying intention while paired with an inherent allegation than when paired with an external claim provided that environmentally friendly behaviours tend to be affected through extrinsically motivation or internally motivation, the finding leads to greater considerate with customers' multiple motives for buying eco-friendly items.

Furthermore, injunctive standard and environmental interest can play a moderating role. Explain why previous research came to conflicting decisions about the effects of

ecological statements in apparel ads.[19,20] In contrast, inherent advertising assertions including intrinsically environmentally friendly goods set off private eco-friendly issue more significantly than external selling claims.

Tertiary, these findings revealed that descriptive norms had little impact on purchasing intentions. Depending on the form of argument, they can differ. In contrast to injunctive standards, factual rules are based on situational rather than individual factors. People are sensitive to other people's actions, regardless of the claim form; hence, the subjective norms caused through a shopping condition influence customers' purchasing plans regardless of the kind of ecological assertion made. By studying others' actions, consumers tend to seek clues from which to view any condition & infer similar social norms.[16]

## CONCLUSION

This study paper focused on pollution control practices in textile dyeing and printing. The majority of factories do not adhere to the laws and regulations. As a result, strict guidelines are required, and the government must monitor whether the companies are abiding by the laws or not. The company generates waste, which must be handled regularly. Furthermore, an additional initiative is needed because air & water emissions as well as soil & noise emissions are on the rise. Action against this is critical, but it sadly lacks in printing & dyeing industries.

The government of India's ministry of environment & forests introduced a programme in 1991, i.e., essentially a system for marking environmentally friendly goods, to raise consciousness regarding the natural effects of goods. It is known as the "Eco Mark" scheme, and it seeks to make it simple to identify environmentally friendly goods. The plan is focused on a "cradle to grave" strategy, which considers a product's effect from raw material production to manufacturing to ultimate disposition. As ethical buyers, it is our duty to buy goods based on their environmental friendliness. Our awareness will force manufacturers to pursue more eco-friendly garment & a healthier and greener world.

Industrial pollution can devastate water systems and lakes, which are important to biodiversity and the lives of billions of people. Harmful contaminants released via Industry have such a wide variety of negative effects, including cancer, hormone disruption, and reproductive system interference. These effects can affect all living things, not just humans. There is a proof that the Textile Industry contributes significantly to water contamination, with the utilize & dismiss of toxic chemicals.

The existence of dangerous substances in the atmosphere demonstrates that the conventional solution to wastewater wastes are ineffective-inefficient water systems cannot handle a wide range of toxic chemicals. These implications for human health & habitats are serious, & hazardous material cleanup is a time-consuming and expensive operation. A new approach to dangerous materials is needed that tackles the issue at the resource rather than in the end. The concepts of "zero discharge" removing each of dangerous chemical dismiss into the marine environment are focused on the realization that acceptable standards for certain hazardous contaminants are difficult to establish. The best solution has proved to be the renovation of commodity & systems to drive out the use & disposal of dangerous chemicals. To keep track of their environmental effects, LCA should be done regularly, particularly for new products and processes. As a result, measures must be taken to minimize emissions and make textile goods and processes more environmentally friendly.

Environmental concerns influence all human activities. Green marketing of the garment industry has become increasingly important as customers become more concerned about the environment and health. It protects the organization's interests and the interests of all of its customers both the buyer and the seller profit from the deal. Enterprises, as the backbone of the garment industry, are actively involved in its operations. Enterprises must implement new green marketing tactics; forego short-term gains in favour of long-term environmental conservation gains. Only by doing so would they maintain their place in the international market and thrive indefinitely. Finally, the garment industry will be able to develop sustainably.

Textiles can be one of the world's least sustainable goods. They can cause serious problems over the complete process, including raw material production or creation from oil to manufacturing, selling, & then disposal. There are advantages at various phases of the sustainable & environmental textile industry, for both suppliers & users, but on a large scale, it's necessary to remember that environmental textile industry & garments can fly halfway across the globe to meet the moral customer. If the market for these environmental clothing grows; new entrepreneurs have a significant opportunity to enter this sector.

## REFERENCES

1. **Eryuruk, S. H. 2012.** Greening of the textile and clothing industry. *Fibres& Textiles in Eastern Europe.* **6A(95):** 22-27.
2. **Hoque, A., & Clarke, A. 2013.** Greening of industries in Bangladesh: pollution prevention practices. *Journal of Cleaner Production*. **51:** 47-56.
3. **Rao, P. 2004.** Greening production: A south-east Asian experience. *International Journal of Operations & Production Management.*
4. **Caniato, F., Caridi, M., Crippa, L., & Moretto, A. 2012.** Environmental sustainability in fashion supply chains: An exploratory case-based research. *International journal of production economics.* **135(2):** 659-670.
5. **Dawson, T. 2012.** Progress towards a greener textile industry. *Coloration Technology*. **128(1):**1-8.
6. **Hole, G., & Hole, A. S. 2019.** Recycling as the way to greener production: A mini review. *Journal of Cleaner Production.* **212:**910-915.
7. **Mukherjee, S. 2015.** Environmental and social impact of fashion: Towards an eco-friendly, ethical fashion. *International Journal of Interdisciplinary and Multidisciplinary Studies*. **2(3):** 22-35.
8. **Mia, R., Selim, M., Shamim, A., Chowdhury, M., & Sultana, S. 2019.** Review various types of pollution problems in the textile dyeing & printing industries of Bangladesh and recommend mitigation recommendations. *Journal of Textile Engineering & Fashion Technology*. **5(4):** 220-226.
9. **Choi, T. M., Lo, C. K., Wong, C. W., Yee, R. W., & Ho, H. P. Y. 2012.** A Five-R analysis for sustainable fashion supply chain management in Hong Kong: a case analysis. *Journal of Fashion Marketing and Management: An International Journal*
10. **Desore, A., & Narula, S. A. 2018.** An overview of corporate responsibility towards sustainability issues in the textile industry. *Environment, Development and Sustainability*. **20(4):** 1439-1459.
11. **Shen, B., Ding, X., Chen, L., & Chan, H. L. 2017.** A low carbon supply chain with energy consumption constraints: Case studies from China's textile industry and a simple analytical model. *Supply Chain Management: An International Journal.*
12. **Thφgersen, J. 1999.** The ethical consumer. Moral norms and packaging choice. *Journal of consumer policy*. **22(4):** 439-460.
13. **D'Souza, C. 2004.** Ecolabel programmes: a stakeholder (consumer) perspective. *Corporate Communications: An International Journal*.
14. **Thφgersen , J. 2009.** The motivational roots of norms for environmentally responsible behavior. *Basic and Applied Social Psychology.* **31(4):** 348-362.
15. **Minton, A. P., & Rose, R. L. 1997.** The effects of environmental concern on environmentally friendly consumer behavior: An exploratory study. *Journal of Business research*. **40(1):** 37-48.
16. **Biel, A., & Thφgersen, J. 2007.** Activation of social norms in social dilemmas: A review of the evidence and reflections on the implications for environmental behaviour. *Journal of economic psychology*. **28(1):** 93-112.
17. **Rivis, A., & Sheeran, P. 2003.** Descriptive norms as an additional predictor in the theory of planned behaviour: A meta-analysis. *Current Psychology.* **22(3):** 218-233.
18. **Cialdini, R. B., Kallgren, C. A., & Reno, R. R. 1991.** A focus theory of normative conduct: A theoretical refinement and reevaluation of the role of norms in human behavior. In Advances in experimental social psychology (Vol. 24, pp. 201-234). Academic Press.
19. **Kim, H. S., & Damhorst, M. L. 1998.** Environmental concern and apparel consumption. *Clothing and Textiles Research Journal*. **16(3):**126-133.
20. **Phau, I., & Ong, D. 2007.** An investigation of the effects of environmental claims in promotional messages for clothing brands. Marketing Intelligence & Planning.

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# Study on the use of *Azadirachta indica* as biological control agent of *Spilosoma obliqua*

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**Abstract-** Agriculture is the key for development in the rise of sedentary human civilization, it is the science of cultivating the soil, growing crops and raising livestock. Commercially available synthetic pesticides damage the entire environment as well as human civilization. Due to the use of synthetic pesticides it led to bioaccumulation in the agriculture field causing increased resistance and reduction in soil biodiversity. Therefore it is time to draw attention towards environmentally friendly natural pesticides and insecticides. *Spilosoma obliqua* is a sporadic pest and polyphagous which damages a large number of plants. In this paper active ingredient of neem (*Azadirachta indica*) plants which exhibit agromedicinal properties are used as biopesticides in control of *Spilosoma obliqua.*

**Key words: Agriculture, *Spilosoma obliqua, Azadirachta indica,* biopesticides, Azadirachtin**

## INTRODUCTION

*Spilosoma obliqua* is a polyphagous moth. It is brown with a 40-50 mm wing span and a red abdomen. The larvae are covered with long yellowish to black hairs. The pupa forms a thin silken cocoon by interwoven shed hairs of the larvae.[1] It feeds on at least 126 species of plant including cereals, pulses, oilseeds, turmeric, vegetables, mulberry, fibre crops and non-cultivated plants and weeds. In India, the insect is a serious pest of fibre crops, sometimes occurring in epidemic outbreaks.[2] It has been found to be serious Lepidopteron nocturnal erebid pest of varieties of weeds & agricultural crops of rural areas of Saharsa. It feeds differently in different stages of life, young larvae feeds gregariously on the under surface of the leaves and cause loss by way of defoliation, pupae damages seeds. It damages almost every part of plant leading to decline in agricultural produce.[3]

Now a day's increase in food production without damaging the environment is a major challenge. Extensive use of commercially available synthetic pesticides damages the entire environment as well as human civilization. Over 85% of the applied pesticides enter the various environmental resources which results to severe health issues. Therefore, growing attention has been given towards the development of alternate environment friendly pesticides/insecticides that would aid an efficient pest management system and also prevent chronic exposures leading to diseases.[4]

In this paper one of such active ingredients has been studied. *Azadirachta indica* commonly known as neem have many different traditional uses. It is known for its

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pesticidal and insecticidal properties, but people also use it in hair and dental products. One of the traditional uses of neem has been for controlling pests of stored products. Farmers usually mix neem leaves with grain before keeping it in storage for several months. A mixture of neem leaves with clay and cow-dung develops pest resistant property so it can be used to make bins for storage of grain.[5] The use of biopesticides could effectively control insect pest complex associated with jute.[6] Azadirachtin is the most prominent constituent and pivotal insecticidal ingredients of neem plant which exhibit agro-medicinal properties. It acts as an antifeedant, repellent, and repugnant agent and induces sterility in insects by preventing oviposition and interrupting sperm production in males.

## MATERIALS & METHODS

In this study areas from rural regions of Saharsa were worked out. During the study cotton fields and jute fields were basically explored. The selected fields were from different areas of Saharsa district of Bihar. Samples were collected by visiting various cotton as well as jute fields. Mature and immature stages of hairy caterpillar were collected from the field and were used for biopesticides treatment. Several male and female adults were collected and released into cage along with jute and cotton plants. For performing this work extract of neem leaves, neem kernels have been procured from the local area.

**Neem Kernal:-** The simple method of preparation of neem kernel aqueous solution has been implemented. For this dried neem seeds were taken, their seed coats were removed with the help of mortar and pistal and after that seeds were grinded and powder was procured. The powder was not the dust it was grinded so, that it gives a tea like powder. The grinded neem kernels were soaked into water in a jar over night and pH neutral adjutant was added. The very next day the solution obtained was filtered and get sprayed on the plants infected with *Spilosoma obliqua*.

**Neem Leaves:-** Preparing biopesticides from leaves of neem (*Azadirachta indica*) is very easy just fresh leaves of neem were gathered and were soaked in fresh water overnight. The very next day the leaves were filtered out of the water and then the liquid extract is sprayed over the infected plants.

## OBSERVATION

The solution obtained by neem leaves and kernel do not usually kill insects directly, but they can affect their behavior in significant ways to reduce pest damage to crops, and reduce their reproductive potential. The larval mortality was observed after 24 hours of sparkling the Biopesticides.

Unlike the chemical insecticides, azadirachtin affects the hormonal system, not the nervous and digestive system. And this affects insect physiology by mimicking a natural hormone. It disrupts or inhibits the resistance in future generation. It sterilizes adults, inhibits the formation of chitin and disrupts mating deterring females from laying eggs. It has been observed that it affect egg production and hatching rates.

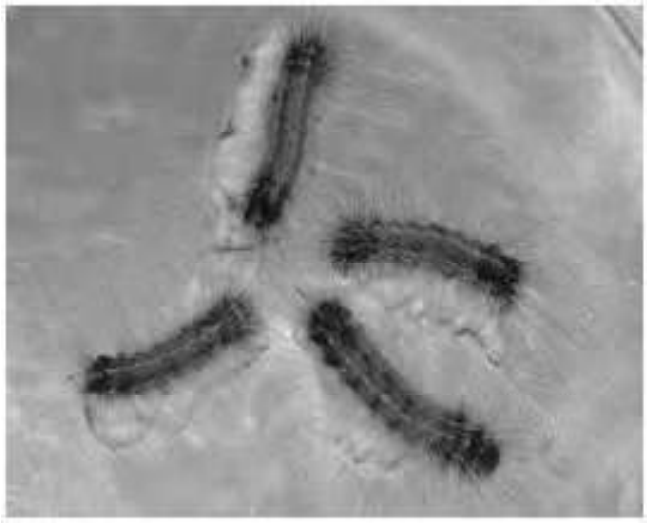


**Fig. 1. *Spilosoma obliqua* feeding on jute plant**



**Fig. 2. Adult of *Spilosoma obliqua***



**Fig. 3. Adult of *Spilosoma obliqua* before placing in cage**

In larvae, azadirachtin can inhibit molting, preventing them from developing into pupae. The potential use of azadirachtin in crop protection, considering the gaps and obstacles associated with the development of sustainable agriculture were observed.

## RESULTS & DISCUSSION

Neem extracts can be made from leaves and other tissues, but the seeds contain the highest concentrations of azadirachtin. Azadirachtin works as an insect repellant and reducing insect feeding. It interferes with the insect hormone systems of insects, so they do not mature correctly or lay eggs. In the present study, it is found that plants infected by *S. obliqua* larvae were when treated by azadirachtin, the larvae become inactive and stop feeding. The larvae become pale yellow and black, flaccid, eventually dies, usually within days. The body contents turn brown to black as they decompose. This prevents growth, metamorphosis and reproduction. Azadirachtin can be quite effective against *Spilosoma obliqua* Bihar hairy caterpillar as part of a comprehensive pest management plan.

Caterpillars can be a big problem for organic gardeners. Depending upon the type of caterpillar, they can destroy ornamental plants, vegetables, and even shrubs and trees. While it's possible to knock caterpillars out with pesticides, the problem is, you also knock out beneficial insects. You run the risk of killing butterfly caterpillars along with destructive moth caterpillars.

Azadirachtin causes a cessation of feeding, delay of moulting and death of larvae in a dose-dependent manner. Microbial activity was enhanced either by causing insect toxicity or by potentiating of the crystal protein. Similar observations were reported by Chowdhury *et al.* (2012)[7]. The use of neem bark as a bio-insecticide is limited, as its pesticidal efficacy is lower than the other components of the neem tree including neem seed and leaves in controlling insect pests.[8]

The toxicity of neem against second instar and resistant effects could be useful for resistant pests as well.[9,10]

## CONCLUSION

In today's world most developing countries relies mainly on the use of imported pesticides. Due to this dependence, the product of agriculture field is decreasing. Instantly for current period when we use those synthetic pesticides the growth is good, but the fertility of land decreases. It may also lead to development of resistant pests. Most of the highly toxic pesticides are readily metabolized and eliminated by the body, however, acute short term exposure can lead to their accumulation. The active ingredients, carriers, solvents, and emulsifiers present in pesticides can cause severe side-effects.[11] To provide controlled release of the active ingredients and to improve stability and sustainability, biopesticides can offer better alternative instead of synthetic pesticides. It enables safe control of pests without hampering the soil fertility rather increases fertility.

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## REFERENCES

1. **Manjula, K.N. and Kotikal, Y.K., 2018.** Studies on major in sect pests of fenugreek and their management. *Annals of Horticulture*. **11(2):**141-149

2. **Sivakumar, G.; Kannan, M.; Babu, V.R.; Mohan, M.; Kumari, Surabhi; Rangeshwaran, R.; Venkatesan, T.; Ballal, Chandish R. 2020.** Characterization and field evaluation of tetrahedral and triangular nucleopolyhedrovirus of *Spilosoma obliqua* (SpobNPV) strain NBAIR1 against jute hairy caterpillar. *Egyptian Journal of Biological Pest Control*. **30** (82).

3. **Gupta, R.K., 1993.** Multipurpose Trees for Agroforestry and Wasteland Utilization. *International Science Publisher*, New York, NY.

4. **Randhawa, N.S. and B.S. Parmar. 1993.** *Neem Research and Development*. New Delhi: Society of Pesticide Science, India

5. **N. Amruta G. Sarika, K. Sravanthi, V. Jyothi Swaroopa. 2015.** Organic Farming: Organic Inputs and Techniques Neem Use and Potential in Organic Agriculture.

6. **Banerjee, P.K., Das, H.P. and Hui, A.K. 2000.** Integrated management strategies of major pests of jute in West Bengal. *Environ. Ecol*., **18:**512–515.

7. **Chowdhury, H., Kar, C.S., Sarkar, S.K. and Tripathi, M.K. 2012.** Feeding inhibitory effect of some plant extracts on jute hairy caterpillar, *Spilosoma obliqua. Indian J. Agric. Sci*. **82:**59–62.

8. **Sirohi A., Tandon P. 2014.** Insecticidal effects of various parts of *Azadirachta indica* on adults of *Aulacophora foveicollis* (Lucas) (Coleoptera: Chrysomelidae). *Trends Biosci*. **7:** 3947–3949

9. **Altre, J.A., Grafius, E.J. and Whalon, M.E. 1996.** Feeding behaviour of CryIIIA-resistant and susceptible Colorado potato beetle (Coleoptera: Chrysomelidae) larvae on *Bacillus thuringiensis* untreated potato foliage. *J. Econ. Entomol*. **89:**311–317.

10. **Trisyono, A. and Whalon, M.E. 1999.** Toxicity of neem applied and in combination with *Bacillus* (Coleoptera: Chrysomelidae). *J. Econ. Entomol*., **92:** 1281–1288.

11. **World Health Organisation. 1990.** Public Health Impact of Pesticides Used in Agriculture. Geneva: World Health Organisation.

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# Comparative study on antimicrobial activity of tulsi (*Ocimum sanctum*) and neem (*Azadirachta indica*) methanal extract

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**Abstract-** This study was focussed on to compare the antimicrobial activity of methanolic leaf extracts of Tulsi and Neem. I chase Tulsi and Neem to compare their antimicrobial activity toward Gram-Positive (*Staphylococcus aureus*) and Gram-negative (*Escherichia coli*) bacteria. Nutrient agar plates were inoculated with the above mentioned microorganisms by spreading bacterial inoculum on the surface of the media. Wells (6mm in diameter) were punched in the agar. The phytochemical extracts of Neem and Tulsi were allowed to diffuse into the medium and after incubation of 24h at 37°C, the zones of inhibition were observed. Statistical analysis showed that Tulsi was more effective toward *E. coli.*

**Key words: Zone of inhibition, Antimicrobial activity, *O. sanctum, A. indica, E. coli, S. aureus,***

## INTRODUCTION

*Azadirachta indica* is commonly known as Neem. Neem is a member of the Meliaceae family. It possesses many constituents that are of great importance for the treatment of various diseases, and the constituents also modulate the genetic pathways or metabolic activities of the host body.[1] Neem is widely used in the field of Pharmaceuticals, Agriculture, as Fertilizer, etc, due to the presence of biologically important phytochemicals.[2]

*Ocimum sanctum*, commonly known as Tulsi has been used since ages in Ayurveda due to its healing property. It is a member of family Lamiaceae. Due to its powerful savor, it is regarded as the elixir in Ayurveda and believed to promote lifespan.[3] Tulsi has been effective in improving the liquid profile and basal metabolic rate.[4] The extract of Tulsi is commonly known for the treatment of Cold, Headache, Stomach disorders, etc.[5] *O. sanctum* extract shows anticancer activity by decreasing cell proliferation, increasing reactive Oxygen species, and by altering mitochondrial membrane potential.[6] Since ages dried leaves of Tulsi have been used for repealing insects in grains.

*Escherichia coli,* Gram-negative bacteria, is the major cause of diarrheal diseases, peritonitis, colitis, and infant mortality.[7] *E. coli* acquires its pathogenicity through virulence factors.[8] *E. coli* responsible for urinary tract infection has become resistant to the drug that has been used to cure it, so need of new and plant extract as a medicine is increasing several folds. *Staphylococcus aureus,* which is a Gram-Positive bacteria, can cause life threatening diseases, such as sepsis, endocarditis, and pneumonia. *S. aureus* is considered notorious due to its ability to become drug resistant and resistant to antibiotics such as penicillin and methicillin.

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## MATERIALS & METHODS

**Plant Leaf Collection**

The leaves of *A. indica* (Neem) and *O. sanctum* were collected. Leaves were manually separated, cleaned, and air-dried for 4-5 days. Subsequently, the leaves were pulverized to a coarse powder through mortar and pestle. Approximately 50 g power of both the leaves was weighted and stored in an airtight bottle.[9]

**Preparation of extracts**

Although 250 ml. of methanol was added in a separate conical flask with 50 g of each leaves power to make a stock solution of 0.2g/ml, 0.3g/ml, 0.4g/ml, 0.5g/ml, 0.6g/ml and 0.7g/ml was prepared n the basis of the formula C1V1=C2V2.

Where,

C1= Concentration of Stack Solution,

C2= Final Concentration of a new Solution.

V1= Volume of Stack Solution.

V2= Final Volume of new Solution.

**Test Organisms:**

Two bacterial cultures were used to evaluate the antimicrobial activity, *E. coli* and *S. aureus*.

**Antimicrobial Activity Test:**

Nutrient agar plates were inoculated with the above mentioned microorganisms by spreading the bacterial inoculum on the surface of the media (Spread plates technique), Wells (6mm in diameter) were punched in the agar.[10] 50 ml. extract of Neem, Tulsi of each concentration is allowed to diffuse out into the agar medium in separate petridishes, after incubation of 24h at 37°C, the zones of inhibition were observed uniformly circular as there was a confluent down of growth. Moreover, finally, the diameter of the zone of inhibition (ZOI) was measured in millimeters by vernier scale.[11]

## RESULT & DISCUSSION

The quantitative assessment for antimicrobial activity of the extracts was performed by measuring the diameter of ZOI at six different concentrations such as 0.2g/ml, 0.3g/ml, 0.4g/ml, 0.5g/ml, 0.6g/ml and 0.7g/ml method extract of both the Neem and Tulsi.

ZOI has been taken as absolute values neglecting I 0.2 mm of standard error.

**ZOI for *S. aureus***

With Neem for the above discussed concentrations, 4mm, 7mm, 8mm. 8mm, 10mm and 12mm diameter, ZOI has been observed. In Tulsi for 0.2g/ml and 0.3g/ml concentrations, n ZOI was observed, whereas for other concentrations, 4mm, 12mm, 17mm and 20mm diameter ZOI was observed.

**Table 1- Neem methanolic leaf extracts at various concentrations (0.2g/ml – 0.7g/ml) and their particular ZOI for *S. aureus***

| Concentration (Neem) (g/ml) | ZOI (*S.aureus*) (mm) |
|---|---|
| 0.2 | 4 |
| 0.3 | 7 |
| 0.4 | 8 |
| 0.5 | 8 |
| 0.6 | 10 |
| 0.7 | 12 |

**Table 2- Tulsi methanolic leaf extracts at various concentrations (0.2g/ml - 0.7g/ml) and their particular ZOI for *S. aureus***

| Concentration (Tulsi) (g/ml) | ZOI (*S.aureus*) (mm) |
|---|---|
| 0.2 | 0 |
| 0.3 | 0 |
| 0.4 | 4 |
| 0.5 | 12 |
| 0.6 | 14 |
| 0.7 | 20 |

**Table 3- Neem methanolic leaf extracts at various concentrations (0.2g/ml - 0.7g/ml) and their particular ZOI for *E. coli***

| Concentration (Neem) (g/ml) | ZOI (*E.coli*) (mm) |
|---|---|
| 0.2 | 2 |
| 0.3 | 5 |
| 0.4 | 8 |
| 0.5 | 12 |
| 0.6 | 16 |
| 0.7 | 18 |

**Table 4- Tulsi methanolic leaf extracts at various concentrations (0.2g/ml - 0.7g/ml) and their particular ZOI for *E. coli***

| Concentration (Neem) (g/ml) | ZOI (*E. coli*) (mm) |
|---|---|
| 0.2 | 0 |
| 0.3 | 0 |
| 0.4 | 6 |
| 0.5 | 12 |
| 0.6 | 14 |
| 0.7 | 20 |

***ZOI for E. coli:***

For the discussed concentration of Neem, 2mm, 5mm, 8mm, 12mm, 16mm and 18mm ZOI has been observed, respectively. In Tulsi for 0.2g/ml and 0.3g/ml concentrations, no ZOI has been observed. Whereas, for other concentrations i.e. 6mm, 12mm, 14mm, and 20mm diameter ZOI has been observed.

## CONCLUSION

The leaf extract of both *A. indica* and *O. sanctum* has shown antimicrobial activity against *E. coli* and *S. aureus.* The methanolic leaf extract of *A. indica* has greater antimicrobial activity against *S. aureus,* whereas the O. *sanctum* was found to be more effective against *E. coli.*

## REFERENCES


1. **Alzohairy M.A. 2016.** Therapeutics role of *Azdirachta indica* and their active constituents in diseases prevention and treatment. *Evid based complement Altern med*. 7382506
2. **Al-Jodidi HS, Hossain N.A. 2015.** Studies on total phenolics, total flavonoids and antimicrobial activity from the leaves crude extracts of Neem traditionally used for the treatment of cough and nausea. *Beri Suef Univ J Basic Appl Sci*. **4:**2314-8535
3. **Krishnan Y, Wong N K. 2015.** Cytotoxicity and antimicrobial properties of Neem leaf extreacts. *Int J Charm Pharm Sci.* **7:**975-1491
4. **Pauche A N, Diwan AD, Chandra SR. 2016.** Flovoroids; *An Nerview. J. Nutr Sci.* **5:**e47
5. **Pattanayak P, Behera P, Das D, Panda S. K. 2010.** *Ocimum sanctum* Linn. A reservoir plant for therapeutic applications : An overview. *Pharmacogn Rev.* **4:**95-105
6. **Sridevi N, John B, Raimini K.A. 2016.** Auti Cancer effect of *Ocimum sanctum* ethanolic extract in non-small cell lung carcinoma cell line. *Int J. Charm Pharm Sci.* **8:**975-1491
7. **Blount Z.O. 2015.** The un-exhausted potential of *E. coli. E life.* **4:**e05826
8. **Taylor EV, Nguyen TA, Machosky K O, Koch E, Satir MJ, Bohm SR. 2013.** Multisote outbreak of *E. coli* 0145 infections associated with romaine lettuce consumption. *J Food Prot*. **76:**939-944
9. **Das S, Borah M, Ahmed S. 2013.** Antibacterial activity of the ethanolic extract of leaves of citrus maxima on *E. coli* and *Pseudomonas acruginosa. Asian J Pharm Clin Res*. **6(4):**136-9
10. **Rajesh H. 2013.** Phytochemical analysis of aqueous extract of *Ocimum sanctum* Linn. *Int J Univ Pharm Bio Sci.* **2:** 462-468
11. **Mariana C, Henrique MC, Massuco J, Tais MB, Luis VS. 2017.** Phytochemical screening of *Azadirachta indica* A. Juss for antimicrobial activity. *Abs. J Microbial Res*. 117-122


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# Study on the histology of the pineal organ of *Channa gachua*

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*Received : 19th January, 2021 ; Revised : 18th February, 2021*

**Abstract-** Pineal organ is thought to be the seat of soul. In early vertebrate the pineal organ is photo sensory organ although during evolution it is transformed into an endocrine gland that secretes a hormone called melatonin. The main cellular component of fish pineal shared many features with the cone photoreceptor of retina. The pineal is located dorsal to the forebrain below or within skull roof. Pineal epithelium consists of photoreceptor cells neurons and ependymal interstitial cells.

**Key words: Histology, Pineal organ, *Channa gachua*, photoreceptor cells, photo sensory organ**

## INTRODUCTION

The pineal gland is part of the endocrine system involved with rhythmic activity in e.g. fish. The main product of the pineal gland is the indole hormone melatonin, synthesised from the amino acid tryptophan. Melatonin is mainly synthesized when it is dark, as light inhibits production. For this reason melatonin is thought to be strongly involved in biological rhythms. The pineal gland is a photoneuroendocrine gland which secretes the hormone melatonin and conveys information to the brain via neural pathway. In fish the entire system (the photodetector, the circadian clock and melatonin synthesizing enzymes) is located in the pineal organ. Several experimental studies indicated that the pineal organ is able to translate environmental information (photoperiod and temperature) into rhythmic messages and the pineal hormone melatonin is the internal chemical messenger of environmental signal or *Zeitgeber* and controls a number of functions, especially reproduction, in vertebrates.

The pineal gland is considered to be part of the system regulating biological rhythmicity, mainly due to its main secretory product, the indole hormone melatonin.[1] In birds and mammals melatonin is strongly involved in the synchronization of diurnal and annual rhythms.

The pineal organ consists of the pineal gland and the parapineal organ. The parapineal organ is present in embryologic stages of development but is lacking in adult fish.[2,3] Whereas the role of melatonin in fish is less clear.[4] Three types of cells are considered the main content of the pineal gland, i.e. pinealocytes (photoreceptor cells), glial (supporting) cells and second order neurons (ganglion cells.[1,2] There are blood vessels supplying blood to all parts of the pineal gland, but they do not penetrate into the parenchyma of the gland.[5,6] The pinealocytes are both photosensitive, containing photopigments, and secretory, producing chemical substances. Pinealocytes have been

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shown to undergo morphological changes in response to changes in photoperiod.[2,3,7]

Several authors reported that gonadal function (wt. and histology) are inhibited with the administration of melatonin. It appears that pineal gland and melatonin are having inhibitory effect to thyroid hormone in fishes during gonadal development and maturation.[8]

The pineal organ develops from the roof of embryonic forebrain and it is a part of central nervous system. Although it was early inferred that the pineal organ of teleost fish is a photo sensory organ.[9,10] However electron microscopic study of main cellular component of fish pineal organ shared many features with the cone photoreceptors of retina that respond to changes in the environmental light conditions. First direct evidence of its photosensitivity was obtained in the rainbow trout (*Onchorhynchus mykiss*).[11] Extensive comparative electron microscopical and neurophysiological studies laid the foundation of our current understanding that during vertebrate evolution the pineal organ has transformed itself from a photosensory organ into an endocrine gland. The pineal organ synthesizes indolamine and the melatonin synthesis is indirectly controlled by the daily dark light cycle. In teleosts, the pineal gland and melatonin have been involved in the control of daily variation of locomotory activity, sleep like state, skin pigmentation or demand feeding and crucial annual function such as growth and reproduction are also influenced by melatonin from pineal gland.

## MATERIALS & METHODS

For microscopic study of pineal gland of *Channa gachua*, fish is collected from local market of Muzaffarpur, Bihar, India. Live fish was dissected and the entire brain was separated from the body and was kept in the aqueous Bouin's fixative for 24 hrs. It was then washed thoroughly in tap water in order to remove the colour of Bouin's fluid completely. The material was then dehydrated in graded alcohol. It was washed by Xylene twice with ten minutes interval. By this time material became translucent and clear. It was then put in Xylol wax in the ratio of 1:1 and incubated at a temperature of 56-58$^{o}$C. Finally the material was infiltrated with wax for twice, each of half hour duration. Block was prepared and sections were cut at 8-10 microns and were subsequently spread and stained. Haemotoxyline-Eosin was used for normal histology. For histochemistry Periodic Acid Schiffs Reagent /Alcian Blue stains were used.

## RESULTS & DISCUSSION

The pineal organ grows to form a relatively large vesicle located dorsal to the forebrain just below or within skull roof. In *Channa gachua* the pineal gland is a club shaped structure which is formed as an evagination from the roof of the Diencephalon located in between the Habenular and posterior commissure. The pineal organ is often differentiated into a proximal slender part, Pineal Stalk (PS) and distal part which is expanded called End Vesicle (EV). Length of Pineal Stalk and End vesicle vary from species to species. Pineal Stalk is half of the End vesicle in length. Through the hollow stalk the end Vesicle communicates with the third ventricle. The End vesicle is situated just below the roof of the brain almost between the lateral eyes in *Channa gachua*. The End Vesicle and Pineal Stalk remain in the same plain. The dorsal end of End vesicle is blunt. The ventral end tapers into Pineal Stalk. The cranium is thick over the pineal End vesicle but the scanty distribution of pigment cells over the pineal apparently renders the area. The lumen of both End vesicle and Pineal Stalk is conspicuous by their presence. At least some part of the content is secreted the lining cells is suggested apparently by PAS and AB positive homogenous coagulum and continuity between the luminal contents and the bordering cells through strand like extensions. In *Channa gachua* the wall of the pineal organ i.e. the pineal epithelium consists of photoreceptor cells (PR), neurons (N), and ependymal interstitial cells (IC). These are often called supportive cells and interstitial cells. Macrophages are found mainly in the central lumen. Oligodendrocytes form myelin sheath surrounding some neural axon are also present. Several forms of pineal photoreceptor cells can be distinguished in fishes. Though not cleare but are differentiated into intermediate or common type. In the apical (receptor) and the basal (effector) poles the distinguishing features are found. In the apical pole, the outer segment contains light sensitive photo pigments and the inner segment contains numerous mitochondria and longitudinal cytoskeletal element. The pineal parenchyma which surrounds the basal pole of the photoreceptor axon forms contact with post synaptic neurons or with the basal lamina. The apical pole may vary in shape and size. The lamellae in the outer segment forms a regular cone shaped

stalk, similar to that of retinal cone photoreceptor. However, more often they form a dome or cup shaped stalk. The stalk usually contain relatively few lamellae. The outer segments are irregular in shape. The outer segment consists of mixture of lamellar and tubular membrane formation. Inner segment is typically short, rounded structure. Inner segment is packed with mitochondria. However, in some photoreceptor cell, it is elongated, when it is usually attached to short cone shaped outer segment. The basal pole is also variable. Short axons are possessed by the large majority of the photoreceptor cells that is clearly presynaptic to intrapineal neurons.

Photoreceptor axon terminal impinging occurs mainly on the dendrites but it can also occur on the cell bodies and axon of intra pineal neurons. They typically contain so called synaptic ribbon have been observed in photoreceptor cell bodies. It may be involved in signal transmission to interstitial cells; such contacts allow photic control of synaptic activity in the interstitial cells. Finally, photoreceptor cells also form gap junction with adjacent photoreceptor cells.

The so called supportive cells or interstitial cells form a large proportion of the cells of the pineal parenchyma. They are located between the photo receptor cells. They are attached by tight junctions which, thus form a diffusion barrier between the CSF (cerebrospinal fluid) of pineal lumen and the extra cellular fluid surrounding the photoreceptor cell bodies and axon and the intra pineal neurons. In most species they appear to have widely extending basal poles that line the basal lamina forming a sheet that separates neural elements and the basal pedicles of most photoreceptors cells from the perivascular space.

Pineal arises as a middorsal neuroectodermal evagination from the roof of the diencephalon. It is one of the circum ventricular organ. The cranium over the end vesicle (EV) may show certain specialization, which has variously been called as pineal spot, fontanelle or pineal foramen in fishes. Pineal window is used as term for defining the translucent region of the head skin in some teleost. It was also observed that the specialization presumably is to allow photic stimuli to reach the pineal organ. Histologically the lumen of which is continuous with ventricle of brain lined by darkly stained epithelial cells. The pineal consists of two parts anterior broader part called as end vesicle and posterior narrow portion called pineal stalk. Fish pineal organ has been shown to be photosensitive in may fishes by behavioral. Fish pineal shows a photo sensory role which is demonstrated by electro physiological studies.

The principle site of synthesis and release of hormone, melatonin is the teleost pineal gland. Both the endocrine and exocrine role of fish pineal is well established. Melatonin is also believed to be a chemical transducer of the photic environment and has been implicated in regulating reproductive function also. A pineal control of gonadal maturation has been shown in fishes. Melatonin is known as good inhibitor which mediates the photosexual responses. The exocrine secretion in form of glycogen, of fish pineal has been shown by many investigators.

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## REFERENCES

1. **Ekström, P. and H. Meissl. 1997.** "The pineal organ of teleost fishes." *Reviews in Fish Biology and Fisheries*. **7:**199-284.
2. **Falcón, J. 1999.** "Cellular circadian clocks in the pineal." *Progress in Neurobiology.* **58:** 121-162.
3. **Holmgren, U. 1959.** On the Structure of the Pineal Area of Teleost Fishes; with special reference to a few deep sea fishes. Göteborgs Kungl. Vetenskaps- och Vitterhets-samhälles Handlingar. Göteborg, Elanders boktryckeri aktiebolag. Sjätte följen.
4. **Kulczykowska, E. 1999.** "Diel changes in plasma arginine vasotocin, isotocin, and melatonin in rainbow trout (*Oncorhynchus mykiss*)." *Fish Physiology and Biochemistry.* **21:**141-146.
5. **Gupta, B. B. P. and Y. Premabati 2002.** "Fish pineal: Structure, function and regulation." *Treatise on Pineal Gland and Melatonin*. 77-102.
6. **Kezuka, H., M. Iigo, K. Furukawa, K. Aida and I. Hanyu. 1992.** Effects of photoperiod, pinealectomy and ophthalmectomy on circulating melatonin rhythms in the goldfish, *Carassius auratus*. *Zoological Science.* **9:**1047-1053.

7. **Kulczykowska, E., H. Kalamarz, J. Warne and R. Balment. 2006.** Day-night specific binding of 2-[I-125] iodomelatonin and melatonin content in gill, small intestine and kidney of three fish species. *Journal of Comparative Physiology Biochemical Systemic and Environmental Physiology.* **176:** 277-285.

8. **Breton B, Mikolajczyk T, Popek W. 1993.** Theneuroendocrine control of the gonadotropin (GtH2) secretion in teleost fish. In: Lahlou B and Vitiello P(eds), *Aquaculture: Fundamental and Applied Research,* pp 199-215, American Geophysical Union, U.S.A.

9. **Dodt E. 1963.** Photosensitivity of pineal organ in teleost, *Salmo irideus* (Gibbons). *Expermientia*. **19:** 642.

10. **Oksche A. 1965.** Survey of the development and comparative morphology of pineal organ. *Progess Brain Res***. 10**.

11. **Morita Y. 1966.** Entladungsmuster pinealer neurone der.Regenbogenforelle (*Salmo irideus*) bei beliebtung des *Zwischenhirns pllugers*. *Arch Histol*. **289:**155.

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# Survey of fluoride content in drinking water in Ranchi City

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*Received : 20th January, 2021 ; Revised : 19th February, 2021*

**Abstract-** Fluoride contamination of groundwater is a growing problem in many parts of the world. Fluoride is an essential element, which is good for the teeth enamel and helps to prevent dental caries. In excessive doses, however, it will lead to a chronic fluoride poisoning, fluorosis. Therefore it is very important to understand the mechanisms of mobilization of fluoride in groundwater in order to mitigate the problem as effectively as possible. The present study involves laboratory testing of various sources of drinking or potable water in Ranchi City of Jharkhand state. 50 water samples were collected and were tested in laboratory for estimating the concentration of fluoride. Fluoride concentrations in the water samples vary between 0.007 mg/L to 3.800 mg/L with highest concentration in Gonda village of Kanke. Potable water available in public hand pumps recorded the highest fluoride concentration. Most of the fluoride contaminated water sources were found in the north western, south western and southern parts of the Ranchi City.

**Key words: Fluoride contamination, fluorosis, mobilization**

## INTRODUCTION

### 1. WATER

Water is essential for life to exist, and access to clean drinking water is a necessity for good health. The World Health Organization (WHO) has acknowledged this and included a part about access to sustainable and clean drinking water in the Millennium Development Goals.[1] However, clean drinking water is not available everywhere. There are reasons for this including water scarcity and pollution of existing water resources. The pollution can be from a natural source or due to anthropogenic activities. This study mainly addresses a specific pollution, namely FLUORIDE CONTENT (contamination) of Groundwater and Surface Water in Ranchi City.

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### 2. RANCHI

Ranchi District lies in the Southern part of Jharkhand State. The district has total area of area of 7698sq. km and is located between 22° 45' - 23°45' north latitude to 84° 45' to 84° 50' east latitude. Area is included under toposheet no.73A, 73B, 73E and 73F.

The district is highly dissected by rivers of varying magnitude. The major water divide in the district runs North to South through Ratu, Lodhma and Khunti. The area is the Eastern part is divided by Subarnarekha and Western part of the divide is drained by South Koel and Karo. The important river basins are Subarnarekha, South Koel, Damodar and the Kharkari.

### 1.2 SOURCES OF DRINKING WATER IN RANCHI CITY

The source of drinking water in Ranchi City includes both Surface Water and Groundwater.

### 1.2.1 SURFACE WATER

Surface Water is the water on the surface of planet. Water supply from surface water for drinking purpose includes:

- Ponds
- Lakes
- Dams

### 1.2.2 GROUNDWATER

Ground Water is the water present beneath Earth's surface in soil pore spaces and in the fractures of rock formations. Water supply from ground water for drinking purpose includes following sources:

- Wells
- Bore wells
- RMC -Ranchi Municipal Corporation (Tap Water)

### 1.3 FLUORIDE AS CONTAMINANT

Fluoride in small doses has remarkable influence on dental system by inhibiting the dental cavities while higher doses it causes a disturbance of enamel structure.

### 1.4 ALLOWABLE FLUORIDE CONTENT IN POTABLE WATER

The permissible limits of fluoride in Drinking water is 1.0 - 1.5 mg/L as F(-). (Manual on Water supply and treatment 1991)US Public work services drinking water standards allow a fluoride concentration in drinking water from 0.8 to 1.7 mg/L dependent on the Annual Average of Maximum Daily Air Temperature of area concerned. (Table 1.4.1)

**Table 1- Recommended limits of fluoride concentration (USPHS drinking water standards, 1962)**

| ANNUAL AVERAGE OF MAY DAILY AIR TEMPERATURE ($^{o}$C) | RECOMMENDED CONTROL LIMITS mg/L | | |
|---|---|---|---|
| | LOWER | OPTIMUM | UPPER |
| 10.0 – 12.1 | 0.9 | 1.2 | 1.7 |
| 12.2 – 14.6 | 0.8 | 1.1 | 1.5 |
| 14.7 – 17.7 | 0.8 | 1.0 | 1.3 |
| 17.8 – 21.4 | 0.7 | 0.9 | 1.2 |
| 21.5 – 26.2 | 0.7 | 0.8 | 1.0 |
| 26.3 – 32.5 | 0.6 | 0.7 | 0.8 |

## MATERIALS & METHOD

### 2.0 AREA OF STUDY

### 2.0.1 RANCHI CITY

The Survey of Fluoride Content in Drinking Water was conducted in Ranchi City, which lies at 23$^{o}$22 N 85$^{o}$ 20 E having municipal area of 175.12 square kilometers, average elevation of 651 m above sea level and a population of 1,126,741 (2011 census); taking into account the different sources of drinking or potable water. The survey was carried out in the month of August. The different sources of water used for drinking purpose in Ranchi City are:-

- Wells
- Handpumps
- Ponds
- Dams ( Getalsud, Rukka, Kanke, Dhurwa)
- Ranchi Municipal Corp. (RMC) Water Supply

The field survey involved the surface water and groundwater sample collection from private wells, household handpumps, public handpumps, ponds and dams mentioned above as well as RMC water supply.

A total of 50 samples were collected and were tested in YUGANTAR BHARATI, Analytical &Environmental Engineering Laboratory, Sidroll, Namkum, Ranchi.

The following are the areas covered for water supply from the respective dams:

- Getalsud- Vikas, Rukka, Booti More, Kokar, Deepatoli, Kantatoli, NamkumChavni, Bahubazar, Siromtoli , Over Bridge, Railway Colony, Chutia, Nivaranpur, Main Road, Lalpur,Vardaman Compound, Hindpiri, Church Road, Morabadi, Bariyatu Road, RIMS, Ratu Road, Pisca More, Harmu Road, Kishore Ganj, Madhukam, Pahari Mandir Area, Upper Bazar and nearby areas.
- Kanke (Gonda)- Kanke Rd. areas on both side, CM House, Raj Bhawan, Vidhan Sabha Adhyaksh Awaas, Chief Justice Awaas, RINPAS and areas nearby are supplied with water.

- Dhurwa- Also called Hatia Dam, supply water to H.E.C, H.E.C Company, Dhurwa, Hatia, Singh More, Latma, Jagganathpur, Birsa Chowk, Doranda, Hinoo, SAIL & Mecon Colony, Shukla Colony, Hawai Nagar and areas nearby.

**2.0.2 FIELD WORK**

Field Work for the present project "Survey of Fluoride Content in Drinking Water in Ranchi City" involved the testing of temperature and pH of the water samples on-site. Out of 50 samples 34 samples were tested for their temperature and pH and recorded.

**2.1 ANALYTICAL METHOD**

Fluoride Content was determined by means of an ION - SELECTIVE ELECTRODE, which makes it possible to measure the total amount of free and complex-bound fluoride dissolved water. The method can be used for water containing at least 20 µg/ litre.[2]

**2.2 ION - SELECTIVE ELECTRODE METHOD**

**2.2.1 GENERAL DISCUSSION**

**a. Principle**

The fluoride electrode is an ion - selective sensor. The key element in the fluoride electrode is the laser type doped lanthium fluoride crystal across which a pontential is established by fluoride solutions of different concentrations. The crystal contacts the sample solution on one face and an internal reference solution at the other. The cell may be represented by:

**Ag | AgCl , Cl - (0.3 M) , F- (0.001 M) | LaF3 | test**

The fluoride electrode can be used with standard calomel reference electrode and almost any modern pH meter having an expanded multi-volt scale. Calomel electrode contains both metallic and dissolved mercury, therefore, dispose of them only in approved sites or recycle. For this reason, the Ag| AgCl reference electrode is preferred.

The fluoride electrode measures the ion activity of fluoride in solution rather than concentration. Fluoride ion activity depends on the solution total ionic strength and pH, and on fluoride complexing species. Adding an appropriate buffer provides a nearly uniform ionic strength background, adjusts pH, and breaks up complexes so that, in effect, the electrode measures concentration.

**b. Interference**

Fluoride forms complexes with several polyvalent cations, notably aluminium and iron. The extent to which complexation takes place depends on solution pH, relative levels of fluoride and complexing species. However, CDTA (Cyclohexylenediaminetetraacetic acid) a component of buffer, preferentially will complex interfering cations and release free fluoride ions. Concentrations of aluminium, the most common interference, upto 3.0 mg/L can be complexed preferentially. In acid solution, F- forms a poorly ionized HF.HF complex but the buffer maintains a pH above 5 to minimize hydrogen fluoride complex formation. In alkaline solution hydroxide ion also can interfere with electrode response to fluoride ion whenever the hydroxide ion concentration is greater than one-tenth the concentration of fluoride ion. At the pH maintained by the buffer, no hydroxide interference occurs.

Fluoborates are widely used in industrial process. Dilute solutions of fluoborates or fluoboric acid hydrolyse to liberate fluoride ion but in concentrated solution, as an electroplating wastes, hydrolysis does not occur completely. Distill such samples or measure fluoroborate with fluoroborate selective electrode. Also distill sample if the dissolved solids concentrations exceed 10000mg/L.

**c. Quality Control (Q.C)**

The QC practices considered to be an integral part of each method.

**2.2.2 APPARATUS**

**a.** Expanded scale or digital pH meter or ion selective meter.

**b.** Sleeve type reference electrode - Do not use fiber - tip reference electrodes because they exhibit erratic behavior in very dilute solutions.

**c.** Fluoride electrode

**d.** Magnetic stirrer

**e.** Timer

**2.2.3 REAGENTS**

**a. Stock fluoride solution -** Dissolve 221.0 mg Anhydrous Sodium Fluoride, NaF, in distilled water and dilute to 1000 ml;

1.00 ml = 100 µg $F^-$

**b. Standard fluoride solution -** Dilute 100 ml stock fluoride solution to 1000 ml with distilled water;

1.00 ml = 10.0 µg $F^-$

**c. Fluoride buffer -** Place approximately 500 ml distilled water in 1-L beaker and add 57 ml glacial acetic acid. 58 g NaCl and 4.0 g 1,2 Cyclohexy lenediaminetetraacetic acid CTDA. Place beaker in a cool water bath and add slowly 6 N NaOH (about 125 ml) with stirring, until pH is between

5.0 and 5.5. Transfer to a 1-L volumetric flask and add distilled water to the mark. This buffer, as well as a more concentrated version, is available commercially. In using the concentrated buffer follow the manufacturer's direction.

**2.2.4 PROCEDURE**

**a. Instrument Calibration:-**

No major adjustment of any instrument normally is required to use electrodes in the range of 0.2 to 2.0 mg F-/ L. For those instruments with zero at center scale adjust calibration control so that the 1.0 mg F-/ L standards reads at the center zero (100 mV) when the meter is in the expanded scale position. This cannot be done on some meters that do not have a millivolt calibration control. To use selective ion meter follow the manufacturer's instructions.

**b. Preparation of fluoride standards:-**

Prepare a series of standards by diluting with distilled water 5.0, 10.0 and 200 ml of standard fluoride solution to 100 ml with distilled water. These standards are equivalent to 0.5, 1.0 and 2.0 mg $F^-$/ L.

**c. Treatment of Standard and Samples:-**

In 100 ml beakers or other convenient containers add by volumetric pipet from 10 to 25 ml standard or sample. Bring standards and samples to same temperature preferentially, room temperature. Add an equal volume of buffer. The total volume should be sufficient to immerse the electrodes and permit operation of stirring bar.

**d. Measurement with electrodes:-**

Immerse electrode in each of the fluoride standards solution and measure developed potential while stirring on magnetic stirrer. Avoid stirring before immersing electrodes because entrapped air around the crystal can produce erroneous readings or needle fluctuations. Let electrodes remain in solution 3 min (or until the reading is constant) before taking a final millivolt reading. A layer of insulating material between stirrer and beaker minimizes solution heating. Withdraw electrodes, rinse with distilled water, and blot dry between readings.

[CAUTIONS -Blotting may poison electrodes if not done gently]. Repeat measurements with samples. When using expanded scale meter or selective-ion meter, frequently recalibrate the electrode by checking the potential reading of the 1.00 mg $F^-$/ L standard and adjusting the calibration control, if necessary, until meter reads as before. If a direct-reading instrument is not used, plot potential measurement of fluoride standards against concentration on two-cycle semi logarithmic graph paper. Plot milligrams $F^-$ per liter on the logarithmic axis (ordinate), with the lowest concentration at the bottom of the graph. Plot millivolts on the abscissa. From the potential measurement for each sample, read the corresponding fluoride concentration from standard curve.

The known-additions method may be substituted for the calibration method described. Follow the direction of the instrument manufacturer.

Selective-ion meters may necessitate using a slightly altered procedure, such as preparing 1.00 and 10.0 mg $F^-$/ L standards or some other concentration.

Follow the manufacturer's directions. Commercial standards, often diluted with buffer, frequently are supplied with the meter. Verify the stated fluoride concentration of these standards by comparing them with standards prepared by the analyst.

**2.2.5 CALCULATION**

$$mg\ F^- / L = \frac{\mu g\ F^-}{mL\ sample}$$

**2.2.6 PRECISION and BIAS**

A synthetic sample containing 0.850 mg F-/ L in distilled water was analyzed in 111 laboratories by electrode method.

## RESULTS

The present study entitled "Survey of Fluoride Content in Drinking Water in Ranchi City" was carried out in Yugantar Bharati, Analytical & Environmental Engineering Laboratory, Sidroll, Namkum, Ranchi. The results obtained during the course of this study are being discussed in the light of literature and being presented in this chapter.

The temperature of the groundwater in Ranchi City ranged from 25.9°C to 31.4°C with an average of 28.2°C. The pH varied from 7.1 to 8.2 with a mean value of 7.3.

**Table 2- The following results were obtained:**

| Sl. No | AREA | TYPE OF SOURCE | ZONE | $F^{-}$ CONC |
|---|---|---|---|---|
| 1 | KARAM TOLI | POND | NORTH EAST | 0.320 ppm |
| 2 | KARAM TOLI | PUBLIC WELL | NORTH EAST | 0.310 ppm |
| 3 | KARAM TOLI | R.M.C | NORTH EAST | 0.450ppm |
| 4 | GETALSUD | DAM | NORTH EAST | 0.360ppm |
| 5 | GETALSUD | TUBE WELL | NORTH EAST | 0.650ppm |
| 6 | RUKKA | DAM | NORTH EAST | 0.330ppm |
| 7 | RUKKA | R.M.C | NORTH EAST | 0.340ppm |
| 8 | RUKKA | TUBE WELL | NORTH EAST | 0.370ppm |
| 9 | SAMLONG | PRIVATE WELL | NORTH EAST | 0.280ppm |
| 10 | SAMLONG | PRIVATE TUBE WELL | NORTH EAST | 0.210ppm |
| 11 | GOSSNER COLLEGE | TUBE WELL | NORTH EAST | 0.280ppm |
| 12 | KANTA TOLI( Bus Stand) | TUBE WELL | NORTH EAST | 0.430ppm |
| 13 | KANTA TOLI | TUBE WELL | NORTH EAST | 0.100ppm |
| 14 | CHUTIA | R.M.C | NORTH EAST | 0.180ppm |
| 15 | LALPUR | TUBE WELL | NORTH EAST | 0.180ppm |
| 16 | SIROM TOLI(MundaChowk) | TUBE WELL | NORTH EAST | 0.043ppm |
| 17 | RANCHI UNIVERSITY | TAP WATER | NORTH EAST | 0.250ppm |
| 18 | RANCHI UNIVERSITY | TUBE WELL | NORTH EAST | 0.270ppm |
| 19 | PATHALKUDWA(Hargari Rd.) | TUBE WELL | NORTH EAST | 0.070ppm |
| 20 | PATHALKUDWA(By Lane) | TUBE WELL | NORTH EAST | 0.074ppm |
| 21 | PATHALKUDWA(Nayatoli Rd.) | TUBE WELL | NORTH EAST | 0.056ppm |
| 22 | OXFORD SCHOOL | TUBE WELL | NORTH EAST | 0.065ppm |
| 23 | GANDHI NAGAR | PRIVATE TUBE WELL | NORTH EAST | 0.740ppm |
| 24 | BAHU BAZAR | R.M.C | NORTH EAST | 0.160ppm |
| 25 | KOKAR (RIMS road) | TUBE WELL | NORTH EAST | 0.290ppm |
| 26 | KOKAR | TUBE WELL | NORTH EAST | 0.380ppm |
| 27 | KHORA TOLI | TUBE WELL | NORTH EAST | 0.290ppm |
| 28 | ST.ANNE'S GIRLS' SCHOOL | TUBE WELL | NORTH EAST | 0.300ppm |
| 29 | BARIATU (RIMS road) | TUBE WELL | NORTH EAST | 0.390ppm |
| 30 | PURULIA ROAD | PRIVATE WELL | NORTH EAST | 0.370ppm |
| 31 | DANGRA TOLI (Peace road) | TUBE WELL | NORTH EAST | 0.360ppm |
| 32 | **KANKE (Gonda)** | **TUBE WELL** | **NORTH WEST** | **3.800ppm** |
| 33 | DORANDA | TUBE WELL | SOUTH EAST | 0.320ppm |
| 34 | RATU ( Garikhana) | TUBE WELL | SOUTH EAST | 0.086ppm |
| 35 | **RANCHI AIRPORT** | **TUBE WELL** | **SOUTH EAST** | **1.200ppm** |
| 36 | DHURWA | DAM | SOUTH WEST | 0.350ppm |
| 37 | **DHURWA** | **R.M.C** | **SOUTH WEST** | **1.600ppm** |
| 38 | DHURWA | TUBE WELL | SOUTH WEST | 0.290ppm |
| 39 | H.E.C COLONY | TUBE WELL | SOUTH WEST | 0.270ppm |
| 40 | H.E.C PLANT | TUBE WELL | SOUTH WEST | 0.110ppm |
| 41 | HULUNDU (Satrangi) | PRIVATE TUBE WELL | SOUTH WEST | 0.250ppm |
| 42 | DIBDIH BYPASS | TUBE WELL | SOUTH WEST | 0.190ppm |
| 43 | **DIBDIH BYPASS** | **PRIVATE WELL** | **SOUTH WEST** | **1.100ppm** |
| 44 | ARGORA | TUBE WELL | SOUTH WEST | 0.400ppm |
| 45 | HATIA (Railway Station) | TUBE WELL | SOUTH WEST | 0.007ppm |
| 46 | HATIA | TUBE WELL | SOUTH WEST | 0.130ppm |
| 47 | TUPUDANA | TUBE WELL | SOUTH WEST | 0.450ppm |
| 48 | FISHERY RESEARCH CENTRE | POND | SOUTH WEST | 0.160ppm |
| 49 | **BIRSA CHOWK** | **TUBE WELL** | **SOUTH WEST** | **2.600ppm** |
| 50 | ASHOK NAGAR ( By road) | TUBE WELL | SOUTH WEST | 0.140ppm |

## DISCUSSIONS

The aim of the present survey was to investigate the fluoride levels in water samples taken from different sources utilized for drinking purpose. Since fluoride plays a vital role in human health therefore it was selected for this study.

From the survey it is clear that Ranchi City is safe from fluoride contamination but high fluoride levels are found in some areas namely Kanke ( Gonda Village), Birsa Chowk, Dhurwa (R.M.C), Ranchi Airport and Dibdih Bypass ranging from 1.100ppm to 3.800ppm. These areas mostly lie in the western and southern parts of ranchi city. Fluoride in high concentration is found in ground water of southern, western and southwestern zones of the Ranchi city.[3] The water is found to be slightly acidic in nature and high in iron concentration in most of the zones. Two out of Fifty samples showed high iron content depicted by their red/brown colour. Also, it was observed that samples with arsenic and iron content had low fluoride levels.

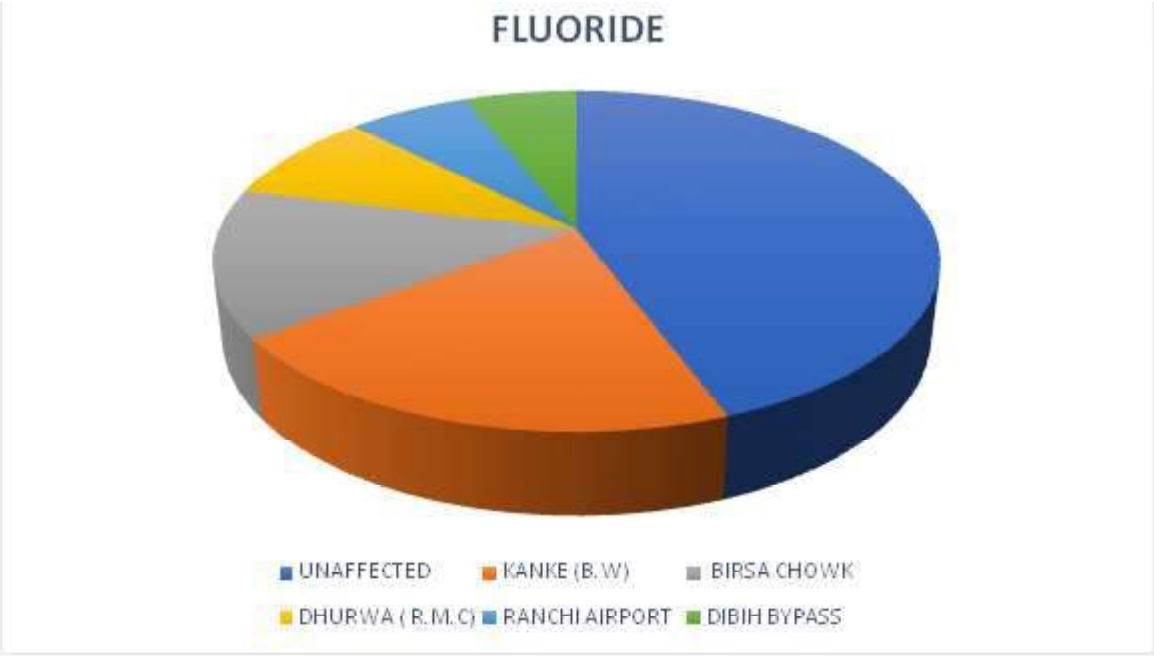
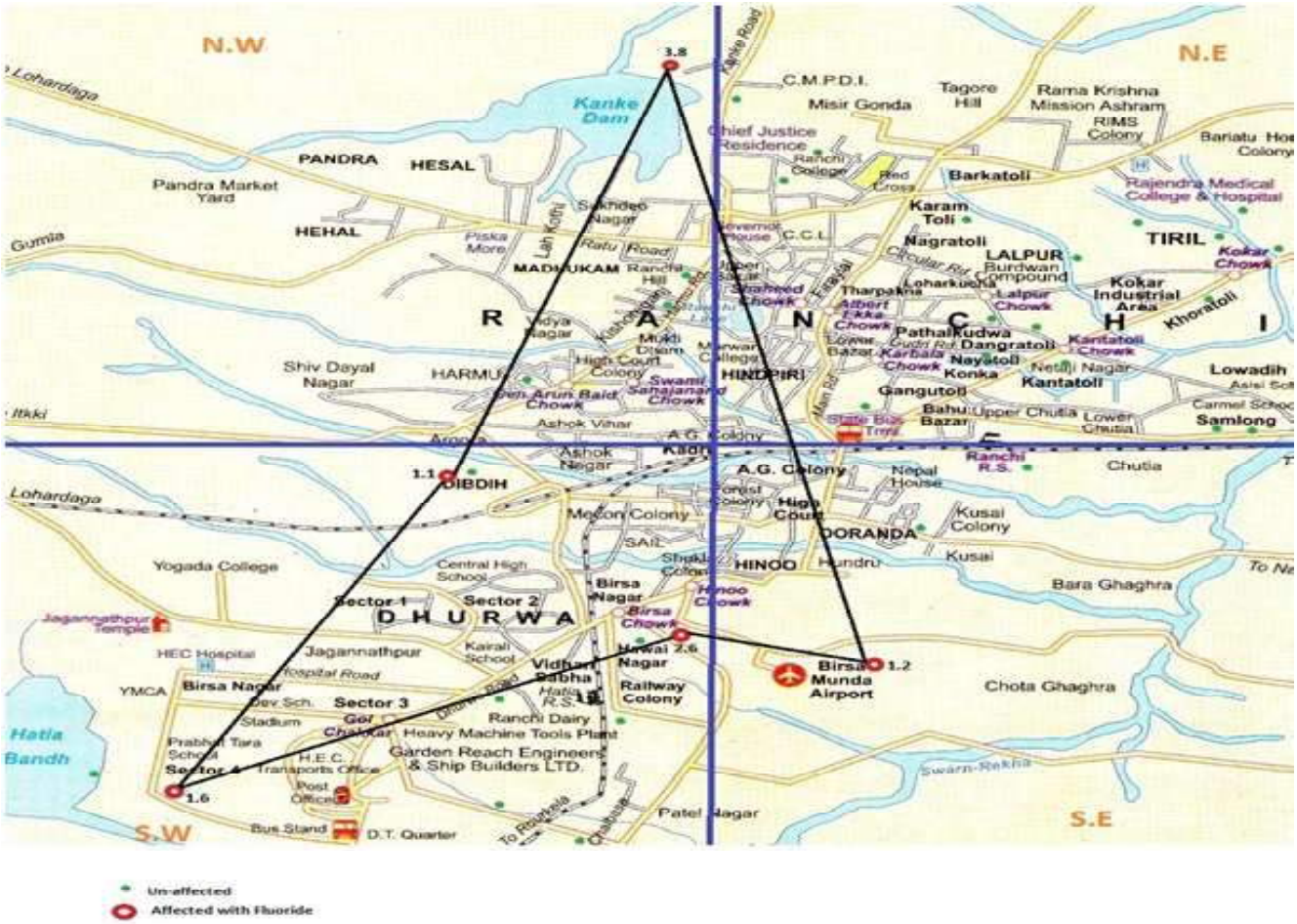



**Chart 1- Showing the fluoride concentrations in affected and unaffected areas**




**Ranchi city map showing affected & unaffected areas. Fluoride in high concentration is found in ground water of southern, western and southwestern zones of the Ranchi city (Priyadarshini. N., 2009).**

High fluoride concentrationsare associated with high TDS, high pH, high Na concentrations, and high sodium absorption ratios (SAR). This suggests that elevated fluoride levels are the result of enhanced fluorite solubility due to Ca depletion and high ionic strength and the release of fluoride from colloid surfaces under high pH conditions.[4] High fluoride concentrations are linked to high sodium contents and pH, and low $Ca^{2+}$ concentrations. The weathering of volcanic rocks increases pH which in turn triggers the dissolution of $CO_2$. The consequent increase in $HCO^{3-}$ and $CO_3^{2-}$ causes oversaturation in the groundwater compared to calcite, leading to the precipitation of this mineral. This precipitation lowers the $Ca^{2+}$ concentration in solution and leads to a sub-saturation for fluorite in the system. As a result, fluorite will dissolve and an increase in $F^-$ concentration is observed.[5]

The following chart shows that most parts of Ranchi City is unaffected by high levels of fluoride whose value ranges from 0.007ppm to 0.740ppm shown in blue. While Kanke (Gonda village) showing the highest level of fluoride detected with 3.800ppm shown in orange, Birsa Chowk with 2.600ppm shown in grey, Dhurwa (R.M.C) with 1.600ppm shown in yellow exceeding the optimum value of fluoride i.e., 1.5 mg/L set by WHO. While Ranchi Airport (dark blue) and Dibdih Bypass (green) bore wells showing fluoride content of 1.2 and 1.1 ppm (mg/L) respectively, though not exceeding the optimum value but still considered high.

## REFERENCE


1. **WHO. 2004a.** Health in Millennium Development. World Health Organization. 1p URL:http://www.who.int/mdg/goals/041222health_ mdgchart_ en.pdf .
2. **Slooff, W., 1988.** Basis document fuoriden, Report no. 758474005. National Institute of Public Health and Environmental Protection, Bilthoven, Netherlands.
3. **Priyadarshini, N. 2009.** URL:nitishpriyadarshini: blogspot.in/2009/09/effect-of-urbanization-groundwater. html
4. **Rafique, T., Naseem, S., Haider, T. 2009.** Geo chemical factors controlling the occurrence of high fluoride groundwater in Nagar Parkar area Sindh, Pakistan. *Journal of Haardous Materials*. **171(1-3):** 424-430.
5. **Coetsiers, M., Kilonzo, F., & Walraevens, K. 2008.** Hydrochemistry and source of high fluoride in groundwater of the Nairobi area, Kenya/Hydrochimie et origine des fortes concentrations en fluorure dans l'eau souterraine de la région de Nairobi, au Kenya. *Hydrological Sciences Journal.* **53(6):**1230-1240.


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# Different varieties of prawn/shrimp leading to increase in the world demand of sea food, market value and growth of small-scale fisheries at East Singhbhum, Jharkhand

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*Received : 20th January, 2021 ; Revised : 19th February, 2021*

**Abstract-** Different varieties of species of prawns/ shrimps that belong to the Penaeidae/ Decapoda family are being surveyed (from August 2019- December 2020) at both lentic and lotic habitat of East Singhbhum, Jharkhand. A total number of four varieties are found in this region. Due to its market value and high demand in the sea food business local people, vendors and fisherman are taking interest in it and started fishing and cultivating it in rivers as well as ponds. This type of cultivation developed to uplift the economic value and growth of people and develop the state in many ways.

**Key words: Decapoda, Penaeidae, lotic,lentic, vendors, cultivation.**

## INTRODUCTION

In India shrimp culture is flourishing day by day which was before limited to the coastal areas, where the existence of varieties of prawn found more easily than the other parts of the country. The brackish waters are channelised in such a way to form different lakes and rivers. These water bodies not only provide different stream of flowing waters to the different parts of the country but also provide many flora and fauna in the freshwater, which are mainly the original existence of marine culture. Chanda & Bhattacharya (2002, 2003, 2004)[1,2,3] gave brief of 3 new species of prawns from Indian waters. From Indian ocean/ seas Palemonid prawns were listed about 46 species by Jayachandran (2005)[4]. Kathirvel *et al.* (2007)[5] provide the most recent review of penaeoid prawns in which 84 species were found & listed of penaeoid prawn. In Jharkhand the aquaculture is not new it is quite common and generally fishes and molluscs of different varieties were being cultivated. As to grow the small-scale fisheries local people started cultivating prawns. Prawns are not new to the area of East Singhbhum because of import and export facilities from the neighboring states, as well as few crustacean species like prawns *Penaeus indicus* which is commonly called as Indian white leg prawn

(*Penaeus indicus*) are generally found in river Subernarekha and many more other varieties of shrimp are being cultivated in small stream or ponds in this region.

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## MATERIALS & METHOD

The samples of prawns/shrimp were collected in the early hours of the morning, with the help of the fisherman at 6 am to 8 am. The collection of the species is done with

the help of a mesh net of about (35 meshes/cm). The net was made of nylon. The collection is done seasonally, when the water is still and constant. These varieties of specimen are being collected and further preserved in 5% of formalin and brought them to the laboratory and further identification is done in the lab.

**STUDY AREA**

There are three specific sites mention below:-

**Site I- Mango, Jamshedpur, (R. Subernarekha) -** In the first site R. Subernarekha flows, which covers a long distance through Ranchi to East Singhbhum. It is a rain fed river and positioned on latitude 21 15' to 23 34'N and longitude 85 8' to 87 32' at an elevation of 610 m (2,000 ft)

**Site II- Galudih, Ghatshila, (R.Subernarekha)-** Here the river is situated in the banks of the region, which flows from Jamshedpur, which is positioned on latitude 22. 60 N and longitude 86.48 E with an elevation of (338 ft).

**Site III- Patamda Block ( Pond)-**Ponds are specially used for the cultivation of fishes, prawns etc for the local people and positioned 85.47'- 34.84' E with an altitude of 710 ft.

## RESULT & DISCUSSION

After the collection of the species sample, they are brought in the laboratory and further identified and analysed in the laboratory. There are four types of prawn/ shrimp species found at three specific sites of East Singhbhum, Jharkhand which include *Peneaus indicus, L.vannamei, P.monodon, M. lammarraei,* all these types are found in the specific site in moderate to most abundant species that are cultivated or reared naturally in the river waters at East Singhhum. In addition, Giant fresh water prawns (*Machrobrachium rosenbergii*) which is also called as Golda chingri in Bengali, they are not cultivated in this region but are being imported from the neighbouring places or states.

The species belong to the family of Peneidae and Decapoda, the species are not new in India as much more varieties are found among the coastline of our country. Earlier nobody was much aware of the cultivation of prawns, so they were being exported and imported to the other parts, but as days are passing people are getting knowledge about how to cultivate and produce it in adequate amount to sell them in the market, in a valuable price as shrimps are expensive species and used to cultivate them in ponds and rivers. Shrimp cultivation is not only flourished the fish farming of the state but also provides to grow economically and establish well the small-scale fisheries.

Further investigation shows that these species depending on the monthly variation vary in size as well as in production and quantity but are found in the market throughout the year. Local fisherman depends on the varieties for their fishing, because these species are costly and sold at a very good price which had a huge demand in the sea food business through which variety of cuisine delicacy are being prepared. The prawns are rich in nutrient value that add more value to it. Hence, we can say that, it's not only making a difference in the economic profress but also benefitted people or small- scale fisheries even more to uplift a step ahead the economic growth of the people and the state.

**Table:1 List of Decapoda (prawn) species found in the East Singhbhum, Jharkhand**

| S.No | Species | Family | Location | Common name | Abundance |
|---|---|---|---|---|---|
| 1. | *Peneaus indicus* | Penaeidae | Site I Mango, JSR | Indian prawn | ++++ |
| 2. | *L.vannamei* | Penaeidae | Site II Galudih | White leg prawn | +++ |
| 3. | *Peneaus monodon* | Penaeidae | Site III Patamda | Tiger prawn | +++ |
| 4. | *M. lammarrei* | Penaeidae | All 3 sites | Kuncho prawn | ++ |

**{ Notation : ++++= most abundant,+++= abundant, ++=moderate}**

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## REFERENCE

1. **Chanda, A& T. Bhattacharya. 2002.** *Melicertus similis*, a new species of prawn. Decapoda: Penaeidae from India. *J. Bombay Nat. Hist. Soc*. **99(3):** 495-498.
2. **Chanda, A & T. Bhattacharya. 2003.** *Fennero penaeus* Konkani, a new species of prawn (Decapoda: Penaedae) from Indian coast. *Sci. & Cult.,* **69:** 229-230.
3. **Chanda, A & T. Bhattacharya. 2004.** A new species of the genus. *parapenaeopsis*. Alock, 1900 (Penaeoidea: Penaedae) from Orissa, India. *Proc. Zool. Soc, Calcutta*. **57:**23-27.
4. **Jayachandran, K.V. 2005.** The biodiversity of Palaemonid prawns from Indian seas (in Hindu). *Aspects of Aquatic Biodiversity. Special Publication* No. 84, CMFRI, Cochin 21-28.
5. **Kathirvel, M.P., Thirumilu and A.Gokul. 2007.** Indian Penaid shrimps- Their biodiversity, economical values. In Editor-Director, National Symposium on conservation & valuation of Marine Biodiversity 2001, India. 161-176.

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# Toxicity and behaviour under Mardo intoxication in a fresh water teleost, *Mystus vittatus* (Bloch)

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*Received : 20th January, 2021 ; Revised : 20th February, 2021*

**Abstract-** Studies on the effect of Mardo (organophosphate pesticide Profenofos, Mol, Wt. 373.6, Molecular formula $C_{11}$ $H_{15}$ Br $Cr_{03}$ PS, obtained from Crop Health products Ltd, Ghaziabad U.P. India) on changes in behavioural of fresh water Teleostean fish, *Mystus vittatus* (Bloch) on the basis of interpolation of the data, the $LC_0$ and $LC_{100}$ Values of Mardo were calculated to be 0.13 mg/l and 0.65 mg/l respectively. Calculated $LC_{50}$ Values has been observed to be 0.490, 0.420, 0.330 and 0.270 mg/l at 24 hours, 48 hrs, 72 hrs, and 96 hrs exposure respectively. The treatment of sub lethal dosages (0.16,0.24, 0.32, 0.40, 0.48 and 0.56 mg/l) of Mardo for 24 hrs, 48 hrs, 72 hr and 96 hrs exposure respectively shows statistically significant decrease in most of the above noted in dices as compared to control. The reason for such changes has been discussed in this research paper.

**Key words: Mardo, Toxicity, Behavioural, Teleostean, Interpolation, Statistically, *Mystus vittatus* (Bloch).**

## INTRODUCTION

The increased application of pesticides to ameliorate the agriculture problems couse pollution of fresh water ponds, lakes and rivers, as these pesticides reach the fresh water system through run off and other means of transportation. The industrial effluents let out by major industries from another source of pollution of water bodies. This severe problem of fresh water bodies has evinced enormous interest among researchers to carry out the investigation towards understanding the effect of a wide spectrum of pesticides on different species of fishes, for careful application of pesticides in agriculture operations. Since the fishes are extremely sensitive to pesticides and they are also used as indicators in water quality management.[1-4] In view of the paucity of information, an attempt is being made to study the toxicity and behaviour under mardo (an organophosphate pesticide) intoxication was grossly dependent on concentration and length of exposure, in an air breathing fish, *Mystus vittatus* (Bloch).

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## MATERIALS & METHODS

Live specimens of fishes *Mystus vittatus* (Bloch) were procured from local fish dealers at Gaya, Bihar. The fishes were transported to the laboratory for proper acclimatization for first Ten days in the laboratory condition. They were fed daily with pieces of goat's liver.

Before Starting any experiment, toxicity values of Mardo an organophosphate pesticide were calculated by the method as described by APHA *et al.* (2012)[5] and the

experiments were conducted at sublethal concentrations (as illustrated in Table-1)

**Table 1- Showing mortality percentage of *Mystus vittatus* (Bloch) exposed to mardo at water temperature 30.+ 1.0$^0$ c, N= 6, Body wt. 35 + 1.0 g.**

| Sl. No. | Concentration (mg/liters) | Time intervals (hrs) | | | |
|---|---|---|---|---|---|
| | | 24hrs | 48hrs | 72hrs | 96 hrs |
| 1. | 0.16 | 10 | 20 | 30 | 40 |
| 2. | 0.26 | 30 | 40 | 40 | 60 |
| 3. | 0.32 | 40 | 60 | 60 | 70 |
| 4. | 0.40 | 60 | 60 | 70 | 80 |
| 5. | 0.48 | 70 | 70 | 80 | 90 |
| 6. | 0.56 | 80 | 80 | 100 | 100 |

## OBSERVATION & RESULT

On the basis of interpolation of the date the $LC_0$ and $LC_{100}$ Values of Mardo were calculated to be 0.13mg/l. and 0.65 mg/l. respectively. Calculated $LC_{50}$ Value has been observed to be 0.490, 0.420, 0.330 and 0.270 mg/l at 24 hrs, 48 hrs, 72 hrs and 96 hrs exposures respectively.

**Table 2- $LC_{50}$ (Median concentration) values (mg/ litre) of Mardo exposed in *Mystus vittatus* at water temperature**

| Group | Pesticide | $LC_{50}$ Values (mg/L) | | | | $LC_0$ | $LC_{100}$ |
|---|---|---|---|---|---|---|---|
| Organo phosphate | Mardo | 24hrs 0.49 | 48hrs 0.420 | 72hrs 0.330 | 96hrs 0.270 | 0.13 | 0.65 |

The behavioural response of fish towards toxicant was grossly dependent on concentration and length of exposed when fishes were suddenly opposed to higher concentration of Mardo hyper excitability, increased aerial excursions and increased opercular movements.[6] Fishes were often observed swimming with jerky movements on the surface of water and tried to jump out of aquaria.[7,8] The higher concentration of Mardo exposure showed white wound patches on the skin surfaces, marked discoloration and depigmentation of the skin.

## CONCLUSION

The present investigation deals with some information about the toxicity of mardo (profenofos). $LC_{50}$ values and behavioural changes after the long time exposure have been taken into consideration on a fresh water edible fish *Mystus vittatus* (Bloch) and the study of aquatic animals is an essential first step concerning to the toxicity and evolution of pesticide impact on the fresh water environment. However, water temperature is one of the most important factors in the environment of an aquatic organism and also play a vital role in the determining their distribution, behaviour metabolism, respiration, blood circulation, growth and reproduction. This pesticide is very toxic but does not last long. It disintegrates fast due to this property. It is considered to be friendlier by farmers, as comparison to organochlorine and other organophosphate group of pesticides.

Since, pollutants induced toxicity in fresh water fishes, is well documented. So, toxicants induced changes in the respiratory behaviour and metabolic rate of fishes that seek the attention of many biologists. It is very interesting to note that all the different groups of pesticides of the same group do not have the same effect on fishes. However, it may be considered as an adaptive mechanism to avoid the stress of pesticide in the aquatic media, a view consistent with the finding of Chaturvedi (2007)[9], Roy and Kumar (2013)[10], Kumar and Keshri (2016)[11] in *Channa punctatus* (Block). Thus, the behavioural response of fish towards toxicant was grossly dependent on concentration and length of exposure.

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## REFERENCES

1. **Holden, A.V. 1973.** Effect of pesticides on fish in:- Environmental pollution by pesticides (eds,C.A Ed ward ), plenum press, New York, pp.213-253.
2. **Chakrawarti, J.B and Claurasia, R.C. 1981.** Toxicity of some organophosphate, chlorinated and carbamate pesticides to some fresh water fishes. *India J.Zool*. **9:**91- 92.
3. **Ranke-Rybicka, R. 1975.** Effect of organophosphate pesticides (Malathion, Foschlor, Dichlorfos) On aquatic organisms, *ROCZN, PZH.* **26:**393-339
4. **Chaudhary, B.P. 1979** Effect of an organophosphate (Malathion) in the metabolism of a fresh water air-breathing fish, *Heteropneustes fossilis* (Bloch). Ph.D thesis T.M. University Bhagalpur, India.

5. **APHA, AWWA and WPCF. 2012.** standard methods for the examination of water and waste water, 22nd edition. *American Public Health Association*, Inc. New York.

6. **Jindal, R. and S.K Jha. 2003.** Toxic effect of monocrotophos on a fresh water teleost *Cyprinus carpio communis* (Linn). 90th session of Indian Science Congress Section. *Animal. Vet And Fishery Sciences* Abs.No. 160. pp-158.

7. **Yasmin, Reshman. 2012.** Phosalone induced changes in metabolic rate in an air breathing fish, *Channa punctatus* (Bloch). *Life Science Bulletin*. **9(1):** 107-108.

8. **Rani Poonam Kumari and A.K pandey, 2009.** Effect of taficide (Herboclin) on bimodal oxygen uptake in an air breathing siluroid fish, *Heteropneustes fossilis* (Bloch). *National Journal of Life Science.* **6(3):** 359-360.

9. **Chaturvedi, M.K. 2007.** Adaptive trends in bimodal oxygen uptake and haematological indices in *Channa punctatus* (Bloch) exposed to pesticide phosalone, Ph.D. Thesis Magadh University, Bodh Gaya.

10. **Roy, B.K and Vijay Kumar. 2013.** Changes in respiratory metabolism in *Clarias batrachus* (Linn) exposed to pesticides, proc. *Zool. Soc. India*. **12(2):** 49-53

11. **Kumar, Paramanand and D.K Keshri. 2016.** Pesticides induced changes in aquatic, aerial and total oxygen consumption in a fresh water facultative air breathing teleost, *Clarias batrachus* (Linn). *Proc. Zool. Soc. India*. **15(2):**19-22

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# Herbal medicines and Indigenous knowledge

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*Received : 21st January, 2021 ; Revised : 21st February, 2021*

**Abstract-** The tribes of Jharkhand are men of forest. They live in the villages and forest and depend mostly upon traditional medicine for their primary health care needs and this therapy involves the use of plant extracts or their active principles. They have great amount of indigenous knowledge for different types of diseases here plants are mentioned which are used for skin diseases.

**Key words: Skin diseases, herbal medicines, tribes, Indigenous knowledge, nature**

## INTRODUCTION

Herbal medicine is based upon the premise that plants contain natural substances that can promote health and alleviate illness modern drugs are often viewed as impersonal, emphasising crisis investigations.[1]

They are not only expensive but many of them bring about side effects which are sometime dangerous than the disease themselves.

**Herbal medicine**

Nowadays the population is increasing so much and side by side pollution is also affecting the nature. 80% population of Jharkhand live in village and depend only upon traditional medicine for their primary healthcare needs and major part of this therapy involves the use of plant extracts or their active principles.[2,3]

The people of Jharkhand suffers from skin disease quite frequently. There are several factors associated to the cause of skin disease.

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- Effect of Sun rays
- Effect of pollen allergy
- Effect of poor nutrients
- Effect of contaminated water
- Effect of environmental factors

The following Herbal plants are used by the people which is found in their native place

**NEEM (*Azadirachta indica*)**

The local people of Jharkhand worship this tree. They plant it in front of their courtyard. There are many reasons to worship this plant which has been scientifically proven. It belongs to the family Meliaceae.

The leaf of *Azadirachta indica* purifies the air. The fresh air is good for health which is scientifically right.

The whole parts of the plants are used as medicine. The mixture of leaf powder is used for treatment in leprosy as per herbal doctor's advice. The pastes of leaves are used in wounds which help in healing. The Neem oil is very useful in skin disease.[4]

At the time of chicken pox the people use its paste as ointment and also a fan for the patients. As precaution against chicken Pox the people use its young leaves and flowers as vegetables.

Neem toothpaste also has medicinal uses. The Gum diseases or Gingivitis is cured by Neem toothpaste and neem brush.

The different parts of plants such as leaf, twigs are also kept in pots of corn for storing corns for long duration as fungal free corn

**BAKAIN CHINA TREE (*Melia azedarach*)**

*Melia azedarach* belongs to the same family meliaceae and is also of same nature. It also used as antifungal for keeping the corn.

The mixture of leaves of *Melia azedarach* is useful in leprosy which is one of the most dangerous skin disease.

**KARANJ, *Millettia pinnata* (L)**

Karanj belongs to the family fabaceae. It is found everywhere particularly in Jharkhand. All parts of the plants are used for medicinal purposes.

The oil of *Millettia pinnata* is much useful in all types of skin diseases. The mixture of camphor and Karanj oil is useful in skin infection. The flowers and Karanj seed powder is an excellent manure as well as antifungal agent. It is a good mosquito repellant.[5]

People using datwan of Karanj as toothbrush need not visit a dentist. The carries tooth and bleeding gum like diseases have been cured simultaneously by this method. Timber of Karanj is also valuable as it is highly antagonistic to white ants

**BRAHMI**

Brahmi booti which is commonly known as beng sag by the local people. The plant is herbaceous and belongs to the family Apiaceae. Botanical name is *Centella asiatica.* This is useful in skin diseases. Normally people eat it as vegetable. The root has the most medicinal value. At the time of jaundice people use it as Sherbat in empty stomach. This herb can purify the blood & increase memory. It is used as antibiotic.

**SHEESHAM**

*Dalbergia latifolia* belongs to the family fabaceae. This plant is planted along the roadside for shade. Oil from the wood and leaves are used in eczema. Leave of sheesham are used in acute stage of gonorrhoea. Sheesham is also a strong wood for timber.

**SINDWAIR**

Sinduwair's botanical name is *Vitex negundo* and belongs to the family lamiaceae. This plant is used for skin diseases like edema etc. The plant is boiled with neem and used for bathing the mother and child post delivery. Seeds and roots are used in treatment of excessive vaginal discharge, pruritus rheumatism etc. *Vitex negundo* is used for treating stored garlic, rice against pests.

The people of Jharkhand village knowingly or unknowingly use these native herbal medicines in their meals which keep them healthy in comparison to town people and they build strong immunity against any diseases.

## REFERENCES

1. **Chopra P.N., Chopral. C, Verma B. S. 1969.** Supplement to glossary of Indian Medicinal plant Publication and Information Directorate New Delhi:
2. **Chopra R.N., Nair S. L., Chopra L.C. 1956.** Glossary of Indian Medicinal plants. Council of Scientific and Industrial Research New Delhi:
3. **Cowen D.V. 1984.** Flowering tree and shrubs in India. Thacker Co. Ltd. Bombay:
4. **Dutta A.C. 1979.** Botany for degree students, Oxford University Press:
5. **Pandey B.P.** Botany for degree students Part l, S. Chand Press

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# Diurnal variations in zooplankton in the waters of Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town,Bihar

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**Abstract-** In the present project, a comprehensive study was done on Diurnal variations in zooplankton in the waters of Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town, Bihar. The pond is situated in the heart of the town and is of utmost importance on account of its historical aspect. Different varieties of fishes are cultured in this pond. Especially carp culture is common. Hajipur town is an important part of Vaishali district. Fresh water of Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town, Bihar have been selected for research work. For analysis of dial variation of zooplankton, samples were collected from two perennial ponds i.e. Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town. In this DIAL study, zooplankton including rotifers, cladocerans, copepods and their larvae show a definite diurnal pattern with a decrease in daylight hours and an increase in morning hours over the three seasons in both ponds. Current data indicate that zooplanktonic organisms avoid bright light and take shelter at the soil-water interface during the day and later climb to the surface at night. In the study we found a total of 15 species of zooplankton and 13 species of zooplankton collected in Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town during the mid-summer season, during the dial cycle of the monsoon month, total 12 species and 11 species of zooplankton in both the ponds were collected and a total of 2 species of copepoda and their nauplii larvae were collected in each of the two ponds during the winter month.

**Key words: Zooplanktons, rotifers, cladocerans, copepods, DIAL**

## INTRODUCTION

At a time when the focus is on conservation and creation of more water bodies for rainwater harvesting and climate resilience across India, many ponds in northern Bihar are disappearing and others are being encroached upon or ruined is being filled. Official records show that by the early 1990s, there were 250,000 ponds in Bihar. Locally known as ponds or puddles, the number of these water bodies has come down to 93,000 today. In the last two decades, we have filled the ponds with garbage, mud and other things. This is set to impact the natural water harvesting, its ability to easily recharge groundwater and provide water security during monsoon. Despite this, Bihar has huge reserves of water resources in the form of fresh water. A great diversity in geomorphological features as well as an equally high variability in monsoon rainfall provides a wide variety of freshwater habitats of great scientific interest besides constituting the country's inland fishery resources. The first limnological studies in India appeared in the early twentieth century. Prasad (1916)[1] conducted the first limnological investigation to unravel the mystery of the water body. Therefore, he can be considered as the foremost activist in limnology of India.

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Pruthy (1933)[2] described the seasonal and India museum tank in Calcutta and Sewell (1934)[3] reported the abundance and seasonality of zooplankton in the same museum tank. As a result, limnology has expanded greatly with the more extensive study of diverse water bodies and its application in a variety of disciplines.

Hajipur town is the headquarters and largest city of Vaishali district within the state of Bihar. It is located at 25.69°N 85.22°E. The city of Hajipur is bounded by the Gandak River in the west and the holy river Ganga in the south. The area looks lush green with plants and trees growing in the semi-tropical monsoon climate. However, the months of May-June are hot, and December-January is cold. Many ecosystems are connected by water and a greater biodiversity of species has been found in ponds than in large freshwater lakes or river systems. As such, ponds are habitats for a wide variety of organisms including plants, amphibians, fish, reptiles, waterfowl, insects and even some mammals. Ponds are used for breeding these species, but also as shelter and even drinking/feeding places for other wildlife. A defining feature of a pond is the presence of standing water that provides habitat for a biological community commonly referred to as pond life. Notable contribution are those of Stephenson (1930)[4], Edmondson (1959)[5], Penac (1978)[6] Singh M (1995)[7], Kant S, Vohra S (1990)[8] and Bosc S.K. & LarKra M.P.(1994)[9]

**STUDY AREA**

The fresh water of Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town, Bihar has been selected for research work. Taj Baj ka pokhra is situated in the heart of Hajipur city whereas Mamu-Bhanja ka pokhra Hajipur is situated slightly away from the main city.



**Fig. 1- Mamu-Bhanja ka pokhra Hajipur**



**Fig. 2- Taj Baj ka pokhra Hajipur**

## MATERIALS & METHODS

For analysis of dial variation of zooplankton, samples were collected from two perennial ponds i.e. Taj Baj ka pokhra and Mamu-Bhanja ka pokhra of Hajipur town, Bihar at monthly intervals from September, 2019 to August, 2020, for comparative study of zooplankton can go of two ecologically distinct waters. To study diurnal variations in zooplankton in different seasons (winter, summer and monsoon), samples were collected at 3 hours intervals in each of the two ponds from 07:00 to 06:00 the next day. Each time 50 liters of surface water was filtered through plankton net made of bolting silk number-21. The zooplanktons were analyzed in the Zoology laboratory of R N College, Hajipur by the same procedure and formula as in the study of monthly changes.

**Benthic macro-invertebrates**

The term benthic fauna refers broadly to the flora and fauna that are associated with any available substrate, including bottom sediments, in an aquatic system. The fauna components of benthos are called macro-benthic organisms, they are generally sessile and slow-moving in nature and their species composition, abundance and distribution in aquatic ecosystems is mainly controlled by water quality. For the analysis of the macro-benthic fauna, five randomly selected samples were taken from each of the two ponds on a continuous monthly basis from September 2019 to August 2020. The benthic organisms were sampled via Ekman's 15.2 × 15.2 cm size dredge. The dredge material was filtered through a 425/µm mesh-screen (brass sieve no.-40 ISS) and the residual organism was preserved in the laboratory of the Department of

Zoology, 5% formalin. In the laboratory, the material was transferred to white enamel trays from which the animals were culled manually following standard laboratory procedures. The taxonomic level was identified. The identity of some enumerated organisms up to the species level was finally confirmed courtesy of Zoological Survey of India and ZSI, Patna Regional Office. The calculations were done as per Jhingaran *et al.* (1969)[10] by converting the average density value of five samples per meter square using formula:

$$N = n/ah,$$

Where:

**N** is the number of macro-benthic organism in square meter,

**n** is the number of macro-benthic organism per sample,

**a** is the area of Ekman's dredge used, and

**h** is the number of hauls taken.

## RESULTS & DISCUSSION

In the DIAL study, zooplankton consisting of rotifers, cladocerans, copepods and their larvae show a definite diurnal pattern of decrease in daylight hours and increase in the early morning hours over the three seasons in both ponds. Current data indicate that zooplanktonic organisms avoid bright light and take shelter at the soil-water interface during the day and later climb to the surface at night. This is in line with the finding of George (1961)[11], Michael (1966)[12], APHA (1985)[13]. The abundance of total zooplankton in Taj Baj ka pokhra showed significant positive correlation with carbonate alkalinity (r = 0.952), pH (r = 0.977) and transparency (r = 0.926) at 0.1 per cent during summer, monsoon and winter, respectively. Whereas in Mamu Bhanja ka pokhra, it shows significant positive correlation at 0.1 percent with bicarbonate alkalinity (r = 0.934) only during winter. During the summer the total zooplankton i.e. rotifers (r = 0.971), cladocerans (r = 0.945) and copepods (r = 0.968) at one percent. During the daily study, rotifers that dominated zooplankton populations reached a surface migration peak in the early morning hours in all three seasons in both ponds. Ahmed & Singh (1993)[14] observed that rotifers were more abundant in nocturnal collections. In this study a large number of *Daphnia carinata*, *Moina dubia* occurred only at night. The abundance of cladocerans and copepods was observed during the night hours by several workers.

### DIAL cycle in summer

In this study, a total of 15 species and 13 species of zooplankton were collected in Taj Baj ka pokhra and Mamu Bhanja ka pokhra, respectively, during the mid-summer season. In Taj Baj ka pokhra, the highest density of total Zooplankton (276 Unit/L) was observed at 03:00 and the lowest was 66 Unit/L at 18:00 hours. Whereas, in Mamu Bhanja ka pokhra their maximum and minimum number of 232 units/l and 40 units/l were recorded at 09:00 and 15:00 respectively. In the present DIAL study, it was observed that the Rotifera were the most diverse group of zooplankton as compared to the other groups and were commonly found in both the water bodies. In Taj Baj ka pokhra, the cladoceran, *Daphnia carinata* showed a maximum population density of 28 units/L at 03:00 and a minimum of 6 units/L at 12:00 hours and 15:00 hours and there were a total of 2 species of Copepoda and their nauplii in the two ponds collected in each.

### DIAL cycle in monsoon

It was found that during the dial cycle of the monsoon month, a total of 11 species and 11 species of zooplankton were collected in both the ponds. A total of 7 species and 6 species (92 units/L) of Rotifera fauna were recorded at 09:00 hours and the lowest 16 units/L at 18:00. Whereas, in other pounds their maximum and minimum densities of 50 units/l and 10 units/l were observed at 06:00 h and 15:00 h, respectively. In the Taj Baj ka pokhra of Hajipur, only 2 species of Copepoda and Naples larvae were collected during the monsoon cycle.

### DIAL cycle in winter

In this DIAL study, a total of 2 species of copepoda and their nauplii larvae were collected in each of the two ponds. In Fist Pond, the Cyclops species showed its highest density of 16 units/l at 09:00 and 24:00 and the lowest at 6 units/l at 15:00. The maximum and minimum densities of nauplii larvae (36 unit/l and 8 unit/l) were recorded at 03:00 hours and 15:00 hours, respectively. Whereas, in Taj Baj ka pokhra, maximum number of Cyclops species (14 units/l) was observed at 03:00 hours and minimum 2 units/l were observed at 15:00 hours and 18:00 hours. The highest and lowest densities of Naupli larvae (16 units/l and 2 units/l) were recorded at 03:00 hours and 15:00 hours, respectively.

## REFERENCES

1. **Prasad, B, 1916.** *J. Asiatic Society, Bengal*. **12:** 142-145

2. **Pruthi, H.S., 1933.** *Int., Revue, ges. Hydrobiol*. **28:** 46-67

2. **Sewell, R.B.S., 1934.** *Int. Revue, Hydrobiol*. **31:** 36

3. **Stephenson, J, 1930.** *Oxford University Press, London*, 978 PP

4. **Edmondson, W.T., 1959.** 2nd Edition, John Wiley & Sons Inc., New York; 1248 PP

5. **Pennak, R.W., 1978.** John Wiley & Sons, New York; 628 PP

6. **Singh M. 1995.** Impact of human activities on the physic-chemical conditions of two fish ponds at Patna, Bihar. *J. Freshwat. Biol*.**7:**13-17.

7. **Kant S. and Vohra S. 1990.** Lakes their management and conservation. In Management of Aquatic Ecosystem. Ed. V.P. Agarwal, B.N Desai and S.A.H. Abidi. *Society of Bioscience*, Muzzafarnagar.155-168.

8. **Bosc S.K. and LarKra M.P.1994.** Studies on the micro-zoobenthos of two freshwater ponds of Ranchi. *J. freshwat. Bio*.**6:** 135-142.

9. **Jhingran, V.G., Natrajan, 1969.** *Bulletin No.-12, Barrackpore*. **28:** 1699-1714

10. **George, M.G. 1961. 18(13):** 265-273

11. **Michael, R.G. 1966.** *J.Fish.* **13:** 48-42

12. **APHA. 1985.** Standard methods for the estimination of water and waste water. Inc New York, 20th edition 29-248, 287, 4134, 53-947 PP

13. **Datta. 1982.** *J.Phy.,Nat. Sci*. **2:** 22-27

14. **Ahmad, S.H. and Singh, A.K., 1993.** *J.Env. Biology.* **14(2):** 95-105

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# Toxic effect of mardo on bimodal oxygen uptake in a fresh water fish, *Labeo rohita* (Ham.)

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**Abstract-** The toxic effect of Mardo (profenofos i.e, pesticide) on bimodal oxygen uptake in a fresh water carps fish *Labeo rohita* (Ham.). The LC50 Values of profenofos (Mardo) were calculated to be 0.450, 0.420, 0.300 and 0.270 mg/L for 24 hrs, 48 hrs, 72 hrs and 96 hrs respectively exposure at 30.0 + 1.0°C . Control group of fishes (35.0 + 1.5g secures 52.480 + 1.72 cc/kg/hr. from gills and 2.810 + 0.32 where as from skin through aerial route i.e, total 55.250 + 1.25 cc/kg/hr Mardo (Profenofos) intoxication (at sub lethal concentrations of 0.20, 0.31 and 0.250 mg/L in *Labeo rohita* caused Significant decrease in aquatic respiration through gills and total oxygen uptake but significant increase in aerial oxygen uptake through skin as compared to control.

**Key words: Mardo, Bimodal $O_2$ uptake, *Labeo rohita* (Ham.).**

## INTRODUCTION

Respiration is one of the most important physiological parameters on which many of the vital functions like growth and reproduction of the fish depends,[1] which in turn, has a direct bearing on the productivity of fresh water ecosystem in terms of fish production per unit area. The fishes are water breathers using their gills and integument for obtaining $O_2$ both through aquatic and aerial route.

It kept moist but few species have developed specialized accessory respiratory organs to extract $O_2$ from air also (ie, bimodal respiration). The indiscriminate and extensive use of pesticides has adversely affected the several non target organisms of aquatic ecosystem. One of the early symptoms of a cute pesticide poisoning is the alteration or failure of respiratory metabolism.[1] Changes in $O_2$ uptake of fishes in response to pesticide exposure are varying in different species of fishes exposed to a variety of pesticides.

The effect of pesticide exposure are varying in different species of fishes exposed to a variety of pesticides. The effect of pesticides on oxygen consumption has been extensively studies in a number of water breathing fishes but the effect of pesticide poisoning on $O_2$ uptake in air breathing has been less studied in siluroid fishes,[2-8] as such the present study has been taken into account in fresh water siluroid fish, *Labeo rohita* (Ham.)

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## MATERIALS & METHODS

Live specimens of *Labeo rohita* (Ham.) were procured from local fish dealers at Dighi pond, Gaya. The fishes were transported to the laboratory for proper

acclimatization for about 10 days. They were Red daily with pieces of goat's liver. The infected or unhealthy fishes were discarded and the rest fishes were washed in 0.1% $KMnO_4$ Solution for 30 minutes to get rid of external infection. Before Starting any experiment, toxicity ($LC_0$, $LC_{50}$ and $LC_{100}$) values of Mardo (Profenofos) were calculated by the Method as described by APHA (2005)[9] APHA *et al.* (2012)[10] and the experiments were conducted at sublethal Concentrations which is illustrated in table-1.

The $LC_{50}$ Values of Mardo (Profemofos) were Computed to be 0.450, 0.420, 0.300, and 0.270 mg/L for 24 hrs, 48 hrs, 72 hr and 96 hrs exposures respectively. The details of the methods employed in the determination of bimodal $O_2$ uptake were those as followed by Ray and Kumar (2013)[3] and Sharma and Kumar (2013)[4]. The difference of significance, if any, between control and experimental animals were calculated by Student's t-test at the level of 5%. The dissolved $O_2$ content of water was measured by winkler's Valumentric method as described by Welch (1948). Aquatic $O_2$ consumption was calculated from the differences between $O_2$ level of the ambient water in the respirometer before and after the experiment and the valume of water inside the respirometer. $O_2$ Consumption from air was calculated from the range of imbalance of the level of the manometric fluid and by the use of combined gas law equation and vapour pressure (Dejours, 1975). PH of ambient water was measured by an electronic PH meter.

## RESULTS & DISCUSSION

The data showing the effect of Mardo, $O_2$ uptake, percentage (%) increase or decrease in $O_2$ consumption in *Labeo rohita* (Ham.) are presented in Table-1 So, that the table indicats that oxygen uptake through aquatic, aerial route and total $O_2$ consumption in control group of fishes were 52.480 + 1.72, 2.820 + 0.32, and 5.5. 250 + 1.25 respectively in control group of fishes. Mardo (profenofos) in toxication (at sublethal con contration of 0.18, 0.21 and 0.215 mg/L) in *Labeo rohita* (Ham.) caused significant decrease both in aquatic and total $O_2$ uptake but significant increase in aerial $O_2$ uptake (through skin) as compared to control.

There are mainly three groups of pesticides such as organochlorine, organophosphate and carbamate which are used for selective killing of pests in biological community. The main thing is that all the different groups of pesticides of the same group do not have some effect on aquatic organism.[11-13] The mode and site of action of different biocides also differ and therefore, effect of different biocides in different species of fishes unless a detailed investigation is carried out. In this investigation Mardo (Profenofos) pesticide brought significant decrease in $O_2$ uptake as compared to control.

The fluctuations in the physico-chemical characteristics of the ambient water, the air breathing fishes are equiped with bimodal gas exchange machinery employing two modes of respiration using highly vascularized air breathing organs to combat the adverse ecological condition of their habitat.[14-17] Therefore, it may be considered as an adaptive mechanism to avoid the stress of pesticide in the aquatic media. However, the increased dependency on respiration through aerial route probably indicates that the fish tried to avoid aquatic medium containing sublethal concentration of any pesticide.

**Table-1: Showing the effect of Mardo on bimodal oxygen uptake in air breathing fish *Labeo rohita* (Ham.) at water temperature 30 + $1^0$C, N.= 6, + = SEM = Significant (P<0.05)**

| Condition | Dose (mg/L) | $O_2$ Consumption (CC/ Kg/hr) | | | % decrease in total $O_2$ uptake | % increase in aerial $O_2$ uptake |
|---|---|---|---|---|---|---|
| | | (Aquatic through Skin + gills) | Aerial | Total | | |
| Control values | 00. | 52.480+1.72* | 2.810+ 0.32 | 55.250+ 1.25* | _ | _ |
| Mardo | 0.20 | 41.061+1.76* | 8.761+ 0.12 | 49.810+ 1.21* | 8.175 | 235.750 |
| Mardo | 0.31 | 42.052 + 1.25* | 5.520+ 0.25 | 47.625+ 1.32* | 12.314 | 103.180 |
| Mardo | 0.250 | 35.152+ 1.56* | 6.582+ 0.18 | 41.810+ 1.21* | 23.125 | 151.250 |

## CONCLUSION

It is the fact that the nutritive nutrients value of each and every species of fishes under this contention is essential. The flora and fauna are vital for aquatic organism that provides nutrients to them and the nutrients absorbed by flora that also transferred to the other tropic levels after utilization. Due to increased application of pesticides to amliorate the problems cause pollution of fresh water ponds, lakes, rivers and show on. The industrial effluents let out by major industries form another sources of water pollution. Since, the fishes are extremely sensitive to pesticides and are also used as indicators in water quality management.

As we know that, fishes having rich sources of animal proteins, carbohydrate and fat. These are cheepr than that of Mutton. These are good sources of vitamins and easily available in the open market. These are maintaining the malnutrients in workers and poorer. At present time, it provides good employment to maintain the economic status for people.

In recent years pollutants induced toxicity in fresh water fishes is well documented, so toxicants induced changes in the respiratory behaviour as well as metabolic activities of fishes, that indicate the attention of many world Biologists. However present study concluded an object to evaluate the toxic impact of Mardo pesticide on the total $O_2$ uptake of air breathing edible fish *Labeo rohita* (Ham).

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## REFERENCES

1. **Holden, A.V. 1973.** Effect of pesticides on fish in: - Environmental pollution by pesticides (eds,C.A Ed ward), plenum press, New York, pp.213-253.
2. **Rani Poonam Kumari and A.K. Pandey. 2009.** Effect of taficide (Herboclin) on bimodal $O_2$ uptake in an air breathing siluroid fish. *Heteropneustes fossilis* (Bloch). *National Journal of life science*. **6(3):** 359-360.
3. **Ray, B.K and Vijay Kumar. 2013.** Changes in respiratory metabolism in *Clarias batrachus* (Linn) exposed to pesticides, proc. *Zool. Soc. India*. **12(2):** 49-53.
4. **Sharma, U.K and Vijay Kumar. 2013.** Keethane induced variation in fish (*Channa punctatus*, Bloch) respiration. *Proc. Zool. Soc. India*. **12(2):** 11-13
5. **Dubey, R. 2018.** Fish diversity in the fresh water lakes, rivers, food plains and swamps of Kenya : ISSN : 0973. **13(1):** 25-30.
6. **Chakrawarti, J.B. & Chourasia, R.C. 1981.** Toxicity of some organophosphate, chlorinated and carbamate pesticides to some fresh water fishes. *India. J. Zool.,* **9:**91-92.
7. **Chaturvedi, M.K. 2007.** Adaptive trends in bimodal oxygen uptake and haematological indices in *Channa punctatus* (Bloch) exposed to pesticide phosalone, Ph.D. Thesis Magadh University, Bodh Gaya.
8. **Chaudhary, B.P. 1979.** Effect of an organophosphate (Malathion ) In the metabolism of a fresh water air-breathing fish, *Heteropneustes fossilis* (Bloch). Ph.D thesis T.M. University Bhagalpur, India.
9. **APHA. 2005.** Standard methods for the examination of water and waste water, 21st editions, *Amer Publ. Heal. Association*; Amer water works Assoc. and water Poll contr. Country Fed Washington D.C.
10. **APHA, AWWA and WPCF. 2012.** Standard methods for the examination of water and waste water, 22nd edition. *American Public Health Association*, Inc. New York.
11. **Capuzzo, J.M. 1977.** The effect of free chlorine and chloramine on growth and respiration rates of larval lobsters (*Homarus americanns*). *Water Res*. **11:** 1021-1024.
12. **Singh, R.K. and N.D. Goswami. 2010.** Changes in bimodal $O_2$ uptake due to furadan intoxication in a facultative air breathing fish, *Clarias batrachus* (Linn.) *Life science Bulletin*. **7(1):** 127-128.
13. **Kumari, Anita. 2015.** Adaptive trends in bimodal oxygen uptake and haematological
14. Make up in a facultative air breathing fish, *Clarias batrachus* (Linn.) exposed to furadan. Ph.D thesis. Magadh University Bodhgaya, Gaya.

15. **Kumar, Paramanand and D.K Keshri. 2016.** Pesticides induced changes in aquatic, aerial and total oxygen consumption in a fresh water facultative air breathing teleost, *Clarias batrachus* (Linn.) *Proc. Zool. Soc. India.* **15 (2):**19-22.

16. **Pant, SC; S and khana, S.S. 1980.** *Comp. physiol. Eco*. **96(3):** 146-149.

17. **Yasmin, Reshman. 2012.** Phosalone induced changes metabolic rate in an air breathing fish, *Channa punctatus* (Bloch). *Life Science Bulletin*. **9(1):** 107-108.

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# Origin, History and Route of Silk

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## ORIGIN

Silk fibre produced by insects silkworm. It is a natural protein fibre, some forms of which can be woven into textiles. The protein fiber of silk is composed mainly of fibroin and is produced by certain insect larvae to form cocoons. The best-known silk is obtained from the cocoons of the larvae of the Mulberry silkworm *Bombyx mori* reared in captivity (sericulture). The shimmering appearance of silk is due to the triangular prism-like structure of the silk fibre, which allows silk cloth to refract incoming light at different angles, thus producing different colours. Several kinds of wild silk, produced by caterpillars other than the mulberry silkworm, have been known and spun in China, South Asia, and Europe since ancient times. However, the scale of production was always far smaller than for cultivated silks. There are several reasons for this: first, they differ from the domesticated varieties in colour and texture and are therefore less uniform; second, cocoons gathered in the wild have usually had the pupa emerge from them before being discovered so the silk thread that makes up the cocoon has been torn into shorter lengths; and third, many wild cocoons are covered in a mineral layer that prevents attempts to reel from them long strands of silk. Thus, the only way to obtain silk suitable for spinning into textiles in areas where commercial silks are not cultivated was by tedious and labour-intensive carding.

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## HISTORY

The discovery of silk has a romantic history. According to the ancient literature, silk industry originated in china about 3000 B.C. The first authentic reference to silk is found in the chronicles of Chou king of China. The king is reported to have pointed out Empress XI Ling Shi, the worms destroying the mulberry trees in his garden. As she tried to gather the cocoons, she accidently dropped one of them into a bowl of hot tea. While trying to recover the cocoon from the hot liquid with a spoon, she discovered that a very fine and long lustrous thread unwound itself from the cocoon. The art of silk preparation was kept as secret as the National Policy and any one, who attempted to send the eggs of Silk worms out of the country, was hanged to deaths. This art later on spread to the rest of the world through several agencies, like Civil war refugees, was prisoners, marriage of royal families etc.

## ROUTE

Chinese used the silk- industry, not only as a new means of employment, but also as the most valuable commodity in the world trade. Throughout, the world caravans were bartered for this precious and valuable fabric. The whole world was mad after this fiber "Queen of Textiles", as it was also exchanged for gold. Innumerable caravans made their route (known as Silk route) across Asia, bringing silk from China to Damascus. Damascus was the centre of the world market business. When China was the source of silk, east and west exchanged their goods. The Persians took the lead of bringing silk from China. As a result they occupied high position in this trade and began to regulate all the silk coming from China. The fabric was sold at such an attractive price that it becomes a legend. But the Romans were the real sufferers, who paid much. The Roman emperor objected to paying but in vein. He tried to find out a trade to China via Constantinople neglecting Persia, but in vein. Then he sent two Nestorian monks to China to smuggle the secret of "Silk -making." The monks got success and they brought the Silk worms, Mulberry seeds as well as the secret of silk in their hollow clubs. This ended the monopoly and supremacy of both China and Persia as producer and trade of silk. But now it has become extremely popular in Spain, Sicily, France and England.

There is a very complicated story of the origin of silk in Japan. Sericulture made its way towards Japan through Korea. But here the Japanese were not familiar with the secret of its production. So, four Chinese concubines were brought to Japan to teach them the making of woven silk. Today Japan is considered to have seconds position in the world in silk- production.

So far the origin of silk in India is concerned; people say that during 400 A.D., a Chinese Princess brought a cocoon of Silk worm with her, hiding it in her headgear. The princess was married to an India prince.

Sericulture was first established in the tract, which lies between the Ganges and Brahmputra rivers. Our epics like the famous Ramayana and Mahabharata also deal with the origin of silk. Anyway sericulture has been a part and parcel of Indian tradition and culture.

## HISTORY

Silk has a long history in India. It is known as Resham in eastern and north India, and Pattu in southern parts of India. Recent Archaeological discoveries in Harappa and Chanhu-daro suggest that sericulture, employing wild silk threads from native silkworm species, existed in South Asia during the time of the Indus Valley Civilization(now in Pakistan) dating between 2450 BC and 2000 BC, while "hard and fast evidence" for silk production in China dates back to around 2570 BC. Shelagh Vainker, a silk expert at the Ashmolean Museum in Oxford, who sees evidence for silk production in China "significantly earlier" than 2500-2000 BC, suggests, "people of the Indus civilization either harvested silkworm cocoons or traded with people who did, and that they knew a considerable amount about silk". India is the second largest producer of silk in the world after China. About 97% of the raw mulberry silk comes from six Indian states, namely, Andhra Pradesh, Karnataka, Jammu and Kashmir, Tamil Nadu, Bihar and West Bengal. North Bangalore, the upcoming site of a $20 million "Silk City" Ramanagara and Mysore, contribute to a majority of silk production in Karnataka. In India first of all Lefroy (1905=1906) started investigation on the silkworm and sericulture at Pusa institute, New Delhi.

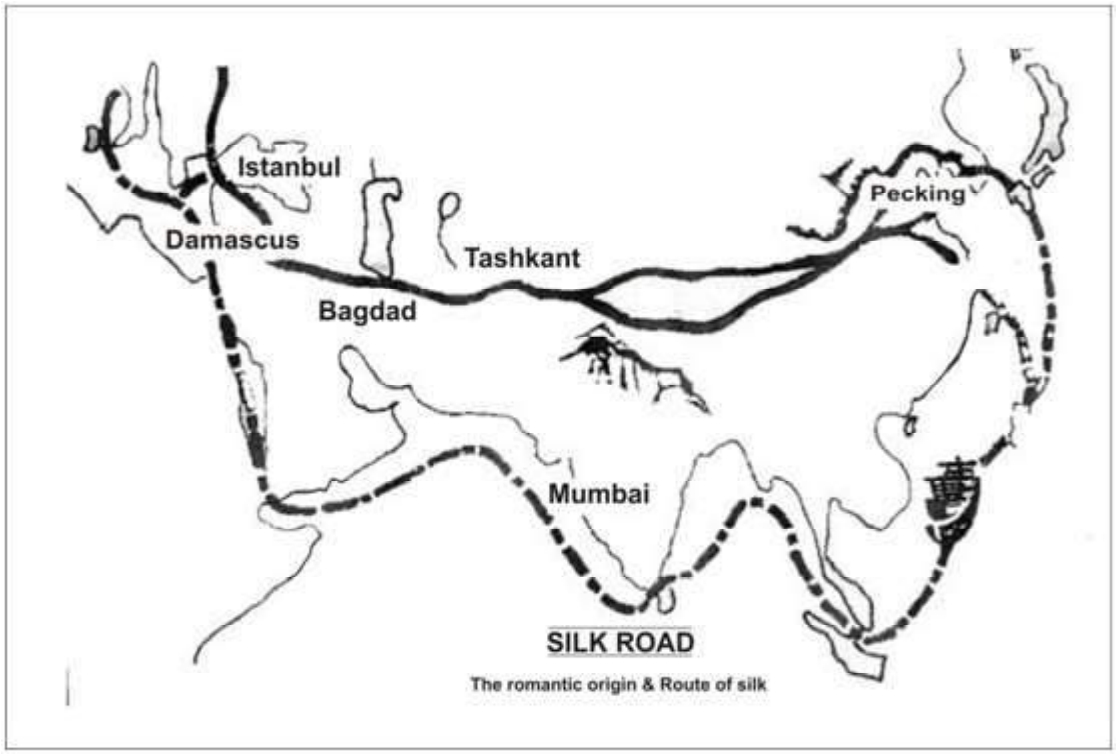
## THE SILK ROAD

The fabulous silks from China and India were carried to European countries all along the 600 miles long road passing through Baghdad Tashkent, Damascus and Istanbul. The 6000 miles long road is historically called as "**The Silk Road**'. The Silk Road was a prestigious network of trade routes linking the West represented by Rome. It was the route used by Buddhist priests and also the trades, who exchanged goods as well as ideas of the two great civilization of the time.

The silk route started in what is now Siam, followed by the Great wall of China to North -west across the Pamir mountains and the Takla Makan deserts to cross Afghanistan to Antioch in Syria, and from thence via Egypt and the Mediterranean ports to Europe. From Mediterranean, it spread to Spain. Latin American and North America. California seems to be the end point of this famous route.

Silk route was hazardous route. Few adventurous people travelled the entire route. The famous Venetian Marco Polo (1256-1323 A.D.) travelled by the Silk route and brought silk from China. Now a day's silk and silk making have become popular and up to date in the whole world. At present sericulture is playing a great role. India,

Japan, China - all have international institution, but China is better than rest of the world, in respect of producing silk. The modern world should be highly obliged and grateful to two unknown monks, who proved the truthfulness of Veni, vidi, vici. (I came. I saw, I conquered) in connection with the secret of silk.




## REFERENCE


1. **Bahn, Paul G. 2000.** The Atlas of World Geology. New York: Checkmark Books. pp. 128. ISBN 978-0-8160-4051-3.
2. **Bag, Sailendra Kumar. 1989.** The Changing Fortunes of the Bengal Silk Industry 1757–1833. Calcutta: Pradip Kumar Banerjee Manasi Press.
3. **Chaudhuri, K. N. 1978.** The Trading World of Asia and the English East India Company 1660–1760. Cambridge: Cambridge University Press.
4. **"The History of Silk". 2007.** The Silk Association of Great Britain. Archived from the original on 2007-09-29. Retrieved 2007-10-23.
5. **Plous, Estelle. "A History of Silk Maps". 2007.** Travel Lady Magazine. Archived from the original on 2007-09-28. Retrieved 2007-05-20.
6. **"History of Silk". 2007.** Silk Road Foundation. Retrieved 2007-03-08.
7. **Hogan, C. Michael. 2008.** "The Megalithic Portal and Megalith Map: Silk Road, North China [Northern Silk Road, North Silk Road] Ancient Trackway". www.megalithic.co.uk. Retrieved 2008-07-05.
8. **Tang, Chi and Miao, Liangyun, "Zhongguo Sichoushi" ("History of Silks in China"). 2007.** Archived 2007-11-23 at the Wayback Machine. Encyclopedia of China, 1st ed.
9. **Vainker, Shelagh. 2004.** Chinese Silk: A Cultural History. Rutgers University Press. pp. 20, 17. ISBN 978-0813534466.


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